

Hiway Gateway Parameter Reference Dictionary

HG09-640

Implementation
Hiway Gateway - 1

Hiway Gateway
Parameter Reference
Dictionary

HG09-640
Release 681
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About This Publication

This publication supports **TotalPlant** Solution (TPS) system network through Release 681. TPS is the evolution of TDC3000.

This is a reference publication for people who need information about the process data point and box data point parameters of the Hiway Gateway (HG). Engineers, supervisors, and operators can use this publication, because the information it contains is useful during the configuration of the HG, and during the operation of the system.

NOTE TO USERS

In this publication, all references to the Data Hiway Port (DHP) also apply to the 620 Hiway Interface Module (620 HIM), except where indicated.

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1 INTRODUCTION

1.1 GENERAL

This publication provides reference information about Hiway Gateway parameters and algorithms. It also provides listings of parameters that are applicable to the various HG data point types.

1.2 USE OF THIS PUBLICATION

Use this publication during configuration and during operation when detailed information about Hiway Gateway parameters is required.

For use in data point configuration, this publication provides definitions for each entry that can be made on the *Data Hiway, Box/Slot, and Data Point Forms* in the *Implementation/Configuration Forms* binder, and in the Parameter Entry Displays at the Universal Station.

For use in process operation, this publication provides information about the parameters that appear for HG box data points and process data points on the displays of Universal Stations that are running with the Operator personality.

1.3 NOTATION

Parameter names that are listed in this publication are the mnemonics that appear within parentheses next to the long names of the parameters on the configuration forms, and on the Universal Station displays. These parameter names consist of up-to-eight alphabetic or alphanumeric characters. The names are in boldface type with all capital letters such as **NAME**, **OUTSSLT**, **SENSSTOP**, etc. When parameter names appear within the definition of another parameter, the parameter names are underlined, such as ASIGNAL LOADDEST, and INPTSSLT.

The algorithm names that are used in this publication identify the Basic Controller (CB), Extended Controller (EC), and Multifunction Controller (MC) algorithms that are enumerations of the ALGIDDAC parameter. Algorithm names are distinguished from the parameter names in that only the first letter is capitalized. The algorithm names are in boldface type. Examples of algorithm names are **Das**, **Pidnorm**, **Switch**, etc.

1.4 REFERENCES

Title	Binder
Engineer's Reference Manual	Implementation/Startup & Reconfiguration - 2
System Control Functions	Implementation/Startup & Reconfiguration - 2
Network Forms	Implementation/Configuration Forms
Network Data Entry	Implementation/Startup & Reconfiguration - 1
Hiway Gateway Control Functions	Implementation/Hiway Gateway - 1
Data Hiway, Box/Slot, and Data Point Form Instructions	Implementation/Hiway Gateway - 1
Data Hiway, Box/Slot, and Data Point Forms	Implementation/Configuration Forms
Hiway Gateway Library Forms	Implementation/Configuration Forms
Basic Controller, Algorithms Engineering Data	Product Manual in the BASIC System bookset
Extended Controller, Algorithms Engineering Data	Product Manual in the BASIC System bookset
Multifunction Controller, Algorithms Engineering Data	Product Manual in the BASIC System bookset
Data Entity Builder Manual	Implementation/Engineering Operations - 1

2 PARAMETER LISTS (PER DATA POINT TYPE)

2.1 GENERAL

This section contains listings of parameters that are applicable to all HG data point types, except regulatory data points. The parameters are listed in a tabular format in the order that they appear on the data-point configuration forms. The data point types that are listed in Tables 2-1 through 2-13 are as follows:

Data Point Type	Table
Analog Composite	Table 2-1
Analog Input	Table 2-2
Analog Output	Table 2-3
Batch Assign	Table 2-4
Counter	Table 2-5
Digital Composite	Table 2-6
Digital Input	Table 2-7
Digital Output	Table 2-8
Flag	Table 2-9
Logic Block	Table 2-10
Numeric	Table 2-11
Process Module	Table 2-12
Timer	Table 2-13

The parameters for the CB, EC, and MC regulatory data points are listed under the names of the algorithms in this manual. Refer to subsection 1.3 (NOTATION) in Section 1 to find out how to look up the parameters that are applicable to each desired algorithm.

Table 2-1 — Analog Composite Data Point Type

DHP	HLPIU	MC	DHP	HLPIU	MC
HG Point Assignment Display			HG Operating Configuration Display		
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM OUTBOXNM PTDISCL SLOTNUM INPTSSLT OUTSLTNM OUTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM OUTBOXNM PTDISCL SLOTNUM INPTSSLT OUTSLTNM OUTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM OUTBOXNM PTDISCL SLOTNUM INPTSSLT OUTSLTNM OUTSSLT LOADDEST	OUTIND PVTV PNTPCTY PNTBOXIN PCADDR1 SPECIF1 RNGCODE1 PNTBOXOT PCADDR01 SPECIF01 RNGCODE2	OUTIND SCANFREQ PIUSMOTH PVTV	OUTIND FLTRCNST PVTV
HG PV Configuration Display			HG Mode Configuration Display		
EUDESC PVFORMAT PVEULO PVEUHI OVERVAL PVRNGOP PVCLAMP CALIBOFF	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI OVERVAL PVRNGOP PVCLAMP CALIBOFF	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$MODESEL NMODE NMODATTR RCASENB MODEPERM MODE	\$MODESEL NMODE NMODATTR RCASENB MODEPERM MODE	\$MODESEL NMODE NMODATTR RCASENB MODEPERM MODE
HG Alarming Display					
\$AUXUNIT \$REALARM ALENBST ALFMT PVLOTP PVHITP DEVLOTP DEVHITP \$ALDLOPT DLYTIME CNFERRPR CCRANK CCPRIPNT EIPPCODE CRITSCAN DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT UNREASPR	\$AUXUNIT \$REALARM ALENBST ALFMT PVLOTP PVHITP DEVLOTP DEVHITP \$ALDLOPT DLYTIME CNFERRPR CCRANK CCPRIPNT EIPPCODE CRITSCAN DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT	\$AUXUNIT \$REALARM ALENBST ALFMT PVLOTP PVHITP DEVLOTP DEVHITP \$ALDLOPT DLYTIME CNFERRPR CCRANK CCPRIPNT EIPPCODE CRITSCAN DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT			

Table 2-2 — Analog Input Data Point Type

CB RV	DHP	HLPIU	LEPIU	LLPIU	MC W/ACC	MC
HG Point Assignment Display						
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM INPTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM INPTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM INPTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM INPTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM AVOPTION PTDISCL SLOTNUM INPTSSLT	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM AVOPTION PTDISCL SLOTNUM INPTSSLT
HG PV Configuration Display						
EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI SP OVERVAL PVRNGOP PVCLAMP CALIBOFF	EUDESC PVFORMAT PVEULO PVEUHI OVERVAL PVRNGOP PVCLAMP CALIBOFF	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI OVERVAL PVRNGOP PVCLAMP CALIBOFF	EUDESC PVCHAR PIUOTDCF INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI OVERVAL PVRNGOP PVCLAMP	EUDESC PVCHAR PIUOTDCF INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVRNGOP PVCLAMP CALIBOFF	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI OVERVAL PVRNGOP PVCLAMP CALIBOFF	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI OVERVAL
HG Alarming Display						
\$AUXUNIT \$REALARM ALENBST ALFMT \$ALDLOPT DLYTIME PVLOTP PVHITP CCRANK CCPRIPNT EIPPCODE CRITSCAN CNFERRPR PVHIPR \$HISELCT PVLOPR \$LOSELCT \$DLSELCT	\$AUXUNIT \$REALARM ALENBST ALFMT \$ALDLOPT DLYTIME PVLOTP PVHITP DEVLOTP DEVHITP CNFERRPR CCRANK CCPRIPNT EIPPCODE CRITSCAN DEVHIPR \$DHSELCT DEVLOPR DEVLOPR PVHIPR \$HISELCT PVLOPR \$LOSELCT UNREASPR	\$AUXUNIT \$REALARM ALENBST ALFMT \$ALDLOPT DLYTIME PVLOTP PVHITP DEVLOTP PVALDB DEVHITP CNFERRPR CCRANK CCPRIPNT EIPPCODE CRITSCAN DEVHIPR \$DHSELCT DEVLOPR DEVLOPR \$DLSELCT OPNTHRPR PVHIPR \$HISELCT PVLOPR \$LOSELCT UNREASPR	\$AUXUNIT \$REALARM ALENBST ALFMT \$ALDLOPT DLYTIME PVLOTP PVHITP DEVLOTP DEVHITP CNFERRPR CCRANK CCPRIPNT EIPPCODE CRITSCAN DEVHIPR \$DHSELCT DEVLOPR DEVLOPR \$DLSELCT OPNTHRPR PVHIPR \$HISELCT PVLOPR \$LOSELCT UNREASPR	\$AUXUNIT \$REALARM ALENBST ALFMT \$ALDLOPT DLYTIME PVLOTP PVHITP DEVLOTP PVALDB DEVHITP CNFERRPR CCRANK CCPRIPNT EIPPCODE CRITSCAN CRITSCAN DEVLOPR \$HISELCT PVLOPR \$LOSELCT \$DHSELCT OPNTHRPR PVHIPR \$HISELCT PVLOPR \$LOSELCT	\$AUXUNIT \$REALARM ALENBST ALFMT \$ALDLOPT DLYTIME PVLOTP PVHITP DEVLOTP DEVHITP CNFERRPR CCRANK CCPRIPNT EIPPCODE CRITSCAN PVHIPR \$HISELCT PVLOPR \$LOSELCT \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT	\$AUXUNIT \$REALARM ALENBST ALFMT \$ALDLOPT DLYTIME PVLOTP PVHITP DEVLOTP DEVHITP CNFERRPR CCRANK CCPRIPNT EIPPCODE CRITSCAN DEVLOPR \$DLSELCT DEVHIPR PVHIPR \$HISELCT PVLOPR \$LOSELCT

Table 2-2 — HG Operating Configuration Display

	PVTV PNTBOXIN PNTPCTY PCADDR1 SPECIF1 RNGCODE1	SCANFREQ PIUSMOTH PVTV	SCANFREQ PIUSMOTH PVTV	SCANFREQ PIUSMOTH PVTV	FLTRCNST AVCONFIG AVCUTOFF STRTSTOP	FLTRCNST PVTV
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Table 2-3 — Analog Output Data Point Type

DHP	HLPIU	MC
HG Point Assignment Display		
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY OUTBOXNM PTDISCL OUTSLTNM OUTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY OUTBOXNM PTDISCL OUTSLTNM OUTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY OUTBOXNM PTDISCL AVOPTION PTDISCL OUTSLTNM OUTSSLT
HG Operating Configuration Display		
OUTIND PNTBOXOT PNTPCTY PCADDR01 SPECIF01 RNGCODE2	OUTIND	OUTIND
HG Mode Configuration Display		
\$MODESEL NMODE NMODATTR RCASENB MODEPERM MODE	\$MODESEL NMODE NMODATTR RCASENB MODEPERM MODE	\$MODESEL NMODE NMODATTR RCASENB MODEPERM MODE

Table 2-4 — Batch Assign Data Point Type

HG Point Assignment Display
NAME UNIT PTDESC PRIMMOD HWYNUM
HG Operating Configuration Display
BH

Table 2-5 — Counter Data Point Type

DHP	HLPIU		MC
	16-BIT	32-BIT	
HG Point Assignment Display			
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM INPTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM INPTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD HWYNUM PNTBOXTY BOXNUM SLOTNUM INPTSSLT LOADDEST	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL
HG Alarming Display			
\$AUXUNIT \$REALARM ALENBST CNFERRPR EIPPCODE CRITSCAN COUNTRPR EIPEVENT EIPENB	\$AUXUNIT \$REALARM ALENBST CNFERRPR EIPPCODE CCRANK CCPRIPNT EIPEVENT CRITSCAN COUNTRPR EIPENB		\$AUXUNIT \$REALARM ALENBST PDEVTP CNFERRPR CRITSCAN PRESETPR PRPRSTPR EIPEVENT EIPENB EIPPCODE
HG Operating Configuration Display			
EUDESC PRESET PNTBOXIN PNTPCTY PCADDRI1 SPECIFI1 OVERVAL	EUDESC AVFORMAT AVCONV PRESET PRESLOCK OVERVAL		EUDESC AVFORMAT AVCONV PRESET PRESLOCK CNTLLOCK RESETCMD STRTSTOP

DHP			HLPIU			MC		
NAME			NAME			NAME		
UNIT			UNIT			UNIT		
PTDESC			PTDESC			PTDESC		
KEYWORD			KEYWORD			KEYWORD		
PRIMMOD			PRIMMOD			PRIMMOD		
HWYNUM			HWYNUM			HWYNUM		
PNTBOXTY			PNTBOXTY			PNTBOXTY		
BOXNUM			BOXNUM			BOXNUM		
OUTBOXNM			OUTBOXNM			OUTBOXNM		
PTDISCL			PTDISCL			PTDISCL		
LOADDEST			LOADDEST			LOADDEST		
SLOTNUM			SLOTNUM			SLOTNUM		
INPTSSLT			INPTSSLT			INPTSSLT		
NMBRINPT			NMBRINPT			NMBRINPT		
OUTSLTNM			OUTSLTNM			OUTSLTNM		
OUTSSLT			OUTSSLT			OUTSSLT		
NMBROUT			NMBROUT			NMBROUT		
STATE2			STATE2			STATE2		
STATE1			STATE1			STATE1		
UBOXCLR			UBOXCLR			UBOXCLR		
LBOXCLR			LBOXCLR			LBOXCLR		
MOOUTIND			MOOUTIND			MOOUTIND		
FBTIME			FBTIME			FBTIME		
OUTIND			OUTIND			OUTIND		
OVERVAL			OVERVAL			OVERVAL		
INPTDIR			INPTDIR			INPTDIR		
PNTPCTY			PNTPCTY			PNTPCTY		
PNTBOXIN			PNTBOXIN			PNTBOXIN		
PCADDR1			PCADDR1			PCADDR1		
PCBIT1			PCBIT1			PCBIT1		
SPECIF1			SPECIF1			SPECIF1		
PCADDR2			PCADDR2			PCADDR2		
PCBIT2			PCBIT2			PCBIT2		
SPECIF2			SPECIF2			SPECIF2		
PNTBOXOT			PNTBOXOT			PNTBOXOT		
PCADDRO1			PCADDRO1			PCADDRO1		
PCBITO1			PCBITO1			PCBITO1		
SPECIFO1			SPECIFO1			SPECIFO1		
PCADDRO2			PCADDRO2			PCADDRO2		
PCBITO2			PCBITO2			PCBITO2		
SPECIFO2			SPECIFO2			SPECIFO2		
NAME			NAME			NAME		
UNIT			UNIT			UNIT		
PTDESC			PTDESC			PTDESC		
KEYWORD			KEYWORD			KEYWORD		
PRIMMOD			PRIMMOD			PRIMMOD		
HWYNUM			HWYNUM			HWYNUM		
PNTBOXTY			PNTBOXTY			PNTBOXTY		
BOXNUM			BOXNUM			BOXNUM		
OUTBOXNM			OUTBOXNM			OUTBOXNM		
PTDISCL			PTDISCL			PTDISCL		
LOADDEST			LOADDEST			LOADDEST		
SLOTNUM			SLOTNUM			SLOTNUM		
INPTSSLT			INPTSSLT			INPTSSLT		
NMBRINPT			NMBRINPT			NMBRINPT		
OUTSLTNM			OUTSLTNM			OUTSLTNM		
OUTSSLT			OUTSSLT			OUTSSLT		
NMBROUT			NMBROUT			NMBROUT		
STATE2			STATE2			STATE2		
STATE1			STATE1			STATE1		
UBOXCLR			UBOXCLR			UBOXCLR		
LBOXCLR			LBOXCLR			LBOXCLR		
MOOUTIND			MOOUTIND			MOOUTIND		
FBTIME			FBTIME			FBTIME		
OUTIND			OUTIND			OUTIND		
OVERVAL			OVERVAL			OVERVAL		
INPTDIR			INPTDIR			INPTDIR		
PNTPCTY			PNTPCTY			PNTPCTY		
PNTBOXIN			PNTBOXIN			PNTBOXIN		
PCADDR1			PCADDR1			PCADDR1		
PCBIT1			PCBIT1			PCBIT1		
SPECIF1			SPECIF1			SPECIF1		
PCADDR2			PCADDR2			PCADDR2		
PCBIT2			PCBIT2			PCBIT2		
SPECIF2			SPECIF2			SPECIF2		
PNTBOXOT			PNTBOXOT			PNTBOXOT		
PCADDRO1			PCADDRO1			PCADDRO1		
PCBITO1			PCBITO1			PCBITO1		
SPECIFO1			SPECIFO1			SPECIFO1		
PCADDRO2			PCADDRO2			PCADDRO2		
PCBITO2			PCBITO2			PCBITO2		
SPECIFO2			SPECIFO2			SPECIFO2		
NAME			NAME			NAME		
UNIT			UNIT			UNIT		
PTDESC			PTDESC			PTDESC		
KEYWORD			KEYWORD			KEYWORD		
PRIMMOD			PRIMMOD			PRIMMOD		
HWYNUM			HWYNUM			HWYNUM		
PNTBOXTY			PNTBOXTY			PNTBOXTY		
BOXNUM			BOXNUM			BOXNUM		
OUTBOXNM			OUTBOXNM			OUTBOXNM		
PTDISCL			PTDISCL			PTDISCL		
LOADDEST			LOADDEST			LOADDEST		
SLOTNUM			SLOTNUM			SLOTNUM		
INPT								

Table 2-7 — Digital Input Data Point Type

DHP	EC	HLPIU	MC
HG Point Assignment Display			
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL LOADDEST SLOTNUM INPTSSLT NMBRINPT	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL LOADDEST SLOTNUM NMBRINPT	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL LOADDEST SLOTNUM INPTSSLT NMBRINPT	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL LOADDEST SLOTNUM INPTSSLT
HG Alarming Display			
\$AUXUNIT \$REALARM ALENBST DIGALFMT DLTIME CHOFSTPR CCRANK CCPRIPNT ZZTEXT EIPPCODE CRITSCAN CNFERRPR OFFNRMMPR \$OFSELCT	\$AUXUNIT \$REALARM ALENBST DIGALFMT DLTIME CHOFSTPR CCRANK CCPRIPNT ZZTEXT EIPPCODE CRITSCAN CNFERRPR OFFNRMMPR \$OFSELCT	\$AUXUNIT \$REALARM ALENBST IPCRDOP DIGALARM DIGALFMT DLTIME CHOFSTPR CCRANK CCPRIPNT CRITSCAN CNFERRPR OFFNRMMPR \$OFSELCT	\$AUXUNIT \$REALARM ALENBST DIGALFMT DLTIME CNFERRPR CCRANK CCPRIPNT ZZTEXT EIPPCODE CRITSCAN OFFNRMMPR \$OFSELCT
HG Operating Configuration Display			
STATE2 STATE1 UBOXCLR LBOXCLR OVERVAL INPTDIR PNTBOXIN PNTPCTY PCADDR1 PCBIT1 SPECIF1 PCADDR2 PCBIT2 SPECIF2	STATE2 STATE1 UBOXCLR LBOXCLR OVERVAL INPTDIR	STATE2 STATE1 UBOXCLR LBOXCLR OVERVAL INPTDIR	STATE2 STATE1 UBOXCLR LBOXCLR OVERVAL INPTDIR

Table 2-8 — Digital Output Data Point Type

DHP	HLPIU	MC
HG Point Assignment Display		
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY OUTBOXNM PTDISCL LOADDEST OUTSLTNM OUTSSLT NMBROUT	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY OUTBOXNM PTDISCL LOADDEST OUTSLTNM OUTSSLT NMBROUT	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY OUTBOXNM PTDISCL LOADDEST OUTSLTNM OUTSSLT NMBROUT
HG Operating Configuration Display		
STATE2 STATE1 UBOXCLR LBOXCLR MOOUTIND OUTIND PNTBOXOT PNTPCTY PCADDRO1 PCBITO1 SPECIFO1 PCADDRO2 PCBITO2 SPECIFO2	STATE2 STATE1 UBOXCLR LBOXCLR PNTPOPOP MOOUTIND PULSEWTH OUTIND	STATE2 STATE1 UBOXCLR LBOXCLR PNTPOPOP MOOUTIND PULSEWTH OUTIND
HG Mode Configuration Display		
\$MODESEL NMODE NMODATTR RCASENB MODEPERM MODE	\$MODESEL NMODE NMODATTR RCASENB MODEPERM MODE	\$MODESEL NMODE NMODATTR RCASENB MODEPERM MODE

Table 2-9 — Flag Data Point Type

HG Point Assignment Display
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM
HG Alarming Display
\$AUXUNIT \$REALARM ALENBST CNFERRPR FLAGPR CCRANK CCPRIPNT EIPPCODE
HG Operating Configuration Display
INTVARNM STATE2 STATE1 UBOXCLR LBOXCLR OVERVAL

Table 2-10 — Logic Block Data Point Type

HG Point Assignment Display
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM
HG Operating Configuration Display
INTVARNM STATE2 STATE1

Table 2-11 — Numeric Data Point Type

HG Point Assignment Display
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM
HG Operating Configuration Display
INTVARNM EUDESC PVFORMAT

Table 2-12 — Process Module Data Point Type

HG Point Assignment Display
NAME UNIT PTDESC PRIMMOD HWYNUM PNTBOXTY BOXNUM SLOTNUM
HG Operating Configuration Display
SEQSLTSZ ACP

Table 2-13 — Timer Data Point Type

HG Point Assignment Display	
	NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM LOADDEST
HG Operating Configuration Display	
	INTVARNM TIMEBASE SP SPLOCK CNTLLOCK EUDESC RESETCMD STRTSTOP

Changing Non-Control Parameters On-Process

With R640, the user can change non-control parameter values of an HG (Data-Hiway) point without making a point inactive by using one of the following tools:

- CL/AM
- Custom Schematic or DATACHNG schematic
- Alter parameters command from the Command display
- Standard Detail display

This will save time and avoid an unnecessary shutdown of an operating loop. In addition, the Data-Hiway box does not have to be in an idle state when changing the values of these parameters.

However, if you are using the DEB to change the value, it will still be necessary to idle the Data-Hiway box.

The following table contains a listing of the applicable non-control parameters.

Table 2-14 — Non-Control Parameters

HG Point Assignment Display	
CDTEXT*	
EUDESC*	
KEYWORD*	
LBOXCLR*	
NCMPTXT1*	
NCMPTXT2*	
PTDESC*	
STATE1*	
STATE2*	SUPPIO
UBOXCLR*	ZZTEXT*

Indicates that the parameter cannot be changed from a standard display (Detail Display, Group Display, or Change Zone). The parameters that do not have the asterisk () can be modified from the Detail Display and their values will be updated on the standard display as they are modified.

CL/AM

CL/AM can change the value of the parameters while the point is active.

The following example shows how CL/AM can change parameter values for an HPM point, an MC point, and an AM point:

```
--Start of CL/AM File
--Modifying the value of KEYWORD of an HPM point,
--LBOXCLR of an MC point
--and OVERVAL of an AM point.
BLOCK CHNGPAR (point AMCUS1; at general)
EXTERNAL HPM27DC4
EXTERNAL MC_R001
EXTERNAL AMREG1
SET HPM27DC4.KEYWORD = "DEVCTRL4"
SET MC_R001.LBOXCLR = BLUE
SET AMREG1.OVERVAL = 21
END CHNGPAR
--End of CL/AM file
```

Displays

When the value of a non-control parameter has been changed, the standard display showing the parameter value should be called up again to see the parameter value changes that are made by a schematic or by the Alter Parameters command.

Alarming

Alarms generated prior to the parameter change will display the previous value. Alarms generated after the parameter change will display the new value.

Journals

When the parameter value is changed by the Alter Parameters command from the Command Display, the change is journaled in the Engineering Change Journal.

When the parameter value is changed by a schematic, the value change is journaled in the Process Change Journal.

3 PARAMETER AND ALGORITHM DEFINITIONS

3.1 FORMAT

This section describes the format that is used to define the parameters and algorithms that are listed in the remaining sections of this publication.

3.1.1 Parameters

Each parameter definition in this manual contains a description of the parameter, the source of the parameter, its default value, the access-lock level, the value type and value range as follows:

Source—For the HG data points, the source of the parameter can be either the User or the System.

User—Parameter is defined during configuration and can or cannot be changed during system operation, depending on the access lock.

System—Parameter value is provided by the system and can or cannot be changed, depending on the access lock.

Default Value—Value that is automatically assigned to the parameter by the system when the user does not supply the value.

Access Lock—Indicates who can "write" to a parameter to change the value of the parameter. Each request to change the value of a parameter carries with it an access level that identifies the requester. If a value is to be changed and the appropriate access level is not included in the access lock, the system will reject the request for a value change.

The following access levels and access locks are available:

Access Level	Used By
Oper	Operator
Supvr	Supervisor
Engr	Engineer
Program	A user-written program, such as a CL program
ContC	Continuous Control
DEB	Data Entity Builder; parameter has been defined during configuration and cannot be changed

Access Lock Access Levels Included

Oper	All
Supvr	All except Oper
Engr	All except Oper and Supvr
Program	All except Oper, Supvr, and Engr
DEB	DEB only
Read Only	Parameter can only be viewed or accessed, but cannot be changed

Value Type—Indicates the value type of the parameter.

Boolean—Value has only two possible entries such as On and Off, or 0 and 1, or True and False.

Entity ID—Value is a tag name, which is the name of a data point in the system.

Enumeration (Enum)—The value for the parameter is chosen from a set of predefined character strings (enumerations). As an example, parameter EMSENB has one of a set of values that are defined in ENBLSTAT. The values in ENBLSTAT are Disable and Enable.

Integer—Value is a whole number and does not contain a decimal point.

Real—Value is a real number that contains a decimal point.

String—Value contains a user-defined string of alphanumeric characters.

Value Range—Value that is entered for this parameter during configuration. Value can be an entity ID, enumeration, integer, real number, or character string as required by the above value types.

3.1.2 Algorithms

Each algorithm definition contains a description of the algorithm and a listing of the parameters that are applicable to the algorithm. The parameter names are listed in exactly the same order as they appear on the configuration forms and the Parameter Entry Displays.

On the HG Data Point Configuration Forms, the algorithm-dependent parameters are marked with an asterisk (*). Refer to the respective algorithm parameter listing in this publication to determine which parameters apply to the selected algorithm.

- \$ -

\$ALDLOPT—Alarm Delay Option (AI, AC and RegCtl)**NOTE**

Changing the \$ALDLOPT value terminates the current alarm timer. The new value will be in effect for the next applicable alarm event.

This R620 parameter, in conjunction with the DLYTIME parameter, is intended to reduce the number of unnecessary alarms reported by analog input points during periods when the process is unstable (startup/shutdown/upsets/spikes).

The \$ALDLOPT parameter can be used to implement an on-delay or an off-delay. The off-delay option delays turning alarms off for the time period specified by the DLYTIME parameter to minimize the effects of an input value oscillating in and out of alarm. The on-delay option delays turning alarms on for the time period specified by the DLYTIME parameter to suppress the reporting of transient alarms, or to suppress alarms which are not important to the operator unless they stay in alarm for a minimum amount of time.

The time of reporting of the alarm/return-to-normal events, which underwent alarm delay, will be the time when the delay timer has expired, not the actual time of occurrence of the event.

REFERENCE: Refer to the definition of the DLYTIME parameter for the time-delay value.

WARNING: Out-of-the-ordinary situations and configurations can cause peculiar results.

Therefore, the alarm delay option with its alarm priority filter can be used in such a way that critical safety related alarms will not be suppressed or delayed by the alarm delay option. The effect of alarm delay in different scenarios should be understood before configuring any point with this option.

Source:	User	Default Value:	None	Access Lock:	Engineer
Value Type:	Enum. of \$ALDLOPT	Value Range:	None : Neither on delay nor off delay is selected OffDlyLo : Off delay selected & alarms with less than or equal to Low priority are subjected to alarm delay functionality. OffDlyHi : Off delay selected & alarms with less than or equal to High priority are subjected to alarm delay functionality. OffDlyEm : Off delay selected & alarms with less than or equal to Emergency priority are subjected to alarm delay functionality. OnDlyLo : On delay selected & alarms with less than or equal to Low priority are subjected to alarm delay functionality. OnDlyHi : On delay selected & alarms with less than or equal to High priority are subjected to alarm delay functionality. OnDlyEm : On delay selected & alarms with less than or equal to Emergency priority are subjected to alarm delay functionality.		

NOTE: **Journal**, **Jnlprnt** and **Printer** have priorities less than **Low**.

\$AUXUNIT

\$AUXUNIT—Auxiliary Unit

The Auxiliary Unit of an alarmable process point. If an Auxiliary Unit ID is specified, alarms from this point go to the Auxiliary Unit instead of the Primary Unit. If the \$AUXUNIT parameter is set to null (- -), alarms go to the Primary Unit. Available in Release 520 and later software.

Source:	System	Default Value:	Null	Access Lock:	Oper.
Value Type:	Enum. of UNIT	Value Range:	A-Z, 0-9 and Null (Any valid unit ID as configured in the NCF)		

\$BXRSTRP—Box Reset Report (R680)

This PSDP parameter allows the user to enable or inhibit the system alarm, when the Hiway box state changes are caused by the point load or change of address related parameters of DHP or PIU. This value is read from the SYSTEM ALARM ON DHP/PIU POINT LOAD option of NCF when HG is rebooted. The user can redefine the value.

Source:	User	Default Value:	Enable	Access Lock:	Engineer
Value Type:	Enumeration	Value Range:	Enable = Annunciates the system alarm. Inhibit = Suppresses the system alarm due to DHP/PIU point load or parameter change.		

\$CDETAIL (All HG/AM/NIM Points)—Custom Detail Display Name

\$CDETAIL contains the 8-character name of the custom schematic that defines the user-supplied Custom Detail Display. When the “Detail” of the point is requested, the schematic identified by the \$CDETAIL parameter is invoked instead of the standard Detail Display. If the \$CDETAIL parameter is blank, the standard Detail Display call is made.

If the \$CDETAIL parameter is not blank, and the schematic specified in that parameter cannot be invoked, then the standard Detail Display will be invoked with an error message that informs the user that the invocation of the Custom Detail Display failed.

Source:	User	Default Value:	8 space character ASCII character 20 HEX	Access Lock:	Engineer
Value Type:	String 8 characters	Value Range:	Any 8-character string, valid for a Schematic name.		

\$DHSELCT

\$DHSELCT – Deviation High Selective Cutout (R681)

This parameter allows or cuts out annunciation of Deviation High alarm of a point even though the point is cutout.

Source:	User	Default Value:	CUTOUT	Access Lock:	Engineer
Value Type:	Enum of the \$xxSELCT	Range:	CUTOUT = Cuts out annunciation of DVHI alarm of a point when cutout state is true. ALLOW = Allows annunciation of DVHI alarm of a point irrespective of the cutout state.		

\$DLBADAL – Provides delaying in Return to Normal conditions (R681)

This parameter provides delaying of Return to Normal conditions for a time configured by the delay time parameter DLYTIME.

Source:	User	Default Value:	NONE	Access Lock:	Engineer
Value Type:	Enum of \$DLBADAL	Value Range:	OFFDELAY = Reporting of Return to Normal is delayed for the time specified in DLYTIME parameter. NONE = Reporting of Return to Normal is not delayed.		

\$DLSELCT – Deviation Low Selective Cutout (R681)

This parameter allows or cuts out annunciation of Deviation Low alarm of a point even though the point is cutout.

Source:	User	Default Value:	CUTOUT	Access Lock:	Engineer
Value Type:	Enum of \$xxSELCT	Value Range:	CUTOUT = Cuts out annunciation of DVLO alarm of a point when the cutout state is true. ALLOW = Allows annunciation of DVLO alarm of a point irrespective of the cutout state.		

\$EIPMAXC

\$EIPMAXC – Maximum number of EIP events in any one second in the current hour (R660)

This PSDP parameter indicates the maximum number of Event Initiated Processing events in any one second in the current hour.

Source:		Default	Access
	System	Value: 0	Lock: View
Value		Value	
Type:	Integer	Range: ≥ 0	

\$EIPMAXP – Maximum number of EIP events in any one second during previous hour (R660)

This PSDP parameter indicates the maximum number of Event Initiated Processing events in any one second in the previous hour.

Source:		Default	Access
	System	Value: 0	Lock: View
Value		Value	
Type:	Integer	Range: ≥ 0	

\$EIPTOTC – Total number of EIP events sent to the LCN in the current hour (R660)

This PSDP parameter indicates the total number of Event Initiated Processing events sent to the LCN in the current hour.

Source:		Default	Access
	System	Value: 0.0	Lock: View
Value		Value	
Type:	Real	Range: ≥ 0.0	

\$EIPTOTP**\$EIPTOTP – Total number of EIP events sent to the LCN in the previous hour (R660)**

This PSDP parameter indicates the total number of Event Initiated Processing events sent to the LCN in the previous hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$EIPTOTS – Total number of EIP events sent to the LCN during previous 15 seconds (R660)

This PSDP parameter indicates the total number of Event Initiated Processing events sent to the LCN in the previous 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$EVTMAXC – Maximum number of events in any one second in the current hour (R660)

This PSDP parameter indicates the maximum number of events in any one second in the current hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	≥ 0		

\$EVTMAXP

\$EVTMAXP – Maximum number of events in any one second during previous hour (R660)

This PSDP parameter indicates the maximum number of events in any one second in the previous hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	≥ 0.0		

\$EVTTOTC – Total number of events sent to the LCN in the current hour (R660)

This PSDP parameter indicates the total number of events sent to the LCN in the current hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$EVTTOTP – Total number of events sent to the LCN in the previous hour (R660)

This PSDP parameter indicates the total number of events sent to the LCN in the previous hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$HHSELECT – PV High High Selective Cutout (R681)

This parameter allows or cuts out annunciation of PV High High alarm of a point even though the point is cutout.

Source:	User	Default Value:	CUTOUT	Access Lock:	Engineer
Value Type:	Enum of \$xxSELECT	Value Range:	CUTOUT = Cuts out annunciation of PVHH alarm of a point when the cutout state is true. ALLOW = Allows annunciation of PVHH alarm of a point irrespective of the cutout state.		

\$HISELCT

\$HISELCT – PV High Selective Cutout (R681)

This parameter allows or cuts out annunciation of PV High alarm of a point even though the point is cutout.

Source:	User	Default Value:	CUTOUT	Access Lock:	Engineer
Value Type:	Enum of \$xxSELCT	Value Range:	CUTOUT = Cuts out annunciation of PVHI alarm of a point when the cutout state is true. ALLOW = Allows annunciation of PVHI alarm of a point irrespective of the cutout state.		

\$LSELCT – PV Low Low Selective Cutout (R681)

This parameter allows or cuts out annunciation of PV Low Low alarm of a point even though the point is cutout.

Source:	User	Default Value:	CUTOUT	Access Lock:	Engineer
Value Type:	Enum of \$xxSELCT	Value Range:	CUTOUT = Cuts out annunciation of PVLL alarm of a point when the cutout state is true. ALLOW = Allows annunciation of PVLL alarm of a point irrespective of the cutout state.		

\$LOSELCT – PV Low Selective Cutout (R681)

This parameter allows or cuts out annunciation of PV Low alarm of a point even though the point is cutout.

Source:	User	Default Value:	CUTOUT	Access Lock:	Engineer
Value Type:	Enum of \$xxSELCT	Value Range:	CUTOUT = Cuts out annunciation of PVLO alarm of point when the cutout state is true. ALLOW = Allows annunciation of PVLO alarm of a point irrespective of the cutout state.		

\$MODESEL — Mode Selection List (R630 and later)

This R630 parameter allows the engineer to remove the MODE/MODATTR selections from the operator's standard-display MODE selection lists. This prevents the data point from being placed in the wrong mode and from having the wrong mode attribute assigned.

This function is implemented by the user configuring a mode selection list to determine the applicable modes and mode attribute. During normal operation, the mode selection list is displayed on the standard detail, faceplate, group, and change zone displays when you select the mode parameter. The operation of the MAN/AUTO/NORM keyboard keys is also affected by the mode selection list choices.

	Default	Access
Source: User	Value: On	Lock: Engineer

Value

Type: For Reg Ctl, AO, AC, DO, DC, Dev Ctl point types: Boolean Array (1...5)

Value**Range:**

- \$MODESEL(1) - On : Allows the user to select Man mode from a display.
 - Off: Does not allow the user to select Man mode from a display.
- \$MODESEL(2) - On : Allows the user to select Auto mode from a display.
 - Off: Does not allow the user to select Auto mode from a display.
- \$MODESEL(3) - On : Allows the user to select Cas mode from a display.
 - Off: Does not allow the user to select Cas mode from a display.
- \$MODESEL(4) - On : Allows the user to select BCas mode from a display.
 - Off: Does not allow the user to select BCas mode from a display.
- \$MODESEL(5) - On : Allows the user to select Prog mode attribute from a display.
 - Off: Does not allow the user to select Prog mode attribute from a display.

\$NODMAXC

\$NODMAXC – Maximum number of node data accesses in any one second in the current hour (R660)

This PSDP parameter indicates the maximum number of node data accesses in any one second in the current hour.

Source:	System	Default Value: 0	Access Lock:	View
Value Type:	Integer	Value Range: ≥ 0		

\$NODMAXP – Maximum number of node data accesses in any one second in the previous hour (R660)

This PSDP parameter indicates the maximum number of node data accesses in any one second in the previous hour.

Source:	System	Default Value: 0	Access Lock:	View
Value Type:	Integer	Value Range: ≥ 0		

\$NODTOTC – Total number of node data accesses during the current hour (R660)

This PSDP parameter indicates the total number of node data accesses during the current hour.

Source:	System	Default Value: 0.0	Access Lock:	View
Value Type:	Real	Value Range: ≥ 0.0		

\$NODTOTP – Total number of node data accesses in the previous hour (R660)

This PSDP parameter indicates the total number of node data accesses in the previous hour.

Source:	System	Default Value: 0.0	Access Lock:	View
Value Type:	Real	Value Range: ≥ 0.0		

\$NODTOTS

\$NODTOTS – Total number of node data accesses in the last 15 seconds (R660)

This PSDP parameter indicates the total number of node data accesses in the last 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0.0		

\$OFSELECT – Off Normal Selective Cutout (R681)

This parameter allows or cuts out annunciation of Off Normal alarm of a point even though the point is cutout.

Source:	User	Default Value:	CUTOUT	Access Lock:	Engineer
Value Type:	Enum of \$xxSELECT	Value Range:	CUTOUT = Cuts out annunciation of OFFNORM alarm of a point when the cutout state is true. ALLOW = Allows annunciation of OFFNORM alarm of a point irrespective of the cutout state.		

\$OPALEN – Output Alarm Enable (R640)

This R640 parameter allows the user to enable or disable the output (OP) alarming function for an HG regulatory point.

Source:	User	Default Value:	Disable	Access Lock:	Engineer
Value Type:	Enum of ENBLSTAT	Value Range:	Disable = Disables output alarming function Enable = Allows user to implement the output alarming function for an HG regulatory point. User must enter values for OPHITP, OPLOTP, OPALDB, OPHIPR, and OPLOPR parameters. In addition, the \$OPSNPRD parameter must be configured for the Data Hiway point to specify the scan period of the output alarm scan.		

NOTE

When the value of \$OPALEN is changed from Disable to Enable, a check is made to ensure that a valid relationship exists between OPHITP/OPLOTP and OPHILM/OPLOLM. If the relationship is valid, store will be successful. If the relationship is not valid, a Limit/Range Error is reported and the value of \$OPALEN will not be changed.

When the value of \$OPALEN is changed from Enable to Disable, relationship checking is not performed.

\$OPOVRC

\$OPOVRC – Output Overrun Current Hour (R640)

This PSDP parameter indicates the number of output alarm scanning overruns in the current hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0		

\$OPOVRP – Output Overrun Previous Hour (R640)

This PSDP parameter indicates the number of output alarm scanning overruns in the previous hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0		

\$OPOVRUN – Output Alarm Task Overrun (R640)

This PSDP parameter displays the cumulative total of output alarm scanning overruns. OP alarm scanning completes the current cycle even if the overrun occurs.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0		

\$OPSNPRD – Output Alarm Scan Period (R640)

This parameter enables the output (OP) alarm scanning function for the Data Hiway and defines the period for the OP alarm scanning cycle. For example, if the period is set to 1 second, the OP alarm scanning will be executed every 1 second. If the period is set to 0, the output alarm scanning function for the Data Hiway is disabled. View this parameter with \$HIWAYnn.\$OPSNPRD. It is also visible on the HG Node Status display. It is configured on the first page of the HG Hiway Point Assignment in the PED of \$HIWAYnn.

Source:	User	Default Value:	0	Access Lock:	Engineer
Value Type:	Integer (in Seconds)	Value Range:	0 – Output alarm scanning function for the Data Hiway is disabled. 1 – 60 – Output alarm scanning is enabled for the Data Hiway. Specifies period of the output alarm scan cycle.		

\$OPTOL (RegCtl, AC, AO)—Output Tolerance Parameter Definition

Tolerance limit for a manually entered OP. The difference between a new OP and a current OP is compared against \$OPTOL. If the tolerance is violated in either a positive or negative direction from the current value of the OP, operator confirmation is required before the value is stored. A value of 0.0 disables this check. A NaN or a negative value is not allowed.

Source:	User	Default Value:	0.0	Access Lock:	Engineer
Value Type:	Real (in Percent)	Value Range:	0.0 to 106.9		

\$PARMAXC – Maximum parameter access rate in any one second in the current hour (R660)

This PSDP parameter indicates the maximum parameter access rate in any one second in the current hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	>=0		

\$PARMAXP

\$PARMAXP – Maximum parameter access rate in any one second in the previous hour (R660)

This PSDP parameter indicates the maximum parameter access rate in any one second in the previous hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	>=0		

\$REALARM (Reg, AC, AI, DC, DI, Counter, Flag) – Re-Alarm Interval for Process Alarms

With R660, acknowledged and unacknowledged process alarms (event-type 21) are rebroadcast if they have not returned to normal (RTN) within the re-arming interval specified by the \$REALARM parameter. The re-alarm feature is applicable for all types of process alarms that can have an RTN event. The user-configured re-alarm interval is the same value for all process alarms associated with the same point.

Re-arming is consistent across all stations in the console in that area. Rebroadcast alarms are also sent to the History Module with a new timestamp, and reported to the RTJ and Process Alarm Journal where they appear as an alarm.

On the alarm displays, the rebroadcast alarm behaves like a new alarm but has a yellow time stamp.

Rebroadcast alarms appear on the GUS Alarm Summary, and on the following GUS/US native-window displays:

- Area Alarm Summary
- Unit Alarm Summary
- Alarm Annunciator
- Point Detail

The re-alarm feature is implemented using the \$REALARM parameter that has a range of 0 through 9999 minutes. The value specified for this parameter determines the interval, in minutes that will elapse before the specified point is re-alarmed.

Source:	User	Default Value:	0 (no re-alarm)	Access Lock:	Engr
Value Type:	Integer	Value Range:	0 through 9999 minutes (value range is from one minute to approximately 167 hours). A value of zero minute disables the rebroadcast alarm function.		

\$RNSELECT – PV Negative Rate of Change Selective Cutout (R681)

This parameter allows or cuts out annunciation of PV Negative Rate of Change alarm of a point even though the point is cutout.

Source:	User	Default Value:	CUTOUT	Access Lock:	Engineer
Value Type:	Enum of \$xxSELECT	Value Range:	CUTOUT = Cuts out annunciation of PVROCN alarm of a point when the cutout state is true. ALLOW = Allows annunciation of PVROCN alarm of a point irrespective of the cutout state.		

\$RPSELECT – PV Positive Rate of Change Selective Cutout (R681)

This parameter allows or cuts out annunciation of PV Positive Rate of Change alarm of a point even though the point is cutout.

Source:	User	Default Value:	CUTOUT	Access Lock:	Engineer
Value Type:	Enum of \$xxSELECT	Value Range:	CUTOUT = Cuts out annunciation of PVROCP alarm of a point when the cutout state is true. ALLOW = Allows annunciation of PVROCP alarm of a point irrespective of the cutout state.		

\$PARSECC – Parameter access rate (running average per second) in the current hour (R660)

This PSDP parameter indicates the parameter access rate in the current hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0.0		

\$PARSECP – Parameter access rate (overall average per second) in the previous hour (R660)

This PSDP parameter indicates the parameter access rate in the previous hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0.0		

\$PRCMAXC

\$PRCMAXC – Maximum number of process alarms in any one second in the current hour (R660)

This PSDP parameter indicates the maximum number of process alarms in any one second in the current hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	≥ 0		

\$PRCMAXP – Maximum number of process alarms in any one second in the previous hour (R660)

This PSDP parameter indicates the maximum number of process alarms in any one second in the previous hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	≥ 0		

\$PRCTOTC – Total number of process alarms sent to the LCN in the current hour (R660)

This PSDP parameter indicates the total number of process alarms sent to the LCN in the current hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$PRCTOTP – Total number of process alarms sent to the LCN in the previous hour (R660)

This PSDP parameter indicates the total number of process alarms sent to the LCN in the previous hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$PRCTOTS

\$PRCTOTS – Total number of process alarms sent to the LCN in the previous 15 seconds (R660)

This PSDP parameter indicates the total number of process alarms sent to the LCN in the previous 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$RDAVGC – Parameter read rate (running average per second) in the current hour (R660)

This PSDP parameter indicates the parameter read rate in the current hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$RDAVGP – Parameter read rate (overall average per second) in the previous hour (R660)

This PSDP parameter indicates the parameter read rate in the previous hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$RDAVGS – Parameter read rate (per second) in the last 15 seconds (R660)

This PSDP parameter indicates the parameter read rate in the last 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

\$RDMAXC

\$RDMAXC – Maximum number of parameters read in any one second in the current hour (R660)

This PSDP parameter indicates the maximum number of parameters read in any one second in the current hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	>=0		

\$RDMAXP – Maximum number of parameters read in any one second in the previous hour (R660)

This PSDP parameter indicates the maximum number of parameters read in any one second in the previous hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	>=0		

\$SEQMAXC – Maximum number of sequence events/messages in any one second in the current hour (R660)

This PSDP parameter indicates the maximum number of sequence events/messages in any one second in the current hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	>=0		

\$SEQMAXP – Maximum number of sequence events/messages in any one second in the previous hour (R660)

This PSDP parameter indicates the maximum number of sequence events/messages in any one second in the previous hour.

Source:	System	Default Value:	0	Access Lock:	View
Value Type:	Integer	Value Range:	>=0		

\$SEQTOTC

\$SEQTOTC – Number of sequence events/messages sent to the LCN in the current hour (R660)

This PSDP parameter indicates the maximum number of sequence events/messages sent to the LCN in the current hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0.0		

\$SEQTOTP – Number of sequence events/messages sent to the LCN in the previous hour (R660)

This PSDP parameter indicates the maximum number of sequence events/messages sent to the LCN in the previous hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0.0		

\$SEQTOTS – Number of sequence events/messages sent to the LCN in the previous 15 seconds (R660)

This PSDP parameter indicates the maximum number of sequence events/messages sent to the LCN in the previous 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0.0		

\$SPTOL (RegCtl)—Setpoint Tolerance Parameter Definition

Tolerance limit for a manually entered SP. The difference between a new SP and a current SP is compared against \$SPTOL. If the tolerance is violated in either a positive or negative direction from the current value of the SP, operator confirmation is required before the value is stored. A value of 0.0 disables this check. A NaN or a negative value is not allowed.

Source:	User	Default Value:	0.0	Access Lock:	Engineer
Value Type:	Real (in Engineering Units)	Value Range:	0.0 to 9999		

\$STRMAXC

\$STRMAXC – Maximum number of parameters stored in any one second in the current hour (R660)

This PSDP parameter indicates the maximum number of parameters stored in any one second in the current hour.

Source:	System	Default	Access
		Value: 0	
Value	Type: Integer	Value	Lock: View
		Range: >=0	

\$STRMAXP – Maximum number of parameters stored in any one second in the previous hour (R660)

This PSDP parameter indicates the maximum number of parameters stored in any one second in the previous hour.

Source:	System	Default	Access
		Value: 0	
Value	Type: Integer	Value	Lock: View
		Range: >=0	

-A-

A—A Input Value (EC Regulatory Data Point)

This parameter is the current value (in percent) of the A signal at an EC regulatory data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	6.86% to 106.86%		

ABHEMSD—Abnormal Handler Emergency Shutdown Enable Option (MC Process Module Data Point)

This parameter allows the operator to enable or disable the emergency shutdown sequence part of the program.

Source:	System	Default Value:	Disable (supplied by DEB)	Access Lock:	Oper.
Value Type:	Enum. of ENBLSTAT	Value Range:	Disable = Emergency shutdown sequence is disabled. Enable = Emergency shutdown sequence is enabled.		

ABHHOLD—Abnormal Handler Hold Enable Option (MC Process Module Data Point)

This parameter allows the operator to enable or disable the hold sequence part of the program.

Source:	System	Default Value:	Disable (supplied by DEB)	Access Lock:	Oper.
Value Type:	Enum. of ENBLSTST	Value Range:	Disable = Hold sequence is disabled. Enable = Hold sequence is enabled.		

ABHRSTR—Abnormal Handler Restart Enable Option (MC Process Module Data Point)

This parameter allows the operator to enable or disable the restart sequence part of the program.

Source:	System	Default Value:	Disable (supplied by DEB)	Access Lock:	Oper.
Value Type:	Enum. of ENBLSTST	Value Range:	Disable = Restart sequence is disabled. Enable = Restart sequence is enabled.		

ABHSHDN—Abnormal Handler Shutdown Enable Option (MC Process Module Data Point)

This parameter allows the operator to enable or disable the shutdown sequence part of the program.

Source:	System	Default Value:	Disable (supplied by DEB)	Access Lock:	Oper.
Value Type:	Enum. of ENBLSTST	Value Range:	Disable = Shutdown sequence is disabled. Enable = Shutdown sequence is enabled.		

ABHSTEP—Abnormal Handler Current Step (MC Process Module Data Point)

This parameter indicates the current step number of the abnormal sequence.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	String				

ABHSTMT—Abnormal Handler Current Statement (MC Process Module Data Point)

This parameter indicates the current statement number of the abnormal sequence.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 to 255		

ACP—Advanced Control Point ID

This parameter defines the tag name of an Advanced Control Data Point in a CG or CM, which this MC Process Module Data Point notifies when the process module point receives a sequence message. Refer to NAME for the valid character set.

Source:	User	Default Value:	All underscores	Access Lock:	DEB
Value Type:	Entity ID	Value Range:	N/A		

ACPRATE— ACP Count (CIU Processor Status Data Points)

This parameter defines the number of ACP turn-on messages during the most recent (full) 1-minute sample period.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	REAL	Value Range:	≥ 0.0		

ACTIVCBL – Active Cable

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 & 1		

ADIGST—A Input Digital State (EC Regulatory Data Point)

This parameter is the current digital state of the A input signal at an EC Regulatory Data Point that has been configured for the **Logic** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Current state of the A input is true. False = Current state of the A input is false.		

ALARMS1—S1 Alarm (EC Regulatory Data Point)**ALARMS2—S2 Alarm****ALARMS3—S3 Alarm****ALARMS4—S4 Alarm**

These parameters are the current states of the S1, S2, S3 and S4 alarms at an EC regulatory data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Alarm is active False = No alarm has been detected		

ALBURST — Alarm Burst (AM Processor Status Data Points)

The maximum number of alarms in any one second since startup.

Source:	System	Default Value:	0.0	Access Lock:	Engr
Value Type:	INTEGER	Value Range:	0-32767		

ALENBST—Point Alarm Enable State**NOTE**

Changing the ALENBST value terminates the current alarm timer. The new value will be in effect for the next applicable alarm event.

This parameter defines whether data-point alarms are to be reported to the History Module (HM) and the Universal Station.

Source:	User	Default Value:	Enable	Access Lock:	Access Lock is configurable as OPER, SUPVR, or ENGR in Network Configuration under System Wide Values
Value Type:	Enum. of ALENBST	Value Range:	Enable = Data point alarms are historized, annunciated, and alarm indicators appear on Group and Detail displays. Disable = Data point alarms are historized and alarm indicators appear on Group and Detail displays, but are not annunciated. Inhibit = Data point alarms are not reported to the HM or to the Universal Station.		

ALFMT—Alarm Format (Analog Data Point)

This parameter defines the type of alarming (PV, deviation, etc.) that is to be used for an analog data point. (The alarm trip point values for the selected alarm format are entered on the configuration form in the lines that follow the ALFMT line.)

WARNING: For a point with aliases, modify the alarm format only when the point is not in alarm.

Source:	User	Default Value:	Alfmt00	Access Lock:	Engr.
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Value Type: Enum. of ALFMT

Value

Range: For a CB RV data point:

No Alarm = No alarms are to be detected by this data point.
Alfmt00 = PV alarming (PVLOTP & PVHITP)

For a DHP, HLPIU, LEPIU, LLPIU, or MC analog data point:

Alfmt00 = PV alarming (PVLOTP & PVHITP)
Alfmt01 = Deviation alarming (DEVLOTP & DEVHITP)

ALFMT (CB)

ALFMT—Alarm Format (CB Regulatory Data Point)

This parameter defines the type of alarming (PV, deviation, etc.) that is to be used for a CB regulatory data point. (The alarm trip-point values for the selected alarm format are entered on the configuration form in the lines that follow the ALFMT line.)

WARNING: For a point with aliases, modify the alarm format only when the point is not in alarm.

Source:	User	Default Value:	Alfmt00	Access Lock:	Engr.
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Value Type: Enum. of ALFMT

For a CB regulatory data point, the alarm formats depend on the selected algorithm as follows:

Algo. No.	Name	Alarm Formats	
		Alfmt00	Alfmt01
00	Das	X	X
01	Pidnorm	X	X
02	Pidratio	X	X
03	Autratio	X	X
04	Autobias	X	X
05	Pidcma	X	X
06	Pidcm	X	X
07	Pidsp	X	X
10	Pidersqg	X	X
11	Pidersqi	X	X
12	Pidgap		X
13	Pidpl50	X	X
14	Pidpl50c	X	X
20	Leadlag	X	(NOTE)
23	Sumrwman	X	
24	Multwman	X	
25	Automan	X	
26	Switch	X	
30	Cbsumr	X	
31	Mult	X	
32	Dividr	X	
33	Sqrt	X	
34	Sqrtprod	X	
35	Sumsqrt	X	
36	Hisel	X	
37	Losel	X	

Alfmt00 = PV alarming (PVLOTP & PVHITP)

Alfmt01 = Deviation alarming (DEVLOTP & DEVHITP)

NOTE

Algorithms 20-37 do not require a selection of an alarm format because only PV alarming (PVLOTP and PVHITP) is applicable to these algorithms.

ALFMT (EC)

ALFMT—Alarm Format (EC Regulatory Data Point)

This parameter defines the type of alarming (PV, deviation, etc.,) that is to be used for a EC regulatory data point. (The alarm trip-point values for the selected alarm format are entered on the configuration form in the lines that follow the ALFMT line.) The S1–S4 states can be viewed using the S1S2S3S4 parameter. Parameter ALMASK is used to select which alarm states are enabled.

WARNING: For a point with aliases, modify the alarm format only when the point is not in alarm.

Source: User **Default Value:** Alfmt00 **Access Lock:** Engr

Value Type: Enum. of ALFMT

For an EC regulatory data point, the alarm formats depend on the selected algorithm as follows (this listing applies to all algorithms except for the **Rampsoak** algorithm; the **Logic** algorithm does not have any alarming capability):

Alarm Format	Name of Alarm Type	Description	Status Output
Alfmt00	PVLOTP	PV Lo Trip Point	S2
	PVHITP	PV Hi Trip Point	S1
Alfmt01	DEVLOTP	Deviation Lo Trip Point	S2
	DEVHITP	Deviation Hi Trip Point	S1
Alfmt02	PVLOTP	PV Lo Trip Point	S3
	DEVLOTP	Deviation Lo Trip Point	S4
	PVHITP	PV Hi Trip Point	S1
	DEVHITP	Deviation Hi Trip Point	S2
Alfmt03	PVLOTP	PV Lo Trip Point	S4
	PVHITP	PV Hi Trip Point	S2
	PVLLTP	PV Lo Lo Trip Point	S3
	PVHHTP	PV Hi Hi Trip Point	S1
Alfmt04	DEVLOTP	Deviation Lo Trip Point	S4
	DEVHITP	Deviation Hi Trip Point	S2
	DEVLLTP	Deviation Lo Lo Trip Point	S3
	DEVHHTP	Deviation Hi Hi Trip Point	S1
Alfmt05	PVLOTP	PV Lo Trip Point	S4
	PVHITP	PV Hi Trip Point	S3
	PVHHTP	PV Hi Hi Trip Point	S2
	PVHHHTP	PV Hi Hi Hi Trip Point	S1
Alfmt06	DEVLOTP	Deviation Lo Trip Point	S4
	DEVHITP	Deviation Hi Trip Point	S3
	DEVHHTP	Deviation Hi Hi Trip Point	S2
	DEVHHHTP	Deviation Hi Hi Hi Trip Point	S1
Alfmt07	PVLOTP	PV Lo Trip Point	S4
	PVHITP	PV Hi Trip Point	S1
	PVLLTP	PV Lo Lo Trip Point	S3
	PVLLLTP	PV Lo Lo Lo Trip Point	S2

ALFMT (EC)

Alarm Format	Name of Alarm Type	Description	Status Output
Alfmt08	DEVLOTP	Deviation Lo Trip Point	S4
	DEVHITP	Deviation Hi Trip Point	S1
	DEVLLTP	Deviation Lo Lo Trip Point	S3
	DEVLLLTP	Deviation Lo Lo Lo Trip Point	S2
Alfmt09	(NOTE 1)		S1-S4
Alfmt10 through Alfmt13	PVLOTP	PV Lo Trip Point	S2
	PVHITP	PV Hi Trip Point	S1
	PVROCTP	PV Rate Of Change Trip Point	S3
			S4 (NOTE 1)
Alfmt14 through Alfmt17	DEVLOTP	Deviation Lo Trip Point	S2
	DEVHITP	Deviation Hi Trip Point	S1
	PVROCTP	PV Rate of Change Trip Point	S3
			S4 (NOTE 1)
Alfmt18	PVHITP	PV Hi Trip Point	S1
	PVLOTP	PV Lo Trip Point	S2
	DEVHITP	Deviation Hi Trip Point	S3
	DEVLOTP	Deviation Lo Trip Point	S4
Alfmt19	DEVHITP	Deviation Hi Trip Point	S1
	DEVLOTP	Deviation Lo Trip Point	S2
	PVHITP	PV Hi Trip Point	S3
	PVLOTP	PV Lo Trip Point	S4

NOTE 1

Mode alarms M, C, A, and P are not supported in alarm formats Alfmt09 through Alfmt17 at the **TotalPlant** Solution (TPS) system Universal Station. If your present system has a BASIC Operator Station, these alarms are reported at the BASIC Operator Station.

The following listing of alarm formats applies to only the **EC Rampsoak** algorithm:

Alarm Format	Name of Alarm Type	Description	Status Output
Alfmt00	PVLOTP	PV Lo Trip Point	S2
	PVHITP	PV Hi Trip Point	S1
Almft01	(NOTE 2)		S1
			S2
			S3 (NOTE 3)
			S4 (NOTE 3)
Alfmt02	PVLOTP	PV Lo Trip Point	S3
	OFFST1TP	Offset 1 Trip Point	S4
	PVHITP	PV Hi Trip Point	S1
	OFFST2TP	Offset 2 Trip Point	S2
Alfmt03	PVLOTP	PV Lo Trip Point	S4
	PVHITP	PV Hi Trip Point	S2
	PVLLTP	PV Lo Lo Trip Point	S3
	PVHHTP	PV Hi Hi Trip Point	S1

ALFMT (EC)

Alarm Format	Name of Alarm Type	Description	Status Output
Alfmt04	OFFST1TP	Offset 1 Trip Point	S3
	OFFST2TP	Offset 2 Trip Point	S2
	PVROCTP	PV Rate of Change Trip Point	S4
			S1 (NOTE 5)
Alfmt05	OFFST2TP	Offset 2 Trip Point	S1
Alfmt08	OFFST1TP	Offset 1 Trip Point	S2
	PVROCTP	PV Rate of Change Trip Point	S3
			S4 (NOTE 3)
Alfmt06	PVHITP	PV Hi Trip Point	S1
Alfmt07	PVLOTP	PV Lo Trip Point	S2
	PVROCTP	PV Rate of Change Trip Point	S3
			S4 (NOTE 3)
Alfmt09	(NOTE 4)		S1-S4
Alfmt10	OFFST1TP	Offset 1 Trip Point	S3
	OFFST2TP	Offset 2 Trip Point	S2
	OFFST3TP	Offset 3 Trip Point	S1
			S4 (NOTE 3)
Alfmt11	OFFST1TP	Offset 1 Trip Point	S3
	OFFST2TP	Offset 2 Trip Point	S2
	OFFST3TP	Offset 3 Trip Point	S1
			S4 (NOTE 3)

NOTE 2

No entries are required for the alarm trip points on the *EC Regulatory Data Point Configuration Form* (see *HG Data Point Forms*) if alarm format Alfmt01 has been entered; however, entries are required for the MARK1SEG-MARK2TT parameters to define the alarm values for status outputs S1 and S2.

NOTE 3

Mode alarms M, A, C, and P are not supported at the Universal Station for the respective status outputs; however, these mode alarms are still supported at the BASIC Operator Station.

NOTE 4

No entries are required for the alarm trip points on the *EC Regulatory Data Point Configuration Form* (see *HG Data Point Forms*) if alarm format Alfmt09 has been entered because this alarm format deals only with mode alarms M, A, C, and P, which are not supported at the Universal Station; however, these mode alarms are still supported at the BASIC Operator Station.

ALFMT (EC)

NOTE 5

For status output S1, make entries in the MARK1SEG-MARK1TT parameters on the *EC Regulatory Data Point Configuration Form* (see *HG Data Point Forms*). These entries determine the alarm values for status output S1.

NOTE 6

Certain alarms are not applicable to EC regulatory points.

For the **Das** algorithm, the following values don't apply to ALFMT: Alfmt02 and Alfmt08. For Alfmt09 there is no DEVHITP alarm. For Alfmt10 through 17, PVROCTP is related to the S4 status output.

For the following algorithms, there are no deviation alarms: **Extrabi**, **Massflow** (except DEVLOTP with Alfmt19), **Selovddc**, **Sqrtprod**, **Ecsumr**, **Multdiv**, **Automan**, **Switch**, **Incrmtl**, **Deadtm**, and **Charact**.

ALFMT does not apply to the **Selovddc** algorithm.

ALFMT (MC)

ALFMT—Alarm Format (MC Regulatory Data Point)

This parameter defines the type of alarming (PV, deviation, etc.) that is to be used for an MC regulatory data point. (The alarm trip point values for the selected alarm format are entered on the configuration form in the lines that follow the ALFMT line.)

WARNING: For a point with aliases, modify the alarm format only when the point is not in alarm.

Source: User **Default Value:** Alfmt00 **Access Lock:** Engr

Value Type: Enum. of ALFMT

For an MC regulatory data point, the alarm formats depend on the selected algorithm as follows:

Algo No	Name	Alarm Formats		
		Alfmt00	Alfmt01	Alfmt02
00	Das	X	X	X
01	Pidnorm	X	X	X
02	Pidratio	X	X	X
03	Autratio	X	X	X
04	Autobias	X	X	X
05	Pidcma	X	X	X
06	Pidcm	X	X	X
07	Pidsp	X	X	X
10	Pidersqg	X	X	X
11	Pidersqi	X	X	X
12	Pidgap	X	X	X
20	Leadlag	X		X
23	Sumrwman	X		X
24	Multwman	X		X
25	Automan	X		X
26	Switch	X		X
30	Cbsumr	X		X
31	Mult	X		X
32	Dividr	X		X
33	Sqrt	X		X
36	Hisel	X		X
37	Losel	X		X

A1fmt00 = PV alarming (PVLOTP & PVHITP)

A1fmt01 = Deviation alarming (DEVLOTP & DEVHITP)

A1fmt02 = PV rate of change negative and positive alarming (PVROCNTP & PVPOCPTP)

ALGIDDAC—Algorithm (CB Regulatory Data Point)

This parameter defines the algorithm that is to be used for a CB regulatory data point. The parameters that are applicable to each algorithm are contained in separate listings, and can be found by looking for the name of the algorithm in this dictionary.

Detailed information about each algorithm is contained in *Basic Controller Algorithm Engineering Data*, in the *BASIC Product Manual* binder.

Source:		Default	Access	
User		Value: Das	Lock:	Engr.
Value				
Type:		Enum. of ALGIDDAC		
Value				
Range:		Das = Data Acquisition, Algo. 00 Pidnorm = PID Normal, Algo. 01 Pidratio = PID Ratio, Algo. 02 Autratio = PID Auto Ratio, Algo. 03 Autobias = PID Auto Bias, Algo. 04 Pidcma = PID CMA, Algo. 05 Pidcm = PID CMA, Algo. 06 Pidspc = PID SPC, Algo. 07 Pidersqg = PID Error Squared on Gain, Algo. 10 Pidersqi = PID Error Squared on Integral, Algo. 11 Pidgap = PID Gap, Algo. 12 Pdp150 = PD + 50%, Algo. 13 (See note) Pdp150C = PD + 50% (CMA), Algo. 14 (See note) Leadlag = Lead Lag, Algo. 20 Overhisl = Override Hi Selector, Algo. 21 Overlosl = Override Lo Selector, Algo. 22 Sumrwman = Summer (with Manual), Algo. 23 Multwman = Multiplier (with Manual), Algo. 24 Automan = Auto-Manual, Algo. 25 Switch = Switch, Algo. 26 Cbsumr = Summer, Algo. 30 Mult = Multiplier, Algo. 31 Dividr = Divider, Algo. 32 Sqrt = Square Root, Algo. 33 Sqrtprod = Double Square Root, Algo. 34 Sumsqrt = Sum of Square Roots, Algo. 35 Hisel = Hi Selector, Algo. 36 Losel = Lo Selector, Algo. 37		

NOTE

The **Pdp150** and **Pdp150c** algorithms are applicable only if revision of CB is 1C. The HG automatically determines CB revision by reading the revision code from the CB box.

This parameter's value is stored in both the HG and the box. If CHNGFLAG contains Detect, the HG detects any changes made to the value in the box from some other source, and the HG updates its value to match that of the box.

ALGIDDAC—Algorithm (EC Regulatory Data Point)

This parameter defines the algorithm that is to be used for an EC regulatory data point. The parameters that are applicable to each algorithm are contained in separate listings, and can be found by looking for the name of the algorithm in this dictionary. Detailed information about each algorithm is contained in *Extended Controller Algorithm Engineering Data*, in the *BASIC Product Manual* binders.

Source:	User	Default Value:	Das	Access Lock:	Engr.
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Value Type: Enum. of ALGIDDAC

Value Range:

- Das = Data Acquisition, Algo. 00
- Pidnorm = PID Normal, Algo. 01
- Pidratio = PID Ratio, Algo. 02
- Autratio = PID Auto Ratio, Algo. 03
- Autobias = PID Auto Bias, Algo. 04
- Pidddc = PID DDC, Algo. 05
- Piddcsp = PID DDC SP, Algo. 06
- Pidnonln = PID Non Linear Gain, Algo. 10
- Pidgap = PID Gap, Algo. 12
- Pidexfb = PID with External Feedback, Algo. 13
- Extratbi = External Ratio Bias, Algo. 14
- Massflow = Mass Flow, Algo. 16
- Leadlags = Lead Lag Summer with DDC, Algo. 20
- Selovddc = Override Selector, Algo. 21
- Sqrtprod = Square Root of Product, Algo. 22
- Ecsuwr = Summer, Algo. 23
- Multdiv = Multiplier/Divider, Algo. 24
- Automan = Auto-Manual Station, Algo. 25
- Switch = Switch, Algo. 26
- Leadlagm = Lead/Lag Multiplier with DDC, Algo. 30
- Midof3 = Middle-of-Three Selector, Algo. 31
- Incrmtl = Incremental, Algo. 32
- Rampsoak = Ramp and Soak, Algo. 34
- Logic = Logic, Algo. 35
- Deadtm = Dead Time, Algo. 36
- Charact = Characterizer, Algo. 37

NOTE

This parameter's value is stored in both the HG and the box. If CHNGFLAG contains Detect, the HG detects any changes made to the value in the box from some other source, and the HG updates its value to match that of the box.

ALGIDDAC—Algorithm (MC Regulatory Data Point)

This parameter defines the algorithm that is to be used for an MC regulatory data point. The parameters that are applicable to each algorithm are contained in separate listings, and can be found by looking for the name of the algorithm in this dictionary. Detailed information about each algorithm is contained in *Multifunction Controller Algorithm Engineering Data*, in the *BASIC Product Manual* binders.

Source:	User	Default Value:	Das	Access Lock:	Engr.
Value Type:	Enum. of ALGIDDAC				
Value Range:	Das = Data Acquisition, Algo. 00 Pidnorm = PID Normal, Algo. 01 Pidratio = PID Ratio, Algo. 02 Autratio = PID Auto Ratio, Algo. 03 Autobias = PID Auto Bias, Algo. 04 Pidcma = PID CMA, Algo. 05 Pidcm = PID CMA, Algo. 06 Pidspc = PID SPC, Algo. 07 Pidersqg = PID Error Squared on Gain, Algo. 10 Pidersqi = PID Error Squared on Integral, Algo. 11 Pidgap = PID Gap, Algo. 12 Leadlag = Lead Lag, Algo. 20 Overhisl = Override Hi Selector, Algo. 21 Overlosl = Override Lo Selector, Algo. 22 Sumrwman = Summer (with Manual), Algo. 23 Multwman = Multiplier (with Manual), Algo. 24 Automan = Auto-Manual, Algo. 25 Switch = Switch, Algo. 26 Cbsumr = Summer, Algo. 30 Mult = Multiplier, Algo. 31 Dividr = Divider, Algo. 32 Sqrt = Square Root, Algo. 33 Hisel = Hi Selector, Algo. 36 Losel = Lo Selector, Algo. 37				

NOTE

This parameter's value is stored in both the HG and the box. If CHNGFLAG contains Detect, the HG detects any changes made to the value in the box from some other source, and the HG updates its value to match that of the box.

ALHIFL—Alarm Hi Fail

This parameter indicates if the point is in the high alarm state.

Source:	System	Default Value:	Off	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	Off = Point is not in high alarm state. On = Point is in high alarm state.		

ALLOFL—Alarm Lo Fail

This parameter indicates if the point is in the low alarm state.

Source:	System	Default Value:	Off	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	Off = Point is not in low alarm state. On = Point is in low alarm state.		

ALMASK—Selection of Alarms Enabled (EC Regulatory Data Point)

This parameter allows the user to select which of the status alarms (S1, S2, S3, or S4) are to be enabled.

Source:	User	Default Value:	Noalarm	Access Lock:	Engr.
Value Type:	Enum. of ALMASK	Value Range:	Noalarm = No alarm statuses are enabled. S1 = S1 alarm status is enabled. S2 = S2 alarm status is enabled. S3 = S3 alarm status is enabled. S4 = S4 alarm status is enabled. S1S2 = S1 and S2 alarm statuses are enabled. S1S3 = S1 and S3 alarm statuses are enabled. S1S4 = S1 and S4 alarm statuses are enabled. S2S3 = S2 and S3 alarm statuses are enabled. S2S4 = S2 and S4 alarm statuses are enabled. S3S4 = S3 and S4 alarm statuses are enabled. S1S2S3 = S1, S2, and S3 alarm statuses are enabled. S1S3S4 = S1, S3, and S4 alarm statuses are enabled. S2S3S4 = S2, S3, and S4 alarm statuses are enabled. S1S2S4 = S1, S2, and S4 alarm statuses are enabled. S1S2S3S4 = S1, S2, S3, and S4 alarm statuses are enabled.		

ALMDIS—Alarm Output (MC Regulatory Data Point)

This parameter selects the alarm output option that determines whether digital outputs SOA and SOB for control slots 1-8 are driven by CL/MC and logic blocks, or by the alarms in control slots 1-8.

Source:	User	Default Value:	Disable	Access Lock:	DEB
Value Type:	Enum. of ENBLSTAT	Value Range:	Enable = Allows alarms in control slot 1-8 to drive digital outputs SOA and SOB. Disable = Allows CL/MC and logic blocks to drive digital outputs SOA and SOB.		

ALPRIOR—Alarm Priority

The value of this parameter will be the composite, that is, the highest priority, of all the separate alarm priorities for all the trip points configured for this point. If a priority is stored into this parameter, it will also be stored into all the separate alarm priorities for all the trip point configured for this point. ALPRIOR can only be updated from CL or from a schematic.

Source:	User	Default Value:	Low	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.
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ANLGCARD—Analog Output Card (MC Regulatory Data Point)

This parameter defines whether the output of this regulatory data point is to be connected to an analog output card. Make an entry in this parameter only if the slot number entered for the SLOTNUM parameter is greater than 8.

Source:	User	Default Value:	Notsel	Access Lock:	DEB/Eng
Value Type:	Enum. of OVRINSS	Value Range:	Notsel = Analog output card is not used. Sel = Analog output card is used.		

NOTE

An Sel entry is valid for only control slots 9-16 if an analog output card is present in the following point card file locations:

Point card file slot 1, subslots 1-4 for control slots 9-12.

Point card file slot 2, subslots 1-4 for control slots 13-16.

AREAALM—Area Alarms (US Processor Status Data Points)

This parameter defines the number of area process alarms reported to this US within the sample period. (Same number as the number of alarms appearing on the Alarm Summary Display.)

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	REAL	Value Range:	≥ 0.0		

ARWNET—Anti-Reset Windup Status on Setpoint

This parameter shows the windup status of the setpoint at the specified data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of WINDUP	Value Range:	Normal = The setpoint is not wound-up. Hi = The setpoint is wound-up high. Lo = The setpoint is wound-up low. Hilo = Setpoint is being initialized or changing the setpoint value does not affect downstream data points.		

ARWOP—Anti-Reset Windup Status on Output

This parameter shows the windup status of the data point's output.

Source:	System	Default Value:	Normal	Access Lock:	Read Only
Value Type:	Enum. of WINDUP	Value Range:	Normal = There is no windup of the output. Hi = The output is wound-up high. Lo = The output is wound-up low. Hilo = The output is being initialized or changing the output value does not affect downstream data points.		

ASIGNAL—A Signal Type (EC Regulatory Data Point)

The A signal types are the same as those listed for the Z SIGNAL parameter except that this signal is applied to the A input rather than the Z input. The tag name of the data point that provides the A signal is defined by ASLTSRC. Refer to the definition of the Z SIGNAL parameter.

ASLTSRC—A Input Source (EC Regulatory Data Point)

This parameter defines the tag name of the data point that provides the signal to the A input of an EC regulatory data point. Refer to the definition of NAME (Tag Name) for the valid character set.

Source:	User	Default Value:	Tag Name from line 1 of the Config. Form	Access Lock:	Engr.
Value Type:	Entity ID	Value Range:	See <u>NAME</u>		

NOTE

To update the value in this parameter, MODE must contain Man, INITCONF must contain Noinit, EMS must contain Disable, and the value in this parameter must be the tag name of a point in this HG.

ASSOCDSP—Associated Display

This parameter specifies a user-configured schematic associated with a point. The specified associated display can be called from a Point Detail display, from any summary display, or from the Group display when the point is selected.

Source:	User	Default Value:	Blank	Access Lock:	Engr.
Value Type:	String_8	Value Range:	N/A		

AUTOBIAS (CB)

Autobias—PID Auto Bias (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Autobias** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Autobias** algorithm is that it is a basic control algorithm with ratio and bias on the remote setpoints except that the bias coefficient is automatically calculated when not in the cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD ASSOCDSP PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPEVENT EIPENB EIPPCODE	SUPPIO CTLACTN CTLEQN (A,B) OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2 RATIO	\$MODESEL NMODE NMODATTR PSTMODE MODEPERM MODE

AUTOBIAS (EC)

Autobias—PID Auto Bias (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Autobias** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Autobias** algorithm is that it is a basic control algorithm with ratio and bias on the remote setpoint except that the bias coefficient is automatically calculated when not in the cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	\$AUXUNIT	SUPPIO	NMODE
UNIT	PVCHAR	ALFMT	CTLACTN	NMODATTR
PTDESC	INPTCOND	\$ALDLOPT	CTLEQN(A,B,C)	PSTMODE
KEYWORD	PVFORMAT	\$DLBADAL	OUTIND	RCASENB
PRIMMOD	PVTEMP	DLYTIME	PVSIGNAL	MODEPERM
ASSOCDSP	PVEULO	\$REALARM	PVSLT SRC	SHEDMODE
HWYNUM	PVEUHI	BADPVPR	SPSIGNAL	EMSENB
PNTBOXTY	OVERVAL	DEVHHPR	SPSLT SRC	MODE
BOXNUM	PVRNGOP	DEVHIPR	PVTRACK	\$MODESEL
PTDISCL	PVCLAMP	\$DHSELCT	INITCONF	
SLOTNUM	PVDSPHI	DEVLLPR	SPHILM	
LOADDEST	PVDSPLO	DEVLOPR	SPLOLM	
ALGIDDAC	CALIBOFF	\$DLSELCT	\$SPTOL	
		PVHHPR	SP	
		\$HHSELCT	SOASIGNL	
		PVHIPR	SOASRC	
		\$HISELCT	SOBSIGNL	
		PVLLPR	SOBSRC	
		\$LLSELCT	MSIGNAL	
		PVLOPR	MSLT SRC	
		\$LOSELCT	ASIGNAL	
		(Alarm	ASLT SRC	
		Trip-point	CSIGNAL	
		Values)	CSLT SRC	
		ALPRIOR	GREENSGL	
		CCRANK	GREENSRC	
		CCPRIPNT	SENSSTOP	
		EIPPCODE	OPLOLM	
		EIPEVENT	OPHILM	
		EIPENB	\$OPTOL	
		ALMASK	ITLOLM	
			ITHILM	
			TFCLLM	
			TFCHLM	
			TD	
			K	
			T1	
			T2	
			RATIO	

AUTOBIAS (MC)

Autobias—PID Auto Bias (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Autobias** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Autobias** algorithm is that it is a basic control algorithm with ratio and bias on the remote setpoints except that the bias coefficient is automatically calculated when not in the cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD CTLACTN CTLEQN (A,B) ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2 RATIO	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

AUTOMAN (CB)**Automan—Auto-Manual (CB Regulatory Data Point)**

The parameters that are applicable to a CB regulatory data point when it is configured for the **Automan** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Automan** algorithm is that it is a single-input auxiliary algorithm that provides auto-(unity gain) or manual-transfer capability.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND PVSIGNAL PVSLTSRC OPLOLM \$OPTOL OPHILM \$OPTOL TD	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

AUTOMAN (EC)

Automan—Auto-Manual (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Automan** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Automan** algorithm is that it is a single-input auxiliary algorithm, the output of which is the result of bias added to the input.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDOPT \$DLBADAL DLYTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHHPR \$HHSELCT PVHIPR \$HISELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK	SUPPIO OUTIND PVSIGNAL PVSLTSRC INITCONF SOASIGNL SOASRC SOBSIGNL SOBSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD	NMODE NMODATTR PSTMODE MODEPERM EMSENB MODE \$MODESEL

AUTOMAN (MC)

Automan—Auto-Manual (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Automan** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Automan** algorithm is that it is a single-input auxiliary algorithm that provides auto-(unity gain) or manual-transfer capability.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCTPT DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC OPLOLM OPHILM \$OPTOL TD	NMODE NMODATTR PSTMODE MODEPERM EMSENB MODE \$MODESEL

AUTRATIO (CB)

Autratio—PID Auto Ratio (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Autratio** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Autratio** algorithm is that it is a basic control algorithm with ratio and bias on the remote setpoints except that the ratio coefficient is automatically calculated, while maintaining a preset bias, when not in the cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO CTLACTN CTLEQN (A,B) OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2 BIAS	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

AUTRATIO (EC)

Autratio—PID Auto Ratio (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Autratio** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Autratio** algorithm is that it is a basic control algorithm with ratio and bias on the remote setpoint except that the ratio coefficient is automatically calculated, while maintaining a preset bias, when not in the cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	\$AUXUNIT	SUPPIO	NMODE
UNIT	PVCHAR	ALFMT	CTLACTN	NMODATTR
PTDESC	INPTCOND	\$ALDLOPT	CTLEQN (A,B,C)	PSTMODE
KEYWORD	PVFORMAT	\$DLBADAL	OUTIND	RCASENB
PRIMMOD	PVTEMP	DLYTIME	PVSIGNAL	MODEPERM
ASSOCDSP	PVEULO	\$REALARM	PVSLTSRC	SHEDMODE
HWYNUM	PVEUHI	BADPVPR	SPSIGNAL	EMSENB
PNTBOXTY	PVDSPHI	DEVHHPR	SPSLTSRC	MODE
BOXNUM	PVDSPLO	DEVHIPR	PVTRACK	\$MODESEL
PTDISCL	OVERVAL	\$DHSELECT	INITCONF	
SLOTNUM	PVRNGOP	DEVLLPR	SPLOLM	
LOADDEST	PVCLAMP	DEVLOPR	SPHILM	
ALGIDDAC	CALIBOFF	\$DLSELECT	\$SPTOL	
		PVHHPR	SP	
		\$HHSELECT	SOASIGNL	
		PVHIPR	SOASRC	
		\$HISELECT	SOBSIGNL	
		PVLLPR	SOBSRC	
		\$LLSELECT	MSIGNAL	
		PVLOPR	MSLTSRC	
		\$LOSELECT	ASIGNAL	
		(Alarm	ASLTSRC	
		Trip-point	CSIGNAL	
		Values)	CSLTSRC	
		ALPRIOR	GREENSGL	
		CCRANK	GREENSRC	
		CCPRIPNT	SENSSTOP	
		EIPPCODE	OPLOLM	
		EIPEVENT	OPHILM	
		EIPENB	\$OPTOL	
		ALMASK	ITLOLM	
		INITCONF	ITHILM	
		SPHILM	TFCLLM	
		SPLOLM	TFCHLM	
		\$SPTOL	TD	
			K	
			T1	
			T2	
			BIAS	

AUTRATIO (MC)

Autratio—PID Auto Ratio (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Autratio** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Autratio** algorithm is that it is a basic control algorithm with ratio and bias on the remote setpoints except that the ratio coefficient is automatically calculated, while maintaining a preset bias, when not in the cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD CTLACTN CTLEQN (A,B,C) ALMDIS CONFTRND OUTIND PVSIGNAL PVSLT SRC SPSIGNAL SPSLT SRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2 BIAS	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

AV—Accumulator Value (DHP & HLPIU Counter Data Points)

This parameter is the current accumulated value at the specified DHP or HLPIU counter data point.

Source:	System	Default Value:	0.0	Access Lock:	Oper.
Value Type:	Real	Value Range:	HLPIU 32-bit counters: 0 to 999999 HLPIU 16-bit counters and DHP counters: 0 to 65535* <u>AVCONV</u>		

AV—Accumulator Value (MC Counter Data Point)

This parameter is the total accumulated counts at the specified MC counter data point.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	Value depends on AVFORMAT value as follows:		
			<u>AVFORMAT</u>	<u>AV Value</u>	
			00	Accumulated value	
			01	(Accumulated Value)/10	
			02	(Accumulated Value)/100	
			03	(Accumulated Value)/1000	

AVCONFIG—Accumulation Configuration (MC Analog Input Data Point with Accumulation)

This parameter defines the time base, scaling, and maximum time during which the PV value is to be accumulated, as related to the PV decimal format (PVFORMAT).

Source:	User	Default Value:	1	Access Lock:	Engr.																										
Value Type:	Integer	Value Range:	<table><tr><td><u>Entry</u></td><td><u>Time Base</u></td><td><u>Scaling (S)</u></td><td><u>Max. Time</u></td></tr><tr><td>1</td><td>Hour</td><td>2</td><td>1 Hour</td></tr><tr><td>2</td><td>Hour</td><td>1</td><td>10 Hours</td></tr><tr><td>3</td><td>Hour</td><td>0</td><td>36 Hours</td></tr><tr><td>4</td><td>Minutes</td><td>1</td><td>10 Minutes</td></tr><tr><td>5</td><td>Minutes</td><td>0</td><td>100 Minutes</td></tr><tr><td>6</td><td>Minutes</td><td>-1</td><td>1000 Minutes</td></tr></table>	<u>Entry</u>	<u>Time Base</u>	<u>Scaling (S)</u>	<u>Max. Time</u>	1	Hour	2	1 Hour	2	Hour	1	10 Hours	3	Hour	0	36 Hours	4	Minutes	1	10 Minutes	5	Minutes	0	100 Minutes	6	Minutes	-1	1000 Minutes
<u>Entry</u>	<u>Time Base</u>	<u>Scaling (S)</u>	<u>Max. Time</u>																												
1	Hour	2	1 Hour																												
2	Hour	1	10 Hours																												
3	Hour	0	36 Hours																												
4	Minutes	1	10 Minutes																												
5	Minutes	0	100 Minutes																												
6	Minutes	-1	1000 Minutes																												
Constraint: Values are to be selected such that DA = D + S where: DA = Accumulator Decimal Point Value 0 through 5 (see AVFORMAT). D = PV Decimal Point Value 0 through 3 (see PVFORMAT). S = Scaling Value where S is not less than -1 or greater than 2.																															

AVCONV—Scale Factor (HLPIU & DHP Counter Data Points)

This parameter defines the scale factor that determines the total number of counts in the AV. For example, if AVCONV=0.01 and PV=100 pulses, the AV is incremented by one count.

Source:	User	Default Value:	1.0	Access Lock:	Engr.
Value Type:	Real	Value Range:	0.00006 to 1.0, and 0		

AVCONV—Scale Factor (MC Counter Data Points)

This parameter defines the scale factor that represents the number of pulses required to increment the MC counter data point by one count.

Source:	User	Default Value:	1.0	Access Lock:	Engr.
Value Type:	Real	Value Range:	1.0 - 1 count requires 1 pulse 0.1 - 1 count requires 10 pulses 0.01 - 1 count requires 100 pulses 0.001 - 1 count requires 1000 pulses		

AVCUTOFF—Accumulation Cutoff (MC Analog Data Point with Accumulation)

This parameter defines the low cutoff value for accumulation.

Source:	User	Default Value:	Cutoff0	Access Lock:	Engr.
Value Type:	Enum. of AVCUTOFF	Value Range:	Cutoff0 = when the nonfiltered current value is > 0%, accumulate it. Cutoff10 = when the nonfiltered current value is > 10%, accumulate it.		

AVEUHI —Accumulator High End of Range (MC Analog Data Point w/Accumulation)
AVEULO—Accumulator Low End of Range

These parameters indicate the predefined high and low ends of the accumulator range.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	<u>AVEUHI</u> is the predefined high end of the accumulator range; <u>AVEUHI</u> = 999999 <u>AVEULO</u> is the predefined low end of the accumulator range; <u>AVEULO</u> = 0.0		

AVFORMAT—Accumulator Decimal Point Position

This parameter defines the position where the decimal point is to be displayed in the accumulated value (AV).

Source: System **Default Value:** N/A

Access

Lock: For MC Analog Input Data Point: Read Only
For HLPIU, DHP, and MC Counter Data Points: Engr

Value Type: Enum. of VALFORMAT **Value Range:** D0 = XXXXXX.
D1 = XXXXX.X
D2 = XXXX.XX
D3 = XXX.XXX
D4 = XX.XXXX
D5 = X.XXXXX

NOTE

Counter data points have formats of D0-D3

AVOPTION—Accumulation Option (MC Analog Data Point with Accumulation)

This parameter selects the accumulation option for this data point.

Source: User **Default Value:** Disable **Access Lock:** DEB

Value Type: Enum. of ENBLSTAT **Value Range:** Disable = Data point does not accumulate the PV value.
Enable = Data point is to accumulate the PV value.

AVP—Accumulator Percent of Range (MC Analog Data Point w/Accumulation)

This parameter is the current accumulated value, which is expressed as a percent of the accumulator range. Accumulator range is from 0.0 to 999999.

Source: System **Default Value:** N/A **Access Lock:** Read Only

Value Type: Real **Value Range:** -6.86% to 106.86%

-B-**BADPVPR—Bad PV Alarm Priority**

When multiple alarms are detected for a data point, this parameter defines the priority of the bad PV alarms. A listing of alarms follows and is arranged in priority order, beginning with the lowest alarm states (Noalarm, Invalid) and ending with the highest alarm (Badpv).

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION = Alarm is not reported to the system. Alarm is not annunciated. JOURNAL = Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays. LOW = Alarm is historized, annunciated, and does appear on Group and Detail displays. HIGH = Alarm is historized, annunciated, and does appear on Group and Detail displays. EMERGNCY = This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity. PRINTER = Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.				
NOTE If PRINTER is chosen, the alarm is lost if the printer is not working.					
	JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.			
Noalarm =	No alarm has been detected.				
Invalid =	PV hi value has been set below the PV lo value.				
Prpreset =	For an MC counter data point, the accumulated value is equal to or greater than the pre-alarm value established by the <u>PDEVTP</u> parameter.				
Prpreset =	For an MC counter data point, the accumulated value is equal to or greater than the pre-alarm value established by the <u>PDEVTP</u> parameter.				

Preset =	For an MC counter data point, the accumulated value is equal to or greater than the alarm trip-point established by the <u>PRESET</u> (Preset Value) parameter.
Offnorm =	For a digital data point, the state of the point has changed from the normal state.
Openther =	Open thermocouple has been detected at an LEPIU or LLPIU for one of the following input/sensor types: Sensr501-506, Sensr401-406.
Flag =	MC flag data point has been set.
Counter =	For an HLPIU counter data point, the accumulated value is equal to or greater than preset value established by the <u>PRESET</u> parameter.
Chngofst =	Change-of-state alarm has been detected at a digital data point; refer to definition of <u>DIGALFMT</u> .
Unreasbl =	Unreasonable alarm condition; low alarm trip point has been set above the high alarm trip point, or a data point has been taken off scan, or the A/D converter has failed.
Cmddis =	Command disagree alarm has been detected at a digital composite data point.
Compshed =	Occurs after a predefined time (TOG1 or TOG2) has passed without a signal from a host computer.

NOTE

The Counter and Chngofst alarm states are not shown on group or detail displays because there is no return-to-normal.

S4logic =	Output of gate 4 is TRUE for an EC regulatory data point that has been configured for the Logic algorithm.
Offset1 =	Offset1 alarm has been detected for an EC regulatory data point that has been configured for the Rampsoak algorithm.
Offset2 =	Offset2 alarm has been detected for an EC regulatory data point that has been configured for the Rampsoak algorithm.
Offset3 =	Offset3 alarm has been detected for an EC regulatory data point that has been configured for the Rampsoak algorithm.
Pvrocn =	PV Rate-Of-Change Negative; applies to only an MC regulatory data point.
Pvrocp =	PV Rate-Of-Change Positive alarm; applies to only an MC regulatory data point.
Pvroc =	PV Rate-Of-Change alarm; applies to only an EC regulatory data point.
Devlo =	Deviation Lo alarm.
Devhi =	Deviation Hi alarm.
Devll =	Deviation Lo Lo alarm; applies to only an EC regulatory data point.
Devhh =	Deviation Hi Hi alarm; applies to only an EC regulatory data point.
Devlll =	Deviation Lo Lo Lo alarm; applies to only an EC regulatory data point.

BADPVPR

Devhhh =	Deviation Hi Hi Hi alarm; applies to only an EC regulatory data point.
Pvlo =	PV Lo alarm.
Pvhi =	PV Hi alarm.
Pvll =	PV Lo Lo alarm; applies to only an EC regulatory data point.
Pvhh =	PV Hi Hi alarm; applies to only an EC regulatory data point.
Pvlll =	PV Lo Lo Lo alarm; applies to only an EC regulatory data point.
Pvhhh =	PV Hi Hi Hi alarm; applies to only an EC regulatory data point.
Badpv =	Indicates that the PV is outside of the configured range.

BATCHID—Batch Identifier

This parameter is the name of the current batch that is executing in the specified MC.

Source:		Default	Access	
	System	Value:	All Underscores	Lock: DEB

Value
Type: String

BH—Batch History Point ID

Reserved for possible future use.

BIAS—Bias (CB, EC, & MC Regulatory Data Points)

This parameter defines the bias value that is added to the setpoint (Y) input value on regulatory points when MODE contains Cas.

Source:	User	Default Value:	0	Access Lock:	Oper
Value Type:	Real	Value Range:	For all algorithms except Extratbi : -99.99 to 99.99		
			For Extratbi algorithm: -100.0 to 100.0		

NOTE

Mode attributes and the control interlocks determine whether and by whom the value in this parameter can be updated. Refer to 4.4.3 in *System Control Functions* in the *Implementation/Startup and Reconfiguration* binder.

BO—BO (Output Bias; EC Regulatory Data Point)

This parameter defines the output bias for an EC regulatory data point that has been configured for the **Automan**, **Ecsumr**, **Leadlagm**, **Leadlags**, **Multdiv**, or **Sqrtprod** algorithm.

Source:	User	Default Value:	0.0	Access Key:	Supvr.
Value Type:	Real	Value Range:	-100 to 100		

BOXASSN—Box Assignment

For an HG, this parameter establishes the relationships between the HGs on the Data Hiway. For a box on the hiway, this parameter defines the box to the HGs. For more detailed information, refer to subsections 1.5.3 and 1.5.4 in *Hiway Gateway Implementation Guidelines* in the *Implementation/Hiway Gateway - 1* binder.

Source:	User	Default Value:	ThisHG	Access Lock:	DEB
Value Type:	Enum. of BOXASSN	Value Range:	ThisHG = For an HG, the box number (address) being configured belongs to this HG, not another HG on the same hiway. For a box, the HG that is configured as ThisHG provides support for all of this box's functions. AddedHG = For an HG, the box number (address) being configured belongs to an added HG on the same LCN as ThisHG. The added HG is used to share the processing load of ThisHG. For a box, the HG that is configured as AddedHG provides support for this box. RemoteHG = For an HG, the box number (address) being configured belongs to an HG on this hiway, but on another LCN. For a box, the HG that is configured as ThisHG provides limited support for this box. This box is fully supported by the HG that is configured as RemoteHG. H4500 = Box only. The HG that is configured as ThisHG provides limited support for this box. The box is shared by a Honeywell 4500 or 45000 process computer or a similar preferred-access device on the hiway.		

BOXBUFFS(n)—Box Report Buffers or Circular Lists (Processor Status Data Point)

An array of n values, where n is the address of each box on the highway served by this HG. Each array value contains the number of report buffers or circular lists that is reported by the hiway box whose address is n, during the last 15 second sample period.

Source:	System	Default Value:	0	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 to the maximum number of report buffers or circular lists.		

BOXINIT—Box Cache Initialization (Box Data Point)

This parameter requests that the HG cache values for the box are to be sent to the HG.

Source:	User	Default Value:	Init	Access Lock:	Open
Value Type:	Enum. of BOXINIT	Value Range:	NoInit = Init =	Default State There is no request to update the cache values for the box. Cached update is requested for the box. The value is set to NoInit after the cache is updated.	

NOTE

You can request cache updates from a schematic or a CL control program. The box functional states can be READ, FULL, or BASIC; the box assignment can be ThisHG, ADDED, RemoteHG, or 4500. You must ensure that the box is in a RUN state. The HG can detect or set change flags. The cache data, which is uploaded to the HG, can include the engineering unit range and characterization, alarm format, alarm mask, and control algorithm data.

BOXCLINK—C-Link Number (MC Box Data Point)

This parameter defines the number of the C-Link to which this MC box is connected.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1 to 8		

BOXCRDST—Box Card Status

This parameter indicates the status of an I/O card in the specified hiway box.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of BOXCRDST	Value Range:	Uncertn = Hiway security tests have not yet been performed to determine card status during startup of the hiway box. After the hiway security tests have been performed, each card will be in one of the following two states. Running = Card is operational (OK). Failed = Card has failed, as detected by the process box.		

BOXFSTAT—Box Functional State

This parameter defines the functional state of a box. BOXFSTAT can be overridden by HWYCTLST (Hiway Control State).

Source:	System	Default Value:	N/A	Access Lock:	Supvr.
Value Type:	Enum. of BOXFSTAT	Value Range:	Read = AM can read data from this box, but cannot write data to this box. Full = AM can read data from this box, and can write data to this box. Basic = AM can read data from this box, but cannot write data to this box. Test = Box is being tested.		

BOXNUM

BOXNUM—Box Number or Input Box Number

The box number defines the physical address of the hiway box (in which this data point resides) on the Data Hiway. The box numbers are initially established in parameter BOXNUM on the HG-Box Point Assignment Display. For an input data point or composite data point, the input box number defines the box in which the input portion of the data point resides. Refer to the applicable box data point configuration form for this information.

Source:	User	Default Value:	00	Access Lock:	DEB
Value Type:	Integer	Value Range:	1-63		

NOTE

For a composite data point, the input boxnumber and the output boxnumber must be the same number.

BOXPROT—Box Protocol (DHP Box Data Point)

This parameter defines the type of protocol (message format) that is to be used by the programmable controllers that are connected to this DHP.

Source:	User	Default Value:	Allenbrd	Access Lock:	DEB
Value Type:	Enum. of BOXPROT	Value Range:	Allenbrd = Allen-Bradley Modicon = Modicon Honywell = Honeywell/ISSC Notconfg = Protocol is not configured		

NOTE

If GE PCs are to be used, select Modicon for this parameter.

BOXRCDA—Reserve Controller Director Box Number

This parameter defines the address of the Reserve Controller Director (RCD) on the Data Hiway.

Source:	System	Default Value:	0	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 to 63		

BOXSIZE

BOXSIZE —DHP or PIU Box Size (Box Data Point)

This parameter defines the number of slots in a box.

Source:	User	Default Value:	Regular	Access Lock:	DEB
Value Type:	Enum. of BOXSIZE	Value Range:	For the DHP: Regular = 15 slots (slots 1-15) Extended = 30 slots (slots 1-15 & 17-31) For the HLPIU, LEPIU, and LLPIU: Regular = 16 slots (slots 1-16) Extended = 32 slots (slots 1-32)		

NOTE

A single 620-HIM can contain up-to-120 slots that are divided among four extended boxes (30 slots per extended box).

BOXSTART—Box Startup (DHP Box Data Point)

This parameter defines the state the box is to enter when it is started up.

Source:	User	Default Value:	Cold	Access Lock:	DEB
Value Type:	Enum. of BOXSTART	Value Range:	Cold = DHP does not enter its normal operating state after startup; it remains in a reset state. Hot = DHP enters its normal processing state after startup.		

BOXSTAT—Box Status

This parameter defines the status of the particular hiway box.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of BOXSTAT				
Value Range:	Uncertn = Box status has not yet been determined by the hiway security tests during box startup. Run = Box is OK and is in the processing state. Failcom = Communication failure has been detected on the hiway; box is not responding. Failhdw = Hiway address error or continued communication error has been detected. Runprtl = A failure has been detected in a card during normal operation; box is still OK. Runsftrl = A soft failure such as A/D drift has been detected during normal operation; box is still OK. Idle = Box is in idle state. Idlesft = A soft failure has been detected while the box is in the idle state. Reset = Box is in reset state. Idleprt = A partial failure has been detected while the box is in the idle state.				

BOXSTRT — Timer Start Array (MC Process Module Data Point)

This parameter allows the operator to start or stop a selected timer data point that is listed on the Process Module Detail Display.

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Enum. of STRTSTOP	Value Range:	Start = Starts timer data point. Stop = Stops timer data point.		

BOXTOG1 — TOG Interval 1 (Box Data Points)
BOXTOG2 — TOG Interval 2

These parameters allow the user to define two time-out intervals (in seconds), either of which can be used as the time-out interval for a particular slot in this box as specified by the TOGINTSL (Slot TOG Interval) parameter. The time-out interval is the length of time between two successive transmissions from a supervisory controller to a particular slot in this hiway box. If the time-out interval is exceeded, the slot sheds from the remote cascade mode to the shed mode.

If you do not want this slot to shed, enter a 0 for this parameter.

Source:	User	Default Value:	0 seconds (No Shed)	Access Lock:	DEB
Value Type:	Integer	Value Range:	0 to 9999 seconds		

BOXTREND — Box Trending (CB and MC Box Data Points)

This parameter defines whether this box has the trend option.

Source:	User	Default Value:	Notrend	Access Lock:	DEB
Value Type:	Enum. of BOXTREN	Value Range:	Notrend = Trend =	Box does not have the trend option. Box has the trend option.	

NOTE

MCs at revision 20 can't have both sequence memory and trend memory. If MCREV contains Rev20 and SOPLMEN contains Sopl, this parameter can't be configured with Trend.

BOXTYPE — Box Type (Box Data Point)

This parameter defines the type of box on the Data Hiway.

Source:	User	Default Value:	Notconfig	Access Lock:	DEB
Value Type:	Enum. of BOXTYPE	Value Range:	Notconfig = This entry is used to delete a box from the hiway. Cb = Basic Controller Au = Analog Unit Hlpiu = High-Level Process Interface Unit Lepiu = Low-Energy Process Interface Unit Llpui = Low-Level Process Interface Unit Cbrcd = Basic Controller Reserve Controller Director Ec = Extended Controller Ecrd = Extended Controller Reserve Controller Director Mc = Multifunction Controller Mrcd = Multifunction Controller Reserve Controller Director Dhp = Data Hiway Port/IPCD 620 Hiway Interface Module Opsta = Operator Station, Basic or Enhanced H4500 = Honeywell 4500 Computer HG = Hiway Gateway GPCI = General Purpose Computer Interface		

NOTE

The values for this parameter include devices (Au, Opsta, H4500, HG, and GPCI) that are not LCN-supported process-connected boxes. These values are included so that the HG is aware of such devices, and can include them on the Hiway Status Display.

BOXUAC — Uninterrupted Automatic Control State

This parameter defines the state of a box in relationship to its Reserve Controller Director (RCD).

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of BOXUAC	Value Range:	None = This controller is not being backed up and testing is not being performed. Softsave = This controller is being backed up because of a soft failure. Saved = This controller is being backed up because it has failed. Tested = This controller is being backed up during a test.		

NOTE

Softsave and Tested can be overridden by a failure of a controller.

BOXVDB — Box Variable Alarm Deadband

This parameter defines whether an LLPIU or DHP box will use the Variable Alarm Deadband option. Variable Alarm Deadband allows the setting of a variable deadband value in percent of engineering units for analog input points instead of using the default setting of 1% of engineering unit range.

The variable deadband value may be entered in the PVALDB parameter. This option requires special variable deadband firmware to be installed in the LLPIU (part#38120017) and/or DHP (part#38120019).

Source:	User	Default Value:	Off	Access Lock:	Eng
Value Type:	Boolean				
Value Range:	Off = Variable alarm deadband processing is turned off. PVALDB will be initialized to 1% when special deadband firmware is present on a box load and BOXVDB is OFF. On = Variable alarm deadband processing is turned on. Allows entry of variable deadband values in PVALDB. Must have BOXDVB set to ON before a box load if PVALDB values are NOT to be initialized. BOXSIZE is required to be EXTENDED and the special variable deadband firmware must be present before BOXVDB can be set ON.				

BX — BX (Bias Constant on X Input; EC Regulatory Data Point)

This parameter defines the bias constant on the X input for an EC regulatory data point that has been configured for the **Multdiv** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-100.0 to 100.0		

BX — BX (Ratio Bias; EC Regulatory Data Point)

This parameter defines the ratio bias for an EC regulatory data point that has been configured for the **Extratbi** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-100.0 to 100.0		

BXACCESS (N) — # of Box Accesses (HG Processor Status Data Points)

This parameter defines the number of box accesses to hiway address N during the last 15-second sample period.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	ARRAY [1...63] of REAL	Value Range:	≥ 0.0		

BY — BY (Bias Constant on Y Input; EC Regulatory Data Point)

This parameter defines the bias constant on the Y input for an EC regulatory data point that has been configured for the **Massflow** or **Multdiv** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	For Massflow algorithm: 0.0 to 100.0 For Multdiv algorithm: -100.0 to 100.0		

BZ — BZ (Bias Bias; EC Regulatory Data Point)

This parameter defines the bias for an EC regulatory data point that has been configured for the **Extratbi** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-100.0 to 100.0		

BZ — BZ (Bias Constant on Z Input); EC Regulatory Data Point)

This parameter defines the bias constant on the Z input for an EC regulatory data point that has been configured for the **Massflow** or **Multdiv** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	For Massflow algorithm: 0.0 to 100.0		
			For Multdiv algorithm: -100.0 to 100.0		

-C-

C — C Input Value (EC Regulatory Data Point)

This parameter is the current value (in percent) of the C signal at the specified EC regulatory data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	-6.86% to 106.86%		

CABLESTS – Cable Status

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 - 4		

CALIBOFF — Calibration Offset

This parameter allows the user to specify a positive or negative offset value (in engineering units) to adjust the value of the PV input received from the field, thereby eliminating minor variations that may exist between the same types of field devices.

Source:	User	Default Value:	0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9999 to 9999		

NOTE

The PV value used for DHP alarming is not adjusted by this offset. Hence a point can be shown in alarm, even though it is displayed (adjusted) PV is within limits.

CASREQ

CASREQ — Remote Cascade Mode Request Flag

This parameter indicates that the remote cascade mode for has been requested for a data point. The remote-cascade mode exists when MODE contains Cas and RCASENB contains On. When a request to change MODE to Cas is received from a US or a program, MODE doesn't immediately change to Cas. Instead, CASREQ contains Request and "-C" appears to the right of the mode indicator on the Group and Detail displays. Then when continuous control in an AM determines that CASREQ contains Request, it requests MODE to go to Cas, and changes CASREQ to Notreq.

Should the point shed while it is in the remote-cascade mode, MODE goes to the state defined in SHEDMODE, and CASREQ goes back to Request.

Source:	System	Default Value:	Notreq	Access Lock:	Program
Value Type:	Enum. of CASREQ	Value Range:	Notreq = Remote cascade mode request flag is not set (operator has not requested cascade mode or <u>RCASENB</u> is false). Request = Remote cascade mode request flag is set (operator has requested cascade mode and <u>RCASENB</u> is true).		

NOTES

CASREQ can be updated only when RCASENB contains On.

If when a request to from a US or program is received while MODE is already Cas, CASREQ does not go to Request because the point is already in the remote-cascade mode.

If CASREQ contains Request ("-C" is to the right of the mode indicator), and a BASIC Operator Station on the hiway changes the mode to "computer," the HG changes CASREQ to Notreq and "-C" is removed from the displays.

When this point's CASREQ value is changed, the HG updates CASREQ in all aliases of this point, to the same value as that of this point.

CASREQS — Remote Cascade Mode Request Source

This parameter indicates whether the cascade mode request flag has been set by the operator, or by the fact that a regulatory data point is shedding from the cascade mode to the shed mode.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of CASREQS	Value Range:	MMI = Cascade mode has been requested by the Operator. Shed = Regulatory data point has shed from the cascade mode.		

CBREV — Basic Controller Revision Number

This parameter defines the revision level of the specified CB box.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of CBREV	Value Range:	Notconfg = Revision level has not been determined because hiway security tests have not yet been performed during box startup. Cb1c = CB is revision 1C. Cb1d = Cb is revision 1D.		

Cbsumr — Summer (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Cbsumr** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Cbsumr** algorithm is that it is a 2-input algorithm whose output is the sum of two independently scaled inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC INITCONF OPLOLM OPHILM \$OPTOL TD K K1 K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Cbsumr — Summer (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Cbsumr** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on MC Regulatory Data Point Configuration Form (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Cbsumr** algorithm is that it is a 2-input algorithm whose output is the sum of two independently scaled inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC INITCONF SPLOLM SPHILM \$SPTOL OPLOLM OPHILM \$OPTOL TD K K1 K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

CCPRIPNT — Primary Point

This parameter specifies the tag name of the primary data point in a contact-cutout chain that is to report the alarm condition to the system. An entry has to be made for this parameter when the selection for the CCRANK parameter is Secondary. Refer to the definition of NAME (Tag Name) for the valid character set.

Source:	User	Default Value:	All Underscores	Access Lock:	DEB/Engr.
Value Type:	Entity ID	Value Range:	See <u>NAME</u>		

NOTE

To update the value in this parameter, CCRANK must contain Secondary and CCRANK of the point named by the CCPRIPNT value in this point must contain Primary.

When updating the default value, underscores cannot be reentered once it has been changed.

CCRANK — Contact-Cutout Rank

This parameter allows the user to specify whether a data point is to be part of a contact-cutout chain, and its position in the chain (primary or secondary). The contact-cutout function allows multiple secondary data points to be assigned to a primary data point (specified by CCPRIPNT) so that if the primary data point goes into an alarm condition, any alarms at the secondary data points are ignored (cut out) by the system. This prevents multiple alarms from being generated because one data point goes into an alarm condition. The primary data point resides in the same HG (on the same Data Hiway) as the secondary data points.

Source:	User	Default Value:	Neither	Access Lock:	DEB
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Value Type: Enum. of CCRANK

Value Range:

- Neither = This data point is not part of a contact-cutout chain.
- Primary = This data point is to report the alarm condition to the system.
- Secondary = This data point is to report the alarm condition to the primary data point specified by CCPRIPNT (Primary Point) that resides in the same HG (on the same Data Hiway) as the primary data point. If Secondary is entered for this parameter during configuration, also make an entry in the CCPRIPNT line to identify the tag name of the primary point.

CDIGST — C Input Digital State (EC Regulatory Data Point)

This parameter is the current digital state of the C input signal at an EC regulatory data point that has been configured for the **Logic** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Current state of the C input is true. False = Current state of the C input is false.		

CDTEXT — Command Disagree Text (Data Hiway Data Point)

This parameter defines a descriptor of up to eight characters that will be displayed for an MC digital composite data point that has been configured for Cmmdis (Command Disagree alarming).

Source:	User	Default Value:	Cmd_Dis	Access Lock:	DEB
Value Type:	String	Value Range:	The permissible character set consists of 11 characters all characters on the Engineer's Keyboard. Basically this set consists of Alphabetics A-Z, Numerics 0-9 and the following special characters: space ! % & ' () * + - / : ; > < = ? __ , . \$		

Charact — Characterizer (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Charact** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Charact** algorithm is that it is an 8-linear-segment signal-characterizing auxiliary algorithm.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHHPR \$HHSELCT PVHIPR \$HISELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM	SUPPIO OUTIND PVSIGNAL PVSLTSRC INITCONF SOASIGNL SOASRC SOBSIGNL SOBSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD SEGTOT OUTPT0	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

CHNGFLAG

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
			X1 OUTPT1 X2 OUTPT2 X3 OUTPT3 X4 OUTPT4 X5 OUTPT5 X6 OUTPT6 X7 OUTPT7 OUTPT8	

CHNGFLAG — Change Flag Detection

This parameter defines whether changes are made in the hiway boxes (except for the MC) by preferred devices or by LCN devices.

Source:	User	Default Value:	Notconfig	Access Lock:	DEB
Value Type:	Enum. of CHNGFLAG	Value Range:	Notconfig = No preferred devices are on this Data Hiway (speeds up communication). Detect = Enables HG to alert LCN devices such as the Universal Stations to changes in box status that originated in preferred devices on the same hiway. Set = Enables LCN devices such as Universal Stations to alert preferred devices to changes it makes in the box on the same Data Hiway.		

NOTE

A preferred device on the Data Hiway can be an operator station, a GPCI, or an H4500.

CHOFSTPR—Change of State Alarm Priority

This parameter defines the priority of the Change of State alarm generated by a data point.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			
		NOTE			
		If PRINTER is chosen, the alarm is lost if the printer is not working.			
	JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.			

CHPINHWY — Checkpoint Hiway Inhibit (Data Hiway Data Point)

This parameter enables the engineer to specify whether checkpointing is allowed for boxes on the hiway. CHPINHWY overrides CHPINOPR.

Source:	User	Default Value:	Enable (supplied by DEB)	Access Lock:	Engr.
Value Type:	Enum. of CHPINDAC	Value Range:	Enable = Checkpointing for the hiway boxes is allowed. Inhibit = Checkpointing for the hiway boxes is not allowed.		

CHPINOPR — Checkpoint Inhibit (Operator Generated)

This parameter allows the engineer to specify whether checkpointing is allowed for this box.

Source:	User	Default Value:	Enable (supplied by DEB)	Access Lock:	Engr.
Value Type:	Enum. of CHPINDAC	Value Range:	Enable = Checkpointing is allowed for this box. Inhibit = Checkpointing is not allowed for this box.		

CLEAR — Automatic Clear of Logic Algorithm Gate Option (EC Regulatory Data Point)

This parameter provides the supervisor with a one-shot initializing entry for resetting a latch-gate or pulse-gate output of an EC regulatory point that is using the Logic algorithm. After the point is next processed, the value in CLEAR returns to zero.

Source:	System	Default Value:	0	Access Lock:	Supvr.
Value Type:	Integer	Value Range:	0 = Clear has not been issued by the supervisor. 1 = Clear has been issued by the supervisor.		

CLINKNUM — C Link Number (MC Box Data Point)

This parameter defines the address of the MC box on the C link.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1 to 8		

CMDDISPR — Command Disagree Alarm Priority

This parameter defines the priority of a command disagree alarm generated by a data point.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
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Value Type:	Enum. of ALPRIOR
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Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.
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CMPSHDPR — Computer Shed Alarm Priority

This parameter defines the priority of a computer shed alarm generated by an EC regulatory data point in cascade mode.

Source:	User	Default Value:	Low	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.
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CNFERRPR — Configuration Error Alarm Priority

This parameter defines the priority of a configuration error alarm generated by a data point.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			
NOTE					
If PRINTER is chosen, the alarm is lost if the printer is not working.					
	JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.			

CNTLLOCK — Operator Control Lock (MC Counter and Timer Data Points)

This parameter defines whether the operator is permitted to start, stop, and reset the counter or timer.

Source:	User	Default Value:	Permit	Access Lock:	Engr.
Value Type:	Enum. of MODEPERM	Value Range:	Permit =	Operator is allowed to start, stop, and reset the counter or timer.	
			Nopermit =	Operator cannot start, stop, and reset the counter or timer.	

CONFMSG — Confirm Message Request (MC Process Module Data Point)

This parameter is used to confirm messages pending in an MC process module data point. Up to two messages can be pending at any given time.

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Integer	Value Range:	0 = Message with sequence bit = 0 is confirmed. 1 = Message with sequence bit = 1 is confirmed.		

CONFTRND — Trend Memory Configuration (MC Box Data Point)

This parameter defines the types of values that are to be used to build trends.

Source:	User	Default Value:	Average	Access Lock:	Engr.
Value Type:	Enum. of CONFTRND	Value Range:	Average = Average values are to be used. Instant = Instantaneous (snapshot) values are to be used.		

COUNTRPR — Counter Overflow Alarm Priority

This parameter defines the priority of a counter overflow alarm generated by a data point.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.
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CPTIMEFL – Time required to checkpoint to a Zip drive (R660)

This PSDP parameter indicates the time required to checkpoint to a Zip drive.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0.0		

CPTIMEHM – Time required to checkpoint to an HM (R660)

This PSDP parameter indicates the time required to checkpoint to an HM.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	>=0.0		

CPUFREE — Central Processing Units' Free Time (All Processor Status Data Points)

This parameter defines the percentage of free CPU time averaged over a 10-second sample period. (Note: Although the calculation of this value takes into account different processor types, it does not distinguish between the various memory board types; therefore, the significance attached to the value should be indicative, rather than absolute.)

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	0.0 to 100.0		

CPUMAX — Central Processing Units' Maximum Free Time (All Processor Status Data Points)

This parameter defines the maximum percentage of free CPU time averaged over a 15-second sample for the current hour. (Note: Although the calculation of this value takes into account different processor types, it does not distinguish between the various memory board types; therefore, the significance attached to the value should be indicative, rather than absolute.)

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	.0 to 100.0		

CpumIn — Central Processing Units' Minimum Free Time (All Processor Status Data Points)

This parameter defines the minimum percentage of free CPU time averaged over a 15-second sample for the current hour. (Note: Although the calculation of this value takes into account different processor types, it does not distinguish between the various memory board types; therefore, the significance attached to the value should be indicative, rather than absolute.)

Source:	System	Default Value:	100.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	0.0 to 100.0		

CRITSCAN — Critical Scanning (at a .5 second rate of up to 50 points)

Allows user to select up through 50 points to be scanned at a .5 second rate AND scanned first on Startup and Failover.

Source:	User	Default Value:	OFF	Access Lock:	Engr/Deb
Value		Value			
Type:	Boolean	Range:	OFF = Will not be scanned every .5 seconds OR first on Startup and Failover.		
			ON = Point will be scanned at a .5 second rate AND		scanned first on Startup and Failover.

CAUTION

Do not configure CRITSCAN for Sequence of Events digital input points because CRITSCAN scanning alters the sequence-of-events time stamp, making it meaningless.

CSIGNAL — C-Signal Type (EC Regulatory Data Point)

The C-signal types are the same as those listed for the Z SIGNAL parameter except that this signal is applied to the C input rather than the Z input. The tag name of the data point that provides the input signal to the C input is specified by the CSLTSRC parameter. Refer to the definition of the Z SIGNAL parameter.

CSLTSRC — C-Input Source (EC Regulatory Data Point)

This parameter defines the tag name of the data point that provides the input signal to the C input of an EC regulatory data point. Refer to the definition of NAME (Tag Name) for the valid character set.

Default

Value: Tag name from Line 1 of configuration form

CTLACTN — Control Action (CB, EC, and MC Regulatory Data Points)

This parameter defines whether this data point has direct or reverse control action.

Source:	User	Default Value:	Direct	Access Lock:	Engr.
Value Type:	Enum. of POLARITY	Value Range:	Direct = Reverse =	The value of the output signal from this data point increases as the PV input signal to this data point increases. The value of the output signal from this data point decreases as the PV input signal to this data point increases.	

NOTE

To update the value in this parameter, MODE must contain Man. For EC Regulatory points, EMS must contain Disable. For CB and MC regulatory points, INIT must contain Noinit, and if ALGIDDAC contains Overhisl or Overlosl, CTLACTN must contain Direct.

CTLEQN — Control Equation (CB, EC, MC Regulatory Data Points)

This parameter defines the control equation that is used for a regulatory data point. Refer to the parameter listing for the particular algorithm (**Pidnorm**, **Pidratio**, etc.) in this dictionary to determine which equations are applicable to the algorithm. The applicable equations are in parenthesis next to the CTLEQN parameter in each algorithm parameter listing.

Source:	User	Default Value:	EQA	Access Lock:	Engr.
Value Type:	Enum. of ALGOEQN	Value Range:	EQA = Equation A EQB = Equation B EQC = Equation C		

NOTE

To update the value in this parameter, MODE must contain Man and INIT must contain Noinit.

-D-

Das — Data Acquisition (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the Das algorithm are contained in the following listing. These parameters are listed in the same order as they appear on CB Regulatory Data Point Configuration Form (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the Das algorithm is that it is used for data point display and/or manual output generation.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND \$OPTOL PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC TD	NMODATTR MODE \$MODESEL

Das — Data Acquisition (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the Das algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry Displays. The outstanding characteristic of the Das algorithm is that it is used for data point display and/or manual output generation.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR	SUPPIO OUTIND \$OPTOL PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC SPCHAR SPFORMAT	NMODATTR MODEPERM MODE \$MODESEL

Das (EC)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
PTDISCL	OVERVAL PVRNGOP	\$DHSELECT DEVLLPR DEVLOPR \$DLSELECT PVHHPR \$HHSELECT PVHIPR \$HISELCT PVLLPR \$LLSELECT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SPEULO	

Das (MC)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
SLOTNUM LOADDEST ALGIDDAC	PVCLAMP CALIBOFF		SPEUHI SOASIGNL SOASRC SOBSIGNL SOBSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP TFCLLM TFCHLM TD	

Das — Data Acquisition (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Das** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Das** algorithm is that it is used for data point display and/or manual-output generation.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSELCT	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND \$OPTOL PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC	NMODATTR MODE \$MODESEL

Das (MC)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
		PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB		

DDTGRAT (N)

DDTGRAT (N) — Rate of Data Definition Tables Get Requests (CIU Processor Status Data Points)

This parameter defines the "Data Definition Table Get Rate." The DDT parameter get rate is based on the number of DDT get requests from the ULP, where

N = 1, count for the most recent (full) 1-minute sample period;

N = 2, count for the previous 1-minute sample period; and

N = 3 to 10, reserved for use by Honeywell IAC.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	ARRAY [1..10] of REAL	Value Range:	≥ 0.0		

DDTSRAT (N) — Data Definition Tables Store Rate (CIU Processor Status Data Points)

This parameter defines the "Data Definition Table Store Rate." The DDT parameter store rate is based on the number of DDT store requests from the ULP, where

N = 1, count for the most recent (full) 1-minute sample period;

N = 2, count for the previous 1-minute sample period; and

N = 3 to 10, reserved for use by Honeywell IAC.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	ARRAY [1..10] of REAL	Value Range:	≥ 0.0		

DEADBAND — Deadband (EC Regulatory Data Point)

This parameter defines the deadband in percent for an EC regulatory data point that has been configured for the **Logic** algorithm. Make an entry for this parameter only if Compare was entered for a GATETYP1-GATETYP3 parameter.

Source:	User	Default Value:	0.0%	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0% to 113.7%		

DEADTIME

DEADTIME — Dead Time (EC Regulatory Data Point)

This parameter defines the dead time in minutes for an EC regulatory data point that has been configured for the **Deadtm** algorithm.

Source:	User	Default Value:	0.0 minutes	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0 to 120 minutes		

Deadtm — Dead Time (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Deadtm** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Deadtm** algorithm is that it is a single input auxiliary algorithm, the output of which is delayed by a user-set dead-time constant.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	\$AUXUNIT	SUPPIO	NMODE
UNIT	PVCHAR	ALFMT	OUTIND	NMODATTR
PTDESC	INPTCOND	\$ALDLOPT	\$OPTOL	PSTMODE
KEYWORD	PVFORMAT	\$DLBADAL	PVSIGNAL	MODEPERM
PRIMMOD	PVTEMP	DLYTIME	PVSLT SRC	MODE
ASSOCDSP	PVEULO	\$REALARM	INITCONF	\$MODESEL
HWYNUM	PVEUHI	BADPVPR	SOASIGNL	
PNTBOXTY	PVDSPHI	DEVHHPR	SOASRC	
BOXNUM	PVDSPLO	DEVHIPR	SOBSIGNL	
PTDISCL	OVERVAL	\$DHSELCT	SOBSRC	
		DEVLLPR	EIPENB	
		DEVLOPR	ALMASK	
		\$DLSELCT		
		PVHHPR		
		\$HHSELCT		
		PVHIPR		
		\$HSELCT		
		PVLLPR		
		\$LLSELCT		
		PVLOPR		
		\$LOSELCT		
		(Alarm		
		Trip-point		
		Values)		
		ALPRIOR		
		CCRANK		
		CCPRIPNT		
		EIPPCODE		

DEADTIME

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
		EIPEVENT		
		EIPENB		
		ALMASK		
		INITCONF		
		SPHILM		
		SPLOLM		
		\$SPTOL		

Deadtm

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
SLOTNUM LOADDEST ALGIDDAC	PVCLAMP CALIBOFF		MSIGNAL MSLTsrc ASIGNAL ASLTsrc CSIGNAL CSLTsrc GREENSGL GREENsrc SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD DEADTIME	

DEV — Deviation

This parameter is the current deviation (PV minus SP) in percent at the specified regulatory data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	-113.7% to 113.7%		

DEVHHHTP

DEVHHHTP — Deviation High High High Trip Point
DEVHHTP — Deviation High High Trip Point
DEVHITP — Deviation High Trip Point
DEVLLLTP — Deviation Low Low Low Trip Point
DEVLLTP — Deviation Low Low Trip Point
DEVLOTP — Deviation Low Trip Point

These parameters define the high and low values (in percent) of the deviation that cause an alarm condition at a data point. These parameters may or may not have to be configured depending on the selection for the ALFMT parameter. DEVHITP and DEVLOTP are applicable to analog and regulatory data points. DEVHHTP, DEVHHHTP, DEVLLTP, and DEVLLLTP are applicable to only EC regulatory data points.

Source:	User	Default Value:	0.0%	Access Lock:	Supvr.
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Value Type:	Real	Value Range:	<u>DEVHITP</u> = 0% to 113.7 <u>DEVLOTP</u> = -113.7% to 0
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Ranges for DEVHHTP, DEVHHHTP, DEVLLTP, and DEVLLLTP are the same as described above for DEVHITP and DEVLOTP.

DEVHHHPR—Deviation High High High Alarm Priority
DEVHHPR—Deviation High High Alarm Priority
DEVHIPR—Deviation High High High Alarm Priority
DEVLOPR—Deviation Low Alarm Priority
DEVLLPR—Deviation Low Low Alarm Priority
DEVLLLPR—Deviation Low Low Low Alarm Priority

These parameters define the priority of the deviation high and low alarms generated by a data point.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
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Value Type:	Enum. of ALPRIOR
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Value		
Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.
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DHMSGSEC(i)

This parameter indicates the message rate (per second) in internal queues (1 to 5) in the last 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	>= 0.0		

DHPARSEC(i)

This parameter indicates the parameter rate (per second) in internal queues (1 to 5) in the last 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	:>= 0.0		

DIGALARM — Alarming Function (HLPIU-Digital Data Points)

This parameter defines whether or not alarm conditions are to be detected by a digital input or digital composite data point. This parameter is to be filled in only when Chngdect or SOE has been entered for the IPCRDOP (Point Card Option).

Source:	User	Default Value:	Enable	Access Lock:	DEB
Value Type:	Enum. of ENBLSTAT	Value Range:	Enable = Alarms are to be detected. Disable = Alarms are not to be detected.		

DIGALFMT — Alarm Format (Digital Data Points and MC Flag Points)

This parameter defines the alarm condition that is to be detected at a digital input data point or a digital composite data point.

For additional information about digital states, state text parameters, and state descriptors, refer to 2.3.1 in *Hiway Gateway Control Functions* in the *Implementation/Hiway Gateway - 1* binder.

WARNING

If you choose the Cmmdis (Command Disagree) alarm format, first set up command disagree alarming at any Basic or Enhanced Operator Station before setting it up at a Universal Station.

WARNING: For a point with aliases, modify the alarm format only when the point is not in alarm.

Source:	User	Default Value:	Chngofst	Access Lock:	Engr.
Value Type:	Enum. of DIGALFMT				

NOTE

For MC Flag points, the DIGALFMT parameter has a fixed value of Flag.

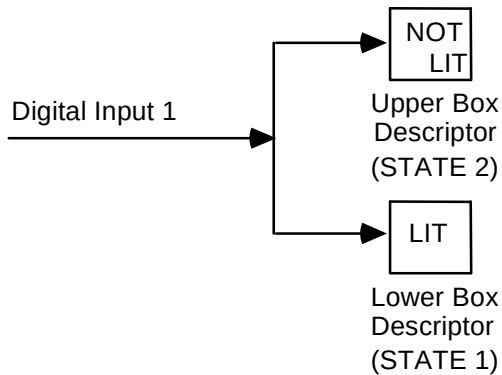
Value Range:	The following valid alarm formats are described using the upper and lower boxes that appear for a digital data point on the Group and Detail Displays. Refer to the configuration form to determine which alarm formats are applicable to the specific digital data point.				
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FOR A SINGLE-INPUT DIGITAL DATA POINT:

Valid alarm formats are Noalarm, State1, State2, Chngofst, and Cmmdis as follows:

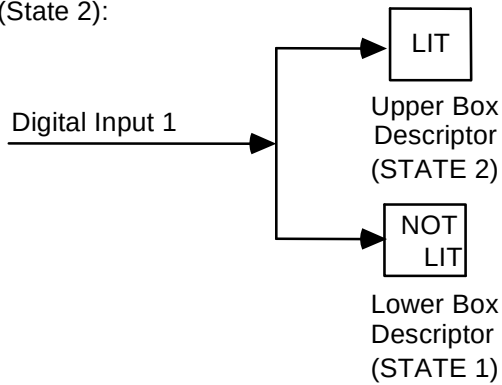
Noalarm = This data point does not have an alarm condition (applies to only MC digital points).

State1 (State 1):



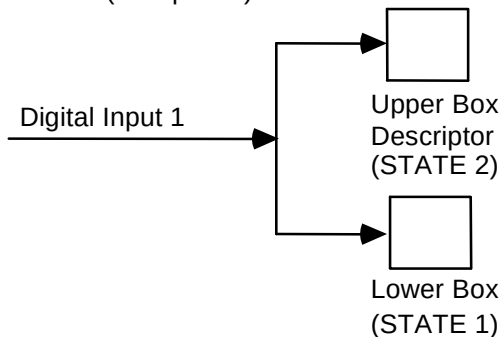
This data point is in an alarm condition when lower box lights because Input 1 contact is open (direct action) or because Input 1 contact is closed (reverse action).

State2 (State 2):



This data point is in an alarm condition when upper box lights because Input 1 contact is closed (direct action) or because Input 1 contact is open (reverse action).

Chngofst (Change of State):
(except MC)



This data point is in an alarm condition when the condition of either lighted box has changed because Input 1 contact has changed from closed to open or from open to closed. This alarm is annunciated to the operator and then is cleared after the alarm is acknowledged by the operator.

continued

Cmmdis (Command Disagree):

NOTES

- 1. This alarm format applies to only a digital-composite data point with the following output options:
HLPIU & MC—Latchout (Single or Dual Output)
HLPIU & MC—Pulseout (Single or Dual Output)
DHP—Latchout (Single or Dual Output)
(Momentary Output is not valid)
- 2. This alarm format is not applicable for an HLPIU-digital data point if the selection for thePIUCRDOP(Slots 1-32 Card Option) parameter during network configuration is Chngdect.
- 3. This alarm format is not applicable for an MC digital data point that has its Critica Scanning(CRITSCAN)parameter set to ON.

This type of digital data point is in an alarm condition when the input from the field drives the upper and lower boxes to a condition where they do not agree with the "commanded" state. The "commanded" state is determined by the action of the

 (Upper State Change) and  (Lower State Change) keys of the Operator's Keyboard as shown below.

OUTPUT

"COMMANDED" STATE

 + 


Upper Box
Descriptor
(STATE 2)

After the feedback time (transient time) has expired, if the actual input state does not drive the upper and lower boxes to the condition as shown, this data point is in an alarm condition.


Lower Box
Descriptor
(STATE 1)

 + 


Upper Box
Descriptor
(STATE 2)

(transient time) has expired, if the actual input state does not drive the upper and lower boxes to the condition as shown, this data point is in an alarm condition.


Lower Box
Descriptor
(STATE 1)

continued

No alarm condition can occur during the configured feedback time (transient time) period after an output command is issued

(by the  or  key and the **ENTER** key) to allow the affected device to respond.

An alarm condition can occur:

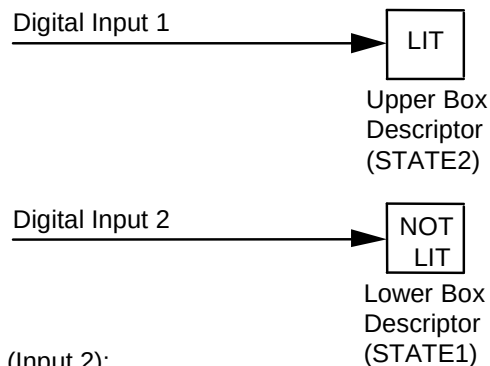
1. When a device does not correctly respond to an output command after the feedback time has expired.
2. Immediately upon a noncommanded change-of-state of the device (if the feedback time period has previously expired).

FOR A DUAL-INPUT DIGITAL DATA POINT:

Valid alarm formats are Noalarm, Input1, Input2, Either, Both, Chngofst, and Cmmdis as follows:

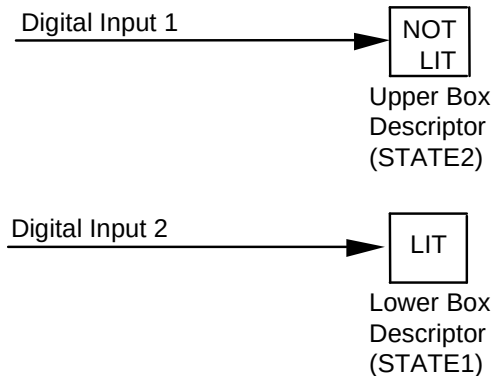
Noalarm This data point does not have an alarm condition (applies to only MC digital points).

Input1 (Input 1):



This data point is in an alarm condition when upper box lights because Input 1 contact is closed (direct action) or because Input 1 contact is open (reverse action).

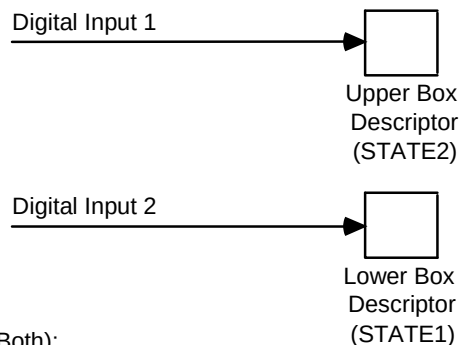
Input2 (Input 2):



This data point is in an alarm condition when lower box lights because Input 2 contact is closed (direct action) or because Input 2 contact is open (reverse action).

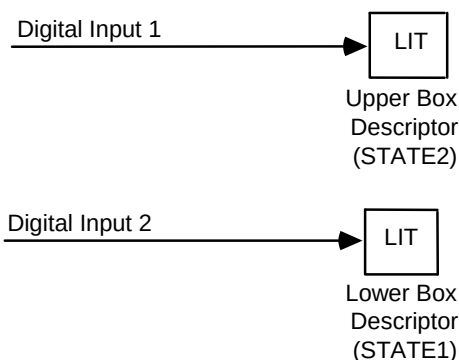
continued

Either (Either):



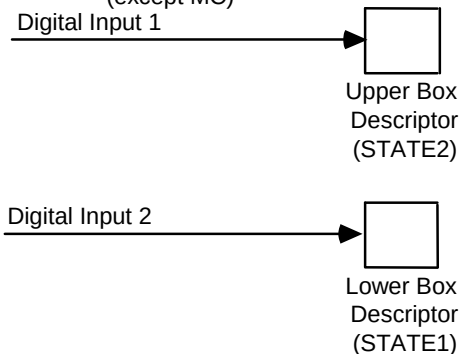
This data point is in an alarm condition when either the upper or lower box lights because the respective Input 1 or Input 2 contact is closed (direct action) or open (reverse action).

Both (Both):



This data point is in an alarm condition when both the upper and lower boxes are lighted because both Input 1 and Input 2 contacts are closed (direct action) or are open (reverse action).

Chngofst (Change of State):
(except MC)



This data point is in an alarm condition when the condition of either lighted box has changed because the Input 1 or Input 2 contact has changed from closed to open or from open to closed. This condition is annunciated to the operator and is acknowledged by the operator.

Cmmdis (Command Disagree):

NOTES

1. This alarm format applies to only a digital-composite data point with the following output options:
 HLPIU & MC—Latchout (Single or Dual Output)
 HLPIU & MC—Pulseout (Single or Dual Output)
 DHP—Latchout (Single or Dual Output)
2. This alarm format is not applicable for an HLPIU-digital data point if the selection for the PIUCRDOP (Slots 1-32 Card Option) parameter during network configuration is Chngdect.

continued

3. This alarm format is not applicable for an MC digital data point that has an alarm priority (ALPRIOR) of Emergency.

This type of digital data point is in an alarm condition when the inputs from the field drives the upper and lower boxes to a condition where they do not agree with the "commanded" state. The "commanded" state is determined by the action of the



(Upper State Change) and



(Lower State Change) keys

of the Operator's Keyboard as shown below.

OUTPUT



+ **ENTER**



+ **ENTER**

"COMMANDED" STATE



Upper Box
Descriptor
(STATE 2)



Lower Box
Descriptor
(STATE 1)

After the feedback time (transient time) has expired, if the actual input states do not drive the upper and lower boxes to the condition as shown, this data point is in an alarm condition.



Upper Box
Descriptor
(STATE 2)



Lower Box
Descriptor
(STATE 1)

After the feedback time (transient time) has expired, if the actual input states do not drive the upper and lower boxes to the condition as shown, this data point is in an alarm condition.

No alarm condition can occur during the configured feedback time (transient time) period after an output command is issued

(by the  or  key and the **ENTER** key) to allow the affected device to respond.

An alarm condition can occur:

1. When a device does not correctly respond to an output command after the feedback time has expired.
2. Immediately upon a noncommanded change-of-state of the device (if the feedback time period has previously expired).

DIGTEXT — Digital State Text (Flag, Digital, and Logic Block Data Points)

This parameter is the current state of the data point. Refer to definitions of STATE1 and STATE2 parameters.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	String				

DIGINTXT — Digital Input State Text (Digital Input and Composite Data Points)

This parameter is the current state of the input at a digital data point. Refer to definitions of STATE1 and STATE2 parameters.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	String				

DIGOUTXT — Digital Output State Text (Digital Output and Composite Data Points)

This parameter is the current state of the output at a digital data point. Refer to definitions of STATE1 and STATE2 parameters.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	String				

Dividr — Divider (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Dividr** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Dividr** algorithm is that it is a 2-input auxiliary algorithm whose output is the quotient of two inputs.

Dividr (MC)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND PVSIGNAL PVSLT SRC SPSIGNAL SPSLT SRC INITCONF OPLOLM OPHILM \$OPTOL TD K K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Dividr — Divider (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Dividr** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Dividr** algorithm is that it is a 2-input auxiliary algorithm whose output is the quotient of two inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROC NTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT	SUPPIO ANLG CARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLT SRC SPSIGNAL SPSLT SRC INITCONF SPLOLM SPHILM \$SPTOL OPLOLM OPHILM \$OPTOL ITLOLM TD	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Dividr (MC)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
		PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	K K2	

DLYTIME — (R600) Delay Time (All HG Digital Input and Digital Composite Points)

NOTE

Changing the DLYTIME value terminates the current alarm timer. The new value will be in effect for the next applicable alarm event.

This R600 parameter enables suppression of redundant alarms caused by chattering digitals by maintaining the original alarm for a specified time interval configured in the DLYTIME parameter.

Source:	User	Default Value:	0 seconds	Access Lock:	Supvr.
Value Type:	Integer (in seconds)	Value Range:	0 to 60 seconds		
Point Types:	HG Digital Inputs and Digital Composites				

NOTE

- A NaN or negative value is not allowed.
- When a point that has alarm suppression currently in progress is deleted, the alarm suppression will abort immediately.
- EIP functionality is not affected by the Delay Timer.

EXCEPTION

For a CHNGOFST alarm, if the PV is the same state when the delay timer expires as it was when the timer was started, future state changes are immediately alarmed. If it is in the opposite state, a second CHNGOFST alarm is produced and the delay timer is restarted.

DLYTIME — (R620) (Analog Input, Analog Composite & Regulatory Control)

NOTE

Changing the DLYTIME value terminates the current alarm timer. The new value will be in effect for the next applicable alarm event.

This R620 parameter enables suppression of redundant alarms caused by chattering analog values.

This parameter specifies the amount of time during which the alarm reporting (in the case of on delay) and return-to-normal reporting (in the case of off delay) is prevented. Alarms that return to normal during the time delay will not be reported. The timer will be restarted on either every transition into alarm (for on delay) or for every transition out of alarm (for off delay).

Source:	User	Default Value:	0 seconds	Access Lock:	Engineer
Value Type:	Integer (in seconds)	Value Range:	0 to 9999 seconds		
Point Types:	Analog inputs, Analog Composites and Regulatory Control				

WARNING: This parameter can be accessed through DATCHNG, Point Detail Display, DEB or AM/CL. MC/CL cannot access DLYTIME parameter since it is resident in HG.

NOTE

- A NaN or negative value is not allowed.
- When a point that has alarm suppression currently in progress is deleted, the alarm suppression will abort immediately.
- EIP functionality is not affected by the Delay option.

-E-

Ecsumr — Summer/Square Rooter (EC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Ecsumr** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Ecsumr** algorithm is that it is a 3-input algorithm whose output is the result of the summation. Optionally, all inputs can be square-rooted and/or PV square rooted.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLYTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHHPR \$HHSELCT PVHIPR \$HISELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLEQN (A,B) OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPCHAR SPFORMAT SPEULO SPEUHI SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC SPDSPHI SPDSPLO ZSIGNAL ZSLTSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD KX KY KZ BO	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE EMSENB MODE \$MODESEL

EIPENB—Enable Event Initiated Processing

Defines whether event-initiated processing (EIP) is enabled or not (see 4.2 in *System Control Functions*).

Source:	User	Default Value:	Enable	Access Lock:	Supvr.
Value Type:	Enum. of ATTRIB	Value Range:	Enable = EIP is enabled. Disable = EIP is disabled.		

EIPEVENT—EIP Event Type

Defines which types of events are reported to the target data point specified in EIPPCODE when event-initiated processing is enabled (see 4.2 in *System Control Functions*).

Source:	User	Default Value:	Any	Access Lock:	Engr.
Value Type:	Enum. of \$EIPST	Value Range:	Any = Any point alarm state change is reported. Alarm = Any change from no alarm state to alarm state Return = Any change from alarm state to no alarm state High = Any change from no high alarm to high alarm High-R = Any change from high alarm to no high alarm is reported. Low = Any change from no low alarm to low alarm is reported. Low-R = Any change from low alarm to no low alarm is reported. Chgstate= Any point alarm state change that resulted from a change in digital PV is reported. State1 = Any change to digital “STATE1” alarm is reported. State2 = Any change to digital “STATE2” alarm is reported.		

EIPPCODE — EIP Point ID

This parameter defines the tag name of the data point in the Application Module or CG/CM that is to be notified when an event is detected by this HG point (see 4.2 in *System Control Functions*). Refer to the definition of NAME (Tag Name) for the valid character set.

Source:	User	Default Value:	All Underscores	Access Lock:	DEB
Value Type:	Entity ID	Value Range:	See <u>NAME</u>		

EIPSTATE—EIP Status

This parameter indicates whether notification of the last EIP event was received by the point named in EIPPCODE (see 4.2 in *System Control Functions*).

Source:	System	Default Value:	Ok	Access Lock:	Read Only
Value Type:	Enum. of \$EIPSTAT	Value Range:	Ok = The last EIP event was received. Error = The last EIP event was not received.		

EMS — External Mode Switch Activation (EC Regulatory Data Point)

This parameter is used to provide external mode switching at an EC regulatory data point. This parameter is used only when external mode switching is configured (EMSENB = Enable).

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Enum. of ENBLSTAT	Value Range:	Enable = Mode switching is active. Disable = Mode switching is not active.		

NOTE

To update the value in this parameter, PVSOURCE must contain Auto.

EMSENB — External Mode Switch (EC Regulatory Data Point)

This parameter defines whether the MAN, AUTO, and CAS modes for an EC regulatory data point can be selected using the M, A, and C input lines to the data point. This parameter is not applicable to the following EC algorithms:

Das	Logic
Deadtm	Pidexfb
Extratbi	Selovddc
Incrmtl	

After this parameter is configured, the EMS parameter on the Detail Display is used to activate/deactivate the external-mode switching function.

Source:	User	Default Value:	Disable	Access Lock:	ENGR
Value Type:	Enum. of ENBLSTAT	Value Range:	Disable = No external mode switching. Enable = External mode switching is allowed.		

NOTE

To update the value in this parameter, MODE must contain Man.

ENT_TYPE—Entity Type

This parameter defines the type of data point and the type of node where the point resides. CL programs can use this parameter to verify types of data points.

Source:	System	Default Value:	System	Access Lock:	View
Value Type:	Enum. Of ENTYTYPE	Value Range:	Anlcmphg = HG analog composite point Anlinhg = HG analog input point Anlouthg = HG analog output point Counthg = HG counter point Ctlcount = HG control counter point Digcmphg = HG digital composite point Diginhg = HG digital input point Digouthg = HG digital output point Flaghg = HG flag point Hiway = Data Hiway configuration point Lgblkhg = HG logic block point Numerc = HG numeric point Prcomodhg = HG process module point Procstat = Processor status point (in all nodes) Timerhg = HG timer point		

EUDESC — E.U. Descriptor

This parameter specifies the name of the engineering units (EU) that are displayed on the Group and Detail Displays for this data point as shown in Figure N-1 (see NAME). In the example shown in the figure, the engineering unit descriptor is LBS/SEC.

Source:	User	Default Value:	All Blanks	Access Lock:	DEB
Value Type:	String	Value Range:	The permissible character set consists of all characters on the Engineer's Keyboard. Basically this set consists of Alphabetics A-Z, Numerics 0-9 and the following special characters: space ! % & ' () * + - / : ; > < = ? , \$		

EVENTPRC — Event Processing (Box Data Points)

This parameter defines whether process events (process alarms, process changes, and operator messages) that pertain to this particular hiway box are to be routed through the HG to the LCN.

Source:	User	Default Value:	Enable	Access Lock:	Engr.
Value Type:	Enum. of ENBLSTAT	Value Range:	Disable = Process events for this particular hiway box are not to be routed through the HG to the LCN. Enable = Process events for this particular hiway box are to be routed through the HG to the LCN.		

EVTRATE— # of Detected Process Changes (HG Processor Status Data Points)

This parameter defines the number of detected process changes during the last 15-second sample period.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

Extratbi — External Ratio and Bias (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Extratbi** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on the *EC Regulatory Data Point Configuration Form* and on the HG data entry displays. The outstanding characteristic of the **Extratbi** algorithm is that it is an auxiliary algorithm that applies ratio and bias to an external signal. Ratio and/or bias can be computed from external signals or from applicable constants.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHHPR \$HHSELCT PVHIPR \$HISELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM	SUPPIO OUTIND PVSIGNAL PVSLTSRC INITCONF SPCHAR SPFORMAT SPEULO SPEUHI SPDSPHI SPDSPLO SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC ZSIGNAL ZSLTSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD RATIO BIAS KX KZ BX BZ	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE ERASENB MODE \$MODESEL

-F-

FBTIME — Feedback Time (HLPIU and MC Digital Composite Data Points)

This parameter defines the duration (in seconds) for which alarming is suppressed following an output command from a digital composite data point that has been configured for command disagree alarming.

Source:	User	Default Value:	0	Access Lock:	Supvr.
Value Type:	Integer	Value Range:	0 to 1023 seconds		

FL — Flag Array (MC Flag Data Points)

This parameter calls up the flag array on the Process Module Detail Display that shows the current state of each of the 256 possible flag data points for a particular MC box. Each flag data point can be in one of the two states listed below.

Source:	System	Default Value:	False	Access Lock:	Oper.
Value Type:	Boolean Array	Value Range:	True = Flag is set. False = Flag is not set.		

FLAGPR — Flag Alarm Priority

This parameter defines the priority of a flag alarm generated by a data point.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
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Value Type:	Enum. of ALPRIOR
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Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.
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FLTRCNST (MC)

FLTRCNST — Filtering Constant Code (MC Analog Data Point)

This parameter defines the filtering time constant (in minutes) for the PV input to this data point.

Source:	User	Default Value:	0 minutes	Access Lock:	Supvr.
Value Type:	Integer	Value Range:	Entry	Filter Time Constant (In Minutes)	
			00		0.0
			01		0.006
			02		0.015
			03		0.028
			04		0.046
			05		0.072
			06		0.109
			07		0.161
			08		0.234
			09		0.339
			10		0.517
			11		1.050
			12		2.117
			13		4.250
			14		8.517
			15		17.05

FRMMD — FRMMD (Form Modifier; EC Regulatory Data Point)

This parameter determines whether the form modifier is 0 or 1 for an EC regulatory data point that is configured for the **Pidnonln** algorithm.

Source:	User	Default Value:	Fmdisab	Access Lock:	Supvr.
Value Type:	Enum. of FRMMD	Value Range:	Fmdisabl = Form modifier is 0. Fmenable = Form modifier is 1		

-G-**GAPHI — Gap High Limit (EC & MC Regulatory Data Points)****GAPLO — Gap Low Limit**

These parameters define the high and low limits of the gap in percent for an EC or MC regulatory data point that has been configured for the **Pidgap** algorithm.

Source:	User	Default Value:	0.0%	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-113.7% to 0% for gap low limit. 0% to 113.7% for gap high limit.		

GATETYP1- GATETYP2 — Gate 1 Type - Gate 3 Type (EC Regulatory Data Point)

These parameters define the type of logic gates for an EC regulatory data point that has been configured for the **Logic** algorithm.

Source:	User	Default Value:	Nogate	Access Lock:	Engr.
Value Type:	Enum. of GATETYPE	Value Range:	Nogate = This logic gate is not configured. And = And gate (NOTE 1) Or = Or gate (NOTE 1) Nor = Nor gate (NOTE 1) Xor = Exclusive-Or gate (NOTE 1) Nand = Nand gate (NOTE 1) Latch = Latch (NOTE 1) Pulse = Pulse generator (NOTE 2) Compare = Comparator (NOTE 3)		

NOTES

1. Make entries in ONDEL (On Delay Time) and OFFDEL (Off Delay Time) lines of the configuration form if And, Or, Xor, Nand, Nor, or Latch has been entered for this parameter.
2. Make an entry in the PULSE line of the configuration form if Pulse has been entered for this parameter.
3. Make an entry in the DEADBAND (Deadband) line of configuration form if Compare has been entered for this parameter.
4. To update the value in this parameter, MODE must contain Man and EMS must contain Disable.

GHISTR (N) — GETHIS Parameters Rate (CIU Processor Status Data Points)

This parameter defines the “Get History Rate.” The GETHIS rate is based on the number of GETHIS value requests from the ULP, where:

- N = 1, count of parameters for the most recent (full) 1-minute sample period; and
- N = 2, count of parameters for the previous 1-minute sample period.
- N = 3 to 10, reserved for use by Honeywell IAC.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
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Value Type:	ARRAY [1..10] OF REAL	Value Range:	≥ 0.0
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GPARGRT(N) — Single Point Parameter Get Rate (CIU Processor Status Data Points)

This parameter defines the single-point-parameter’s get rate. The get rate is based on the number of single point parameter get requests from the ULP, where

- N = 1, number of parameters for the most recent (full) 1-minute period;
- N = 2, number of parameters for the previous 1-minute period;
- N = 3, number of elements for the most recent 1-minute period;
- N = 4, number of elements for the previous 1-minute period; and
- N = 5 to 10, reserved for use by Honeywell IAC.

(Counts for parameters and elements are both available, because a parameter that is an array consists of a number of elements, each of which is a separate value. For example, a request for a parameter that is a 15- element array increases the count of parameters by one and the count of elements by 15.)

Source:	System	Default Value:	0.0	Access Lock:	Read Only
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Value Type:	ARRAY [1..10] of REAL	Value Range:	≥ 0.0
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GREENSGL — Green Pointer Signal Type (EC Regulatory Data Point)

This parameter defines the type of signal that is to be applied to the green pointer (setpoint pointer) of an Analog Display Unit. The tag name of the data point that provides the signal is specified by the GREENSRC parameter.

Source:	User	Default Value:	Notconfig	Access Lock:	Engr.
Value Type:	Enum. of GREENSGL	Value Range:	Y = Y signal (local setpoint) is applied to the green pointer. Ratio = Ratio signal is applied to the green pointer. Bias = Bias signal is applied to the green pointer. Notconfig = The green pointer is not configured for this data point.		

GREENSRC — Green Pointer (EC Regulatory Data Point)

This parameter defines the tag name of the data point that provides the input signal to the green pointer (setpoint) pointer of an Analog Display Unit. Refer to the definition of NAME (Tag Name) for the valid character set.

Source:	User	Default Value:	*	Access Lock:	Engr.
Value Type:	Entity ID	Value Range:	See <u>NAME</u>		

-H-**HEAPBCN2**

This parameter indicates the current number of free heap memory blocks in pool 2.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	>=0.0		

HEAPBCNT

This parameter indicates the number of free heap blocks in pool 1.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	>=0.0		

HEAPFRA2

This parameter indicates the percent of pool 2 fragmentation. A value above 2 indicates problems in heap.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 - 100 0 = No fragmentation 100 = Totally fragmented		

HEAPFRAG

This parameter indicates the percent of pool 1 fragmentation.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 - 100		

HEAPFRE2

HEAPFRE2

This parameter indicates the current amount of free heap memory (in words) for pool 2 in the node.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

HEAPFREE

This parameter indicates the current amount of free heap (in words) in pool 1.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

HEAPMIN2

This parameter indicates the minimum pool 2 heap (in words) that has reached from the time the node is started.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

HEAPMIN

This parameter indicates the minimum pool 1 heap (in words) that has reached from the time the node is started.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

HEAPTOT2

This parameter indicates the total amount of heap memory (in words) in pool 2.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

HEAPTOTL

This parameter indicates the total amount of heap memory (in words) in pool 1.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

HYSTARTS

This parameter indicates the number of Input Output Control Block chain, which starts on Hiway.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

HIGHAL — Highest Alarm State

When multiple alarms are detected for a data point, this parameter determines the priority of the alarms. This listing is arranged in priority order, beginning with the lowest alarm states (Noalarm, Invalid) and ending with the highest alarm (Badpv).

Source: System		Default Value: N/A	Access Lock: Read Only
Value Type:	Enum. of ALMTYPE	Value Range:	Noalarm = No alarm has been detected.
Value Range:	Noalarm = Invalid = Prpreset = Preset = Offnorm = Openther = Flag = Counter = Chngofst = Unreasbl = Cmddis = Compshed =	No alarm has been detected. PV hi value has been set below the PV lo value. For an MC counter data point, the accumulated value is equal to or greater than the pre-alarm value established by the <u>PDEVTP</u> parameter. For an MC counter data point, the accumulated value is equal to or greater than the alarm trip-point established by the <u>PRESET</u> (Preset Value) parameter. For a digital data point, the state of the point has changed from the normal state. Open thermocouple has been detected at an LEPIU or LLPIU for one of the following input/sensor types: Sensr501-506, Sensr401-406. MC flag data point has been set. For an HLPIU counter data point, the accumulated value is equal to or greater than preset value established by the <u>PRESET</u> parameter. Change-of-state alarm has been detected at a digital data point; refer to definition of <u>DIGALFMT</u> . Unreasonable alarm condition; low alarm trip point has been set above the high alarm trip point, or a data point has been taken off scan, or the A/D converter has failed. Command disagree alarm has been detected at a digital composite data point. Occurs after a predefined time (TOG1 or TOG2) has passed without a signal from a host computer.	

NOTE

The Counter and Chngofst alarm states are not shown on group or detail displays because there is no return-to-normal.

S4logic = Output of gate 4 is TRUE for an EC regulatory data point that has been configured for the **Logic** algorithm.
 Offset1 = Offset1 alarm has been detected for an EC regulatory data point that has been configured for the **Rampsoak** algorithm.
 Offset2 = Offset2 alarm has been detected for an EC regulatory data point that has been configured for the **Rampsoak** algorithm.
 Offset3 = Offset3 alarm has been detected for an EC regulatory data point that has been configured for the **Rampsoak** algorithm.
 Pvrocn = PV Rate-Of-Change Negative; applies to only an MC regulatory data point.
 Pvrocp = PV Rate-Of-Change Positive alarm; applies to only an MC regulatory data point.
 Pvroc = PV Rate-Of-Change alarm; applies to only an EC regulatory data point.
 Devlo = Deviation Lo alarm.
 Devhi = Deviation Hi alarm.
 Devll = Deviation Lo Lo alarm; applies to only an EC regulatory data point.
 Devhh = Deviation Hi Hi alarm; applies to only an EC regulatory data point.
 Devlll = Deviation Lo Lo Lo alarm; applies to only an EC regulatory data point.
 Devhhh = Deviation Hi Hi Hi alarm; applies to only an EC regulatory data point.
 Pvlo = PV Lo alarm.
 Pvhi = PV Hi alarm.
 Pvll = PV Lo Lo alarm; applies to only an EC regulatory data point.
 Pvhh = PV Hi Hi alarm; applies to only an EC regulatory data point.
 Pvlll = PV Lo Lo Lo alarm; applies to only an EC regulatory data point.
 Pvhhh = PV Hi Hi Hi alarm; applies to only an EC regulatory data point.
 Badpv = Indicates that the PV is outside of the configured range.
 Oplo = OP Lo alarm
 Ophi = OP Hi alarm

HIGHALPR—High Alarm Priority

This parameter defines the priority of a high alarm generated by a data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.
------------	--

HISEL — High Selection Value (EC Regulatory Data Point)

This parameter is the difference (in percent) between the output value and the value of the selected input ($HISEL = SELINP$ minus OP). HISEL is applicable to only an EC regulatory data point that has been configured for the **Selovddc** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	0.0% to 113.7%		

Hisel — Hi Selector (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Hisel** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Hisel** algorithm is that it is a 2-input Hi-signal selector.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF OPLOLM OPHILM \$OPTOL TD	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Hisel — Hi Selector (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Hisel** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on the *MC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Hisel** algorithm is that it is a 2-input Hi-signal selector.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTsrc SPSIGNAL SPSLTsrc PVTRACK INITCONF SPLOLM SPHILM \$SPTOL OPLOLM OPHILM \$OPTOL TD	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

HIWAYWRD(0)-HIWAYWRD(1023) — Diagnostic Hiway Addressing

Each of these parameters accesses the contents of a memory location for a specified Multifunction Controller. This information is used for diagnostic purposes.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer				

NOTE

See Table 9 in the BASIC System Interface Manual, SY-09-02, (BASIC bookset) for descriptions of memory contents.

HWY — Hiway Input

This parameter is the current value (in percent) that a regulatory data point in a hiway box is receiving from the hiway (otherwise known as the hiway location). This parameter is used to accomplish setpoint control (SPC) or direct digital control (DDC) depending on the algorithm. For SPC-type algorithms, the hiway location is used as the Y-input when the mode is remote cascade (Mode = Cas and RCASENB = On). For DDC-type algorithms, the output value is written to the hiway location when in the remote cascade mode. In both cases, HWY is a Read Only parameter.

For EC algorithms that do not support the remote cascade mode, HWY is used as a "free variable" and can be changed at the Universal Station. HWY is a "free variable" for the following EC algorithms:

Automan
Charact
Das
Deadtm
Leadlagm
Leadlags
Logic
Massflow
Midof3
Selovddc
Switch

Source:	System	Default Value:	N/A	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-6.86% to 106.86%		

HWYACT — Active Hiway Indicator

This parameter indicates which data hiway cable (A or B) is presently being used for transmitting information.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of HWYACT	Value Range:	Nonact = Neither hiway cable is active. Acta = Hiway cable A is active. Actb = Hiway cable B is active.		

HWYASWST — Hiway Automatic Swap Option

If the HG is acting as the HTD, the parameter HWYASWST can be set to ENABLE or DISABLE. If the HG is redundant, and the primary HG experiences a failure, is shutdown, or reset, then the HWYASWST parameter value will automatically be changed from ENABLE to DISABLE.

Note: You could implement an AM/CL program that checks the (\$HIWAYxx.HWYASWAST)PERIOD parameter so that if it goes to DISABLE, the program would send one of the following messages to the engineer:

- The PERIOD parameter was set to DISABLE, or that
- The engineer should set the PERIOD parameter to ENABLE.

Source:	System	Default Value:	Enable (supplied by DEB)	Access Lock:	Engr.
Value Type:	Enum. of ENBLSTAT	Value Range:	Enable = Automatic hiway switching is permitted. Set to enable if the HG is doing HTD functions. Disable = Automatic hiway switching is not permitted. Set to disable if the HG is not doing HTD functions.		

NOTE

If Enable is entered for this parameter, automatic hiway switching may not occur depending on the condition of the boxes on the hiway, as determined by the hiway swap algorithm.

HWYCTLST — Hiway Control State

This parameter defines the functional state of a Data Hiway.

Source:	System	Default Value:	Basic (supplied by DEB)	Access Lock:	Supvr.
Value Type:	Enum. of BOXFSTAT	Value Range:	Read = AM can read data from the boxes, but cannot write data into the boxes on this hiway. Full = AM can read data from and write data into the boxes on this hiway. Basic = AM can read data from the boxes, but cannot write data into the boxes on this hiway. Test = A box is being tested.		

NOTE

The Hiway Control State supersedes the Box Functional State. For example, if the HWYCTLST = BASIC and the BOXFSTAT = FULL then the BOXFSTAT = BASIC until the HWYCTLST = FULL.

HWYDEVEN — Hiway Preferred Device Option

When the HG serves as the HTD, it can give the hiway grant to different devices. This parameter indicates which devices are active preferred devices.

Source:	System	Default Value:	None	Access Lock:	Read Only
Value Type:	Enum. of HWYDEVEN	Value Range:	None = There are no preferred devices. DeviceA = Normally, the redundant HGs. DeviceB = Another HG or an operator station. DeviceAB = Both devices together.		

HWYERR (N) — # of Errors for Box N at each Hiway (HG Processor Status Data Points)

This parameter defines the number of errors for Box N on each hiway during the last 15-second sample period.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	ARRAY [1..63] of REAL	Value Range:	≥ 0.0		

HWYHSAST — Hiway Swap Algorithm State

This parameter indicates the state of the hiway swap algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of ENBLSTAT	Value Range:	ON = The hiway swap algorithm passes. OFF = The hiway swap algorithm fails.		

NOTE

This parameter corresponds with the

SWAP HIWAY

 target on the Hiway Network Status display.

HWYHTD — HTD Assignment (Data Hiway Data Point)

This parameter specifies the device that performs the function of a Hiway Traffic Director (HTD).

Source:	User	Default Value:	Htd	Access Lock:	DEB
Value Type:	Enum. of HWYHTD	Value Range:	ThisHG = The HG assigned to this hiway acts as the HTD on this physical hiway. RemoteHG = The HG on another LCN acts as the HTD on this physical hiway. AddedHG = The HG on this LCN, but assigned to a different hiway number acts as the HTD. Htd = Physical HTD is in use.		

NOTE

If Thishg is entered, also make entries in HWYSWINT (Switch Interval) and HWYSWBAS (Switch Base Time) parameters on the Data Hiway Data Point Configuration Form.

HWYNUM — Hiway Number

The hiway number defines the Data Hiway that is attached to the hiway box in which a data point resides.

Source:	User	Default Value:	01	Access Lock:	DEB
Value Type:	Integer	Value Range:	01-20		

HWYREDST — Hiway Redundancy State

This parameter indicates which HG (NCF configured as primary or secondary) is active on a Data Hiway.

Source:	System	Default Value:	Primary	Access Lock:	Read Only
Value Type:	Enum. of HWYREDST	Value Range:	Inactive = HG redundancy state is presently inactive (transient state that occurs during the upgrade procedure). Primary = HG is the primary HG (also the state returned for a nonredundant HG). Secndry = HG is the redundant HG.		

HWYSTAT — Hiway Status

This parameter indicates the current status of a Data Hiway.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of HWYSTAT	Value Range:	Uncertain = An HG has not finished startup or there is no HG box configured on the hiway. Init = Hiway is being initialized. Run = Hiway is operating properly. Failed = HG cannot communicate with any of the boxes on the hiway.		

HWYSWBAS — Switch Base Time (Data Hiway Data Point)

This parameter defines the time of day (in minutes) during which Data Hiway switching is to take place.

Source:	User	Default Value:	0 (12:00 AM midnight)	Access Lock:	DEB
Value Type:	Integer	Value Range:	0 to 1440 minutes Example: 1:30 AM would be entered as 90 Example: 3:00 PM would be entered as 900		

NOTE

An entry is to be made for the HWYSWBAS parameter if Thishg has been entered for the HWYHTD (HTD Assignment) parameter on the configuration form.

HWYSWINT — Switch Interval (Min) (Data Hiway Data Point)

This parameter defines the amount of time (in minutes) between automatic switching of the primary and backup data hiways.

Source:	User	Default Value:	1440 minutes	Access Lock:	DEB
Value Type:	Integer	Value Range:	0 to 1440 minutes (1440 is the number of minutes in one day).		

NOTE

An entry is to be made for this parameter if Thishg has been entered for the HWYHTD (HTD Assignment) parameter on the configuration form.

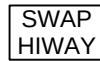
HWYSWST — Hiway Swap State

When the HG is the HTD, this parameter indicates if the hiways have yet to be swapped (based on the hiway swap algorithm).

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of ENBLSTAT	Value Range:	Enable = Hiways have not yet been swapped. Disable = Hiways have been swapped.		

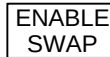
NOTE

To correct cable trouble: fix the cable, invoke the HG Network Status display, then set the Hiway Swap Algorithm State (HWYHSAST) parameter to ON by selecting the



target, then re-enable the the Hiway Swap State (HWYSWST) parameter

by selecting the



target.

-I-**Incrmtl — Incremental (EC Regulatory Data Point)**

The parameters that are applicable to an EC regulatory data point when it is configured for the **Incrmtl** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Incrmtl** algorithm is that it is a 6-input incremental summer algorithm.

Incrmtl (EC)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM PVCLAMP ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP ALMASK CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLYTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELECT DEVLLPR DEVLOPR \$DLSELECT PVHHPR \$HHSELECT PVHIPR \$HISELECT PVLLPR \$LLSELECT PVLOPR \$LOSELECT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC INITCONF SPCHAR SPFORMAT SPTEMP SPEULO SPEUHI SPDSPHI SPDSPLO SP SOASIGNL SOASRC SOBSIGNL SOBSRC ZSIGNAL ZSLTSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TFCLLM TFCHLM TD KX KY KZ KM KA KC OVB	NMODE NMODATTR PSTMODE RSCASENB MODEPERM SHEDMODE MODE \$MODESEL

INITCONF (CB/EC/MC)

INITCONF — Initialization Configuration (CB, EC, & MC Regulatory Data Points)

This parameter defines whether this point is to participate in initialization or not. Generally, each secondary point in a cascade control strategy does participate and the ultimate primary point in a cascade does not. For a description of the initialization process, refer to the appropriate *Algorithm Engineering Data* in the *Basic System Product* binder.

Source:	User	Default Value:	Noinit	Access Lock:	Engr.
Value Type:	Enum. of INITCONF	Value Range:	Noinit = Initialization is not to be performed. Init = Initialization is to be performed.		

NOTE

To update the value in this parameter, MODE must contain Man, and for EC Regulatory points, EMS must contain Disable.

INITMAN — Initialization Manual Bit

This parameter indicates whether the output of the specified regulatory data point is being initialized by a downstream regulatory data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Initialization is currently being performed. "INIT" is displayed above the data point mode at the Universal Station. False = Initialization is not being performed. Blanks are displayed above the data point mode at the Universal Station.		

INPTCOND — Input Conditioning

This parameter defines the type of input conditioning required for the PV input of analog and regulatory data points.

Source:	User	Default Value:	Linear	Access Lock:	Engr.
Value Type:	Enum. of INPTCOND	Value Range:	Linear = Linear Conditioning Sqrroot = Square Root Conditioning		

NOTE

Do not make an entry for this parameter if a Temperature Display (TD) input/sensor type has been selected for an HLPIU-analog point.

To update the value in this parameter, MODE must contain Man; for CB and MC points, INITCONF must contain Noinit; and for EC points, EMS must contain Disable.

INPTDIR — Input Direction

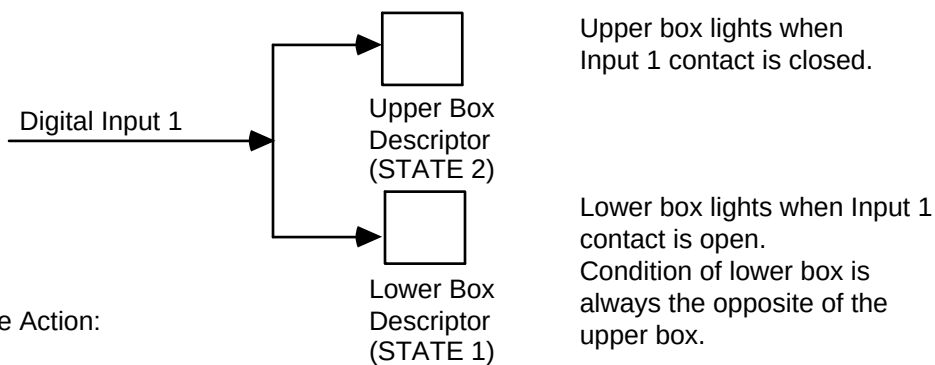
This parameter defines the contact condition required to light the upper or lower box on a Group and Detail Display for a digital input data point or a digital composite data point.

Source:	User	Default Value:	Direct	Access Lock:	DEB
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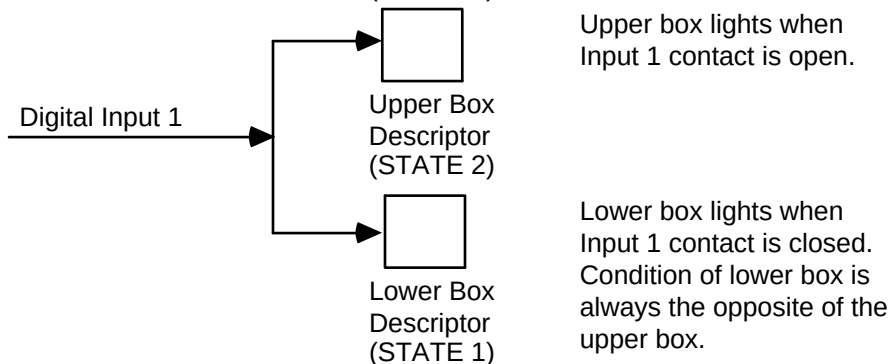
Value Type:	Enum. of POLARITY	Value Range:	Input direction can be Direct or Reverse as follows:
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FOR A SINGLE-INPUT DIGITAL DATA POINT

Direct Action:



Reverse Action:

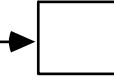


continued

FOR A DUAL-INPUT DIGITAL DATA POINT:

Direct Action:

Digital Input 1



Upper box lights when Input 1 contact is closed.

Upper Box
Descriptor
(STATE2)

Digital Input 2

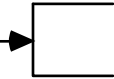


Lower box lights when Input 2 contact is closed.

Lower Box
Descriptor
(STATE1)

Reverse Action:

Digital Input 1



Upper box lights when Input 1 contact is open.

Upper Box
Descriptor
(STATE2)

Digital Input 2



Lower box lights when Input 2 contact is open.

Lower Box
Descriptor
(STATE1)

INPTSSLT —Subslot Number or Input Subslot Number (DHP, HLPUI, LEPIU, LLPIU, MC)

This parameter specifies which of the inputs (subslots) on the selected card (slot) is to be used as the input to this analog, digital, or counter data point.

For a dual-input digital input or digital composite data point, this parameter defines the input (Input 1) that lights the upper box on the Group and Detail Displays. The subslot number of the second input (Input 2), which lights the lower box on the displays, is automatically assigned by the system.

The number that is assigned by the system is the next consecutive number that follows the number entered for the INPTSSLT parameter. For example, if a 3 was entered for the INPTSSLT parameter, then the subslot number that is automatically assigned by the system for the second input (Input 2) is 4.

The parameter that determines the number of inputs for a data point is NMBRINPT (Number of Inputs). If a 2 is entered for the NMBRINPT parameter (to specify that a digital data point has two inputs), the system automatically assigns the number of the second subslot as described above.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer follows:	Value Range:	The subslot number ranges for the various data points are as follows:		

Data Point Type	Subslot Number Range
DHP AI & AC	1-6
DHP DI & DC	1-16 (single-input data point) 1-15 (for the first input of a dual-input data point)
DHP Counter	1-8
HLPIU AI & AC	1-8

NOTE

The following subslot addresses are not available for use in an HLPIU when analog input cards are installed in these slots: slot 16 subslot 8, and slot 32 subslot 8.

HLPIU DI & DC	1-16 (single-input data point) 1-15 (for the first input of a dual-input data point)
HLPIU Counter	1-8
HLPIU Control Counter	1-4
LEPIU AI	1-8

NOTE

The following subslot addresses are not available for use in an LEPIU: Slot 16 subslot 8 (Remote Box 8 last point), and slot 32 subslot 8 (Remote Box 16 last point).

LLPIU AI	1-8
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NOTE

The following subslot addresses are not available for use in an LLPIU: Slot 1 subslots 1-4, slot 16 subslot 8, and slot 32 subslot 8.

MC AI & AC	1-8
MC DI & DC	1-16 (single-input data point) 1-15 (for the first input of a dual-input data point)
MC Counter	1-4
AI = Analog Input	DI = Digital Input
AC = Analog Composite	DC = Digital Composite

INTVARNM (MC)

INTVARNM — Flag Number (MC Flag Data Point)

This parameter defines the flag number of a flag data point in the set of 256 flag data points in each MC.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1-256		

INTVARNM — Logic Block Number (MC Logic Block Data Point)

This parameter defines the number of the logic block data point in the set of 128 logic-block data points in each MC.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1-128		

INTVARNM — Numeric Number (MC Numeric Data Point)

This parameter defines the number of the numeric data point in the set of 88 numeric data points in each MC.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1-88		

INTVARNM (MC)

INTVARNM — Timer Number (MC Timer Data Point)

This parameter defines the number of the timer data point in the set of 32 timer data points in each MC.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1-32		

IOCBS — # of Hiway IOCBs (HG Processor Status Data Points)

This parameter defines the number of hiway IOCBs during the last 15-second sample period.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

IPCRDOP — Input Card Option

This parameter is used for the following two purposes:

1. During off-process configuration, it is used for configuring the input card option for HLPIU digital data points.
2. During on-process operation, it is used for displaying any mismatches between the configured input card option and the actual input card option installed in the same hiway box slot.

For off-process configuration:

This parameter specifies the point card option for an HLPIU digital input or digital composite data point. Refer to the *HLPIU Box Data Point Configuration Form* that has been filled out during network configuration to ensure that the option selected here is the same as the option selected on the configuration form for the same HLPIU digital slot. (Refer to the PIUCDROP parameter on the form.) If Chngdect or SOE is entered for this parameter, also make an entry for the DIGALARM (Alarming Function) parameter.

Source:	User	Default Value:	Notifyst	Access Lock:	DEB
Value Type:	Enum. of PIUCRDOP	Value Range:	Notifyst = Chngdect = SOE =	Notify Status. Change Detect. Sequence of Events.	

For on-process operation:

This parameter specifies that the input card option configured for this slot is not the same as the input card option that is installed in the hiway box slot. IPCRDOP represents the input card option as it exists in the hiway box slot. The configured input card option is also displayed and the engineer should be aware of any mismatches.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of PIUCRDOP	Value Range:	Notifyst = Notify Status (DHP and HLPIU digital input data points). Chngdect = Change Detect (HLPIU digital input data points). SOE = Sequence of Events (HLPIU digital input data points). Countr16 = 16-Bit Counter (DHP, HLPIU, and MC counter data points). Countr32 = 32-Bit Counter (HLPIU control counter data points). Status = Status digital input (MC digital input data points). Latchin = Latched digital input (MC digital input data points). None = Card option is undefined.		

IPCRDTY — Input Card Type

This parameter indicates that a conflict exists between the card type that has been configured for a slot and the card type that is installed in the slot. IPCRDTY represents the card type that is currently installed in the slot. The configured card type is also displayed and the operator should be aware of any mismatches.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of PIUCRDY	Value Range:	None = This card type does not have an input. Analogin = Analog Input Card. Digin = Digital Input Card. Counter = Counter Card.		

NOTE

When PVCHAR is set to Sensr300, Sensr400, or Sensr500 for a PIU, IPCRDTY is displayed as None so that it is consistent with the BASIC or Enhanced Operator Station which display "Undef" in these cases.

ITHILM (CB/EC/MC)

ITHILM — Integral High Limit (CB, EC, & MC Regulatory Data Points)

ITLOLM — Integral Low Limit

These parameters define the high and low limits of the integral in percent.

Source:	User	Default Value:	-5.0% for low limit 105.0% for high limit	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-6.86% to 106.86%		

-K-**K — K (Gain Constant; CB, EC, & MC Regulatory Data Points)**

This parameter defines the gain constant for PID algorithms in the CB, EC, and MC, and for Lead/Lag algorithms in the CB and MC. If the gain constant is specified as 0, the gain function is disabled.

Source:	User	Default Value:	0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.1 to 99.99 and 0		

K — K (Scale Factor; CB and MC Regulatory Data Points)

This parameter defines the scale factor for an MC regulatory data point that has been configured for one of the following algorithms; **Sumrwman**, **Cbsumr**, **Multwman**, **Mult**, **Dividr**, **Sqrt**, **Sqrtprod** (CB Only), and **Sumsqrt** (CB only) For algorithms 23-33 in hiway-based systems, K is referred to as KA (Y scaling factor).

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.999		

K1 — K1 (Auxiliary Constant on X Input; MC Regulatory Data Point)

This parameter defines the auxiliary constant on the X input for an MC regulatory data point that has been configured for the **Sumrwman**, **Cbsumr**, or **Sqrt** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.999		

K1 (CB)

K1 — K1 (X Scale Factor; CB Regulatory Data Point)

This parameter defines the X scale factor for a CB regulatory data point that has been configured for one of the following algorithms: **Sumrwman**, **Cbsumr**, **Sqrt**, or **Sumsqrt**.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.999		

K2 — K2 (Auxiliary Bias Coefficient; CB Regulatory Data Point)

This parameter defines the auxiliary bias value for a CB regulatory data point that has been configured for one of the following algorithms: **Sumrwman**, **Cbsumr**, **Multwman**, **Dividr**, **Sqrtprod**, **Sqrt**, **Sumsqrt**, or **Mult**.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-99.99 to 99.99		

K2 — K2 (Auxiliary Bias Coefficient; MC Regulatory Data Point)

This parameter defines the auxiliary bias coefficient for an MC regulatory data point that has been configured for one of the following algorithms: **Sumrwman**, **Cbsumr**, **Mult**, **Dividr**, or **Sqrt**.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-99.99 to 99.99		

KA — KA (Scaling Coefficient on A Input; EC Regulatory Data Point)

This parameter defines the scaling coefficient on the A input for an EC regulatory data point that has been configured for the **Incrmtl** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.999		

KC — KC (Scaling Coefficient on C Input; EC Regulatory Data Point)

This parameter defines the scaling coefficient on the C input for an EC regulatory data point that has been configured for the **Incrmtl** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.999		

KEYWORD — Point Keyword

The point keyword may be used to describe an important aspect of this particular data point, and it appears on the Group and Detail Displays for the data point. For example, in Figure N-1 (see NAME) the keyword for the data point is REFLUX.

Source:	User	Default Value:	All Blanks	Access Lock:	DEB
Value Type:	String	Value Range:	The point keyword can consist of up to eight characters, and the permissible character set is as follows: Alphabetics A through Z (uppercase only); Numerics 0 through 9 (an all numeric keyword is not permissible); Underscore () cannot be used as the first character or last character in a keyword, and consecutive underscores are not allowed.		

KGAP — KGAP (Gain Reduction Factor; EC Regulatory Data Point)

This parameter defines the gain reduction factor of the gap for an EC regulatory data point that has been configured for the **Pidgap** algorithm.

Source:	User	Default Value:	0.1	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.01 to 0.999		

KL — KL (Gain Constant; EC Regulatory Data Point)

This parameter defines the gain constant for an EC regulatory data point that has been configured for the **Leadlagm** or **Leadlags** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.1 to 99.99 and 0		

KM — KM (Scaling Coefficient on M Input; EC Regulatory Data Point)

This parameter defines the scaling coefficient on the M input for an EC regulatory data point that has been configured for the **Multdiv**, **Sqrtprod**, or **Incrmtl** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.999		

KMF — KMF (EC Regulatory Data Point)

This parameter defines the overall gain for the **Massflow** algorithm as follows, where:

$$\text{KMF} = (\text{Overall Gain}) \times (\text{Input Scaling for X}) \times (\text{Orifice Constant}).$$

Source:	User	Default Value:	0.1	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.1 to 2.0		

KNL — KNL (Nonlinear Factor; EC Regulatory Data Point)

This parameter defines the nonlinear factor for an EC regulatory data point that is configured for the **Pidnonln** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.1 to 9.999 and 0.0		

KPP (EC)

KPP — Effective Gain in Pidgap and Pidnonln Algorithms (EC Regulatory Data Point)

KPP is the current calculated value of the modified gain at an EC regulatory data point that has been configured for the **Pidgap** or **Pidnonln** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	0.0 to 99.99		

KX — KX (Ratio Gain; EC Regulatory Data Point)

This parameter defines the ratio gain for an EC regulatory data point that has been configured for the **Extratbi** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.999		

KX — KX (Scaling Coefficient on X Input; EC Regulatory Data Point)

This parameter defines the scaling coefficient on the X input for an EC regulatory data point that has been configured for the **Multdiv** or **Ecsumr** algorithm.

Source:	User	Default Value:	0.1	Access Lock:	Supvr.
Value Type:	Real	Value Range:	For Ecsumr : -9.999 to 9.999 For Multdiv : 0.1 to 9.999		

KX — KX (Scaling Coefficient on X-Input Increment; EC Regulatory Data Point)

This parameter defines the scaling coefficient on the X-input increment for an EC regulatory data point that has been configured for the **Incrmtl** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.999		

KY — KY (Scaling Coefficient on Y-Input Increment; EC Regulatory Data Point)

This parameter defines the scaling coefficient on the Y-input increment for an EC regulatory data point that has been configured for the **Incrmtl** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-99999 to 999999		

KY — KY (Scaling Coefficient on Y Input; EC Regulatory Data Point)

This parameter defines the scaling coefficient on the Y input for an EC regulatory data point that has been configured for the **Massflow**, **Multdiv**, or **Ecsumr** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	For Ecsumr : -9.999 to 9.999 For Massflow : 0.1 to 9.999 and 0.0 For Multdiv : 0.1 to 9.999		

KZ — KZ (Bias Gain; EC Regulatory Data Point)

This parameter defines the bias gain for an EC regulatory data point that has been configured for the **Extratbi** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.999		

KZ — KZ (Scaling Coefficient on Z Input; EC Regulatory Data Point)

This parameter defines the scaling coefficient on the Z input for an EC regulatory data point that has been configured for the **Ecsumr**, **Leadlagm**, **Leadlags**, **Massflow**, or **Multdiv** algorithm.

Source:	User	Default Value:	0.1	Access Lock:	Supvr.
Value Type:	Real	Value Range:	For Ecsumr , Leadlagm , or Leadlags : -9.999 to 9.999 For Massflow : 0.1 to 1.0 For Multdiv : 0.1 to 9.999		

KZ — KZ (Scaling Coefficient on Z-Input Increment; EC Regulatory Data Point)

This parameter defines the scaling coefficient on the Z-input increment for an EC regulatory data point that has been configured for the **Incrmtl** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9.999 to 9.99		

-L-

LB — Logic Block Array (MC)

This parameter defines the set of logic block values on the Process Module Detail Display. It shows the current states of the 128 possible logic blocks. Each logic block is in one of the two states listed below.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = State of logic block is true. False = State of logic block is false.		

LBOXCLR — Lower Box Color

This parameter defines the color of the lower (STATE 1) box that is used to show the status of the input(s) of a digital input data point or digital composite data point.

Source:	User	Default Value:	Red	Access Lock:	DEB
Value Type:	Enum. of BOXCOLOR	Value Range:	Red Green White Black Cyan Yellow Blue		

Leadlag — Lead Lag (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Leadlag** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on the *CB Regulatory Data Point Configuration Form* and on the HG data entry displays. The outstanding characteristic of the **Leadlag** algorithm is that it is a single-input auxiliary algorithm providing adjustable time constants for signal delay or signal anticipation.

Leadlag (MC)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATI ON DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HSELECT PVLOPR \$LOSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND PVSIGNAL PVSLTSRC OPLOLM OPHILM \$OPTOL TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Leadlag — Lead Lag (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Leadlag** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Leadlag** algorithm is that it is a single-input auxiliary algorithm providing adjustable time constants for signal delay or signal anticipation.

Leadlag (MC)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVDSPLO PVEUHI PVDSPHI OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSELCT PVROCPPR \$RPSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	EIPENB SUPPIO CTLACTN CTLEQN ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2 GAPLO GAPHI RATIO BIAS	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Leadlagm (EC)

Leadlagm — Lead/Lag Multiplier with DDC (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Leadlagm** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Leadlagm** algorithm is that it is a multipurpose 3-input algorithm, with provision for direct digital control of the output by a TPS module, and it can perform lead/lag feedforward compensation and multiplication.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	\$AUXUNIT	SUPPIO	NMODE
UNIT	PVCHAR	ALFMT	CTLEQN (A,B)	NMODATTR
PTDESC	INPTCOND	\$ALDLOPT	OUTIND	PSTMODE
KEYWORD	PVFORMAT	\$DLBADAL	PVSIGNAL	RCASENB
PRIMMOD	PVTEMP	DLYTIME	PVSLTSRC	MODEPERM
ASSOCDSP	PVEULO	\$REALARM	SPSIGNAL	SHEDMODE
HWYNUM	PVEUHI	BADPVPR	SPSLTSRC	EMSENB
PNTBOXTY	PVDSPhi	DEVHHPR	PVTRACK	MODE
BOXNUM	PVDSPL0	DEVHIPR	INITCONF	\$MODESEL
PTDISCL	OVERVAL	\$DHSELCT	SPCHAR	
SLOTNUM	PVRNGOP	DEVLLPR	SPFORMAT	
LOADDEST	PVCLAMP	DEVLOPR	SPTemp	
ALGIDDAC	CALIBOFF	\$DLSELCT	SPEULO	
		PVHHPR	SPEUHI	
		\$HHSELCT	SPDSPhi	
		PVHIPR	SPDSPL0	
		\$HSELCT	SPLOLM	
		PVLLPR	SPHILM	
		\$LLSELCT	\$SPTOL	
		PVLOPR	SP	
		\$LOSELCT	SOASIGNL	
		(Alarm	SOASRC	
		Trip-point	SOBSIGNL	
		Values)	SOBSRC	
		ALPRIOR	ZSIGNAL	
		CCRANK	ZSLTSRC	
		CCPRIPNT	MSIGNAL	
		EIPPCODE	MSSLTSRC	
		EIPEVENT	ASIGNAL	
		EIPENB	CSIGNAL	
		ALMASK	CSLTSRC	
		INITCONF	GREENSGL	
		SPHILM	GREENSRC	
		SPLOLM	SENSSTOP	
			OPLOLM	
			OPHILM	
			\$OPTOL	
			TFCLLM	
			TFCHLM	
			TD	
			KL	
			T1	
			T2	
			T3	
			T5	
			KZ	
			BO	

Leadlags (EC)

Leadlags — Lead/Lag Summer with DDC (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Leadlags** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Leadlags** algorithm is that it is a multipurpose 3-input algorithm, with provision for direct digital control of the output by a TPS module, and it can perform lead/lag feedforward compensation and summation.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	\$AUXUNIT	SUPPIO	NMODE
UNIT	PVCHAR	ALFMT	CTLEQN (A,B,C)	NMODE
PTDESC	INPTCOND	\$ALDLOPT	OUTIND	NMODATTR
KEYWORD	PVFORMAT	\$DLBADAL	PVSIGNAL	PSTMODE
PRIMMOD	PVTEMP	DLYTIME	PVSLTSRC	RCASENB
HWYNUM	PVEULO	\$REALARM	SPSIGNAL	MODEPERM
PNTBOXTY	PVEUHI	BADPVPR	SPSLTSRC	SHEDMODE
BOXNUM	PVDSPHI	DEVHHPR	PVTRACK	EMSENB
PTDISCL	PVDSPLO	DEVHIPR	INITCONF	MODE
SLOTNUM	OVERVAL	\$DHSELCT	SPCHAR	\$MODESEL
LOADDEST	PVRNGOP	DEVLLPR	SPFORMAT	
ALGIDDAC	PVCLAMP	DEVLOPR	SPTEMP	
	CALIBOFF	\$DLSELCT	SPEULO	
		PVHHPR	SPEUHI	
		\$HHSELCT	SPDSPHI	
		PVHIPR	SPDSPLO	
		\$HSELCT	SPLOLM	
		PVLLPR	SPHILM	
		\$LLSELCT	\$SPTOL	
		PVLOPR	SP	
		\$LOSELCT	SOASIGNL	
		(Alarm	SOASRC	
		Trip-point	SOBSIGNL	
		Values)	SOBSRC	
		ALPRIOR	ZSIGNAL	
		CCRANK	ZSLTSRC	
		CCPRIPNT	MSIGNAL	
		EIPPCODE	MSLTSRC	
		EIPEVENT	ASIGNAL	
		EIPENB	ASLTSRC	
		ALMASK	CSIGNAL	
		INITCONF	CSLTSRC	
		SPHILM	GREENSGL	
		SPLOLM	GREENSRC	
			SENSSTOP	
			OPLOLM	
			OPHILM	
			\$OPTOL	
			TFCLLM	
			TFCHLM	
			TD	
			KL	
			T1	
			T2	
			T3	
			T5	
			KZ	
			BO	

LGCS1DES (EC)

LGCS1DES — Logic Algorithm S1 Descriptor (EC Regulatory Data Point)

LGCS2DES — Logic Algorithm S2 Descriptor

LGCS3DES — Logic Algorithm S3 Descriptor

These parameters are the descriptors that have been configured in the HG Library to label the three gate outputs of an EC regulatory data point that has been configured for the **Logic** algorithm.

Source:	System	Default Value:	All blanks	Access Lock:	Read Only
Value Type:	String				

LIBNUMEC — Library For Descriptors (EC Regulatory Data Point)

This parameter defines the HG Library that contains the descriptors for an EC regulatory data point that has been configured for the **Logic** algorithm.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1-4		

LIBRYIND — Library Index (EC Regulatory Data Point)

This parameter defines the location of the first of three descriptors that are taken in sequential order from the HG Library to label status bits S1-S3 for an EC regulatory data point that has been configured for the **Logic** algorithm. The library index numbers for the second and third descriptors are not required; the system automatically selects three sequential descriptors starting with the library index of the first descriptor.

Source:	User	Default Value:	001	Access Lock:	DEB
Value Type:	Integer	Value Range:	001 to 576		

LIBRYNUM — Library (Library entity in HG)

This parameter defines the HG Library that is used by this MC. For more information on HG libraries, refer to 3.8.9 in *Hiway Gateway Control Functions* in the *Implementation/Hiway Gateway - 1* binder.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1 to 4		

LIBRYTXT — Library Text (Library entity in HG)

This parameter is used to write text into and read text from the HG Library. The library text can be a sequence word or descriptor. Sequence words are used by Control Language/ Multifunction Controller (CL/MC) sequences (formerly known as SOPL). Descriptors are the three descriptors that are used to label the S1, S2, and S3 status outputs for an EC regulatory data point that has been configured for the **Logic** algorithm. During configuration, the sequence words and descriptors are assigned index numbers 001-576.

Source:	User	Default Value:	All Blanks	Access Lock:	DEB
Value Type:	String	Value Range:	The valid character set consists of all characters on the Engineer's Keyboard.		

LM — Loop Manual

This parameter specifies the state of the Loop Manual mode in the slot.

Source:	System	Default Value:	Off	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	Off = Loop Manual mode not enabled On = Loop Manual mode enabled		

LOADDEST — Load Destination

This parameter determines whether the configuration information for this particular data point is to be loaded into only the HG, or into the HG and the hiway boxes that are connected to the HG. If the configuration information is to be loaded into only the HG, the user does not have to reconfigure all the parameters that are already residing in a hiway box. When you use the DEB to reconstitute such an HG point, the HG returns the HG-resident parameters with the parameters previously stored in the box, making a complete data base for this point.

Source:	User	Default Value:	Hg	Access Lock:	DEB
Value Type:	Enum. of LOADDEST	Value Range:	Hg = The configuration parameters are to be loaded into only the HG. Hg_Hiway = The configuration parameters are to be loaded into the HG and into the hiway box whose number has been specified by the <u>BOXNUM</u> (Box Number) parameter for this data point. The selection of the Hg and Brief entries for <u>LOADDEST</u> and <u>PTDISCL</u> parameters allows the user to fill in the absolute minimum number of lines when configuring a data point. The following is a combined listing of all the parameters that have to be filled in when Hg and Brief are selected. This listing contains parameters for all types of data points for all hiway boxes. On an average, the user may have to fill in only 15 to 20 of the listed parameters to configure a particular data point.		

LOADDEST

Parameter Identifier	Name
ALFMT	Alarm Format
ALGIDDAC	Algorithm
ALPRIOR	Alarm Priority
AVOPTION	Accumulation Option
BH	Batch History
BOXNUM	Box Number
BOXTYPE	Box Type
CNTLLOCK	Operator Control Change
DIGALFMT	Alarm Format
EUDESC	E.U. Descriptor
FBTIME	Feedback Time
HWYNUM	Hiway Number
INPTSSLT	Input Subslot
INTVARNM	Flag Number
INTVARNM	Numeric Number
INTVARNM	Timer Number
KEYWORD	Keyword
LBOXCLR	Lower Box Color
MODEPERM	Operator Mode Change
MOOUTIND	Momentary O/P Indication
NAME	Tag Name
NMBRINPT	Number of Inputs
NMBROUT	Number of Outputs
NMODATTR	Normal Mode Attribute
NMODE	Normal Mode
OUTBOXNM	Output Box Number
OUTSLTNM	Output Slot Number
OUTSSLT	Output Subslot Number
OVERVAL	Overview Value
PNTBOXTY	Box Type
PNTOPOP	Output Type
PNTPCTY	PC Type
PRESLOCK	Operator Preset Change
PRIMMOD	Primary Module Point ID
PTDESC	Point Descriptor
PTDISCL	Point Disclosure
SEQSLTSZ	Sequence Slot Size
SLOTNUM	Slot Number
SPLOCK	Operator Preset Change
STATE1	State 1 Text
STATE2	State 2 Text
UBOXCLR	Upper Box Color
UNIT	Unit ID
ZZTEXT	Non-Comp State 00 Text

NOTE

Internal parameter PVSOURCE is automatically set to Auto and ALENBST (Alarm Enable State) to Enable for the applicable points when the configuration parameters are loaded into the HG.

When a reconstitute procedure for LOADDEST is performed at a Universal Station, the value Hg_Hiway is returned into the Parameter Entry Display, regardless of which value was last loaded.

Logic — Logic (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Logic** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Logic** algorithm is that it provides logic computations that yield true/false responses.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	PVCHAR	\$AUXUNIT	SUPPIO	NMODE
UNIT	INPTCOND	ALFMT	CTLEQN (A,B,C)	NMODATTR
PTDESC	PVFORMAT	\$ALDLOPT	OUTIND	PSTMODE
KEYWORD	PVTEMP	\$DLBADAL	\$OPTOL	MODEPERM
PRIMMOD	PVEULO	DLYTIME	PVSIGNAL	MODE
ASSOCDSP	PVEUHI	\$REALARM	PVSLTSRC	\$MODESEL
HWYNUM	PVDSPhi	BADPVPR	SPSIGNAL	
PNTBOXTY	PVDSPL0	DEVHHPR	SPSLTSRC	
BOXNUM	OVERVAL	DEVHIPR	SPCHAR	
PTDISCL	PVRNGOP	\$DHSELECT	SPFORMAT	
SLOTNUM	PVCLAMP	DEVLLPR	SPTEMP	
LOADDEST	CALIBOFF	DEVLOPR	SPEULO	
ALGIDDAC		\$DLSELECT	SPEUHI	
		PVHHPR	SPDSPHI	
		\$HHSELECT	SPDSPLO	
		PVHIPR	SOASIGNL	
		\$HISELCT	SOASRC	
		PVLLPR	SOBSIGNL	
		\$LLSELECT	SOBSRC	
		PVLOPR	ZSIGNAL	
		\$LOSELCT	ZSLTSRC	
		(Alarm	MSIGNAL	
		Trip-point	MSLTSRC	
		Values)	ASIGNAL	
		ALPRIOR	ASLTSRC	
		CCRANK	CSIGNAL	
		CCPRIPNT	CSLTSRC	
		EIPPCODE	GREENSGL	
		EIPEVENT	GREENSRC	
		EIPENB	SENSSTOP	
			TFCLLM	
			TFCHLM	
			TD	
			GATETYP1	
			GATETYP2	
			GATETYP3	
			PULSE	
			ONDEL	
			OFFDEL	
			DEADBAND	
			LIBNUMEC	
			LIBRYIND	
			STATE1	
			STATE2	

Losel (CB)

Losel — Lo Selector (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Losel** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Losel** algorithm is that it is a 2-input LO-signal selector.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF OPLOLM OPHILM \$OPTOL TD	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Losel — Lo Selector (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Losel** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Losel** algorithm is that it is a 2-input LO-signal selector.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL OPLOLM	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Losel (MC)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
		\$HSELCT PVLOPR \$LOSELCT PVROCNPR \$RNSLCT PVROCPPR \$RPSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	OPHILM \$OPTOL TD	

LSP — Local Setpoint (Regulatory Data Points)

This parameter represents the local setpoint value of a regulatory data point. The LSP location is typically used as the Y input when the mode is AUTO.

Certain algorithms do not use this LSP location. In this case, the LSP is referred to as a "free variable" and may be used at the discretion of the user. When the LSP is a "free variable," its value is displayed in percent, and it may be changed through the supervisor or engineer access lock. When LSP is not a "free variable," its value is displayed in engineering units and it has an access lock of Read Only.

LSP is a "free variable" for the following EC algorithms:

Automan
Charact
Das
Deadtm
Incrmtl
Logic
Selovddc
Switch

Source: System		Default Value: 0.0	Access Lock: Supvr.
Value Type: Real		Value Range: -6.86% to 106.86% when <u>LSP</u> is a "free variable." -6.86 to 106.86 percent of scale in engineer units when <u>LSP</u> is not a "free variable."	

-M-**M — M Input Value (EC Regulatory Data Point)**

This parameter is the current value (in percent) of the M signal at an EC regulatory data point that has been configured for the **Incrmtl**, **Logic**, or **Selovddc** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	-6.86% to 106.86%		

MARK1SEG-MARK2SEG — Mark1seg-Mark2seg (EC Regulatory Data Point)

These parameters define the segment numbers for time marks 1 and 2, respectively, for an EC regulatory data point that has been configured for the **Rampsoak** algorithm. These parameters are used in conjunction with the MARK1TF-MARK2TF and MARK1TT-MARK2TT parameters to define the alarm trip-point reference levels, time elapsed from soak-segment start in minutes, segment number and offset. These parameters are to be filled in only if an alarm format of Alfnt01 has been selected for this algorithm.

Source:	User	Default Value:	1	Access Lock:	Supvr.
Value Type:	Integer	Value Range:	1-6		

MARK1TF-MARK2TF — Mark1tf-Mark2tf (EC Regulatory Data Point)**MARK1TT-MARK2TT — Mark1tt-Mark2tt**

These parameters define the time in minutes after which time marks 1 and 2 are set to false (tf = time false) or set to true (tt = time true) as required for an EC regulatory data point that has been configured for the **Rampsoak** algorithm. These parameters are used in conjunction with the MARK1SEG-MARK2SEG parameters to define the alarm trip-point reference levels, time elapsed from soak-segment start in minutes, segment number, and offset. These parameters are to be filled in only if an alarm format of Alfnt01 has been selected for this algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0 to 960.0 minutes		

Massflow (EC)

Massflow — Mass Flow (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Massflow** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Massflow** algorithm is that it is a 3-input auxiliary algorithm that provides mass-flow computation.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHP DEVHIPR \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHHP \$HHSELCT PVHIPR \$HSELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO OUTIND (A,B) PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC SPCHAR SPFORMAT SPTEMP SPEULO SPEUHI SPDSPHI SPDSPLO SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC ZSIGNAL ZSLTSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD KMF KY KZ BY BZ	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE EMSENB MODE \$MODESEL

MCREV — MC Revision (MC Box Data Point)

This parameter defines the revision level of an MC box.

Source:	User	Default Value:	No Default	Access Lock:	DEB
Value Type:	Enum. of MCREV	Value Range:	Rev20 = Revision level 20; MC does not support UAC system Rev30 = Revision level 30; MC supports UAC system Rev31 = Revision level 31; A-MC; MC supports UAC system and mode attribute tracking. For more information on mode attribute tracking, refer to 4.4.3, 4.4.4, and 4.4.5 in <i>System Control Functions</i> in the <i>Implementation/Startup & Reconfiguration - 2</i> binder.		

MDIGST — M Input Digital State (EC Regulatory Data Point)

This parameter is the current digital state of the M input signal at an EC regulatory data point that has been configured for the **Logic** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = M input is true. False = M input is false.		

MESSPEND — Message Pending Flag

This parameter indicates whether a message is pending for the operator.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of MESSPEND	Value Range:	Notpend = No message is pending. Pending = A message is pending.		

Midof3 (EC)

Midof3 — Middle of Three Selector (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Midof3** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Midof3** algorithm is that it yields an output based on an input-selection formula that is user-configurable and an input-survivorship formula should one, two, or three inputs go bad.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELECT DEVLLPR DEVLOPR \$DLSELECT PVHHPR \$HHSELECT PVHIPR \$HISELECT PVLLPR \$LLSELECT PVLOPR \$LOSELECT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLEQN (A,B,C) OUTIND PVSIGNAL PVSLT SRC SPSIGNAL SPSLT SRC INITCONF SP SOASIGNL SOASRC SOBSIGNL SOBSRC ZSIGNAL ZSLT SRC MSIGNAL MSLT SRC ASIGNAL ASLT SRC CSIGNAL CSLT SRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMORE EMSENB MODE \$MODESEL

MIOU — Manually Incremented Output (EC Regulatory Data Point)

This parameter is used to manually increment the output when the mode is AUTO, CAS, or BCAS at an EC regulatory data point that has been configured for the **Incrmtl** algorithm. The value entered for MIOU is added to the output when the output indication (OUTIND) is direct; otherwise it is subtracted. After the addition, MIOU is reset to zero.

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Real	Value Range:	-100.0% to 100.0%		

MODATTR — Current Mode Attribute

This parameter defines the attribute of the current mode of the data point. The attribute denotes who has the authority to change certain parameters of a data point. At a Universal Station, the mode attribute is displayed next to the mode of the data point. If the mode attribute is Program, a "P-" is displayed to the left of MODE. If the mode attribute is Operator, blanks are displayed to the left of MODE.

Source:	System	Default Value:	Operator (supplied by DEB)	Access Lock:	Oper.
Value Type:	Enum. of MODATTR	Value Range:	Operator = Operator can supply setpoint, output value, mode, ratio, and bias for a data point (operator-access level). Program = Program can supply setpoint, output value, mode, ratio, and bias for a data point (program access level). Normal = When <u>MODATTR</u> is changed to normal, the value for <u>MODATTR</u> is set to whatever was configured as the normal mode attribute (<u>NMODATTR</u>).		

NOTE

An access level of Ccont (Continuous Control) allows supervisory control from a module such as the AM to change the setpoint, output value, and mode of an HG data point.

MODATTR is displayed in reverse video on the Detail Display when the value in NMODATTR differs from that in MODATTR (except when NMODE contains None).

MODE — Operating Mode

On a configuration form, this parameter does not contain any entries and serves only to remind the user that this data point is in the manual mode after the data point is loaded during configuration data entry.

During operation, this parameter represents the current operating mode of a data point.

Source:	User	Default Value:	Man.	Access Lock:	Value Oper.	Type: Enum.
Value Range:	Man =	Manual; this mode provides the operator or the discontinuous program with direct control over the output value of the data point regardless of any automatic control strategy. Manual mode is applicable to regulatory data points, analog output and analog composite data points, and digital output and digital composite data points. HG points must be in Manual mode before they can be deleted.				
	Auto =	Automatic; in this mode, the output value is computed by the slot's configured algorithm (<u>ALGIDDAC</u>) and the setpoint comes from the local setpoint (<u>LSP</u>) location in the controller slot. An operator or a discontinuous program can change the setpoint value. Automatic mode applies to only regulatory data points.				
	Cas =	Cascade; in this mode, the data point receives its setpoint value from a primary data point.				

When the entry for RCASENB is On (true), the data point is in the remote cascade mode. This means that the setpoint value comes from a primary data point that is at a higher level of control, such as from a program or in the AM. This mode corresponds to the hiway-based computer mode.

When an operator or discontinuous control program selects the cascade mode for a data point, the data point does not immediately go into the cascade mode. Instead, a cascade request (CASREQ) is generated. This condition is displayed at the Universal Station as "-C" to the right of the current operating mode, and it also indicates to the primary data point that it may take over control of the data point. Later, when the AM requests the cascade mode, the cascade request is removed and the data point goes into the remote cascade mode.

When RCASENB is Off (false), the data point is in the local cascade mode. This means that the data point is to receive the setpoint value from a primary data point that is in the same hiway box. This corresponds to the cascade mode in hiway-based systems and it is applicable to only regulatory data points. When the operator or a discontinuous control program selects the cascade mode, the data point immediately goes into the cascade mode because the cascade request function does not apply.

The cascade mode applies regulatory data points, analog output and analog composite data points, and digital output and digital composite data points.

Beas = Backup cascade; this is the local cascade mode where a data point receives its setpoint from a primary data point that is in the same hiway box even though RCASENB is On. This mode corresponds to the cascade mode in hiway-based systems. The Beas mode cannot be requested if RCASENB is Off.

NOTES

1. If the selected algorithm is Das, only the manual mode is allowed.
2. The Bcas mode is applicable to only the following CB and MC algorithms:

CB	MC
Pidcma	Pidcma
Pidcm	Pidcm
Pidspc	Pidspc
Pidpl50C	
3. The Cas mode (when RCASENB = On) and the Bcas are not applicable to the following EC algorithms.
 Automan
 Charact
 Deadtm
 Logic
 Massflow
 Midof3
 Switch
4. The Cas mode (when RCASENB = Off) is not applicable to the following EC algorithms:
 Automan
 Charact
 Deadtm
 Logic
 Midof3
 Switch
5. When changing the mode of a data point to a mode other than manual, PVSOURCE must be Auto, and REDTAG must be Off.
6. For CB/MC regulatory data points; if the requested mode is Cas with RCASENB being Off, and the selected algorithm is other than **Overhisl**, **Overlosl**, or **Switch**, then SPSIGNAL must be configured to an entity other than itself if SPSLTSRC is Lsp. (The Y input cannot be configured to its own Lsp.)
7. For analog or digital data points, the requested mode must be either Man or Cas.
8. Updates to the value in this parameter depend on mode attributes, control interlocks, and the normal mode and normal-mode attribute. Refer to 4.4.3 and 4.4.4 in *System Control Functions* in the *Implementation/Startup and Reconfiguration* binder.
9. MODE is displayed in reverse video on the Detail Display when the value in NMODE differs from that in MODE (except when NMODE contains None).

MODEAPPL — Array of Applicable Modes

When retrieved, this parameter is a set of four Boolean values that correspond to the following four operating modes: MAN, AUTO, CAS, and BCAS. The values listed below (True and False) indicate which mode is currently allowed for the data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean Array	Value Range:	True = Mode is allowed. False = Mode is not allowed.		

MODEPERM — Operator Mode Change (Operator Keylock-Access-Level Change Permission)

This parameter defines whether the Operator Keylock-Access Level is allowed to change the control mode of the data point.

Source:	User	Default Value:	Permit	Access Lock:	Engr.
Value Type:	Enum. of MODEPERM	Value Range:	Permit = Operator key level can change the control mode for this data point. Nopermit = Operator key level cannot change the control mode of this data point.		

NOTE

Although the external name that is shown on HG Mode Configuration Displays is Nopermit, the internal name that is used for exception building and other purposes is Notperm.

MOOUTIND — Momentary O/P (Output) Indication

This parameter specifies whether the latched single or dual outputs from a digital-output data point or a digital-composite data point are to be momentary ("doorbell" action) or nonmomentary.

Source:	User	Default Value:	Nomoment	Access Lock:	DEB
Value Type:	Enum. of MOOUTIND	Value Range:	Nomoment = Action is to be nonmomentary. Moment = Action is to be momentary.		

MPV (EC)

Value Nomoment = Output from the data point is held in the commanded state
Range: (energized or de-energized) until another command is issued

by the  (Upper state change) or  (Lower state change)
and the  Keys.

Moment = Output from the data point is held in the commanded state

(energized or de-energized) until the  or  key is released. If Moment is entered, do not make any entry for the RCASENB parameter. Refer to the definition of the OUTIND (Output Indication) parameter for more information.

NOTE

If DIGALFMT=Cinddis, you must ensure that the MOOUTIND parameter contains Nomoment.

MPV — Manual PV in EC Lead/Lag Algorithms

This parameter is the current value of the PV (in percent) that is manually entered by the operator for a regulatory data point that has been configured for an EC lead/lag algorithm. MPV allows the operator to bumplessly switch the X input from the configured X input signal to the manually-entered PV (MPV). MPV can be changed only when SELINP is set to Sel.

Source:	System	Default Value:	0	Access Lock:	Oper.
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Value		Value	
Type:	Real	Range:	-6.86% to 106.86%

MSIGNAL — M Signal Type (EC Regulatory Data Point)

The M-signal types are the same as those listed for the ZSIGNAL parameter except that this signal is applied to the M input rather than the Z input. The tag name of the data point that supplies the M-input signal is specified by the MSLTSRC parameter. Refer to the definition of the ZSIGNAL parameter.

MSLTSRC — M Input Source (EC Regulatory Data Point)

This parameter defines the tag name of the data point that provides the input signal to the M input of an EC regulatory data point. Refer to the definition of NAME (Tag Name) for the valid character set.

Default

Value: Tag Name from Line 1 of configuration form

NOTE

To update the value in this parameter, MODE must contain Man, INITCONF must contain NOINIT, and EMS must contain Disable.

Mult — Multiplier (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Mult** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Mult** algorithm is that it is a 2-input auxiliary algorithm whose output is the product of two inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC INITCONF OPLOLM OPHILM \$OPTOL TD K K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Mult (MC)

Mult — Multiplier (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Mult** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Mult** algorithm is that it is a 2-input auxiliary algorithm whose output is the product of two inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC INITCONF SPLOLM SPHILM \$SPTOL OPLOLM OPHILM \$OPTOL TD K K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Multdiv (EC)

Multdiv — Multiplier/Divider (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Multdiv** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Multdiv** algorithm is that it is a 3-input auxiliary algorithm whose output is the result of multiplication or division.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHPR DEVHIPR \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHPR \$HHSELCT PVHIPR \$HSELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLEQN (A,B,C) OUTIND PVSIGNAL PVSLT SRC SPSIGNAL SPSLT SRC PVTRACK INITCONF SPCHAR SPFORMAT SPTEMP SPEULO SPEUHI SPDSPHI SPDSPLO SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC ZSIGNAL ZSLT SRC MSIGNAL MSLT SRC ASIGNAL ASLT SRC CSIGNAL CSLT SRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD KM KX KY KZ BO BX BY BZ	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE EMSENB MODE \$MODESEL

Multwman (CB)

Multwman — Multiplier (with Manual) (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Multwman** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Multwman** algorithm is that it is a 2-input auxiliary algorithm whose output is the product of two inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND PVSIGNAL PVSLSRC SPSIGNAL SPSLSRC INITCONF OPLOLM OPHILM \$OPTOL TD K K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Multwman (MC)

Multwman — Multiplier (with Manual) (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Multwman** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Multwman** algorithm is that it is a 2-input auxiliary algorithm whose output is the product of two inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC INITCONF SPLOLM SPHILM \$SPTOL OPLOLM OPHILM \$OPTOL TD K K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

-N-

NAME — Tag Name

The Tag Name defines the name of the data point in the system. It is used to identify the data point in all displays, listings, messages, alarms, etc., throughout the system. For example, Figure N-1 shows the location of Tag Name FC100T1 on the Group and Detail Displays.

Source: User	Default Value: All Underscores	Access Lock: DEB
Value Type: String	Value Range: The Tag Name can consist of up-to-eight characters, or if long point names are configured on the system, the name can be up to 16 characters. The permissible character set is as follows: Alphabetics A through Z (upper case only). Numerics 0 through 9 (an all numeric tag name is not permissible). Underscore () cannot be used as the first character or the last character in the tag name, and consecutive underscores are not allowed. Embedded space characters are not allowed.	

NCMPTXT1 — Text 1 (Data Hiway Data Point)

This parameter defines a descriptor of up to eight characters that is to be displayed when both inputs to a digital data point are noncomplementary (00 or 11), as defined by the ZZTEXT (Non-Comp State 00 Text) parameter. If Text1 is entered for ZZTEXT, this descriptor is displayed when both inputs are false (00). If Text2 is entered for ZZTEXT, this descriptor is displayed when both inputs are true (11).

Source: User	Default Value: Badpv	Access Lock: DEB
Value Type: String	Value Range: The permissible character set consists of all characters on the Engineer's Keyboard. Basically this set consists of Alphabetics A-Z, Numerics 0-9 and the following special characters: space ! % & ' () * + - / : ; > < = ? _ , . \$	

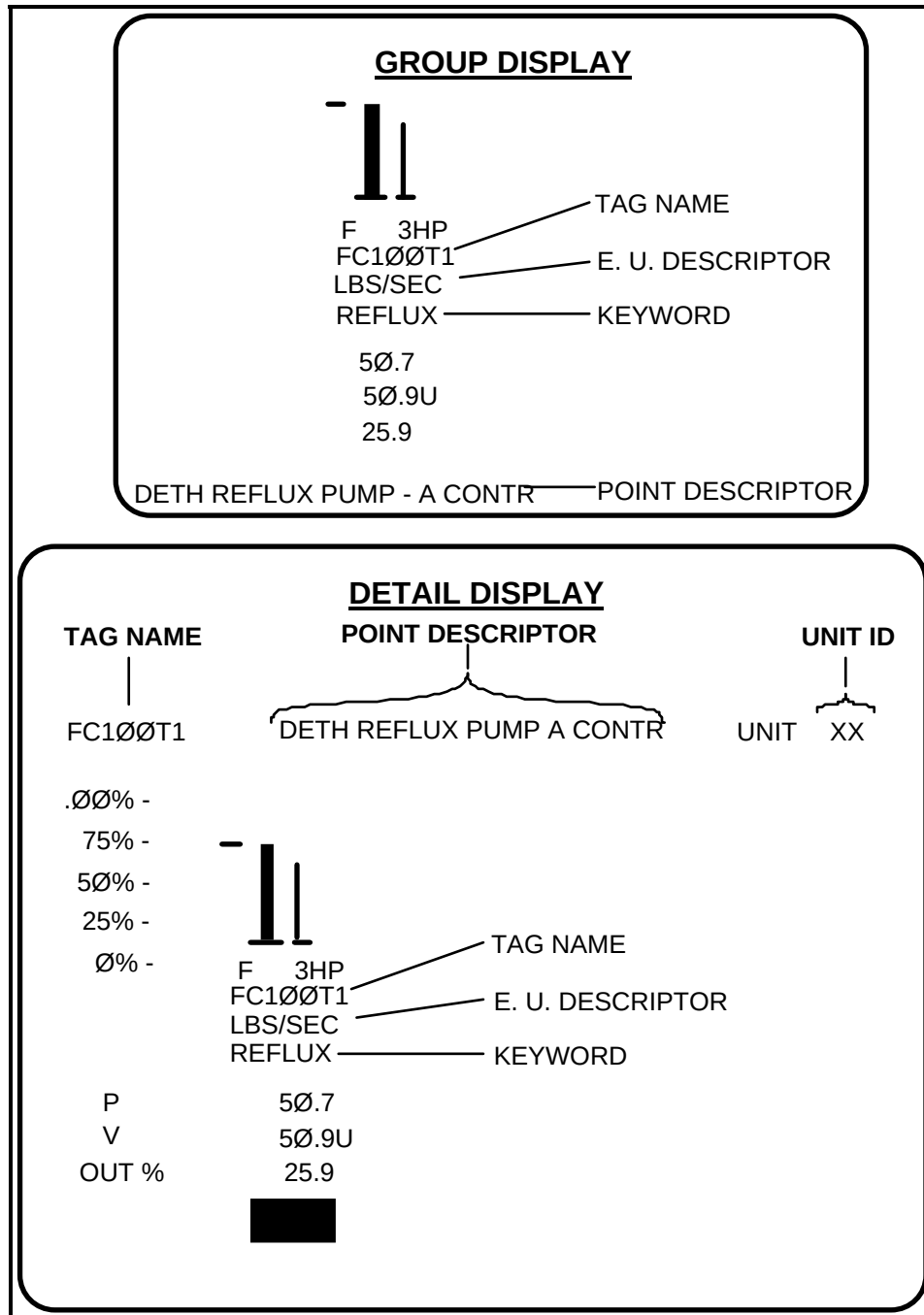


Figure N-1 — Location of Terms on Group and Detail Displays

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NCMPTXT2 — Text 2 (Data Hiway Data Point)

This parameter defines a descriptor of up to eight characters that is to be displayed when both inputs to a digital data point are noncomplementary (00 or 11) as defined by the ZZTEXT (Non-Comp State 00 Text) parameter. If Text1 is entered for ZZTEXT, this descriptor is displayed when both inputs are true (11). If Text2 is entered for ZZTEXT, this descriptor is displayed when both inputs are false (00).

Source:	User	Default Value:	Inbetwn	Access Lock:	DEB
Value Type:	String	Value Range:	The permissible character set consists of all characters on the Engineer's Keyboard. Basically this set consists of Alphabetics A-Z, Numerics 0-9 and the following special characters: space ! " % & ' () * + - / : ; > < = ? _ , . \$		

NMACIDP — # of ACIDP Points (CIU Processor Status Data Points)

This parameter defines the number of ACIDP points built on a node.

Source:	System	Default Value:	0	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 - 32767 (Subject to CG Point build limitations)		

NMBRBCFL — Number of I/O Files (MC Box Data Point)

This parameter defines the number of I/O files that are attached to this MC.

Source:	User	Default Value:	0	Access Lock:	DEB
Value Type:	Integer	Value Range:	0, 1, or 2		

NOTES

1. If no I/O files are to be configured, skip the PIUCRDY (Slot Card Type) and PIUCRDOP (Slot Card Option) parameters and continue with the BOXASSN (Box Assign) parameter.
2. If one I/O card file is to be configured, make entries in the PIUCRDY and PIUCRDOP parameter for slots 1-8.
3. If two I/O card files are to be configured, make entries in the PIUCRDY and PIUCRDOP parameter for slots 1-16.
4. If reconfiguring NMBRBCFL from 2 to 1, all points built in slots 9-16 must first be deleted.
5. If reconfiguring NMBRBCFL from 2 to 0 or from 1 to 0, all points built in those slots must first be deleted.

NMBRINPT — Number of Inputs

This parameter defines the number of inputs for a digital input or digital composite data point.

If a 2 is entered for this parameter (to specify that a digital data point has two inputs), the system automatically assigns the number of the second subslot as described for the INPTSSLT (Input Subslot) parameter.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1 = Digital data point has one input. 2 = Digital data point has two inputs.		

NMBROUT — Number of Outputs

This parameter defines the number of outputs for a digital output or digital composite data point.

If a 2 is entered for this parameter (to specify that a digital data point has two outputs), the system automatically assigns the number of the second subslot as described for the OUTSSLT (Output Subslot) parameter.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1 = Digital data point has one output. 2 = Digital data point has two outputs.		

NMCHPTS — # of Continuous History Points (HM Processor Status Data Points)

This parameter defines the number of history points currently loaded in continuous history groups.

Source:	System	Default Value:	0	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 - 32767 (Subject to history configuration limitations.)		

NMCRDP — # of CRDP Points (CIU Processor Status Data Points)

This parameter defines the number of CRDPs built on a node.

Source:	System	Default Value:	0	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 - 32767 (Subject to CG point build limitations.)		

NMODATTR— Normal Mode Attribute

This parameter defines the normal mode attribute. The normal mode attribute is the attribute that is copied into MODATTR when the NORM button is pressed at a Universal Station (also at that time, the content of NMODE is copied into MODE). This then becomes the mode attribute for this point. Refer to 4.4.4 in *System Control Functions* in the *Implementation/Startup and Reconfiguration* binder.

Source:	User	Default Value:	None	Access Lock:	Engr.
Value Type:	Enum. of MODATTR	Value Range:	Operator = Operator supplies the mode, setpoint, and output value. Program = Program supplies the mode, setpoint, and outpoint value. None = The normal mode does not have an attribute.		

NMODE — Normal Mode (CB, DHP, EC, HLPIU, MC)

This parameter defines the normal mode. The normal mode is the mode that is copied into MODE when the NORM button is pressed at a Universal Station (also at that time, the content of NMODATTR is copied into MODATTR). This then becomes the mode for this point. Refer to 4.4.4 in *System Control Functions* in the *Implementation/Startup and Reconfiguration - 2* binder.

This parameter is applicable to analog and digital output data points, analog and digital composite data points, and regulatory data points.

Source:	User	Default Value:	None	Access Lock:	Engr.
Value Type:	Enum. of MODE	Value Range:	None = Normal mode is not configured. Man = Normal mode is manual. Auto = Normal mode is automatic. Cas = Normal mode is cascade.		

(continued)

NOTES

1. MODE/MODATTR is displayed in reverse video when either conflicts with NMODE/NMODATTR except if NMODE is None.
2. When NMODE = None is selected when the point is built, NMODATTR defaults to None. When an NMODE selection other than None is chosen, the possible values for NMODATTR are None, Operator, and Program.
3. The user can set NMODATTR to None and NMODE to something other than None. In this case, pressing the NORM key results in MODE being changed to the configured NMODE and MODATTR staying the same.
4. For MC Rev31 data points:
 - a. If NMODE is None, NMODATTR is automatically None.
 - b. When NMODE is changed to something other than None, NMODATTR is automatically set to Operator. You can change NMODATTR to Program at that point, if desired.
 - c. You cannot configure NMODE to something other than None and NMODATTR to None.

NMSCHPR — # of Schematic Parameters Requested (US Processor Status Data Points)

This parameter defines the number of parameters requested on the last schematic display call-up.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

NN — Numerical Variable Array (MC)

This parameter contains the 88 possible numeric variables of an MC that are shown on a Process Module Detail Display. Each numeric can have a value that is in the range listed below, and the numeric value can be changed by the operator as required.

Source:	System	Default Value:	0.0	Access Lock:	Oper.
Value Type:	Real	Value Range:	-999999 to 999999		

NUMBBASE — PC Number Base (DHP)

This parameter indicates the number base for addresses in the programmable controllers that are attached to a DHP.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of NUMBBASE	Value Range:	Decimal = Decimal number base. Hex = Hexadecimal number base. Octal = Octal number base.		

-O-

OFFDEL — Off Delay; Refer to definition of **ONDEL**.

OFFNRMPR—Off Normal Alarm Priority

This parameter defines the priority of an off normal alarm generated by a data point.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.
------------	--

OFFST1PR—Offset 1 Alarm Priority**OFFST2PR—Offset 2 Alarm Priority****OFFST3PR—Offset 3 Alarm Priority**

This parameter defines the priority of an offset alarm generated by a data point.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			
		NOTE			
		If PRINTER is chosen, the alarm is lost if the printer is not working.			
	JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.			

OFFST1TP — Offset 1 Trip Point (EC Regulatory Data Point)**OFFST2TP — Offset 2 Trip Point****OFFST3TP — Offset 3 Trip Point**

These parameters define the offset trip points (in percent) that cause an alarm condition to be generated at an EC regulatory data point that has been configured for the **Rampsoak** algorithm. These parameters are to be filled in depending on the alarm format that is selected.

Source:	User	Default Value:	0.0%	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0% to 113.7%		

OII — Ordinary Input Identification for Switch Algorithm (EC Regulatory Data Point)

This parameter is a tuning constant at an EC regulatory data point that has been configured for the **Switch** algorithm. This parameter allows the operator or a TPS module to select one of three inputs (X, Y, or Z) that is to be routed to the output of the switch.

Source:	System or User	Default Value:	0	Access Lock:	Oper.
Value Type:	Enum. of OII	Value Range:	None = Selects no inputs; output holds at previous selection. Selectx = Selects X input. Selecty = Selects Y input. Selectz = Selects Z input.		

NOTE

If the data point was previously configured as a PID point (ALGIDDAC = PID), it is necessary to change the tuning constants—K, T1, and T2—to zero (0.0) prior to modifying the ALGIDDAC parameter. Otherwise, it will not be possible to change OII.

ONDEL — On Delay (EC Regulatory Data Point)**OFFDEL — Off Delay**

These parameters define the on and off delay times in minutes-of-time-remaining for an EC regulatory data point that has been configured for the **Logic** algorithm. Entries are to be made in these parameters only if one of the following gate types was entered for the GATETYP1-GATETYP3 parameters: And, Or, Xor, Nand, Nor, or Latch.

Source:	User	Default Value:	0.0 minutes	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0 to 9.992 minutes		

OP — Output Value for Analog and Regulatory Data Points

This parameter is the current output value (in percent) of the specified analog or regulatory data point. Output value can also be supplied by the operator, by a TPS module, or by CL/MC (if it is an MC data point) depending on the mode of the data point.

When the value of OUTIND is set to REVERSE, the parameter value is displayed as follows:
Displayed OP value = 100.0% – Actual OP value

Source:	System or User	Default Value:	0.0	Access Lock:	Oper.
Value Type:	Real	Value Range:	-6.86% to 106.86%		

NOTE

To update the value in this parameter, unless BOXSTAT contains Idle or Reset, MODE must contain Man. If a program or continuous control is to update the value in this parameter, HWYCTLST (for this hiway) and BOXFSTAT (for this box) must contain Full.

OP — Output Value for Digital Data Points

This parameter is the current output state of the specified digital output or digital composite data point. Output state can also be supplied by the operator, by a TPS module, or by CL/MC (if it is an MC data point) depending on the mode of the data point.

Source:	System or User	Default Value:	0	Access Lock:	Oper.
Value Type:	Self-Defining Enum.	Value Range:	For a data point with a single output: 0 = <u>STATE 1</u> 1 = <u>STATE 2</u>		

For a data point with dual outputs:
 0 = Neither STATE 1 nor STATE 2
 1 = STATE 1
 2 = STATE 2
 3 = Both STATE 1 and STATE 2

NOTE

To update the value in this parameter, unless BOXSTAT contains Idle or Reset, MODE must contain Man. If a program or continuous control is to update the value in this parameter, HWYCTLST (for this hiway) and BOXSTAT (for this box) must contain Full.

OPALDB – Output Alarm Deadband (R640)

Defines the deadband for the OP alarms in output units.

Source:	User	Default Value:	5 Output Units	Access Lock:	Engineer
Value Type:	Real	Value Range:	0-25 Output Units		

OPCRDOP — Output Card Option

This parameter is shown on the Detail Display for a data point to indicate that a conflict exists between the card option that was configured for the slot, and the actual card option that is installed in the hiway box slot. OPCRDOP specifies the card option that is installed in the hiway box slot. The configured card option is also displayed and the user should be aware of any mismatches.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of PIUCRDOP	Value Range:	None = Output slot option is not configured. Pulseout = Pulsed output. Latchout = Latchout = Latched Output.		

OPCRDTY — Output Card Type

This parameter is shown on the Detail Display for a data point to indicate that a conflict exists between the output card type that was configured for the slot, and the actual output card type that is installed in the hiway box slot. OPCRDTY specifies the output card that is installed in the hiway box slot. The configured output card is also displayed and the user should be aware of any mismatches.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of PIUCRDTY	Value Range:	None = This card type does not have an output. Analogot = Analog Output Card. Digout = Digital Output Card.		

OPHAFL – Output High Alarm Flag (R640)
OPLAFL – Output Low Alarm Flag (R640)

These parameters indicate whether the output value for the specified data point has reached or has exceeded the configured high or low output alarm trip point. With R670, if the output alarming is enabled and the OUTIND parameter is set to Reverse, the output indications for these parameters are swapped.

Source:	System	Default Value:	Off	Access Lock:	View
Value Type:	Logical	Value Range:	On Off If OUTIND = Reverse (R670): Displayed Output High Alarm flag value = Actual Output Low Alarm flag value Displayed Output Low Alarm flag value = Actual Output High Alarm flag value		

OPHIFL — Output Hi Limit Exceeded Flag
OPLOFL — Output LO Limit Exceeded Flag

These parameters indicate whether the output value for the specified data point has reached or has exceeded the configured high or low output limit. When the value of OUTIND is set to REVERSE, the value of the parameter is swapped.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Configured limit has been reached or exceeded. False = Configured limit has not been reached or exceeded. If OUTIND = Reverse: Displayed Output High Limit Exceeded flag = Actual Output Low Limit Exceeded flag value Displayed Output Low Limit Exceeded flag = Actual Output High Limit Exceeded flag value		

OPHILM — Output High Limit (CB, EC, & MC Regulatory Data Points)
OPLOLM — Output Low Limit

These parameters define the output limits (in percent) for a regulatory data point. The output limits do not apply when the data point is in the manual mode. However, they may apply when the point is in P-Manual mode. When the value of OUTIND is set to REVERSE, the values of the parameters are reversed.

Source:	User	Default Value:	-5.0% for low limit 105.0% for high limit	Access Lock:	Supvr.
Value Type:	Real	Value Range:	IF \$OPALEN=DISABLE (R640): OPHILM must be between -5.0% to 105.0% and must be $\geq$ OPLOLM. OPLOLM must be between -5.0% to 105.0% and must be $\leq$ OPHILM. IF \$OPALEN=ENABLE (R640): OPHILM must be between -5.0% to 105.0% and must be $\geq$ OPLOLM and must be $\geq$ OPHITP . OPLOLM must be between -5.0% to 105.0% and must be $\leq$ OPHILM and must be $\leq$ OPLOTP. If OUTIND = Reverse: Displayed Output High Limit value = 100.0% – Actual Output Low Limit value Displayed Output Low Limit value = 100.0% – Actual Output High Limit value		

OPHIPR – Output High Priority (R640)
OPLOPR – Output Low Priority (R640)

These parameters define the priority of an OP alarm for a regulatory data point. The alarm priority determines how the alarm is handled by the system. With R670, if the output alarming is enabled and the OUTIND parameter is set to Reverse, the output indications for these parameters are swapped.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engineer
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated. Event initiated Processing is NOT deactivated by this selection.			
	JOURNAL =	Alarm is historized but is not reported to the US RTJ and not annunciated. Alarm indicator appears on the Group and Detail displays.			
	LOW =	Alarm is historized, reported to the US RTJ, annunciated, and displayed on the Unit Alarm Summary, Group, and Detail displays.			
	HIGH =	Alarm is historized, reported to the US RTJ, annunciated, and displayed on the Unit and Area Alarm Summary displays, and the Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on the capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized, annunciated or displayed.			
	JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.			
		If OUTIND = Reverse (R670):			
		Displayed Output High Alarm Priority value = Actual Output Low Alarm Priority value			
		Displayed Output Low Alarm Priority value= Actual Output High Alarm Priority value			

OPHITP – Output High Trip Point (R640)
OPLOTP – Output Low Trip Point (R640)

These parameters define the high and low trip point values of the OP alarms. These parameters are displayed only if \$OPALEN is set to ENABLE. With R670, if the output alarming is enabled and the OUTIND parameter set to Reverse, the output indications for these parameters are reversed.

Source:	User	Default Value:	0.0% for low limit 100.0% for high limit	Access Lock:	Supvr.
Value Type:	Real	Value Range:	For OPHITP: OPLOTP to OPHILM $(OPHITP \geq OPLOTP + (2 * OPALDB))$ For OPLOTP: OPLOLM to OPHITP $(OPLOTP \leq OPHITP - (2 * OPALDB))$ If OUTIND = Reverse (R670): Displayed Output High Trip Point value = 100.0% – Actual Output Low Trip Point value Displayed Output Low Trip Point value = 100.0% – Actual Output High Trip Point value		

OPNTHRPR—Open Thermocouple Alarm Priority

This parameter defines the priority of an open thermocouple alarm generated by a data point. The alarm priority determines how the alarm is handled by the system. This parameter does not affect data point alarm detection and alarm indicator visibility on Group and Detail displays.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
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Value Type:	Enum. of ALPRIOR
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Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.
	EMERGENCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but does appear on Group and Detail displays.
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OSI — One-Shot Initializing Option (EC Regulatory Data Point)

This parameter provides the user with a means of initializing the delay table used in the EC **Deadtm** algorithm. By setting **OSI** to 1, the user prompts the initialization of the table. The **OSI** parameter is then set back to 0 by the EC. This can be requested only when the regulatory data point is in the MAN mode.

Source:	User	Default Value:	0	Access Lock:	Supvr.
Value Type:	Integer	Value Range:	0 = No initialization action. 1 = Initializes the delay table.		

NOTE

To update the value in this parameter, MODE must contain Man.

OUTBOXNM — Output Box Number or Box Number

This parameter defines the physical address of the hiway box (in which the output portion of this data point resides) on the Data Hiway. Refer to **BOXNUM** for the actual box numbers.

NOTE

For a composite data point, the input boxnumber and the output boxnumber must be the same number. The input and output portions of the data point must reside in the same box.

OUTIND — Output Indication (Analog Data Points)

This parameter defines how the output of an analog data point is to be displayed at the Universal Station.

Source:	User	Default Value:	Direct	Access Lock:	Engr. for regulatory data points. DEB for analog data points.
Value Type:	Enum. Of POLARITY	Value Range:	Direct = A 4 mA output from this data point is displayed as 0% at the Universal Station. A 20 mA output is displayed as 100%. Reverse = A 20 mA output from this data point is displayed as 0%. A 4 mA output is displayed as 100%.		

OUTIND — Output Indication (Regulatory Data Points)

This parameter defines how the output of a regulatory data point is to be displayed at the Universal Station. When the value of this parameter is set to Reverse, the value of output (OP) and the values of the output limits (OPHILM and OPLOLM) are reversed. The output indications for output limit exceeded flags (OPHIFL and OPLOFL) are swapped as shown in the listing.

Regulatory Control Point Parameters	If OUTIND = Direct, the value of the parameter is displayed as	If OUTIND = Reverse, the value of the parameter is displayed as
OP (Output)	Output value	Displayed OP = 100.0% – Actual OP value
OPHILM (OP High Limit)	Output High Limit value	Displayed OPHILM value = 100% - Output Low Limit value
OPLOLM (OP Low Limit)	Output Low Limit value	Displayed OPLOLM value = 100% - Output High Limit value
OPHIFL (OP High Flag)	Output High Limit Exceeded Flag value	Output Low Limit Exceeded Flag value
OPLOFL (OP Low Flag)	Output Low Limit Exceeded Flag value	Output High Limit Exceeded Flag value

With R670, if the output alarming is enabled and the OUTIND parameter is set to Reverse, the output indication for the output alarm trip points (OPHITP and OPLOTP) is reversed. The output indication for the output alarm priorities (OPHIPR and OPLOPR) and the output alarm flags (OPHAFL and OPLAFL) are swapped as shown in the following listing.

Regulatory Control Point Parameters	If OUTIND = Direct, the value of the parameter is displayed as	If OUTIND = Reverse, the value of the parameter is displayed as
OPHITP (OP High Trip Point)	Output High Trip Point value	Displayed OPHITP value = 100.0% - Output Low Trip Point value
OPLOTP (OP Low Trip Point)	Output Low Trip Point value	Displayed OPLOTP value = 100.0% - Output High Trip Point value
OPHIPR (OP High Alarm Priority)	Output High Alarm Priority value	Output Low Alarm Priority value
OPLOPR (OP Low Alarm Priority)	Output Low Alarm Priority value	Output High Alarm Priority value
OPHAFL (OP High Alarm Flag)	Output High Alarm Flag value	Output Low Alarm Flag value
OPLAFL (OP Low Alarm Flag)	Output Low Alarm Flag value	Output High Alarm Flag value

OUTIND — Output Indication (Regulatory Data Points) Continued....

Source:	User	Default Value:	Direct	Access Lock:	Engr. for regulatory data points. DEB for analog data points.
Value Type:	Enum. of POLARITY	Value Range:	Direct = A 4 mA output from this data point is displayed as 0% at the Universal Station. A 20 mA output is displayed as 100%. Reverse = A 20 mA output from this data point is displayed as 0%. A 4 mA output is displayed as 100%.		

NOTE

If the OUTIND parameter for the Regulatory Control point is set to Reverse, only the visual indication for the OP% and the OP% bar of the Regulatory Control point is reversed. The actual output is not reversed. The Analog Output OP% and OP% bar is different from the OP% and the OP% bar of the Regulatory Control point.

If the OPTDIR parameter is also set to Reverse, the output from the Analog Output is reversed. The OP% and the OP% bar on the Regulatory Control display matches the OPFINAL% value of the Analog Output point.

CAUTION

When using the OUTIND parameter, Honeywell recommends you to carefully design the logic of the affected parameters as described below:

For nodes on the LCN:

Before you can set the value of OUTIND to Reverse, you must modify your AM/CL programs, schematics, and TPS applications to accommodate the modified behavior of these parameters when they are displayed or used by other LCN nodes.

For controllers on the same Data Hiway:

CL/MC programs and connections within a box directly interact with HG and the actual values of the parameters. Therefore, there is a difference between the actual values used in the Data Hiway controllers and the displayed values of the affected parameters that are shown on the LCN nodes.

OUTIND — Output Indication (Digital Output and Digital Composite Data Points)

This parameter is used in conjunction with the PNTOPOP (Output Type) and MOOUTIND (Momentary Output Indication) parameters to specify the output action and the type of output that is to be provided by the single and dual outputs of a Digital Output Data Point or Digital Composite Data Point. The OUTIND parameter determines whether the output is direct or reverse acting, while the entries for the PNTOPOP and MOOUTIND parameters determine the type of output (latched, momentary, or pulsed) as shown in the following chart.

OUTPUT TYPE PARAMETER	ENTRY FOR PNTOPOP PARAMETER	ENTRY FOR MOOUTIND
Latched Momentary Pulsed	Latchout Latchout Pulseout	Nomoment Moment Not Applicable

The pulsewidth for the Pulsed Output Type is specified by the PULSEWTH (Pulsewidth) parameter.

Source: User **Default Value:** Direct **Access Lock:** DEB

Value Type: Enum. of POLARITY

Value Range: Refer to the following paragraphs that describe the possible entries (Direct, Reverse) for the output action as these keys are pressed:



(Upper-State Change),



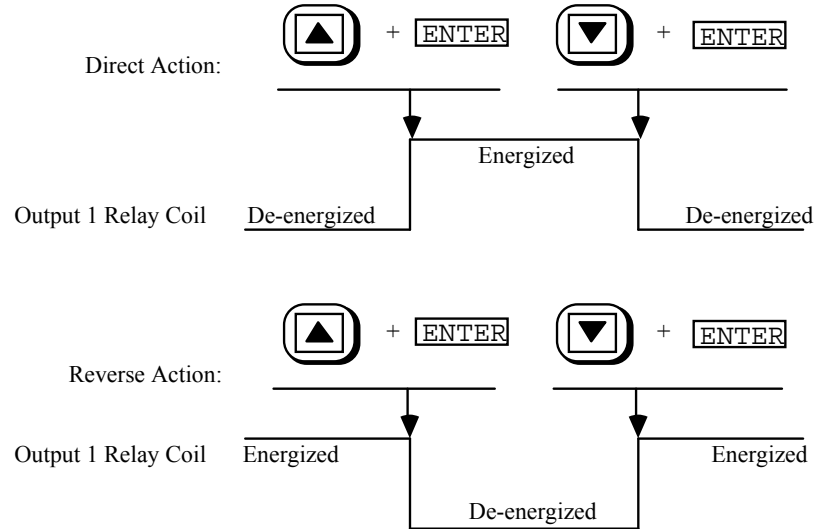
(Lower-State Change),

ENTER

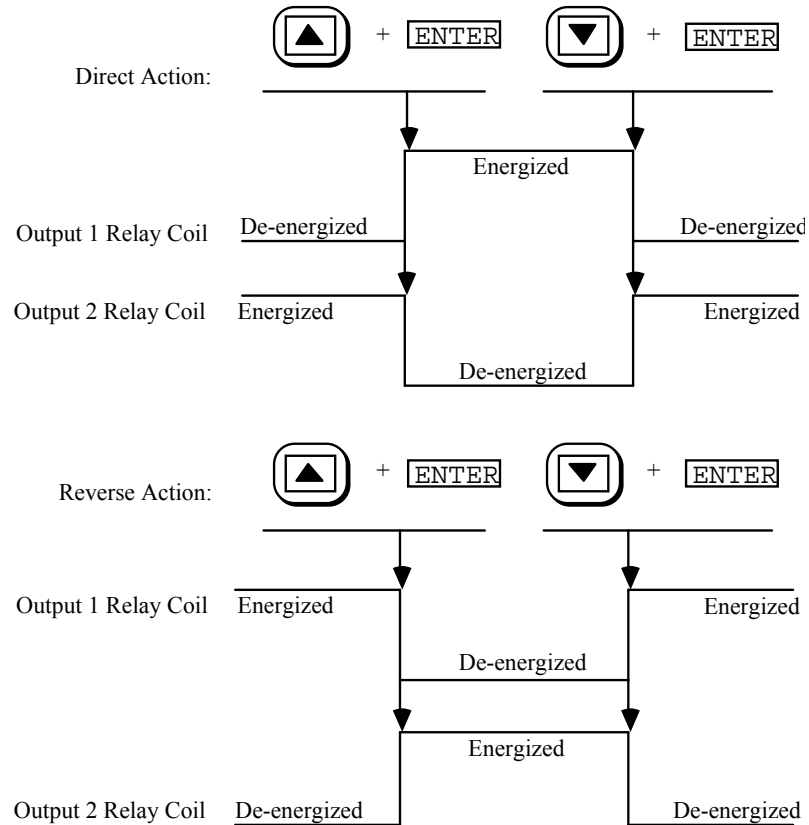
You should note that in these paragraphs, the references to the state of the Output Relay Coil (energized, de-energized) must be interpreted differently for the HLPIU, MC, and DHP as follows:

- HLPIU & MC— The relay is an integral part of the Output Card in the slot. The terminal panel has connections to both "Normally Open" and "Normally Closed" relay contacts.
- DHP— The DHP is a communications device that sends messages to the "configured coil" of the associated programmable controller (or other device) to energize or de-energize the coil. Because the DHP supports the interface to a wide variety of devices, the user should determine how to implement the coils in the device by reading the documentation provided by the PC manufacturer.

FOR A LATCHED SINGLE DIGITAL OUTPUT:



FOR A LATCHED DUAL DIGITAL OUTPUT:

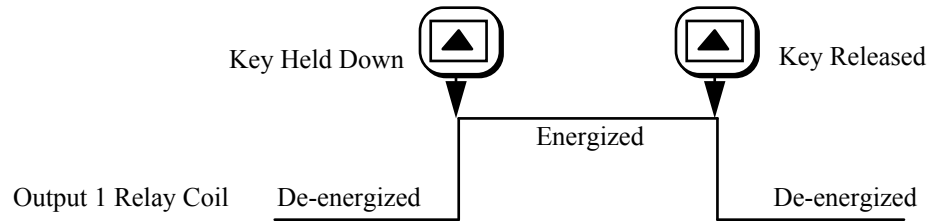


continued

FOR A MOMENTARY SINGLE DIGITAL OUTPUT:

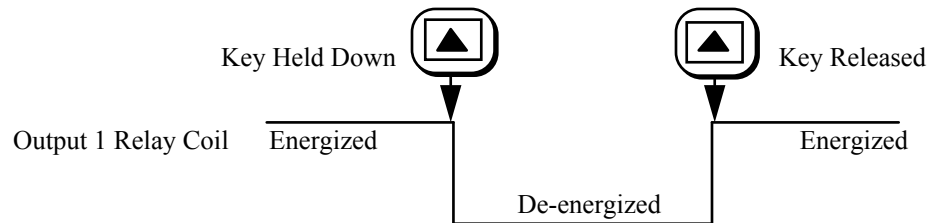
Direct Action:

Output 1 Relay Coil remains energized during the time this key is held down.



Reverse Action:

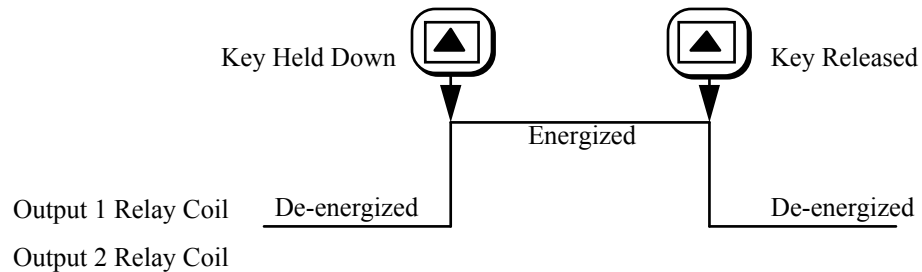
Output 1 Relay Coil remains de-energized during the time this key is held down.



FOR A MOMENTARY DUAL DIGITAL OUTPUT:

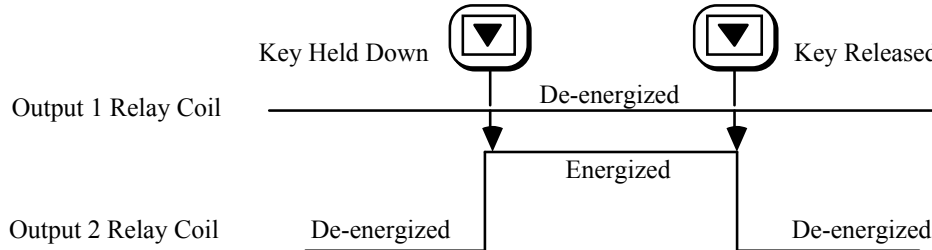
Direct Action:

Output 1 Relay Coil remains energized during the time this key is held down. Output 2 Relay Coil remains de-energized.



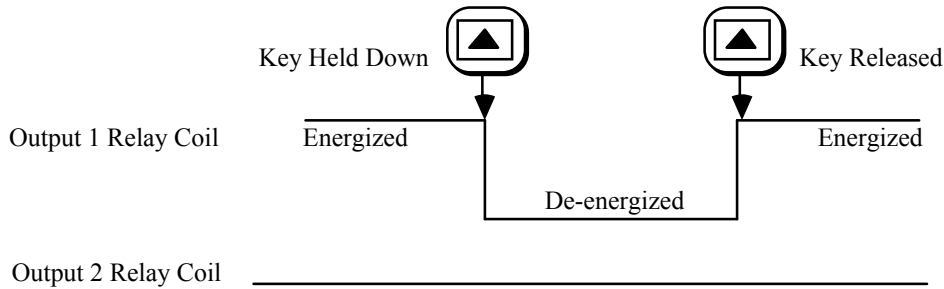
continued

Output 2 Relay Coil remains energized during the time this key is held down. Output 1 Relay Coil remains de-energized.

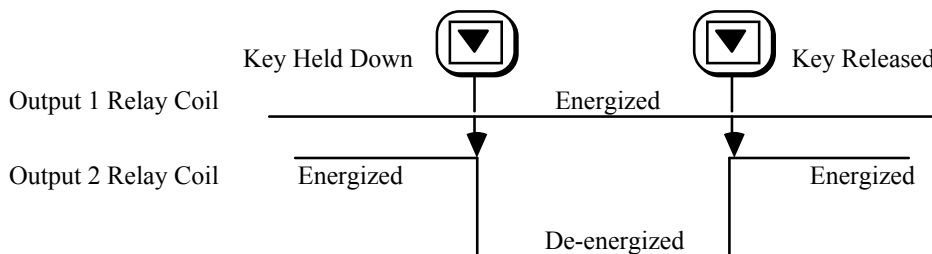


Reverse Action:

Output 1 Relay Coil remains de-energized during the time this key is held down. Output 2 Relay Coil remains energized.



Output 2 Relay Coil remains de-energized during the time this key is held down. Output 1 Relay Coil remains energized.

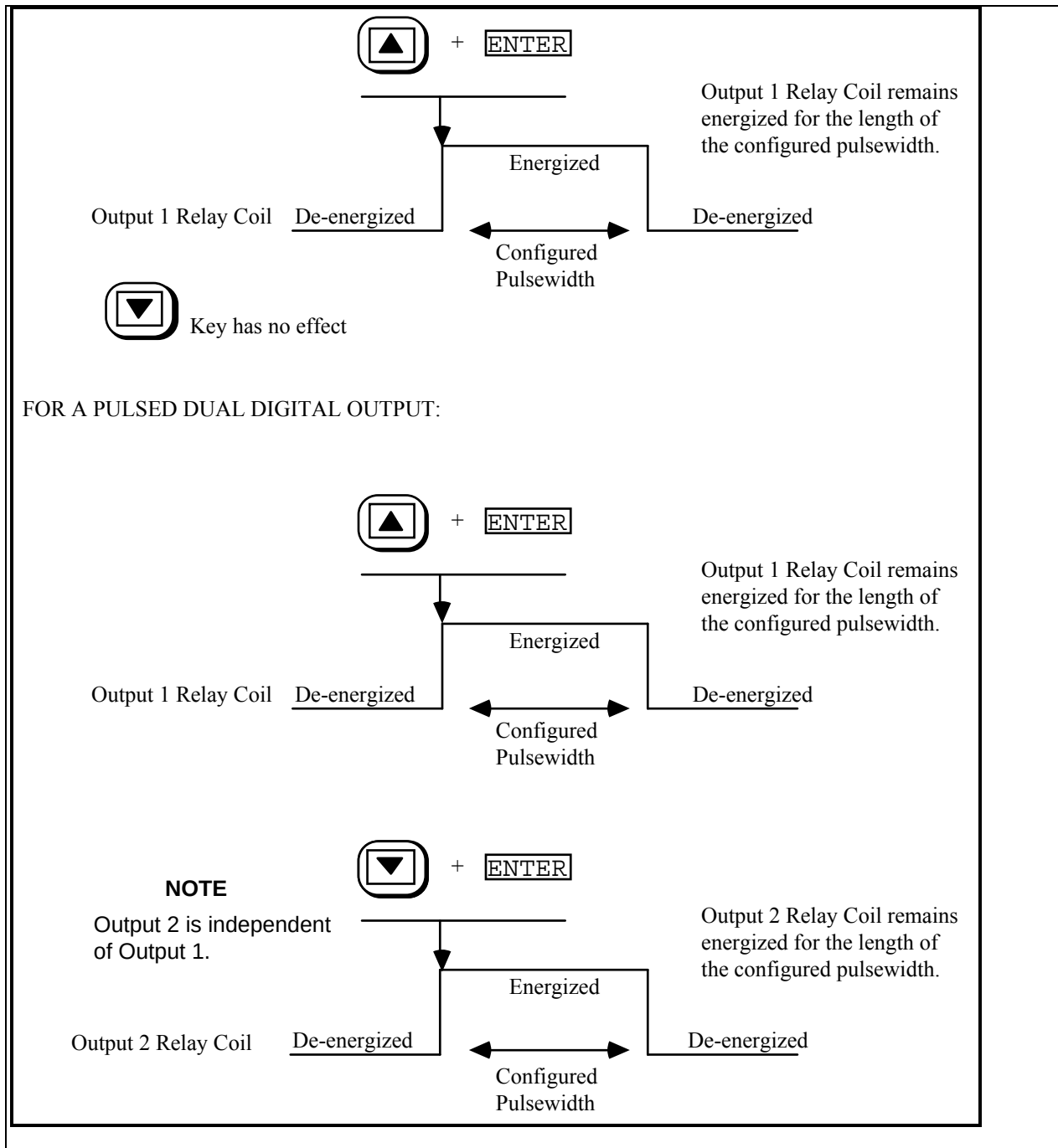


FOR A PULSED SINGLE DIGITAL OUTPUT:

NOTE

The OUTIND (Output Indication) parameter does not apply to Pulsed Digital Outputs. The DHP does not support pulsed digital outputs.

continued



OUTPT0 (EC)

OUTPT0-OUTPT8 — Outpt0-Outpt8 (Output 1-Output 8; EC Regulatory Data Point)

These parameters define the Output 0 - Output 8 coordinates that correspond to the following X coordinates for an EC regulatory data point that has been configured for the **Charact** algorithm:

Output 0; Output value corresponding to -6.86% of X input.
Output 1; X1
Output 2; X2
Output 3; X3
Output 4; X4
Output 5; X5
Output 6; X6
Output 7; X7
Output 8; Output value corresponding to 106.86% of X input.

Source:	User	Default Value:	0%	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-6.86% to 106.86%		

OUTSLTNM — Slot Number or Output Slot Number (DHP, HLPIU, MC)

This parameter specifies the slot number where the output of this data point resides in the file. (For the DHP, it specifies the software slot rather than the hardware slot.)

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	For DHP: 1-15 when entry for <u>BOXSIZE</u> parameter is Regular; 1-15 and 17-31 when entry is Extended. The 620-HIM has 30 slots (1-15 and 17-31) for each box address (and box address +32). For HLPIU: 1-16 when entry for <u>BOXSIZE</u> parameter is Regular; 1-32 when entry is Extended. For MC: 1-16 when entry for <u>NMBRBCFL</u> (Number of I/O Files) is 1; 1-32 when entry is 2.		

OUTSSLT — Subslot Number or Output Subslot Number (DHP, HLPIU, MC)

This parameter specifies which of the outputs (subslots) on the selected card (slot) is to be used as the output from this analog or digital data point.

For a dual output digital output or digital composite data point, this parameter defines the output (Output 1) that is controlled by the



(Upper State Change Key). The subslot number of the second output



(Output 2), which is controlled by the (Lower State Change Key), is automatically assigned by the system. The number that is assigned by the system is the next consecutive number that follows the number entered for the OUTSSLT parameter. For example, if a 3 is entered for the OUTSSLT parameter, the subslot number that is automatically assigned by the system for the second output (Output 2) is 4.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1, 2, 3, or 4 for analog points. 1, 2, 3, 4, 5, 6, 7 or 8 for digital points.		

OVB - (EC Regulatory Data Point)

The parameter that determines the number of outputs for a data point is NMBROUT (Number of Outputs). If a 2 is entered for the NMBROUT parameter (to specify that a digital data point has two outputs), the system automatically assigns the number of the second subslot as described above.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	The subslot number ranges for the various data points are as follows:		

Data Point Type	Subslot Number Range
DHP AO & AC	1-4
DHP DO & DC	1-8 (single-output data point) 1-7 (for the first output of a dual-output data point)
HLPIU AO & AC	1-4
HLPIU DO & DC	1-8 (single-output data point) 1-7 (for the first output of a dual-output data point)
MC AO & AC	1-4
MC DO & DC	1-8 (single-output data point) 1-7 (for the first output of a dual-output data point)
AO = Analog Output	DO = Digital Output
AC = Analog Composite	DC = Digital Composite

OVB (EC)

OVB — Ovb (Output Value Bias Limit on Output; EC Regulatory Data Point)

This parameter defines the output-value bias in percent for an EC regulatory data point that has been configured for the **Incrmtl** algorithm.

Source:	User	Default Value:	0.0%	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0% to 100.0%		

Overhisl — Override Hi Selector (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Overhisl** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for OVERHISL and OVERLOSL algorithms), and on the HG data entry displays. The outstanding characteristic of the **Overhisl** algorithm is that it is an 8-input Hi-signal selector that provides an internal feedback signal to nonselected input algorithms.

POINT ASSIGNMENT DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	SUPPIO	NMODE
UNIT	OUTIND	NMODATTR
PTDESC	OVINSS1	PSTMODE
KEYWORD	OVINSS2	MODEPERM
PRIMMOD	SGNLTY12	MODE
HWYNUM	OVINSS3	\$MODESEL
PNTBOXTY	OVINSS4	
BOXNUM	SGNLTY34	
PTDISCL	OVINSS5	
SLOTNUM	OVINSS6	
LOADDEST	SGNLTY56	
ALGIDDAC	OVINSS7	
	OVINSS8	
	SGNLTY78	
	OPLOLM	
	OPHILM	
	\$OPTOL	
	TD	

Overhisl (MC)

Overhisl — Override Hi Selector (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Overhisl** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for OVERHISL and OVERLOSL algorithms), and on the HG data entry displays. The outstanding characteristic of the **Overhisl** algorithm is that it is an 8-input Hi-signal selector that provides an internal feedback signal to nonselected input algorithms.

POINT ASSIGNMENT DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	SUPPIO	NMODE
UNIT	ANLGCARD	NMODATTR
PTDESC	ALMDIS	PSTMODE
KEYWORD	CONFTRND	MODEPERM
PRIMMOD	OUTIND	MODE
HWYNUM	\$OPTOL	\$MODESEL
PNTBOXTY	OVINSS1	
BOXNUM	OVS GSS1	
PTDISCL	OVINSS2	
SLOTNUM	OVS GSS2	
LOADDEST	OVINSS3	
ALGIDDAC	OVS GSS3	
	OVINSS4	
	OVS GSS4	
	OVINSS5	
	OVS GSS5	
	OVINSS6	
	OVS GSS6	
	OVINSS7	
	OVS GSS7	
	OVINSS8	
	OVS GSS8	
	OVINSS9	
	OVS GSS9	
	OVINSS10	
	OVS GSS10	
	OVINSS11	
	OVS GSS11	
	OVINSS12	
	OVINSS13	
	OVS GSS13	
	OVINSS14	
	OVS GSS14	
	OVINSS15	
	OVS GSS15	
	OVINSS16	
	OVS GSS16	
	OPLOLM	
	OPHILM	
	\$OPTOL	
	TD	

Overlosl (CB)

Overlosl — Override Lo Selector (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Overlosl** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for OVERHISL and OVERLOSL algorithms), and on the HG data entry displays. The outstanding characteristic of the **Overlosl** algorithm is that it is an 8-input Lo-signal selector that provides an internal feedback signal to nonselected input algorithms.

POINT ASSIGNMENT DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	SUPPIO OUTIND OVINSS1 OVINSS2 SGNLTY12 OVINSS3 OVINSS4 SGNLTY34 OVINSS5 OVINSS6 SGNLTY56 OVINSS7 OVINSS8 SGNLTY78 OPLOLM OPHILM \$OPTOL TD	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Overlosl (MC)

Overlosl — Override Lo Selector (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Overlosl** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for OVERHISL and OVERLOSL algorithms), and on the HG data entry displays. The outstanding characteristic of the **Overlosl** algorithm is that it is an 8-input Lo-signal selector that provides an internal feedback signal to nonselected input algorithms.

POINT ASSIGNMENT DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	SUPPIO	NMODE
UNIT	ALMDIS	NMODATTR
PTDESC	ANLGCARD	PSTMODE
KEYWORD	CONFTRND	MODEPERM
PRIMMOD	OUTIND	MODE
HWYNUM	OVINSS1	\$MODESEL
PNTBOXTY	OVSIGSS1	
BOXNUM	OVINSS2	
PTDISCL	OVSIGSS2	
SLOTNUM	OVINSS3	
LOADDEST	OVSIGSS3	
ALGIDDAC	OVINSS4	
	OVSIGSS4	
	OVINSS5	
	OVSIGSS5	
	OVINSS6	
	OVSIGSS6	
	OVINSS7	
	OVSIGSS7	
	OVINSS8	
	OVSIGSS8	
	OVINSS9	
	OVSIGSS9	
	OVINSS10	
	OVSIGSS10	
	OVINSS11	
	OVSIGSS11	
	OVINSS12	
	OVSIGSS12	
	OVINSS13	
	OVSIGSS13	
	OVINSS14	
	OVSIGSS14	
	OVINSS15	
	OVSIGSS15	
	OVINSS16	
	OVSIGSS16	
	OPLOLM	
	OPHILM	
	\$OPTOL	
	TD	

OVERPHAS (MC)

OVERPHAS — Override Phase Request (MC Process Module Data Point)

This parameter allows an operator to override the current phase of a process module data point.

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Enum. of JUMPDIR	Value Range:	Forward = Upper state-change button on US is used to move forward to the desired phase. Backward = Lower state-change button on US is used to move backward to the desired phase.		

NOTE

To update the value in this parameter, SEQEXEC must contain Pause, Fail, or Error; or PROCMOD must contain Ready.

OVERSTAT — Override Statement Request (MC Process Module Data Point)

This parameter allows an operator to override the current statement that is executing in a process module data point.

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Enum. JUMPDIR	Value Range:	Forward = Upper state-change button on US is used to move forward to the desired statement. Backward = Lower state-change button on US is used to move backward to the desired statement.		

NOTE

To update the value in this parameter, SEQEXEC must contain Pause, Fail, or Error; or PROCMOD must contain Redy.

OVERSTEP (MC)

OVERSTEP — Override Step Request (MC Process Module Data Point)

This parameter allows an operator to override the current step that is executing in a process module data point.

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Enum. JUMPDIR	Value Range:	Forward = Backward =	Upper state-change button on US is used to move forward to the desired step. Lower state-change button on US is used to move backward to the desired step.	

NOTE

To update the value in this parameter, SEQEXEC must contain Pause, Fail, or Error; or PROCMOD must contain Redy.

OVERVAL — Overview Value (Analog Input, Analog Composite, Counter, and Regulatory Data Points)

This parameter defines the amount of deviation (in percent) that causes the PV to reach the overview limit on the Overview Display. As shown in Figure O-1, the baseline represents the normal operating value for this PV.

Source:	User	Default Value:	0%	Access Lock:	Engr.
Value Type:	Integer	Value Range:	0% to 100%		

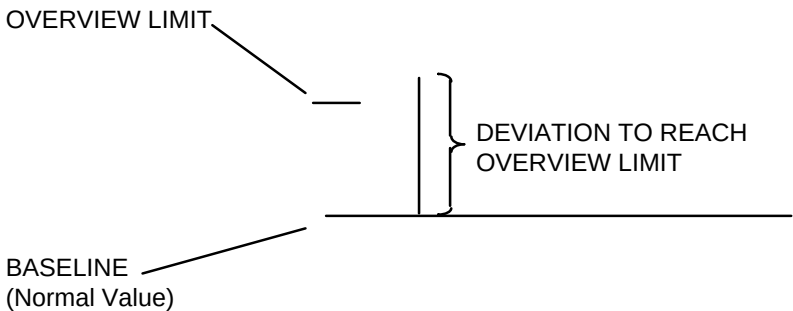


Figure O-1 — Overview Limit

M1302

OVERVAL — Overview Value (Digital Data Point and Flag Data Point)

This parameter defines whether an alarm condition for a digital input, digital composite, or flag data point is or is not to be shown on the Overview Display.

Source:	User	Default Value:	Off	Access Lock:	Engr. for digital data point. Oper. for flag data point.
Value Type:	Enum. of BOOLEAN	Value Range:	Off = Alarm condition is not to be displayed. On = Alarm condition is to be displayed.		

OVINSS1-OVINSS8 — Slot 1 Selection to Slot 8 Selection (CB Regulatory Data Point)

This parameter defines which CB slots provide input signals to the override selector (a regulatory data point that has been configured for the **Overhisl** or **Overlosl** algorithm). The override selector can handle a maximum of eight inputs. These inputs are the remote variables or output signals from the respective eight CB slots. The type of input signal that is to be applied to each slot is specified by the SGNLTY12-SGNLTY78 (Signal Type Slots 1&2 - Signal Type Slots 7&8) parameters.

Source:	User	Default Value:	Nosel	Access Lock:	Engr.
Value Type:	Enum. of OVRINSS	Value Range:	Nosel = This slot does not provide an input signal to the override selector. Sel = This slot provides an input signal to the override selector.		

NOTE

To update the value in this parameter, MODE must contain Man and INITCONF must contain Noinit.

OVINSS1-OVINSS16 (MC)

OVINSS1-OVINSS16 — Slot 1 Selection to Slot 16 Selection (MC Regulatory Data Point)

This parameter defines which MC slots provide input signals to the override selector (a regulatory data point that has been configured for the **Overhisl** or **Overlosl** algorithm). The override selector can handle a maximum of eight inputs. These inputs are the PVs or output signals from any eight of the 16 MC slots. The type of input signal that is to be applied to each pair of slots is specified by the OVSGSS1-OVSSG16 parameters.

Source:	User	Default Value:	Nosel	Access Lock:	Engr.
Value Type:	Enum. of OVRINSS	Value Range:	Nosel = This slot does not provide an input signal to the override selector. Sel = This slot provides an input signal to the override selector.		

NOTE

To update the value in this parameter, MODE must contain Man and INITCONF must contain Noinit.

OVSGSS1-OVSGSS16 — Signal Type (MC Regulatory Data Point)

These parameters define the types of signals that are applied to the inputs of the override selector (an MC regulatory data point that has been configured for the **Overhisl** or **Overlosl** algorithm).

Source:	User	Default Value:	Output	Access Lock:	Engr.
Value Type:	Enum. of OVSGSS	Value Range:	Output = Output signal from the respective slot is to be the input signal to the override selector. PV = PV signal from the respective slot is to be the input to the override selector.		

NOTE

To update the value in this parameter, MODE must contain Man and INITCONF must contain Noinit.

-P-

PARSEC — # Of Parameters Per Second (All Processor Status Data Points)

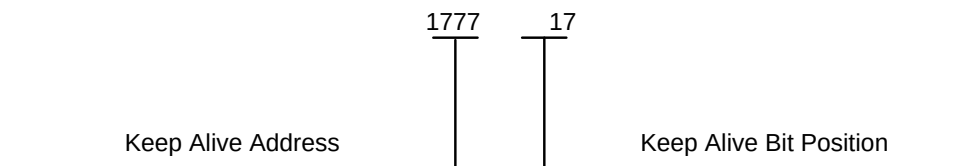
This parameter defines the number of parameters fetched from this node.

Source:	System	Default Value:	0.0	Access Lock:	Operator
Value Type:	Real	Value Range:	≥ 0.0		

PC1ALIVE-PC8ALIVE — Keep Alive Address (DHP Box Data Point)

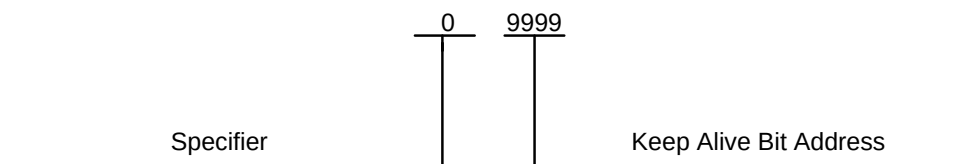
These parameters define the keep-alive addresses for the PCs. The keep-alive address is the address within the PC that must be periodically updated by the DHP to inform the PC that it is "alive." If the DHP is not "alive," all output data points (if the PC is configured appropriately) are switched to PC ladder logic control.

The 620-HIM requires only one keep-alive address; PC1ALIVE for each box data point. For an Allen-Bradley PC, this address is divided into two parts as follows (using address 177717 as an example):



The four most-significant digits are the keep alive address (in octal form) and are to be entered for this parameter. The two-least significant digits are the keep alive bit position (also in octal form) and are to be entered for the appropriate PC1ALVBT-PC8ALVBT parameter.

For a Modicon or Honeywell PC, this address is divided into two parts as follows (using address 09999 as an example):



The most-significant digit is the specifier and is entered in the appropriate PC1ALVSP- PC8ALVSP parameter (the specifier bit is always 0 for the keep alive address). The four least-significant digits are the keep alive address (in decimal form) and are to be entered for this parameter.

Source:	User	Default Value:	0	Access Lock:	DEB
Value Type:	Integer	Value Range:	For Allen-Bradley: 0 to 1777 (Octal). For Modicon or Honeywell: 0 to 9999 (Decimal).		

PC1ALVBT-PC8ALVBT — Keep Alive Bit Position (DHP Box Data Point)

This parameter defines the keep alive bit position for the keep alive address of an Allen-Bradley PC.

Source:	User	Default Value:	0	Access Lock:	DEB
Value Type:	Integer	Value Range:	00 to 17 Octal		

PC1ALVSP-PC8ALVSP — Specifier (DHP Box Data Point)

This parameter defines the specifier code for the keep alive address of a Modicon or Honeywell PC (or the 620 HIM). No entry is required for this parameter because the default value is zero, which is the required specifier code.

Source:	User	Default Value:	0	Access Lock:	DEB
Value Type:	Integer				

PC1PORT- PC8PORT — Port Number (DHP Box Data Point)

This parameter defines which of the four DHP ports is assigned to this PC. Use the default value for a 620-HIM.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1 to 4		

PC1PORTA-PC8PORTA — Port Address (DHP Box Data Point)

This parameter defines the address of the PC on a DHP port. A zero (0) entry indicates that the PC is not configured. Use the default value for a 620 HIM.

Source:	User	Default Value:	0	Access Lock:	DEB
Value Type:	Integer	Value Range:	0 = Undefined. 1 to 377 (Octal) for Allen-Bradley. 1 to 247 (Decimal) for Modicon or Honeywell.		

PC1TYPE-PC8TYPE — PC1-PC8 Model Type (DHP Box Data Point)

This parameter defines the model type of this PC. No entry is required in this parameter for a 620 HIM.

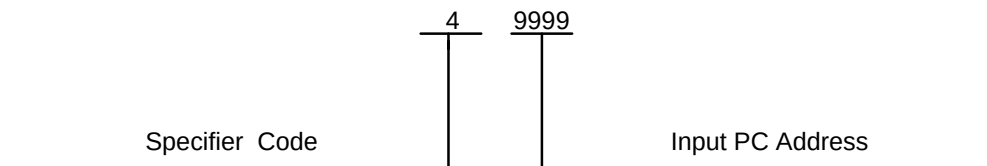
Source:	User	Default Value:	Notconfig	Access Lock:	DEB
Value Type:	Enum. of PCTYPE	Value Range:	If Modicon has been entered for the <u>BOXPROT</u> parameter: Notconfig = Not configured M384 = Model Type 384 (or GE 6/60) M484 = Model Type 484 (or GE 6/600) M584 = Model Type 584 (or GE 6/600) Select M584 if a Modicon 884 is to be used. If Allenbrd has been entered for the <u>BOXPROT</u> parameter: Notconfig = Not configured Aplc = Model Type PLC Aplc2 = Model Type PLC2 Aplc215 = Model Type PLC215 Aplc220 = Model Type PLC220 Aplc230 = Model Type PLC230 Aminplc2 = Model Type Mini PLC2 If Honeywell has been entered for the <u>BOXPROT</u> parameter: Notconfig = Not configured Ipc620 = DHP supports all Honeywell PC models and the 620 HIM types. The model type does not have to be specified.		

PCADDRI1 (DHP)

PCADDRI1 — Input PC Address (DHP Analog or Counter Data Point)

For an analog input, analog composite, or counter data point, this parameter defines the address in the PC of the current input value for this data point.

Source:	User	Default Value:	0	Access Lock:	Engr.
Value Type:	Integer	Value Range:	If Allenbrd has been entered for the <u>PNTPCITY</u> (PC Type) parameter, input PC Address has a range from 1 to 1777 (octal). If Modicon or Honeywell has been entered for the <u>PNTPCITY</u> (PC Type) parameter, input PC address has a range of 1 to 9999 (decimal). The <u>PCADDRI1</u> parameter works with Specifier Code parameter <u>SPECIFI1</u> to determine the address location in the PC as follows:		



Specifier code is 3 or 4 as defined by SPECIFI1 parameter in a separate line on the configuration form.

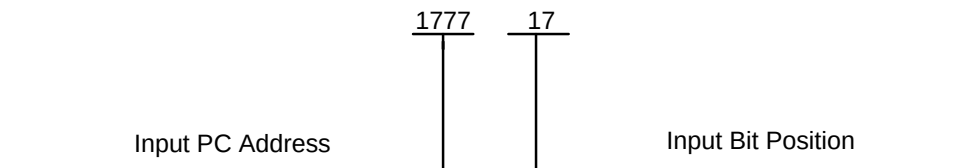
For Modicon 384 and 484, input PC address range is from 1 to 999 (decimal). For Modicon 584 and Honeywell 620 LCS, Input PC address range is from 1 to 9999 (decimal). The selected address is entered for this parameter.

PCADDRI1 (DHP)

PCADDRI1 — Input 1 PC Address (DHP Digital Data Point)
PCADDRI2 — Input 2 PC Address

These parameters define the addresses in the PC (or the 620 HIM) of the current states of input 1 and input 2 for a digital input or digital composite data point. For a digital data point with a single input or dual inputs, the PC address of the first input (Input 1) is defined by the PCADDRI1 (Input 1 PC Address) parameter. For a digital data point with dual inputs, the PC address of the second input (Input 2) is defined by the PCADDRI2 (Input 2 PC Address) parameter.

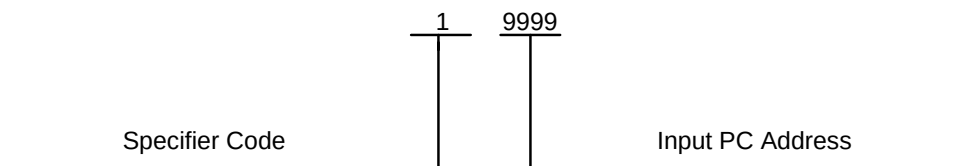
Source:	User	Default Value:	0	Access Lock:	Engr.
Value Type:	Integer	Value Range:	If Allenbrd has been entered for the <u>PNTPCITY</u> (PC Type) parameter, Input PC Address has a range from 100 to 177717 (octal). This address is divided into two parts as follows (using address 177717 as an example):		



The four most-significant digits (1777) represent the portion of the Input PC Address that is to be entered for the PCADDRI1 or PCADDRI2 parameter. Range is from 1 to 1777, octal.

The two least-significant digits (17) represent the portion of the Input PC Address that is to be entered for the PCBITI1 (Input 1 Bit Position) or PCBITI2 (Input 2 Bit Position) parameter. This portion of the Input PC Address is entered on another line of the configuration form. Range is from 00 to 17, octal.

If Modicon or Honeywell has been entered for the PNTPCITY (PC Type) parameter, output PC address has a range of 1 to 9999 (decimal). The PCADDRI1 parameter works with Specifier Code parameter SPECIFI1 to determine the address location in the PC as follows:



Specifier code is 0 or 1 as defined by SPECIFI1 parameter in a separate line on the configuration form. SPECIFI2 works in conjunction with PCADDRI2.

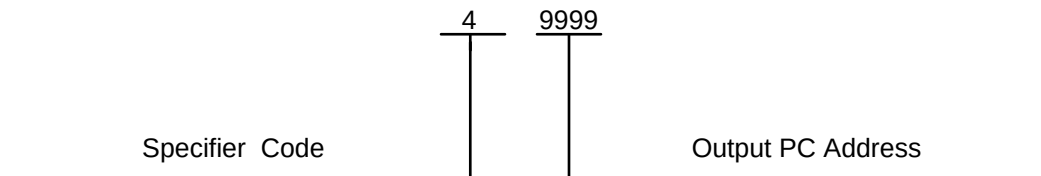
For Modicon 384 and 484, input PC address range is from 1 to 999 (decimal). For Modicon 584 and Honeywell 620 LCS, Input PC address range is from 1 to 9999 (decimal).

PCADDRO1 (DHP)

PCADDRO1 — Output PC Address (DHP Analog Data Point)

For an analog output or analog composite data point, this parameter defines the address in the PC (or 620 HIM) of the current output value for this data point.

Source:	User	Default Value:	0	Access Lock:	Engr.
Value Type:	Integer	Value Range:	If Allenbrd has been entered for the <u>PNTPCTY</u> (PC Type) parameter, output PC Address has a range from 1 to 1777 (octal). If Modicon or Honeywell has been entered for the <u>PNTPCTY</u> (PC Type) parameter, output PC address has a range of 1 to 9999 (decimal). The <u>PCADDRO1</u> parameter works with Specifier Code parameter <u>SPECIF01</u> to determine the address location in the PC as follows:		



Specifier code is 4 for analog output PC addresses as defined by SPECIF01 parameter in a separate line on the Configuration Form.

For Modicon 384 and 484, output PC address range is from 1 to 999 (decimal).
For Modicon 584 and Honeywell 620 LCS, output PC address range is from 1 to 9999 (decimal).

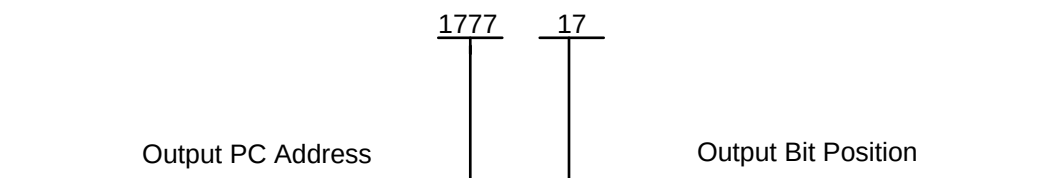
PCADDRO1 (DHP)

PCADDRO1 — Output 1 PC Address (DHP Digital Data Point)

PCADDRO2 — Output 2

For a digital output or digital composite data point, this parameter defines the address in the PC (or 620 HIM) of the current state of Output 1. For a digital data point with dual outputs, the PC (or 620 HIM) address of Output 2 is defined by the PCADDRO2 (Output 2 PC Address) parameter.

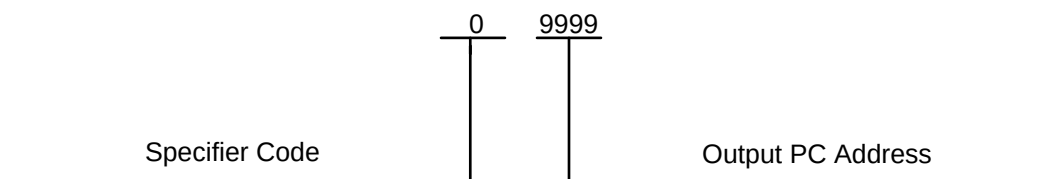
Source:	User	Default Value:	0	Access Lock:	Engr.
Value Type:	Integer	Value Range:	If Allenbrd has been entered for the <u>PNTPCY</u> (PC Type) parameter, output PC Address has a range from 1 to 1777 (octal).		



The four most-significant digits (1777) represent the portion of the Output PC Address that is to be entered for the PCADDRO1 or PCADDRO2 parameter. Range is from 1 to 1777(octal).

The two least-significant digits(17) represent the portion of the Output PC Address that is to be entered for the PCBITO1 (Output 1 Bit Position) or PCBITO2 (Output 2 Bit Position) parameter. This portion of the Output PC Address is entered on another line of the configuration form. Range is 00 to 17 (octal).

If Modicon or Honeywell has been entered for the PNTPCY (PC Type) parameter, output PC address has a range of 1 to 9999 (decimal). The PCADDRO1 parameter works with Specifier Code parameter SPECIFO1 to determine the address location in the PC as follows:



Specifier code is 0 for digital output PC addresses as defined by SPECIFO1 parameter in a separate line on the Configuration Form. SPECIFO2 is used with PCADDRO2.

For Modicon 384 and 484, output PC address range is from 1 to 999 (decimal). For Modicon 584 and Honeywell 620 LCS, output PC address range is from 1 to 9999 (decimal).

PCBITI1 (DHP)

PCBITI1 — Input 1 Bit Position (DHP Digital Data Point)

PCBITI2 — Input 2 Bit Position

These parameters define the two least-significant digits (in octal form) of the Input PC Address when Allenbrd (Allen-Bradley) has been entered for the PCNTPCTY (PC Type) parameter. PCBITI1 is used in conjunction with PCADDRI1 (Input 1 PC Address); PCBITI2 is used in conjunction with PCADDRI2 (Input 2 PC Address).

Source:	User	Default Value:	00	Access Lock:	Engr.
Value Type:	Integer	Value Range:	00-17 (Octal)		

PCBITO1 — Output 1 Bit Position (DHP Digital Data Point)

PCBITO2 — Output 2 Bit Position

These parameters define the two least-significant digits (in octal form) of the Output PC Address when Allenbrd (Allen-Bradley) has been entered for the PCNTPCTY (PC Type) parameter. PCBITO1 is used in conjunction with PCADDRO1 (Output 1 PC Address); PCBITO2 is used in conjunction with PCADDRO2 (Output 2 PC Address).

Source:	User	Default Value:	00	Access Lock:	Engr.
Value Type:	Integer	Value Range:	00-17 (Octal)		

PCCONFI1 — PC Bad Configuration Status for Input 1 (DHP)

PCCONFI2 — PC Bad Configuration Status for Input 2

These parameters indicate whether the input or inputs to a DHP data point have been configured properly. Parameter PCCONFI1 is applicable to all DHP analog and digital data points; PCCONFI2 is applicable to only the second input of digital data points that have dual inputs. The values of PCCONFI1 and PCCONFI2 are set by the DHP.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of PCCONFG	Value Range:	Notcnfg = This input has not been configured. Uncertn = Configuration of this input has not been checked. Bad = This input has been configured incorrectly. Ok = This input has been configured correctly.		

PCCONFO1 (DHP)

PCCONFO1 — PC Bad Configuration Status for Output 1 (DHP)

PCCONFO2 — PC Bad Configuration Status for Output 2

These parameters indicate whether the output or outputs of a DHP data point have been properly configured. Parameter PCCONFO1 is applicable to DHP analog and digital data points; PCCONFO2 is applicable to only the second output of digital data points that have dual outputs.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of PCCONFG	Value Range:	Notcnfg = This output has not been configured. Uncertn = Configuration of this output has not been checked. Bad = This output has been configured incorrectly. Ok = This output has been configured correctly.		

PDEVTP — Prealarm Value (MC Counter Data Point)

This parameter defines the first alarm trip point for an MC counter data point. When the accumulated value is greater than or equal to the preset value minus the prealarm value, this alarm occurs as a warning.

When the accumulated value (AV) exceeds the preset value (PRESET), an additional alarm will occur.

Source:	User	Default Value:	0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0 to 9999		

PFPSHLDB — Hold Breath Count (All Processor Status Data Points)

This parameter defines the "hold breath" count based on the Data Access timeout in accessing data, cumulative since node startup or reset action by engineer. (This is intended to be a measure of the number of failovers, particularly HG failovers.)

Source:	System	Default Value:	0.0	Access Lock:	Engineer
Value Type:	Real	Value Range:	≥ 0.0		

PHASE — Phase Name (MC Process Module Data Point)

This parameter contains the name of the current phase that is executing in a process module.

Source:	System	Default Value:	All Blanks	Access Lock:	Read Only
Value Type:	String				

PHASEAL — Phase Alarm (MC Process Module Data Point)

This parameter indicates whether the phase time (PHASETIM) has expired.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of Boolean	Value Range:	Off = Phase alarm is not active; phase time has not yet expired. On = Phase alarm is active; phase time has expired.		

PHASEIDX— Phase Index (MC Process Module Data Point)

This parameter contains the text-library index of the current phase that is executing in a process module.

Source:	System	Default Value:	0	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0, 128 through 255, and 512 through 576		

PHASETIM — Phase Time (MC Process Module Data Point)

This parameter indicates the amount of time remaining in minutes until phase execution has been completed. PHASETIM is set by a phase statement and is decremented by one each minute while the sequence execution state is Run, Pause, Fail, or Error. When the value of PHASETIM is decremented to 0, a phase alarm is generated and the PHASEAL bit is set to 1.

When the sequence program has no phase time or alarm time specified in the phase statement, PHASETIM is set to 0 and a phase alarm cannot be generated.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 to 9999 minutes		

Pidcm — PID CM (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidcm** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidcm** algorithm is that a TPS module directly generates the output, and that the data point can shed to the manual mode for backup.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO CTLACTN CTLEQN (A,B) OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidcm — PID CM (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Pidcm** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidcm** algorithm is that a TPS module directly generates the output, and that the data point can shed to manual backup.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSLCT PVROCPPR \$RPSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO CTLACTN CTLEQN (A,B) ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidcma — PID CMA (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidcma** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidcma** algorithm is that a TPS module directly generates the output, and that the data point can shed to the automatic or backup cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	DEVHITP	SUPPIO	NMODE
UNIT	PVCHAR	\$AUXUNIT	CTLACTN	NMODATTR
PTDESC	INPTCOND	ALFMT	CTLEQN (A,B)	PSTMODE
KEYWORD	PVFORMAT	PVLOTP	OUTIND	MODEPERM
PRIMMOD	PVTEMP	PVHITP	PVSIGNAL	MODE
ASSOCDSP	PVEULO	DEVLOTP	PVSLTSRC	\$MODESEL
HWYNUM	PVEUHI	\$REALARM	SPSIGNAL	
PNTBOXTY	PVDSPHI	DEVHIPR	SPSLTSRC	
BOXNUM	PVDSPLO	\$DHSELCT	PVTRACK	
PTDISCL	OVERVAL	DEVLOPR	INITCONF	
SLOTNUM	PVRNGOP	\$DLSELCT	SP	
LOADDEST	PVCLAMP	PVHIPR	OPLOLM	
ALGIDDAC	CALIBOFF	\$HISELCT	OPHILM	
		PVLOPR	\$OPTOL	
		\$LOSELCT	ITLOLM	
		ALPRIOR	ITHILM	
		CCRANK	TD	
		CCPRIPNT	K	
		EIPPCODE	T1	
		EIPEVENT	T2	
		EIPENB		

Pidcma — PID CMA (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Pidcma** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidcma** algorithm is that a TPS module directly generates the output, and that the data point can shed to the automatic or backup cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	\$AUXUNIT	SUPPIO	NMODE
UNIT	PVCHAR	ALFMT	CTLACTN	NMODATTR
PTDESC	INPTCOND	\$ALDLOPT	CTLEQN (A,B)	PSTMODE
KEYWORD	PVFORMAT	\$REALARM	ANLGCARD	MODEPERM
PRIMMOD	PVTEMP	PVLOTP	ALMDIS	MODE
ASSOCDSP	PVEULO	PVHITP	CONFTRND	\$MODESEL
HWYNUM	PVEUHI	DEVLOTP	OUTIND	
PNTBOXTY	PVDSPHI	DEVHITP	PVSIGNAL	
BOXNUM	PVDSPLO	PVROCNTP	PVSLTSRC	
PTDISCL	OVERVAL	PVROCPTP	SPSIGNAL	
SLOTNUM	PVRNGOP	DEVHIPR	SPSLTSRC	
LOADDEST	PVCLAMP	\$DHSELCT	PVTRACK	
ALGIDDAC	CALIBOFF	DEVLOPR	INITCONF	
		\$DLSELCT	SPLOLM	
		PVHIPR	SPHILM	
		\$HISELCT	\$SPTOL	
		PVLOPR	SP	
		\$LOSELCT	OPLOLM	
		PVROCNPR	OPHILM	
		\$RNSLCT	\$OPTOL	
		PVROCPPR	ITLOLM	
		\$RPSELCT	ITHILM	
		ALPRIOR	TD	
		CCRANK	K	
		CCPRIPNT	T1	
		EIPPCODE	T2	
		EIPEVENT		
		EIPENB		

Pidddc — PID /Direct Digital Control (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Pidddc** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Pidddc** algorithm is that it is a basic control algorithm that has direct digital control of the algorithm's output by a TPS module. Should the TPS module become unavailable, the algorithm gracefully sheds to the MANUAL, AUTO, or BCAS mode, or no shed in accordance with the user-configured SHEDMODE parameter.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLYTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHHPR \$HHSELCT PVHIPR \$HISELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLACTN CTLEQN (A,B,C) OUTIND \$OPTOL PVSIGNAL PVSLT SRC SPSIGNAL SPSLT SRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC MSIGNAL MSLT SRC ASIGNAL ASLT SRC CSIGNAL CSLT SRC GREENSGL GREENSRC SENSSTOP ITLOLM ITHILM TFCLLM TFCHLM TD K T1 T2	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE EMSENB MODE \$MODESEL

Pidddcsp (EC)

Pidddcsp — PID /Direct Digital Control with Preset Setpoint (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Pidddcsp** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Pidddcsp** algorithm is identical to that of the **Pidddc** algorithm except that its local setpoint (LSP) does not track the PV while in CAS mode in order that, if the TPS module becomes unavailable, the local setpoint can become the fallback setpoint reference. The TPS module can set the local setpoint when the module has control of the output, if desired.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	\$AUXUNIT	SUPPIO	NMODE
UNIT	PVCHAR	ALFMT	CTLACTN	NMODATTR
PTDESC	INPTCOND	\$ALDLOPT	CTLEQN (A,B,C)	PSTMODE
KEYWORD	PVFORMAT	\$DLBADAL	OUTIND	RCASENB
PRIMMOD	PVTEMP	DLYTIME	\$OPTOL	MODEPERM
ASSOCDSP	PVEULO	\$REALARM	PVSIGNAL	SHEDMODE
HWYNUM	PVEUHI	BADPVPR	PVSLT SRC	EMSENB
PNTBOXTY	PVDSPHI	DEVHHPR	SPSIGNAL	MODE
BOXNUM	PVDSPLO	DEVHIPR	SPSLT SRC	\$MODESEL
PTDISCL	OVERVAL	\$DHSELCT	PVTRACK	
SLOTNUM	PVRNGOP	DEVL LPR	INITCONF	
LOADDEST	PVCLAMP	DEVLOPR	SPLOLM	
ALGIDDAC	CALIBOFF	\$DLSELCT	SPHILM	
		PVHHPR	\$SPTOL	
		\$HHSELCT	SP	
		PVHIPR	SOASIGNL	
		\$HISELCT	SOASRC	
		PVLLPR	SOBSIGNL	
		\$LLSELCT	SOBSRC	
		PVLOPR	MSIGNAL	
		\$LOSELCT	MSLT SRC	
		(Alarm	ASIGNAL	
		Trip-point	ASLT SRC	
		Values)	CSIGNAL	
		ALPRIOR	CSLT SRC	
		CCRANK	GREENSGL	
		CCPRIPNT	GREENSRC	
		EIPPCODE	SENSSTOP	
		EIPEVENT	ITLOLM	
		EIPENB	ITHILM	
		ALMASK	TFCLLM	
		INITCONF	TFCHLM	
		SPHILM	TD	
		SPLOLM	K	
		\$SPTOL	T1	
			T2	

Pidersqg — PID Error Squared on Gain (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidersqg** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidersqg** algorithm is that it is a nonlinear PID algorithm, its gain is function of error (X-Y), and that integration is proportional to linear gain and error.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO CTLACTN OUTIND \$OPTOL PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidersqg — PID Error Squared on Gain (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Pidersqg** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidersqg** algorithm is that it is a nonlinear PID algorithm, its gain is function of error (X-Y), and that integration is proportional to linear gain and error.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSELCT PVROCPPR \$RPSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO CTLACTN ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSLTSRC SPSIGNAL PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidersqi (CB)

Pidersqi — PID Error Squared on Integral (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidersqi** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidersqi** algorithm is that it is a nonlinear PID algorithm, and that the integral time constant is a function of error (X-Y).

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM ALGIDDAC LOADDEST	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO CTLACTN OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidersqi (MC)

Pidersqi — PID Error Squared on Integral (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Pidersqi** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidersqi** algorithm is that it is a nonlinear PID algorithm, and that the integral time constant is a function of error (X-Y).

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPhi PVDSPLO OVERVAL PVRNGOP	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSELCT PVROCPPR \$RPSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO CTLACTN ANLGCARD ALMDIS CONFTRND OUTIND \$OPTOL PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidexfb — PID with External Reset Feedback (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Pidexfb** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Pidexfb** algorithm is that it is a PID controller that prevents reset windup with the use of an external reset feedback (Z) signal.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELECT DEVLLPR DEVLOPR \$DLSELECT PVHHPR \$HHSELECT PVHIPR \$HISELECT PVLLPR \$LLSELECT PVLOPR \$LOSELECT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLACTN CTLEQN (A,B,C) OUTIND \$OPTOL PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC ZSIGNAL ZSLTSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TFCLLM TFCHLM TD K T1 T2	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE MODE \$MODESEL

Pidgap (CB)

Pidgap — PID Gap (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidgap** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidgap** algorithm is that it is a nonlinear PID algorithm, and that control action ceases if an error is within a preset gap.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO CTLACTN CTLEQN (A,B) OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidgap (EC)

Pidgap — (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Pidgap** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Pidgap** algorithm is that it is a nonlinear PID algorithm, and that gain is reduced by the preset factor KGAP if an error is within a preset gap.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM ALGIDDAC LOADDEST	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELECT DEVLLPR DEVLOPR \$DLSELECT PVHHPR \$HHSELECT PVHIPR \$HISELECT PVLLPR \$LLSELECT PVLOPR \$LOSELECT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLACTN CTLEQN (A,B,C) OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TFCLLM TFCHLM TD K T1 T2 KGAP GAPLO GAPHI	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE EMSENB MODE \$MODESEL

Pidgap (MC)

Pidgap — (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Pidgap** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidgap** algorithm is that it is a nonlinear PID algorithm, and that control action ceases if an error is within a preset gap.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSELCT PVROCPPR \$RPSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO CTLACTN CTLEQN (A,B,) ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2 KGAP GAPLO GAPHI	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidnonln — PID Nonlinear Gain (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Pidnonln** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Pidnonln** algorithm is that it is a nonlinear PID algorithm whose gain is a function of the absolute value of the error (X-Y).

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPhi PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHP DEVHHP \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHHP \$HHSELCT PVHIPR \$HISELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLACTN CTLEQN (A,B,C) OUTIND PVSIGNAL PVSLTsrc SPSIGNAL SPSLTsrc PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC MSIGNAL MSLTsrc ASIGNAL ASLTsrc CSIGNAL CSLTsrc GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TFCLLM TFCHLM TD K T1 T2 KNL FRMMD	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE EMSENB MODE \$MODESEL

Pidnorm (CB)

Pidnorm — PID Normal (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidnorm** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidnorm** algorithm is that it is a basic control algorithm.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO CTLACTN CTLEQN (A,B) OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

NOTE

Modifying algorithm of existing CB regulatory point from PIDNORM to PIDGAP needs modification of the ALFMT parameter. If Data Entity Builder is used to modify the algorithm type, then ensure that the ALMFMT is also modified manually as this parameter does not automatically modify on the Data Entity builder. Building such points without any modification in the ALMFMT would result in corrupting the point record. Instead, use point detail display with ENGR access level to modify the algorithm from PIDNORM to PIDGAP.

Pidnorm (EC)

Pidnorm — PID Normal (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Pidnorm** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Pidnorm** algorithm is that it is a basic control algorithm.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPhi PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHPR DEVHPR \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHPR \$HHSELCT PVHPR \$HISELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLACTN CTLEQN (A,B,C) OUTIND PVSIGNAL PVSLTsrc SPSIGNAL SPSLTsrc PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC MSIGNAL MSLTsrc ASIGNAL ASLTsrc CSIGNAL CSLTsrc GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TFCLLM TFCHLM TD K T1 T2	NMODE NMODATTR PSTMODE RCASENB SHEDMODE EMSENB MODE \$MODESEL

Pidnorm (MC)

Pidnorm — PID Normal (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Pidnorm** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidnorm** algorithm is that it is a basic control algorithm.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	\$AUXUNIT	SUPPIO	NMODE
UNIT	PVCHAR	ALFMT	CTLACTN	NMODATTR
PTDESC	INPTCOND	\$ALDLOPT	CTLEQN (A,B)	PSTMODE
KEYWORD	PVFORMAT	\$REALARM	ANLGCARD	MODEPERM
PRIMMOD	PVTEMP	PVLOTP	ALMDIS	MODE
ASSOCDSP	PVEULO	PVHITP	CONFTRND	\$MODESEL
HWYNUM	PVEUHI	DEVLOTP	OUTIND	
PNTBOXTY	PVDSPHI	DEVHITP	PVSIGNAL	
BOXNUM	PVDSPLO	PVROCNTP	PVSLTSRC	
PTDISCL	OVERVAL	PVROCPTP	SPSIGNAL	
SLOTNUM	PVRNGOP	DEVHIPR	SPSLTSRC	
LOADDEST	PVCLAMP	\$DHSELCT	PVTRACK	
ALGIDDAC	CALIBOFF	DEVLOPR	INITCONF	
		\$DLSELCT	SPLOLM	
		PVHIPR	SPHILM	
		\$HISELCT	\$SPTOL	
		PVLOPR	SP	
		\$LOSELCT	OPLOLM	
		PVROCNPR	OPHILM	
		\$RNSLCT	\$OPTOL	
		PVROCPPR	ITLOLM	
		\$RPSELCT	ITHILM	
		ALPRIOR	TD	
		CCRANK	K	
		CCPRIPNT	T1	
		EIPPCODE	T2	
		EIPEVENT		

Pidpl50 — PD + 50% (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidpl50** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO CTLACTN CTLEQN (A,B) OUTIND PVSIGNAL PVSLTsrc SPSIGNAL SPSLTsrc PVTRACK INITCONF OPLOLM OPHILM \$OPTOL TD K T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidpl50c — PD + 50% (CMA) (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidpl50c** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT CCPRIPNT	SUPPIO CTLACTN CTLEQN (A,B) OUTIND PVSIGNAL PVSLTsrc SPSIGNAL SPSLTsrc PVTRACK INITCONF OPLOLM OPHILM \$OPTOL TD K T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidpl50 (CB)

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
		EIPPCODE EIPEVENT		

Pidratio (CB)

Pidratio — PID Ratio (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidratio** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidratio** algorithm is that it is a basic control algorithm with ratio and bias on the remote setpoint.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO CTLACTN CTLEQN (A,B) OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2 RATIO BIAS	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidratio (EC)

Pidratio — PID Ratio (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Pidratio** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Pidratio** algorithm is that it is a basic control algorithm with ratio and bias on the remote setpoint.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELECT DEVLLPR DEVLOPR \$DSELECT PVHHPR \$HHSELECT PVHIPR \$HSELECT PVLLPR \$LLSELECT PVLOPR \$LOSELECT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLACTN CTLEQN(A,B,C) OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TFCLLM TFCHLM TD K T1 T2 RATIO BIAS	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE EMSENB MODE \$MODESEL

Pidratio (MC)

Pidratio — PID Ratio (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Pidratio** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidratio** algorithm is that it is a basic control algorithm with ratio and bias on the remote setpoint.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HSELECT PVLOPR \$LOSELECT PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO CTLACTN CTLEQN (A,B) ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2 RATIO BIAS	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidspc — PID SPC (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Pidspc** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidspc** algorithm is that a TPS module generates the setpoint, and that the data point can shed to the automatic or backup cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO CTLACTN CTLEQN (A,B) OUTIND \$OPTOL PVSIGNAL PVSLOTSRC SPSIGNAL SPSLOTSRC PVTRACK INITCONF SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Pidspc — PID SPC (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Pidspc** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Pidspc** algorithm is that a TPS module generates the setpoint, and that the data point can shed to the automatic or backup cascade mode.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSELCT PVROCPPR \$RPSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO CTLACTN CTLEQN (A,B) ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLT SRC SPSIGNAL SPSLT SRC PVTRACK INITCONF SPLOLM SPHILM \$SPTOL SP OPLOLM OPHILM \$OPTOL ITLOLM ITHILM TD K T1 T2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

PIUCRDOP (HLPIU)

PIUCRDOP — Slot Card Option (HLPIU Box Data Point)

This parameter defines the card option for an I/O slot. The value range of the card option depends on the type of card that was configured for the I/O slot as shown below.

Source:	User	Default Value:	No Options	Access Lock:	DEB
Value Type:	Enum. of PIUCRDOP	Value Range:	If Digin has been entered for <u>PIUCRDTY</u> : Notifyst = Notify Status. Chngdect = Change Detect (Latched). SOE = Sequence of Events. If Digout has been entered for <u>PIUCRDTY</u> : Pulseout = Pulsed Output. Latchout = Latched Output. If Counter has been entered for <u>PIUCRDTY</u> : Countr16 = 16-bit pulse counter. Countr32 = 32-bit control counter.		

PIUCRDOP — Slot Card Option (MC Box Data Point)

This parameter defines the card option for an I/O slot. The value range of the card option depends on the type of card that was configured for the I/O slot as shown below.

Source:	User	Default Value:	No Options	Access Lock:	DEB
Value Type:	Enum. of PIUCRDOP	Value Range:	If Digin has been entered for <u>PIUCRDTY</u> : Status = Status. Latchin = Change Detect (Latched). If Digout has been entered for <u>PIUCRDTY</u> : Pulseout = Pulsed Output. Latchout = Latched Output.		

PIUCRDTY (DHP & HLPIU)

PIUCRDTY — Card (Slot) Type (DHP & HLPIU Box Data Points)

This parameter defines the type of card in the particular slot. For the DHP, it defines the type of slot rather than the card type because the slots are software slots rather than hardware slots.

Source:	User	Default Value:	None	Access Lock:	DEB
Value Type:	Enum. of PIUCRDTY	Value Range:	None = This slot is not used. Digin = Digital Input Data Points. Digout = Digital Output Data Points. Analogin = Analog Input Data Points. Analogot = Analog Output Data Points. Counter = Counter Data Points.		

NOTES

1. For the DHP: if Regular is entered for the BOXSIZE (DHP Box Size) parameter, make entries on the Configuration Form for slots 1-15. If Extended is entered, make entries for slots 1-15 and 17-31. Slot 16 is reserved for DHP/PC configuration and box status; slot 32 is reserved for future use.
2. For the HLPIU: if None, Counter, Analogin, or Analogot is entered for a particular slot during configuration, skip the Slot Card Option lines on the configuration form and go to the next Slot Card Type line for the next slot.
3. When re-entering box/slot configuration, if a card type changes for a slot, all points must first be deleted from that slot before the HG will allow the change.

PIUCRDTY — Card (Slot) Type (LEPIU & LLPIU Box Data Points)

This parameter defines the type of card in the particular slot.

Source:	User	Default Value:	None	Access Lock:	DEB
Value Type:	Enum. of PIUCRDTY	Value Range:	None = This slot is not used. Analogin = Analog Input Data Points.		

PIUCRDTY — Card (Slot) Type (MC Box Data Point)

This parameter defines the type of card in the particular slot.

Source:	User	Default Value:	None	Access Lock:	DEB
Value Type:	Enum. of PIUCRDTY	Value Range:	None = This slot is not used. Digin = Digital Input Data Points. Digout = Digital Output Data Points. Analogin = Analog Input Data Points can be specified for only slots 7 and 8). Analogot = Analog Output Data Points. Counter = Counter Data Points.		

NOTE

If None, Analogin, Analogot, or Counter is entered for a particular slot during configuration, skip the Slot Option Card lines on the Configuration Form and go to the next Slot Card Type line for the next slot.

PIUOTDCF — Open Thermocouple Detection Flag (LEPIU & LLPIU)

This parameter determines whether an open thermocouple condition is to be detected. This parameter applies to only the following LEPIU and LLPIU sensor types:

LEPIU: Sensr501-Sensr506
 LLPIU: Sensr401-Sensr406

Source:	User	Default Value:	Disable	Access Lock:	Engr.
Value Type:	Enum. of ENBLSTAT	Value Range:	Disable = Open thermocouple condition is not to be detected. Enable = Open thermocouple condition is to be detected.		

PIUSMOTH — Smoothing Coefficient (HLPIU, LEPIU, LLPIU)

This parameter defines the smoothing coefficient that is used to calculate the filter time constant as follows:

$$T_f = \frac{T_s(1-SMC)}{SMC}$$

where:

T_f = filter time in seconds,
 T_s = scan period in seconds, and
 SMC = selected smoothing coefficient.

Source:	User	Default Value:	None	Access Lock:	Engr.
Value Type:	Enum. of PIUSMOTH	Value Range:	None = No smoothing (1.000) Code01 = 0.7500 Code02 = 0.5312 Code03 = 0.3750 Code04 = 0.2656 Code05 = 0.1875 Code06 = 0.1328 Code07 = 0.0938 Code08 = 0.0664 Code09 = 0.0469 Code10 = 0.0332 Code11 = 0.0234 Code12 = 0.0166 Code13 = 0.0117 Code14 = 0.0083		

PNTBOXIN — PC Box Index or PC Box Index (Input) (DHP)

This parameter defines the address of the PC that contains the input portion of an analog or digital input or composite data point, or a counter data point.

Use the default value 1.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1-8		

NOTE

These numbers correspond to the PC1-PC8 numbers that are used to define up to eight PCs on the DHP Box Data Point Configuration Form.

All PLCs of DHP subslots must be identical.

PNTBOXOT — PC Box Index or PC Box Index (Output) (DHP)

This parameter defines the address of the PC that contains the output portion of an analog or digital output or composite data point.

Use the default value 1.

Source:	User	Default Value:	1	Access Lock:	DEB
Value Type:	Integer	Value Range:	1-8		

NOTE

These numbers correspond to the PC1-PC8 numbers that are used to define up to eight PCs on the DHP Box Data Point Configuration Form.

All PLCs of DHP subslots must be identical.

PNTBOXTY — Box Type

This parameter defines the type of hiway box in which the data point specified by NAME (Tag Name) resides.

Source:	User	Default Value:	Undefined	Access Lock:	DEB
Value Type:	Enum. of BOXTYPE	Value Range:	Cb = Basic Controller. Dhp = Data Hiway Port or 620 HIM. Ec = Extended Controller. Hlpiu = High-Level Process Interface Unit. Lepiu = Low-Energy Process Interface Unit. Llpiu = Low-Level Process Interface Unit. Mc = Multifunction Controller.		

PNTOPOP — Output Type

This parameter defines the output type (pulsed output or latched output) of an HLPIU or MC digital output or digital composite data point. Refer to the description of the OUTIND (Output Indication) parameter for the relationship between PNTOPOP and the MOOUTIND (Momentary Output Indication), PULSEWTH (Pulsewidth), and OUTIND (Output Indication) parameters.

Source:	User	Default Value:	Pulseout	Access Lock:	DEB
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Value Type:	Enum. of PIUCRDOP	Value Range:	Pulseout = Pulsed Output Type. Latchout = Latched Output Type Notcnfg.
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NOTE

If Pulseout has been entered, also make an entry for the PULSEWTH parameter to specify the duration of the pulsewidth. For MC digital output and digital composite data points do not make an entry in OUTIND parameter if Pulseout is entered.

If Latchout has been entered, also make an entry for the MOOUTIND parameter to specify whether the output is to be momentary ("doorbell" action) or nonmomentary.

PNTPCTY — PC Type (DHP)

This parameter defines the manufacturer of the PCs that are connected to this DHP. For a 620 HIM, circle Honeywell.

Source:	User	Default Value:	Allenbrd	Access Lock:	DEB
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Value Type:	Enum. of BOXPROT	Value Range:	Allenbrd = Allen-Bradley. Modicon = Modicon (if GE Series 6 PCs are to be used, select Modicon as the entry for this parameter). Honeywell = Honeywell.
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PNTSTATE — Point State

This parameter provides an indication of the current point state. For the most part, the point state is the same as the box state.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of PNTTYPE	Value Range:	Uncertain = No hiway security tests have been performed or hiway status is uncertain. Ok = Data point is operating properly. Failed = Box has failed. Reset = Box is in reset state. Idle = Box is in idle state (MC only). Softfail = Soft failure such as A/D drift has been detected in box (EC and MC only). Test = Box is being tested.		

PNTTYPE — Point Type

This parameter defines the type of data point.

Source:	System	Default Value:	Null	Access Lock:	Read Only
Value Type:	Enum. of PNTTYPE	Value Range:	Null = This is not a data point. Analgin = Analog Input Data Point. Analgout = Analog Output Data Point. Analgcom = Analog Composite Data Point. Digin = Digital Input Data Point. Digout = Digital Output Data Point. Digcom = Digital Composite Data Point. Reglatry = Regulatory Data Point. Logic = Logic Data Point. Custom = Custom Data Point. Switch = Switch Data Point. Counter = Counter Data Point. Timer = Timer Data Point. Flag = Flag Data Point. Numeric = Numeric Data Point. (continued)		

POLLRES

Procmndl = Process Module Data Point.
Behhist = Batch History Data Point.
Advctrl = Advanced Control Data Point.
Logicblk = Logic Block Data Point
Bchhistp = Batch History Prototype Data Point.
Procstat = Processor Status Data Point.
Upidp = Upper Level Processor (CM60) Interface Data Point
(sometimes referred to as the Advanced Control Interface Data Point - ACIDP).
Upcrdp = Upper Level Processor (CM60) Calculated Results Data Point (sometimes referred to as the Calculated Results Data Point - CRDP).

POLLRES — # of Poll Responses (HG Processor Status Points)

This parameter defines the number of hiway poll responses of DHI board data during the last 15-second sample period.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

POLLS — # of Hiway Polls (HG Processor Status Data Points)

This parameter defines the number of hiway polls of DHI board data during the last 15-second sample period.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

PONOF1 — Clock Time for Pulse Gate 1 (EC Regulatory Data Point)

PONOF2 — Clock Time for Pulse Gate 2

PONOF3 — Clock Time for Pulse Gate 3

These parameters indicate the amount of time remaining in minutes for each of the Gate 1, 2, and 3 pulses at an EC regulatory data point that has been configured for the **Logic** algorithm. This parameter is applicable only if the corresponding Gate type (GATETYP1, GATETYP2, GATETYP3) is And, Or, Xor, Nand, or Nor.

Source:	System	Default Value:	0.0 minutes	Access Lock:	Read Only
Value Type:	Real	Value Range:	0.0 to 9.991 minutes		

PRESET — Preset Value (Counter Data Point)

This parameter specifies the value at which a counter data point goes into an alarm condition (accumulated value exceeds the preset value). The counter starts over when the counter value equals the preset value.

Source:	User	Default Value:	0	Access Lock:	Oper.
Value Type:	Real	Value Range:	For DHP Counter Data Point: 0 to 65535 For HLPIU Counter Data Point: 0 to 65535 For MC Counter Data Point, range depends on <u>AVFORMAT</u> selection as follows:		

<u>AVFORMAT</u>	<u>Range</u>
D0	0 to 999999
D1	0 to 99999.9
D2	0 to 9999.99
D3	0 to 999.999

If the value entered exceeds the maximum value, the HG clamps it to the maximum value.

PRESETPR—Preset Alarm Priority

This parameter defines the priority of a preset alarm generated by a data point. The alarm priority determines how the alarm is handled by the system. This parameter does not affect data point alarm detection and alarm indication visibility or Group and Detail displays.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			
		NOTE			
		If PRINTER is chosen, the alarm is lost if the printer is not working.			
	JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.			

PRESLOCK — Operator Preset Change (HLPIU & MC Counter Data Points)

This parameter defines whether the operator is allowed to change the preset value of a counter data point.

Source:	User	Default Value:	Permit	Access Lock:	Engr.
Value Type:	Enum. of MODEPERM	Value Range:	Permit = Operator is allowed to change the preset value. Notperm = Operator cannot change the preset value.		

PRIMMOD — Primary Module Point ID

The primary module point identifier is the Tag Name of any point in the HG to which a data point can be assigned. It is used for grouping alarms in Event History. Refer to the definition of NAME (Tag Name) for the valid character set.

Source:	User	Default Value:	All Dashes	Access Lock:	Engr.
Value Type:	Entity ID	Value Range:	See <u>NAME</u>		

PROCMOD — Process Module Operational State (MC Process Module Data Point)

This parameter indicates the current operational state of a program in a sequence slot (a Process Module data point has a one-to-one relationship with an MC sequence slot).

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Enum. of PROCMOD	Value Range:	Off = Initial state used for loading a program into the sequence slot. Redy = Ready; program has been loaded and is ready for execution. Norm = Normal; sequence is executing or is finished. Rstr = Restart; transient state from one of the shutdown states to the normal state. Indicates that restart part of the sequence is being executed. Hold = Hold; indicates that the hold part of the sequence is being executed or is finished. Shdn = Shutdown; indicates that the shutdown part of the sequence is being executed or is finished. Emsd = Emergency Shutdown; indicates that the emergency shutdown part of the sequence is being executed or is finished. Strt = Start; initially starts the sequence or restarts the sequence at the current position. Stop = Stop; immediately stops the execution of the sequence.		

PRPRSTPR—Pre-Preset Alarm Priority

This parameter defines the priority of a pre-preset alarm generated by a data point. The alarm priority determines how the alarm is handled by the system. This parameter does not affect data point alarm detection and alarm indication visibility or Group and Detail displays.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION = Alarm is not reported to the system. Alarm is not annunciated. JOURNAL = Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays. LOW = Alarm is historized, annunciated, and does appear on Group and Detail displays. HIGH = Alarm is historized, annunciated, and does appear on Group and Detail displays. EMERGENCY = This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity. PRINTER = Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.				
NOTE					
If PRINTER is chosen, the alarm is lost if the printer is not working.					
	JNLPRINT = Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.				

PSTMODE — Past Mode Recall (Regulatory Data Point)

This parameter defines whether this data point is to be placed in its previous mode after a power-down condition has been corrected. This parameter is not applicable when the regulatory data point has been configured for the **Das** algorithm for the CB, EC, or MC. This parameter valid for slots 1-8 in the EC.

Source:	User	Default Value:	Nopastmr	Access Lock:	Engr.
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Value Type:	Enum. of PSTMODE	Value Range:	Nopastmr = No past mode recall. Pastmr = Past mode recall.
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PTDESC — Point Descriptor

The point descriptor is used to describe the data point, and it appears on the Group and Detail Displays for this data point. In the example shown in Figure N-1 (see [NAME](#)), the point descriptor is DETH REFLUX PUMP A CONTR. The point descriptor appears on the Group Display only when the data point is selected.

Source:	User	Default Value:	All Blanks	Access Lock:	DEB
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Value Type:	String	Value Range:	The point descriptor can consist of up-to-24 characters, and the permissible character set consists of all characters on the Engineer's Keyboard. Basically this set consists of Alphabetics A-Z, Numerics 0-9, space, and the following special characters: ! % & ' () * + - / : ; > < = ? _ , . \$
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PTDISCL — Point Disclosure

This parameter determines whether the Data Entity Builder's parameter entry display is to show the complete set of user-configurable parameters or a smaller set of the parameters, leaving the parameters not displayed at the default values.

Source:	User	Default Value:	Brief	Access Lock:	DEB
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Value Type:	Enum. of PTDISCL	Value Range:	Brief = The brief set of configurable parameters is displayed on the parameter-entry display. If this entry is selected, default values are supplied by the DEB for parameters that do not appear on the parameter-entry display. Full = The full set of configurable parameters is displayed on the parameter-entry display.
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PTEXECST — Point Execution State

This parameter indicates the execution state of the specified data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of PTEXECST	Value Range:	Notconfg = This is a transient state; configuration has not been completed for this data point. Active = Data point is running. Inactive = Data point is fully configured but is not running. Ready = Applies to batch history data point. Done = Applies to batch history data point. Waiting = Applies to batch history data point.		

PTINAL — Point in Alarm

This parameter indicates whether an alarm condition exists on a point.

Source:	System	Default Value:	Off	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	Off = Point is not in alarm. On = Point is in alarm.		

PTORST — Point Override Status

This parameter indicates the current status of this data point during override initialization.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of ORSTATUS	Value Range:	Notcon = Not connected; this data point is not part of the override strategy, or it is currently being initialized. Sel = This data point is part of the override strategy and is currently selected. Notsel = This data point is part of the override strategy but is not currently selected.		

PTSESTBL — # of Established Process Points (All Processor Status Data Points)

This parameter defines the number of process points currently established in this node.

Source:	System	Default Value:	0	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0-32767		

PULSE — Pulse Time (EC Regulatory Data Point)

This parameter defines the length of the pulsewidth in minutes if Pulse has been entered as the gate type for an EC regulatory data point that has been configured for the **Logic** algorithm.

Source:	User	Default Value:	0.0 minutes	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0 to 9.992 minutes		

PULSEWTH — Pulsewidth (HLPIU & MC)

This parameter defines the pulsewidth (in milliseconds) of the single or dual outputs from a digital output or digital composite data point if Pulseout has been entered for the PNTOPOP (Output Type) parameter.

Source:	User	Default Value:	16 milliseconds	Access Lock:	Supvr.
Value Type:	Real	Value Range:	HLPIU = 16 to 4096 milliseconds, in 16 millisecond intervals. MC = 16 to 992 milliseconds, in 16 millisecond intervals.		

NOTE

If a value is entered that is not evenly divisible by 16, it is rounded off to the nearest multiple of 16.

PV — Current State (PV) (Digital Data Points)

This parameter provides the current state of the specified data point.

Source:	System	Default Value:	N/A	Access Lock:	Supvr.
Value Type:	Self-Defining Enum.	Value Range:	For a single input digital data point: 0 = <u>STATE1</u> (lower box lit) 1 = <u>STATE2</u> (upper box lit) For a dual input digital data point: 0 = <u>NCMPTXT1</u> (neither box lit) 1 = <u>STATE1</u> (only lower box lit) 2 = <u>STATE2</u> (only upper box lit) 3 = <u>NCMPTXT2</u> (both boxes lit)		

PV — Current State (PV) (MC Flag and Logic Data Point)

This parameter provides the current state of the specified data point.

Source:	System	Default Value:	N/A	Access Lock:	Oper. for flag data point. Supvr. for logic data point.
Value Type:	Self-Defining Enum.	Value Range:	0 = <u>STATE1</u> (lower box lit) 1 = <u>STATE2</u> (upper box lit)		

PV — Current Value of Counter (MC Counter Data Point)

This parameter provides the current value at the specified counter data point. The current accumulated value is shown by AV.

Source:	System	Default Value:	N/A	Access Lock:	Supvr.
Value Type:	Integer	Value Range:	0 to 65535		

PV — Current Value of Numeric (MC Numeric Data Point)

This parameter provides the current value of the specified numeric data point.

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Real	Value Range:	-999999.0 to 999999.0		

PV — Current Value of PV (Analog and Regulatory Data Points)

This parameter provides the current value of the PV input (in engineering units) at the specified data point.

Source:	System	Default Value:	N/A	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-6.86% to 106.86% percent of scale in engineering units.		

PV — Current Value of Timer (MC Timer Data Point)

This parameter provides the current time counts from the time that the timer has been started.

Source:	System	Default Value:	N/A	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0 to 9999.0 seconds or minutes, as configured.		

PVALDB — PV Alarm Deadband (LLPIU and DHP Analog Input Data Points)

This parameter defines the alarm deadband for the PV high limit and PV low limit for LLPIU and DHP analog input points. A value of 0.0 causes the deadband value to revert to the standard deadband value of 1% of full scale engineering units. To use this parameter, a DHP or LLPIU must have special firmware and the box parameter BOXVDB must be turned on in an EXTENDED box.

To initialize PVALDB to 1% of full scale engineering units (required when upgrading to special firmware), install firmware and load the box BEFORE turning BOXVDB box parameter on. After load of box is complete turn BOXVDB on and save the box data to a checkpoint file. PVALDB will NOT be initialized during a load of the box if BOXVDB is turned on and the special firmware is present.

Source:	User	Default Value:	0.0	Access Lock:	Engr./DEB.
Value Type:	Real	Value Range:	0.00347% to 7.11% of full scale in engineering units		

PVAUTO — PV Automatic

This parameter is the current PV value (in engineering units) that is read from the selected hiway box. This parameter is used in conjunction with the PV parameter when the PVSOURCE is manual or substituted. For these values of PVSOURCE, the PV is read/stored to the HG database; it is not read from the process box (with the exception of MC digital input and digital composite data points and PIU mailbox PVs). This allows the user to see the manual/substituted PV value (PV) as well as the PV value from the process box (PVAUTO).

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	-6.86% to 106.86% of scale in engineering units.		

PVCHAR — Characterization (General Note)**NOTE**

The value for PVCHAR is stored both in the HG and in the box. When a change in the PVCHAR value occurs, the values for the following parameters (also stored both in the HG and in the box) are updated in the HG: PVEUHI, PVEULO and PVTEMP or PVFORMAT (depending on the value in PVCHAR—linear types use PVFORMAT and temperature type use PVTEMP).

Tables P-1 and P-2 contain the temperature ranges for the PVCHAR parameter that are supported by the HG.

Table P-1 — Temperature Ranges for PVCHAR Parameter Entries for CB, EC, HLPIU, and MC

ENTRY	DEGREES F		DEGREES C	
	LOW	HIGH	LOW	HIGH
Btherm	1472.0	3092.0	800.0	1700.0
Etherm	-328.0	1832.0	-200.0	1000.0
Jtherm or Sensr317	-100.0	2000.0	-73.3	1093.3
Ktherm or Sensr318	700.0	2500.0	371.1	1371.1
Rtherm	-58.0	3200.0	-50.0	1760.0
Rptherm	32.0	3218.0	0.0	1770.0
Stherm or Sensr320	0.0	3200.0	-17.8	1760.0
Ttherm or Sensr319	-300.0	750.0	-184.4	398.9
Burnsrtd or Sensr322	-300.0	1200.0	-184.4	648.9
Dinrtd	-350.0	1200.0	-212.2	648.9
Jisrtd	-328.0	1166.0	-200.0	630.0
Radiamat or Sensr323	1100.0	3200.0	593.3	1760.0
Sensr322				

PVCHAR (CB)**Table P-2 — Temperature Ranges for PVCHAR Parameter Entries for LEPIU and LLPIU**

ENTRY	DEGREES F		DEGREES C	
	LOW	HIGH	LOW	HIGH
Sensr401 or Sensr501	-328.0	2012.0	-200.0	1100.0
Sensr402 or Sensr502	-101.2	2501.6	-74.0	1372.0
Sensr403 or Sensr503	-328.0	752.0	-200.0	400.0
Sensr404 or Sensr504	-58.0	3200.0	-50.0	1760.0
Sensr405 or Sensr505	-328.0	1832.0	-200.0	1000.0
Sensr406 or Sensr506	-58.0	3200.0	-50.0	1760.0
Sensr408	-4.0	176.0	-20.0	80.0
Sensr409	-20.2	219.2	-29.0	104.0
Sensr410	-59.8	480.2	-51.0	249.0
Sensr411 or Sensr511	-300.0	1200.0	-184.4	648.9
Sensr415 or Sensr515	1099.4	3200.0	593.0	1760.0

PVCHAR — Characterization (CB)

This parameter defines the type of display characterization for the CB, RV, and CB regulatory data points. Refer to the general note under PVCHAR for a listing of the supported temperature ranges.

Source:	User	Default Value:	Linear	Access Lock:	Engr.
Value Type:	Enum. of VALCHAR	Value Range:	Linear = Linear. Jtherm = Type J Thermocouple. Ktherm = Type K Thermocouple. Ttherm = Type T Thermocouple. Stherm = Type S Thermocouple. Sqrroot = Square Root. BurnsRTD = Burns Resistance Temperature Device. Radiamat = RH Radiamatic.		

NOTE

If Linear or Sqrroot is entered for this parameter, make entries in the INPTCOND (Input Conditioning) and PVFORMAT (Decimal Format) lines of the Configuration Form. If an entry other than Linear or Sqrroot is entered, make entries in the INPTCOND (Input Conditioning) and PVTEMP (Temp Display) lines.

The value for PVCHAR is stored both in the HG and in the box. When a change in the PVCHAR value occurs, the values for the following parameters (also stored both in the HG and in the box) are updated in the HG: PVEUHI, PVEULO, and PVTEMP or PVFORMAT (depending on the value in PVCHAR—linear types use PVFORMAT and temperature type use PVTEMP).

PVCHAR — Characterization of X (EC Regulatory Data Point)

This parameter defines the type of display characterization for an EC regulatory data point. Refer to the general note under PVCHAR for a listing of the supported temperature ranges.

Source:	User	Default Value:	Linear	Access Lock:	Engr.
Value Type:	Enum. of VALCHAR	Value Range:	Linear = Linear Jtherm = Type J Thermocouple Ktherm = Type K Thermocouple Ttherm = Type T Thermocouple Stherm = Type S Thermocouple Sqrroot = Square Root Burnsrtld = Burns Resistance Temperature Device Radiamat = RH Radiamatic Btherm = Type B Thermocouple Etherm = Type E Thermocouple Rtherm = Type R Thermocouple Rptherm = Type R' Thermocouple		

NOTE

If Linear or Sqrroot is entered for this parameter, make entries in the INPTCOND (Input Conditioning) and PVFORMAT (Decimal Format of X or Y) lines of the configuration form. If an entry other than Linear or Sqrroot is entered for this parameter, make entries in the INPTCOND (Input Conditioning) and PVTEMP (Temp Display of X or Y) lines.

PVCHAR — Characterization (HLPIU)

This parameter defines the type of display characterization for an HLPIU analog input or analog composite data point. Refer to the general note under PVCHAR for a listing of the supported temperature ranges.

Source:	User	Default Value:	Undefined	Access Lock:	Engr.
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Value Type: Enum. of VALCHAR

Value Range: Sensr300 = Unimplemented (Note 1 & 6) Linearization or Display Types are as follows:

Sensr308 = 0 to 5 volts
($\pm$ A/D) Linear (LO)
(Note 2)

Sensr309 = -5 to 5 volts
(+A/D) Linear (LO)
(Note 3)

LO—Linear Only
LS—Linear or Square Root
TD—Temperature Display

Sensr317 = Type J T/C
(Note 4) (TD)

Sensr318 = Type K T/C
(Note 4) (TD)

Sensr319 = Type T T/C
(Note 4) (TD)

Sensr320 = Type S T/C
(Note 4) (TD)

Sensr321 = Square Root
(Note 4) (LS)

Sensr322 = Platinum RTD
(Note 4) (TD)

Sensr323 = RH Radiamatic
(Note 4) (TD)

Sensr324 = +1 to 5 volts
(+A/D) Linear (LS)
(Note 2)

Sensr325 = +1 to 5 volts
($\pm$ A/D) Linear (LS)
(Note 3)

T/C = Thermocouple

NOTES

1. Sensr300—this code causes the PIU firmware to stop scanning the selected board, effectively taking the board out of use, and enabling the PIU to run with the board removed.
2. Sensr308 & Sensr324—Preferred operation (unipolar) with jumper option on ADC card in this position.
3. Sensr309 & Sensr325—may be required if "-5 to 0 volts" or "-5 to 5 volts" ranges are measured.
4. Sensr317-Sensr323—These ranges can be measured with the A/D pinned for either Unipolar or Bipolar.
5. If an LO or LS input/sensor type has been entered for this parameter, also make entries in INPTCOND (Input Conditioning) and PVFORMAT (Decimal Format) lines of the configuration form. If a TD (Temperature Display) input/sensor type was entered for this parameter, also make entries in the PVFORMAT (Decimal Format) and PVTEMP (Temperature Display) lines of the Configuration Form.
6. Internal parameter IPCRDTY is displayed as None when PVCHAR = Sensr300. This is compatible with the TPS Basic and Enhanced Operator Stations that display UNDEF in this case.

PVCHAR — Characterization (LEPIU)

This parameter defines the type of display characterization for an LEPIU analog input data point. Refer to the general note under PVCHAR for a listing of the supported temperature ranges.

Source: User **Default Value:** Undefined **Access Lock:** Engr.

Value Type: Enum. of VALCHAR

Value Range: Sensr500 = Unimplemented (Note 1 & 3) Linearization or Display Types are as follows:

Sensr501 = Type J T/C (FC) LO—Linear Only
 Sensr502 = Type K T/C (FC) LS—Linear or Square Root
 Sensr503 = Type T T/C (FC) FC—Degrees F or Degrees C
 Sensr504 = Type S T/C (FC)

Sensr505 = Type E T/C (FC)
 Sensr506 = Type R T/C (FC)
 Sensr511 = RTD 100 Ohms at 0 Deg. C - Platinum (FC)

Sensr515 = Radiamatic Sensor (FC)
 Sensr516 = 1 to 5 Volt (Normalized) (LS)

Sensr520 = -5 mV to +20 mV (Normalized) (LO)
 Sensr521 = -10 mV to +40 mV (Normalized) (LO)
 Sensr522 = -20 mV to +80 mV (Normalized) (LO)

T/C = Thermocouple

NOTES

1. Sensr500—this code effectively causes the PIU firmware to stop scanning the selected board, taking the board out of use, and enabling the PIU to run with the board removed.
2. If a Linear Only (LO) or Linear or Square Root (LS) input/sensor type has been selected for this parameter, also make entries in INPTCOND (Input Conditioning and PVFORMAT (Decimal Format) lines of the Configuration Form.

 If a Degrees F or Degrees C (FS) input/sensor type has been selected for this parameter, also make entries in PVTEMP (Temp Display) and PVFORMAT (Decimal Format) lines of the Configuration Form.
3. Internal parameter IPCRDY is displayed as None when PVCHAR = Sensr500. This is compatible with TPS Basic and Enhanced Operator Stations that display UNDEF in this case.

PVCHAR (LLPIU)

PVCHAR — Characterization (LLPIU)

This parameter defines the type of display characterization for an LLPIU analog input data point. Refer to the general note under PVCHAR for a listing of the supported temperature ranges.

Source:	User	Default Value:	Undefined	Access Lock:	Engr.
Value Type:	Enum. of VALCHAR				
Value Range:	Sensr400 = Unimplemented (Notes 1 & 4) Linearization or Display Types are as follows: Sensr401 = Type J T/C (FC) LO—Linear Only Sensr402 = Type K T/C (FC) LS—Linear or Square Root Sensr403 = Type T T/C (FC) FC—Degrees F or Degrees C Sensr404 = Type S T/C (FC) Sensr405 = Type E T/C (FC) Sensr406 = Type R T/C (FC) Sensr408 = CJR RTD (FC) (Cabinet Temp.) Sensr409 = RTD 10 ohms at 25 Deg. C - Copper (FC) Sensr410 = RTD 120 ohms @ 0 Deg. -Nickel Sensr411 = RTD 100 ohms @ 0 Deg. - Platinum Sensr415 = Radiamatic Sensor (FC) Sensr416 = 2.5 to 12.5 mV (Normalized) (Note 2) (LS) Sensr417 = 5 to 25 mV (Normalized) (Note 2) (LS)				

NOTES

1. Sensr400—this type causes the PIU firmware to stop scanning the selected board, effectively taking the board out of use and enabling the PIU to run with the board removed.
2. Sensr416 through Sensr427 can be used with attenuator daughter boards that can be scaled for the range of 1 V to 5 V.
3. If a Linear Only (LO) or Linear or Square Root (LS) input/sensor type was selected for this parameter, also make entries in INPTCOND (Input Conditioning) and PVFORMAT (PV Decimal Format lines) of the Configuration Form.

If Degrees F or Degrees C (FC) input/sensor type was selected for this parameter, also make entries in PVTEMP (Temp Display) and PVFORMAT (Decimal Format) lines of the Configuration Form.

PVCHAR (LLPIU)

Sensr418 = 10 to 50 mV (Normalized) (Note 2) (LS)	4. Internal parameter <u>IPCRDTY</u> is displayed as None when <u>PVCHAR</u> = Sensr400. This is compatible with the TPS Basic and Enhanced Operator Stations that display UNDEF in this case.
Sensr419 = 20 to 100 mV (Normalized) (Note 2) (LS)	
Sensr420 = ± 12.5 mV (Normalized) (Note 2) (LO)	
Sensr421 = ± 25 mV (Normalized) (Note 2) (LO)	
Sensr422 = ± 50 mV (Normalized) (Note 2) (LO)	
Sensr423 = ± 100 mV (Normalized) (Note 2) (LO)	

The following sensor type values are applicable only if the "PV Alarm Deadband Enhancement" option has been purchased. This enhancement requires special firmware in the LLPIU. For the LLPIU, the enhancement provides a configurable PV Alarm Deadband, as well as the following zero-based sensor types. Contact a Honeywell salesman or Project Manager to request a quote for this option from the Engineering Services group.

Sensr424 = 0 to 12.5 mV (Normalized) (Note 2) (LS)
Sensr425 = 0 to 25 mV (Normalized) (Note 2) (LS)
Sensr426 = 0 to 50 mV (Normalized) (Note 2) (LS)
Sensr427 = 0 to 100 mV (Normalized) (Note 2) (LS)

PVCHAR — Characterization (MC)

This parameter defines the type of display characterization for MC analog and regulatory data points. Refer to the general note under PVCHAR for a listing of the supported temperature ranges.

Source:	User	Default Value:	Linear	Access Lock:	Engr.
Value Type:	Enum. of VALCHAR	Value Range:	Linear = Linear Jtherm = Type J Thermocouple Ktherm = Type K Thermocouple Ttherm = Type T Thermocouple Etherm = Type E Thermocouple Rtherm = Type R Thermocouple Rptherm = Type R' Thermocouple Stherm = Type S Thermocouple BurnsRTD = Burns Resistance Temperature Device Radiamat = RH Radiamatic DinRTD = DIN Resistance Temperature Device JisRTD = JIS Resistance Temperature Device		

NOTE

If Linear is entered for this parameter, make entries in the INPTCOND (Input Conditioning) and PVFORMAT (Decimal Format) lines on the configuration form. If an entry other than Linear is entered for this parameter, make entries in the INPTCOND and PVTEMP (Temp Display) lines on the Configuration Form.

Linear is the only valid value when AVOPTION=Enable is specified for analog points.

PVCLAMP — Clamp Value Option

This parameter defines whether the PV value is clamped when it is outside the configured range established by PVEULO and PVEUHI. If the PV is clamped, it is clamped to the nearest end-point of the extended range as specified by the PV range-option parameter PVRNGOP. Refer to Tables P-3, P-4, and P-5.

Source:	User	Default Value:	Noclamp	Access Lock:	Engr.
Value Type:	Enum. of PVCLAMP as "-----" (Not A Number).	Value: Range:	Noclamp = PV values outside the configured range are not clamped to the specified extended range. PV is displayed Clmp = Configured range is clamped to the specified extended range and the clamped value is displayed for the PV.		

Table P-3 — Clamp Value Option for Basic Controller 1C (with Input Conditioning Configured for Sqrroot)

PV RANGE OPTION SELECTED	CLAMP VALUE OPTION SELECTED	PV VALUE RECEIVED FROM HIWAY (%)	PV VALUE DISPLAYED (%)
None Fullrng Clmpzero	Clamp Clamp Clamp	Not Applicable Not Applicable PV = 0 0 < PV <= 102.9 PV > 102.9	Not Applicable Not Applicable 0.0 PV Value 102.9
None Fullrng Clmpzero	Noclamp Noclamp Noclamp	Not Applicable Not Applicable PV = 0 0 < PV <= 102.9 PV > 102.9	Not Applicable Not Applicable "-----" PV Value "-----"

NOTE: This table applies only when PVSOURCE = Auto.

Table P-4 — Clamp Value Option for all other Data Hiway Boxes Except PIUs

PV RANGE OPTION SELECTED	CLAMP VALUE OPTION SELECTED	PV VALUE RECEIVED FROM HIWAY (%)	PV VALUE DISPLAYED (%)
None	Noclamp	PV < -6.86 -6.86 ≤ PV ≤ 106.86 PV > 106.86	-6.86 PV Value 106.86
Fullrng	Noclamp	PV < -2.9 -2.9 ≤ PV ≤ 102.9 PV > 102.9	"-----" PV Value "-----"
Clmpzero	Noclamp	PV < -2.9 -2.9 ≤ PV < 1 0 ≤ PV ≤ 102.9 PV > 102.9	"-----" 0.0 PV Value "-----"
None	Clamp	PV < -6.86 -6.86 ≤ PV ≤ 106.86 PV > 106.86	-6.86 PV Value 106.86
Fullrng	Clamp	PV < -2.9 -2.9 ≤ PV ≤ 102.9 PV > 102.9	-2.9 PV Value 102.9
Clmpzero	Clamp	PV < -2.9 -2.9 ≤ PV < 1 0 ≤ PV ≤ 102.9 PV > 102.9	0.0 0.0 PV Value 102.9

NOTE

This table applies only when PVSOURCE = Auto.

The "PV VALUE DISPLAYED (%)" column in the table above assumes that PVCHAR is NOT set to Sqrroot. If PVCHAR is equal to "Sqrroot", then values less than 0 (as shown in the "PV VALUE DISPLAYED" column) will result in a BAD PV "-----".

Refer to Table P-6 for the "Over range" and "Under range" values listed in Table P-5.

Table P-5 — Clamp Value Option for PIUs

PV RANGE OPTION SELECTED	CLAMP VALUE OPTION SELECTED	PV VALUE RECEIVED FROM HIGHWAY (%)	PV STATUS DISPLAYED	PV VALUE DISPLAYED (%)
None	Noclamp	PV < Under range Under range <= PV <= Over range PV > Over range	Bad Normal Bad	"-----" PV Value "-----"
Fullrng	Noclamp	PV < Under range Under range <= PV <= Over range PV > Over range	Bad Normal Bad	"-----" PV Value "-----"
Clmpzero	Noclamp	PV < Under range Under range <= PV <1 1 <= PV <= Over range PV > Over range	Bad Normal Normal Bad	"-----" 0 PV Value "-----"
None	Clamp	PV < Under range Under range <= PV <= Over range PV > Over range	Uncertn Normal Uncertn	"-----" PV Value "-----"
Fullrng	Clamp	PV < Under range Under range <= PV <= Over range PV > Over range	Uncertn Normal Uncertn	"-----" PV Value "-----"
Clmpzero	Clamp	PV < Under range Under range <= PV <1 1 <= PV <= Over range PV > Over range	Uncertn Normal Normal Uncertn	"-----" 0 PV Value "-----"

NOTE

For the PIU, "Over range" and "Under range" values are set according to sensor type (refer to Table P6); therefore the PV Range Option, when applied to PIU points, has significance only when the Clmpzero option is selected and the PV value is between zero and the "Under range" value.

Table P-6 — PIU Over Ranges and Under Ranges

PIU OPERATION	OVER RANGE	UNDER RANGE
For HLPIU: Bipolar (1 to 5 Volt range) Bipolar (-5 to +5 Volt range) Unipolar (1 to 5 Volt range) Unipolar (0 to 5 Volt range)	102.875% 101.150% 102.905% 102.350%	-2.875% -1.175% -2.900% 0.0%
For LLPIU: 1 to 5 Volt range (with 1 to 5V attenuator attenuator or 4 to 20 mA current-to-voltage attenuator) Full scale (all gain ranges)	102.63% 101.18%	-2.63% -1.18%
For LEPIU: 1 to 5 Volt range All other gain ranges	106.86% 100.0%	-6.86% 0.0%

PVDSPHI — PV Display Range High

PVDSPLO — PV Display Range Low

These parameters are present only if the HG Display Range option has been purchased for the system. These parameters allow an engineer to specify for this point, a different display range (0% to 100%) for the PV value on Group and Detail displays than the PV range defined by PVEUHI and PVEULO. If the option is not present or if NaN is entered in these parameters, the display range is defined by PVEUHI and PVEULO.

Source:	User	Default Value:	NaN (- -)	Access Lock:	Engr.
Value Type:	Real	Value Range:	Within the range defined by <u>PVEULO</u> and <u>PVEUHI</u> , but not exactly equal to either value. See description of these parameters for further details.		

PVEUHI — PV Range High
PVEULO — PV Range Low

These parameters define 0% and 100% of range in engineering units for display of this PV for analog and regulatory data points.

Source:	User	Default Value:	100.0 for <u>PVEUHI</u> 0.0 for <u>PVEULO</u>	Access Lock:	Engr.
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Value Type:	Real	Value Range:	-9999 to 9999 when <u>PVCHAR</u> = Linear, Sqrroot, or when SensrXXX is an LO or LS type.
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When PVCHAR is nonlinear, refer to the minimum and maximum temperatures for that characterization. Refer to the general note under the PVCHAR parameter entry in this manual.

NOTE

The values for PVEUHI and PVEULO are stored both in the HG and in the box. When a change in the PVEUHI or PVEULO value occurs, the values for the following parameters (also stored both in the HG and in the box) are updated in the HG: PVEUHI, PVEULO, and PVTEMP or PVFORMAT (depending on the value in PVCHAR—linear types use PVFORMAT and temperature type use PVTEMP).

These parameters represent the high and low ends of the EU range for the PV. The maximum, total range allowed is 10000.0 units. For example, if PVEULO contains -10000.0, PVEUHI can be no higher than 0.0 units. Similarly, if PVEULO contains 0.0, PVEUHI can be no higher than 10000.0 units.

Because PVEUHI represents the span of the PV range, a change to the value in PVEULO can affect the value in PVEUHI. For example, if PVEULO initially contains 0.0 and PVEUHI contains 1000.0, and then an engineer changes PVEULO to 100.0, PVEUHI is automatically adjusted to 1100.0, to retain the span of 1000.0 units. The span can be changed only by updating the value in PVEUHI.

PVEXHIFL — PV Extension High Flag
PVEXLOFL — PV Extension Low Flag

These parameters indicate that the current PV value has exceeded the configured high or low range (PVRNGOP).

Source:	System	Default Value:	N/A	Access Lock:	Read Only
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Value Type:	Boolean	Value Range:	True = High or low range value has been exceeded. False = High or low range value has not been exceeded.
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PVFORMAT — Decimal Format

PVFORMAT — Decimal Format of X (EC Regulatory Data Point)

This parameter defines the position of the decimal point for values in analog, regulatory, and numeric data points.

Source:	User	Default Value:	D0	Access Lock:	Engr.
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Value Type:	Enum. of VALFORMAT	Value Range:	For all data points except numeric data points: D0 = <u>+</u> XXXX D1 = <u>+</u> XXX.X D2 = <u>+</u> XX.XX D3 = <u>+</u> X.XXX
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For an MC numeric data point:
D0 = +XXXXXX
D1 = +XXXXX.X
D2 = +XXXX.XX
D3 = +XXX.XXX
D4 = +XX.XXXX
D5 = +X.XXXXX

NOTE

To update the value in this parameter, MODE must contain Man; for EC and CB points, INITCONF must contain Noinit; and for EC points, EMS must contain Disable.

The value for this parameter is stored both in the HG and in the box. When a change to this value occurs, the values in the following parameters are also updated: PVEUHI, PVEULO, and PVCHAR.

PVHHHPR — PV High High High Priority
PVHHPR — PV High High Priority
PVHIPR — PV High Priority
PVLLLPR — PV Low Low Low Priority
PVLLPR — PV Low Low Priority
PVLOPR — PV Low Priority

These parameters define high and low priority levels for PV alarms.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.												
Value Type:	Enum. of ALPRIOR																
Value Range:	<table><tr><td>NOACTION =</td><td>Alarm is not reported to the system. Alarm is not annunciated.</td></tr><tr><td>JOURNAL =</td><td>Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.</td></tr><tr><td>LOW =</td><td>Alarm is historized, annunciated, and does appear on Group and Detail displays.</td></tr><tr><td>HIGH =</td><td>Alarm is historized, annunciated, and does appear on Group and Detail displays.</td></tr><tr><td>EMERGNCY =</td><td>This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.</td></tr><tr><td>PRINTER =</td><td>Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.</td></tr></table>					NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.
NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.																
JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.																
LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.																
HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.																
EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.																
PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.																
NOTE																	
If PRINTER is chosen, the alarm is lost if the printer is not working.																	
JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.																

PVHHHTP

PVHHHTP — PV High High High Trip Point (EC)

PVHHTP — PV High High Trip Point (EC)

PVHITP — PV High Trip Point

PVLLLTP — PV Low Low Low Trip Point (EC)

PVLLTP — PV Low Low Trip Point (EC)

PVLOTP — PV Low Trip Point

These parameters define the high and low trip-point values (in engineering units) of the PV that cause an alarm condition to be generated. (PVHITP and PVLOTP are applicable to analog and regulatory data points. PVHHHTP, PVHHTP, PVLLTP, and PVLLLTP are applicable to only an EC regulatory data point.)

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	<u>PVHITP</u> = -6.86% to 106.86% of full scale in engineering units. <u>PVLOTP</u> = -6.86% to 106.86% of full scale in engineering units. Engineering unit range is from -9999 to 9999. Full scale in engineering units is defined by values assigned to <u>PVEULO</u> and <u>PVEUHI</u> . Ranges for <u>PVHHHTP</u> , <u>PVHHTP</u> , <u>PVLLTP</u> and <u>PVLLLTP</u> are the same as described above for <u>PVHITP</u> and <u>PVLOTP</u> .		

NOTE

PVLOTP cannot be set higher than PVHITP; otherwise, an unreasonable value alarm condition can exist.

PVP — PV Percent

This parameter provides the PV as a percent of range which is used in generating the bar graph on the group and detail displays for the data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	-6.86% to 106.86%		

PVRAW — Raw PV Value

This parameter provides the current value of the PV after A/D conversion, digital filtering and PV square-rooting.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	-6.86 to 106.86%		

PVRNGOP — PV Range Option

This parameter defines the range of the PV input for analog and regulatory data points.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of PVRNGOP	Value Range:	None = -6.86% to 106.86% (no clamping). Fullrng = -2.9% to 102.9% (no clamping). Clmpzero = 0.0% to 1.0% (clamped at zero) AND Clmpzero >= 102.9% (clamped at 102.9%).		

NOTE

Fullrng is not valid for a CB1C controller slot if INPTCOND = Sqrroot is specified.

PVROCPR — PV Rate-of-Change Alarm Priority

This parameter defines rate-of-change priority levels for PV alarms.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGENCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.
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PVROCNPR — PV Rate-of-Change Negative Alarm Priority
PVROCPPR — PV Rate-of-Change Positive Alarm Priority

This parameter defines positive or negative rate-of-change priority levels for PV alarms.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION = Alarm is not reported to the system. Alarm is not annunciated. JOURNAL = Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays. LOW = Alarm is historized, annunciated, and does appear on Group and Detail displays. HIGH = Alarm is historized, annunciated, and does appear on Group and Detail displays. EMERGNCY = This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity. PRINTER = Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.				
NOTE If PRINTER is chosen, the alarm is lost if the printer is not working.					
	JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.			

PVROCNTTP — PV Rate-of-Change Negative Trip Point (MC)
PVROCPTP — PV Rate-of-Change Positive Trip Point

These parameters define the positive or negative rate-of-change trip point in the PV (in % per minute) that causes an alarm to be generated at an MC regulatory data point.

Source:	User	Default Value:	1.0 % per minute	Access Lock:	Supvr.
Value Type:	Real	Value Range:	<u>PVROCNTTP</u> = -999.9 to -1.0 in % per minute. <u>PVROCPTP</u> = 1.0 to 999.9 in % per minute.		

PVROCTP — PV Rate-of-Change Trip Point (EC)

This parameter defines the rate-of-change trip point in the PV (in % per minute) that causes an alarm to be generated at an EC regulatory data point.

Source:	User	Default Value:	1.0 % per minute	Access Lock:	Supvr.
Value Type:	Real	Value Range:	1% to 999.0% per minute.		

PVSIGNAL — PV Signal Type (CB Regulatory Data Point)

This parameter determines which of the signals at the data point (specified by PVSLTSRC) is to be applied to the PV input of this regulatory data point.

Source:	User	Default Value:	Pv	Access Lock:	Engr.
Value Type:	Enum. of SIGNAL	Value Range:	Output = PV input to this data point is the output from the data point specified by <u>PVSLTSRC</u> . Pv = PV input to this data point is the PV at the data point specified by <u>PVSLTSRC</u> . Rv = PV input to this data point is the remote variable at the data point specified by <u>PVSLTSRC</u> . Lsp = PV input to this data point is the local setpoint at the data point specified by <u>PVSLTSRC</u> .		

NOTE

To update the value in this parameter, MODE must contain Man and INITCONF must contain Noinit. If you specify ALGIDDAC = Das, this parameter must contain Pv.

PVSIGNAL — PV Signal Type (EC Regulatory Data Point)

This parameter determines which of the signals at the data point (specified by PVSLTSRC) is to be applied to the PV input of this regulatory data point.

Source:	User	Default Value:	Pv	Access Lock:	Engr.
Value Type:	Enum. of SIGNAL	Value Range:	Output = PV input to this data point is the output from the data point specified by <u>PVSLTSRC</u> . Pv = PV input to this data point is the PV at the data point specified by <u>PVSLTSRC</u> . Hwy = PV input to this data point is the output from a data point that resides in a TPS module as specified by <u>PVSLTSRC</u> . Lsp = PV input to this data point is the local setpoint at the data point specified by <u>PVSLTSRC</u> . S1 = PV input to this data point is the S1 status output from the data point specified by <u>PVSLTSRC</u> . S2= PV input to this data point is the S2 status output from the data point specified by <u>PVSLTSRC</u> . S3 = PV input to this data point is the S3 status output from the data point specified by <u>PVSLTSRC</u> . S4 = PV input to this data point is the S4 status output from the data point specified by <u>PVSLTSRC</u> . Si = PV input to this data point is the SI (hardware input) at the data point specified by <u>PVSLTSRC</u> . S5 = PV input to this data point is the S5 status output from the data point specified by <u>PVSLTSRC</u> . Notconfg = PV input is not configured for this data point.		

NOTE

To update the value in this parameter, MODE must contain Man and INITCONF must contain Noinit and EMS must contain Disable.

PVSIGNAL — PV Signal Type (MC Regulatory Data Point)

This parameter determines which of the signals at the data point (specified by PVSLTSRC) is to be applied to the PV input of this regulatory data point.

Source:	User	Default Value:	Pv	Access Lock:	Engr.
Value Type:	Enum. of SIGNAL	Value Range:	Output = PV input to this data point is the output from the data point specified by <u>PVSLTSRC</u> . Pv = PV input to this data point is the PV from the data point specified by <u>PVSLTSRC</u> . Ai = PV input to this data point is the analog input at the data point specified by <u>PVSLTSRC</u> . Lsp = PV input to this data point is the local setpoint at the data point specified by <u>PVSLTSRC</u> . Sqv = PV input to this data point is the sequence variable of the data point specified by <u>PVSLTSRC</u> .		

NOTE

To update the value in this parameter, MODE must contain Man and INITCONF must contain Noinit.

PVSLTSRC — PV Input Source (CB, MC, & EC Regulatory Data Points)

This parameter defines the Tag Name of the data point that supplies the signal to the PV input of this data point. Refer to the definition of NAME (Tag Name) for the valid character set. This parameter is not applicable to the CB and MC **Overhisl** and **Overlosl** algorithms.

Default Value: Tag Name from Line 1 of the Configuration Form.

NOTE

The data point that supplies the input signal must be in the same hiway box as this regulatory data point. The type of signal that is supplied is determined by the PVSIGNAL (PV Signal Type) parameter.

If the PV input source is from field wiring, do not make any entries in the PVSLTSRC and PVSIGNAL lines of the Configuration Form.

To update the value in this parameter, MODE must contain Man and INITCONF must contain Noinit, and for EC points, EMS must contain Disable.

The Tag Name in PVSLTSRC must be that of an HG point in the same HG as this point.

For MC points, if PVSIGNAL contains Ai, the tag name in PVSLTSRC must be that of an analog input point or an analog I/O composite point; and if PVSIGNAL contains Sqv, the Tag Name in PVSLTSRC must be that of a regulatory point.

PVSOURCE — PV Source (Analog, Digital, and Regulatory Data Points)

This parameter indicates the source of the PV value and which access levels, if any, are permitted to change the PV value.

Source:	User	Default Value:	Auto	Access Lock:	Supvr.
Value Type:	Enum. of PVSOURCE	Value Range:	Auto = Automatic; the PV value is obtained from the hiway box. Changes to the PV value are not allowed. Man = Manual; operator, supervisor, or engineer access level can be used to change the PV. Sub = Substituted; program and continuous control access levels in an AM can be used to change the PV value of this data point.		

NOTE

When this parameter contains Man or Sub, you can still see the value from the process in the PVAUTO parameter. When the value in PVSOURCE is changed from Man or Sub to Auto, the values in PVAUTO and in PV should be made the same, to avoid a "bump."

PVSTS — PV Value Status

This parameter provides the current status of the PV value for the specified data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of PVVALST	Value Range:	Normal = PV value is OK. Uncertain = PV value is uncertain. Bad = PV value is bad.		

PVTEMP — Temp Display**PVTEMP — Temp Display of X (EC Regulatory Data Point)**

This parameter defines the temperature scale to be used for the PV input to analog and regulatory data points.

Source:	User	Default Value:	DegreesF	Access Lock:	Engr.
Value Type:	Enum. of TEMPTURE	Value Range:	For all analog and regulatory data points except the EC regulatory data point: DegreesF = Degrees Fahrenheit DegreesC = Degrees Centigrade For an EC regulatory data point: DegreesF = Degrees Fahrenheit DegreesC = Degrees Centigrade DegreesR = Degrees Rankin DegreesK = Degrees Kelvin		

NOTE

To update the value in this parameter, MODE must contain Man; for EC and CB points, INITCONF must contain Noinit; and for EC points, EMS must contain Disable.

The value for this parameter is stored both in the HG and in the box (except for LLPIU analog inputs, which are stored in the LLPIU only). When a change to this value occurs, the values in the following parameters are also updated: PVEUHI, PVEULO, and PVCHAR.

PVTRACK — PV Tracking (CB, EC, & MC Regulatory Data Points)

This parameter defines whether PV tracking is to be used for a controller-type regulatory data point. If PV tracking is selected, when MODE = Man, the slot's LSP tracks its own X (PV) input.

PV tracking is used primarily with PID type algorithms. PVTRACK does not have meaning for non-PID algorithms in Basic Controllers (CBs) or Multifunction Controllers (MCs), however, PVTRACK does apply to the following non-PID algorithms in Extended Controllers (ECs): Ecsumr, Extrabi, Leadlagm, Leadlags, Multdiv, Sqrtprod, Switch, and Rampsoak.

Source:	User	Default Value:	Notrack	Access Lock:	Engr.
Value Type:	Enum. of TRACKING	Value Range:	Notrack = No PV tracking. Track = PV Tracking.		

NOTE

To update the value in this parameter, MODE must contain Man and INITCONF must contain Noinit, and for EC points, EMS must contain Disable.

PVTV — Target (DHP, HLPIU, LEPIU, LLPIU, MC)

This parameter defines the target (setpoint) value in engineering units for analog input data points and for the input portion of analog composite data points.

Source:	User	Default Value:	0.0	Access Lock:	Oper.
Value Type:	Real	Value Range:	Target value can have a range from -6.86% to 106.86%. Value that is entered is defined by the configured range established by <u>PVEULO</u> and <u>PVEUHI</u> .		

PVTVP — Process Variable Target Value Percent

In the HG, PVTVP is the same as the setpoint percent parameter SPP. Refer to the definition of SPP.

-R-

Rampsoak — Ramp and Soak Generator (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Rampsoak** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on the *EC Regulatory Data Point Configuration Form* and on the HG data entry displays. The outstanding characteristic of the **Rampsoak** algorithm is that it is a setpoint programmer (ramp and soak sequencer) and/or timed-event switcher.

POINT ASSIGNMEN T DISPLAY	PV CONFIGURATIO N DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY		MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLYTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELCT DEVLPR DEVLOPR \$DLSELCT PVHHPR \$HHSELCT PVHIPR \$HSELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLEQN (A,B,C) OUTIND \$OPTOL PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK SPCHAR SPFORMAT SPTEMP SPEULO SPEUHI SPDSPHI SPDSPLO SPHILM SP SOASIGNL SOASRC SOBSIGNL SOBSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC	CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD MARK1SEG MARK1TF MARK1TT MARK2SEG MARK2TF MARK2TT RATE1 SOAKV1 SOAKT1 RATE2 SOAKV2 SOAKT2 RATE3 SOAKV3 SOAKT3	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE EMSENB MODE \$MODESEL

RATE1-RATE3

RATE1-RATE3 — Rate1-Rate3 (Ramp Rates 1-3; EC Regulatory Data Point)

These parameters define the ramp rates during ramp segments for an EC regulatory data point that has been configured for the **Rampsoak** algorithm. For hiway-based operator stations, these parameters are referred to as RR1-RR3.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-100 to 100	(Negative ramp rates yield decreasing output values.)	

RATIO — Ratio (CB, EC, & MC Regulatory Data Points)

This parameter contains the ratio value that is used when ALGIDDAC contains Pidratio, Autoratio, Autobias, or Extratbi. When MODE contains Cas, the configured Y input (SP) is multiplied by the value in RATIO.

Source:	User	Default Value:	1.0	Access Lock:	Oper.
Value Type:	Real	Value Range:	For all algorithms except Extratbi : 0.1 to 9.999 For Extratbi algorithm: -9.999 to 9.999		

NOTE

To update the value in this parameter when ALGIDDAC contains Autoratio, RCASCENB must contain Disable and MODE must contain Cas.

RCASENB — Remote Cascade Enable

This parameter specifies whether supervisory control from a TPS module can be used for this data point. **RCASENB** does not affect the local cascade mode. This parameter is not applicable to the following EC algorithms:

Automan	Logic
Charact	Massflow
Das	Midof3
Deadtm	Pidexfb
Extratbi	Switch

This parameter is also not applicable to a digital data point if Moment has been entered for the **MOOUTIND** (Momentary Output Indication) parameter. This parameter is automatically set for CB and MC regulatory data points that have been configured for the **Pidcma**, **Pidcm**, or **Pidspc** (05, 06, or 07) algorithm, because it uses the hiway-based computer mode.

Refer to **CASREQ** for more information about the use of **RCASENB**.

Source:	User	Default Value:	Off	Access Lock:	Engr. for EC regulatory data points. Read Only for CB and MC regulatory data points.
Value Type:	Boolean	Value Range:	Off = Supervisory control cannot be used. On = Supervisory control can be used.		

The following comparison of the TPS versus hiway-based modes shows the effect of **RCASENB** on supervisory control:

RCASENB	TPS Mode	Hiway-based Mode
On	MAN AUTO BCAS CAS	MAN AUTO CAS COMP
Off	MAN AUTO BCAS CAS	MAN AUTO Not allowed CAS

NOTE

To update the value in this parameter, **MODE** must contain Man and **CASREQ** must contain Notreq, and for EC points, **EMS** must contain Disable.

RCASSHED — Remote Cascade Mode Request Source

This parameter indicates whether the cascade mode request flag has been set by the operator, or by the fact that a regulatory data point is shedding from the cascade mode to the shed mode.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of CASREQS	Value Range:	On = Cascade mode has been requested by the operator. Off = Regulatory data point has shed from the cascade mode.		

READASC — Read ASCII Text

This parameter is used to read up to 30 words from an HLPIU box database, which are converted to 60 ASCII characters. When this parameter is requested, an index must be specified. This index is used as an offset into the HLPIU database.

The index can have a value from 0 to 994. Since the total number of words in the HLPIU database is 1023 and READASC reads up to 30 words, 994 is the highest index allowed (because 30 words still follow in the database).

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	String				

READINT — Read Integer

This parameter is used to read one word from an HLPIU box database in integer form. When READINT is requested, an index must be specified. The index is used as an offset into the HLPIU database.

The index can have a value from 0 to 1023, because this is the total number of words in the HLPIU database.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer				

READREAL

READREAL — Read Real

This parameter is used to read one or two words in real format from an HLPIU box database. When READREAL is requested, an index must be specified. The index is used as an offset into the HLPIU database.

The index can have a value from 0 to 1022. Since the total number of words in the HLPIU database is 1023 and READREAL can read one or two words, 1022 is the highest index allowed (because two words still follow in the HLPIU database).

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	String				

REDTAG — Red Tag

Specifies if a "red tag" is in effect on this data point. If so, the OP value for this point does not move because of changes made from the LCN. It is still possible that changes to OP could be caused by functions on the Data Hiway. Refer to 3.3.12 in *HG Control Functions* in the *Implementation/Hiway Gateway - 1* binder.

Source:	User	Default Value:	Off	Access Lock:	Supvr./Engr.
Value Type:	Enum. of REDTAG	Value Range:	Off = Data point is in service. On = Data point is out of service.		

NOTES

1. When a data point is red-tagged, all aliases of the data point are automatically red-tagged by the HG.
2. To update the value in this parameter, MODE must contain Man and MODATTR must contain Operator.

REQUESTS

This parameter indicates the LCN messages received by the node (per second) in the previous 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	>= 0.0		

RESETCMD

RESETCMD—Reset Command (MC Analog Input Data Points, MC & PIU Counter Data Points)

This parameter allows the operator to reset a counter data point, timer data point, or the accumulator (of an MC analog data point with accumulation).

Source:	User	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Enum. of RESET	Value Range:	Reststrt = Accumulator, counter, or timer is running. Reset = Accumulator, counter, or timer is stopped.		

NOTE

The value in this parameter can be changed by a US, by a CL/AM program, or by an ACP program to only Reset, and it remains with that value until the counter or accumulation is restarted.

RNGCODE1 — PC Range Code for DHP Analog Inputs

RNGCODE2 — PC Range Code for DHP Analog Outputs

These parameters are used for analog input, analog output, and analog composite data points to specify the range that is to be used by the DHP when it converts PC values to B12E format. RNGCODE1 is used for analog input data points, and for the input portion of analog composite data points; RNGCODE2 is used for analog-output data points, and for the output portion of analog composite data points.

Source:	User	Default Value:	0	Access Lock:	DEB
Value Type:	Integer	Value Range:	0 = 0-999 1 = 0-4095 2 = 0-9999 3-7 = Defaults to 0-9999		

RNUMREAD

This parameter indicates the LCN messages the node created (for read) in the previous 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	>= 0.0		

RNUMSTOR

This parameter indicates the LCN messages the node created (for write) in the previous 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

RPARSEC — # of Read Parameter Requests (All Processor Status Data Points)

This parameter indicates the number of parameters THIS node is requesting FROM other nodes on the LCN. The value is based on Intermediate Data Base (IDB) requests made to any LCN node during the previous 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	Engr.
Value Type:	Real	Value Range:	≥ 0.0		

RREQUEST

This parameter indicates the LCN messages the node created (per second) in the previous 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	≥ 0.0		

RSPBGP\$\$ — Ramp/Soak Percent Bar Graph Parameter (EC Regulatory Data Point)

This parameter is an internal parameter that is the value of the bar graph for an EC regulatory data point that has been configured for the **Rampsoak** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	0.0 to 100.0		

-S-

S1S2S3S4 — Alarm Status Summary (EC Regulatory Data Point)

This parameter provides the user with the current states of status bits S1–S4 (refer to ALFMT) using four digits, with each digit representing one of the status bits. The status bits that are displayed are labeled S1–S4 from left to right.

With R670, the default value is changed to NOACTION.

Source:	System	Default Value:	NOACTION	Access Lock:	Read Only
Value Type:	String	Value Range:	0000 to 1111 where the most-significant digit is the state of status bit S1, and the least-significant digit is the state of status bit S4. 0 indicates that the status bit is false; 1 indicates that it is true.		

S4LOGCPR—Logic Gate 4 Alarm Priority

This parameter defines the priority of a preset alarm generated by a data point. The alarm priority determines how the alarm is handled by the system. This parameter does not affect data point alarm detection and alarm indication visibility on Group and Detail displays.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			

NOTE

If PRINTER is chosen, the alarm is lost if the printer is not working.

JNLPRINT = Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.

SCANFREQ — Scan Frequency (HLPIU)

This parameter defines the rate at which this analog input or analog composite data point is sampled. In the HLPIU, the maximum scan rate is 400 points/second. The number of data points in a scan class must be less than the maximum allowed for the scan class. Refer to *PIU General Description* in the *Product Manual* in the *BASIC System* bookset. If too many points are assigned to the faster scan classes, this could cause a processing-overload condition to occur.

Source:	User	Default Value:	Offscan	Access Lock:	Engr.
Value Type:	Enum. of SCANFREQ	Value Range:	Offscan = No scanning Scan60 = 60 seconds Scan15 = 15 seconds Scan5 = 5 seconds Scan1 = 1 second Scan25 = 0.25 second Scan05 = 0.05 second Reserved = Mailboxing is a technique used to set up a PIU slot as an information transfer buffer so that a computer (which can interface to more than one hiway) can take information from one device on one hiway and write it to a PIU located on another hiway where it can be viewed by another operator station which does not have access to the original device.		

SCANFREQ — Scan Frequency (LEPIU)

This parameter defines the rate at which this analog input data point is sampled. In the LEPIU, the maximum scan rate is 16 points/second. The number of data points in a scan class must be less than the maximum allowed for the scan class. Refer to *PIU General Description* in the *Basic System Product Manual* binders. If too many points are assigned to the faster scan classes, this could cause a processing-overload condition to occur.

Source:	User	Default Value:	Offscan	Access Lock:	Engr.
Value Type:	Enum. of SCANFREQ	Value Range:	Offscan = No scanning Scan60 = 60 seconds Scan15 = 15 seconds Scan5 = 5 seconds		

SCANFREQ (LLPIU)

SCANFREQ — Scan Frequency (LLPIU)

This parameter defines the rate at which this analog input data point is sampled. In the LLPIU, the maximum scan rate is 160 data points/second without OTD and 73 data points/second with OTD. The number of data points in a scan class must be less than the maximum allowed for the scan class. Refer to *PIU General Description*. If too many points are assigned to the faster scan classes, this could cause a processing-overload condition to occur.

Source:	User	Default Value:	Offscan	Access Lock:	Engr.
Value Type:	Enum. of SCANFREQ	Value Range:	Offscan = No scanning Scan60 = 60 seconds Scan15 = 15 seconds Scan5 = 5 seconds Scan1 = 1 second Scan25 = 0.25 second		

SCANTIME — Scan Time (Sec) (DHP Box Data Point)

This parameter defines the scan time (in seconds) during which the DHP updates all the PC data points in the DHP database. A scan time of 0 seconds is considered as being free-running.

Source:	User	Default Value:	0 seconds	Access Lock:	DEB
Value Type:	Integer	Value Range:	0 to 15 seconds		

SEGMENT

SEGMENT — Segment Number for Ramp/Soak Algorithm (EC Regulatory Data Point)

This parameter specifies the current segment number for an EC regulatory data point that has been configured for the **Rampsoak** algorithm.

Source:	System	Default Value:	1	Access Lock:	Oper.
Value Type:	Enum. of CURSEGID	Value Range:	Ramp1 = Ramp segment 1 Soak1 = Soak segment 1 Ramp2 = Ramp segment 2 Soak2 = Soak segment 2 Ramp3 = Ramp segment 3 Soak3 = Soak segment 3		

NOTE

To update the value in this parameter, MODE must contain Man.

SEGTOT — Segtot (Number of Segments; EC Regulatory Data Point)

This parameter defines the number of linear segments in an EC regulatory data point that has been configured for the **Charact** algorithm.

Source:	User	Default Value:	1	Access Lock:	Supvr.
Value Type:	Integer	Value Range:	1-8		

SEGTYPE — Current Segment Type (EC Regulatory Data Point)

This parameter indicates the current segment that is in progress for the specified EC regulatory data point that has been configured for the **Rampsoak** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of SEGTYPE	Value Range:	Uncertn = Segment type is uncertain. Ramp = Ramp segment is being performed. Soak = Soak segment is being performed.		

SELINP — Selected Input (EC Regulatory Data Point; Leadlagm and Leadlags Algorithms)

This parameter indicates whether the manual PV (MPV) can be changed.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of OVRINSS	Value Range:	Sel = <u>MPV</u> can be changed. Nosel = <u>MPV</u> cannot be changed.		

SELINP — Selected Input (EC Regulatory Data Point; Switch*, Midof3 and Selovddc Algorithms)

This parameter indicates whether the X, Y, or Z input has been selected.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of SELINPT	Value Range:	For Switch and Midof3 algorithm: None = No input has been selected. Selectx = X input has been selected. Selecty = Y input has been selected. Selectz = Z input has been selected. For Selovddc algorithm: None = No input has been selected. Selectx = X input has been selected. Selecty = Y input has been selected. Selectz = Z input has been selected. Selectm = M input has been selected. Selecta = A input has been selected. Selectc = C input has been selected.		

*Before selecting a value for SELINP, refer to OII.

Selovddc — Selector/Override/DDC (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Selovddc** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Selovddc** algorithm is that it is a 6-input algorithm the output of which is the maximum or the minimum selection of the six inputs, with each unselected input constrained by override initialization. Directdigital control of the output by a TPS module is optional.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVEULO PVEUHI PVDSPhi PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELCT DEVLLPR DEVLOPR \$DLSELCT PVHHPR \$HHSELCT PVHIPR \$HISELCT PVLLPR \$LLSELCT PVLOPR \$LOSELCT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO CTLEQN (A,B,C) OUTIND PVSIGNAL PVSLSRC SPSIGNAL SPSLSRC INITCONF SPCHAR SPFORMAT SPTEMP SPEULO SPEUHI SPDSPHI SPDSPLO SOASIGNL SOASRC SOBSIGNL SOBSRC ZSIGNAL ZSLSRC MSIGNAL MSLSRC ASIGNAL ASLSRC CSIGNAL CSLSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE MODE \$MODESEL

SENSSTOP — Sense of SO Drivers (EC Regulatory Data Point)

This parameter defines the normal state of the status outputs.

Source:	User	Default Value:	Closed	Access Lock:	Engr.
Value Type:	Enum. of SENSSTOP	Value Range:	Closed = Normally closed. Open = Normally open.		

NOTE

To update the value in this parameter, MODE must contain Man and EMS must contain Disable.

SEQERR — Sequence Error (MC Process Module Data Point)

This parameter indicates that a SOPL sequence error or failure has been detected in the sequence that is executing in the process module data point. Sequence errors are 1 through 8; sequence failures are 9 through 23 as shown below.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
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Value Type: Integer

Value Range:

- 0 = No sequence error or failure has been detected.
- 1 = Computational operation overflow or denominator of arithmetic division is 0 (zero).
- 2 = RAM abnormality in the MC box has caused an invalid intermediate code.
- 3 = Write operation was attempted at a memory location that contains a Read Only variable.
- 4 = Disagreement exists between the actual cards installed in the point card file and the cards specified for the same file during configuration. Applies to only digital input, digital output, counter input, and analog output cards.
- 5 = Unconfigured analog input data point has been accessed, or that a disagreement exists between the analog inputs cards installed in the point card file and the analog input cards specified for the same file during configuration.
- 6 = Attempt was made to access parameters that are not applicable to an analog input data point.
- 7 = Attempt was made to access parameters that are not applicable to a regulatory data point.
- 8 = Attempt was made to access a parameter that is not applicable to a digital output data point configured for feedback.

SEQERR

- 9 = Sequence has stopped because operator has selected the STOP target on a module group display.
- 10 = Attempt has been made to jump to an abnormal sequence that has not been enabled.
- 11 = READ/WRITE statement or START statement could not be accomplished because of a failure in the SBLA (C-Link Adapter) card.
- 12 = READ/WRITE statement or START statement could not be accomplished because of a failure in the C-Link or because of a failure in another MC box.
- 13 = Time-out condition has occurred during the feedback time.
- 14 = Time-out condition has occurred on a conditional wait statement.
- 15 = Attempt has been made to access box variables that are in a partial failure state.
- 16 = Sequence program has tried to change the output state of a digital output data point that is not in the sequence mode.
- 17 = Sequence program has tried to change the output value of an analog output data point that is not in the sequence mode.
- 18 = Sequence program has tried to change the setpoint of a regulatory data point that is not in the sequence automatic mode.
- 19 = Sequence program has tried to change the output value of a regulatory data point that is not in the sequence manual or automatic mode.
- 20 = Attempt has been made to access the bias constant of a regulatory data point (which has been configured for the **Autobias** algorithm) that is not in the cascade or sequence cascade mode.
- 21 = Attempt has been made to access the ratio constant of a regulatory data point (which has been configured for the **Autoratio** algorithm) that is not in the cascade or sequence cascade mode.
- 22 = Incorrect mode has been specified for a regulatory, analog output, analog composite, digital output, or digital composite data point.
- 23 = START statement has been issued to a process module that has an operating status of Off.

SEQEXEC — Sequence Execution State (MC Process Module Data Point)

This parameter provides an indication of the current execution state of the CL/MC sequence in the selected process module data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of SEQEXEC	Value Range:	NI = Initial state in which the sequence has not been assigned. This state is also entered from SEQEXEC, the D11 state, when loading has been aborted. (NI = Not Loaded) D11 = Transient state during which the loading of the sequence program into the selected MC is in progress. (D11 = Down-Line Loading) Loaded = Sequence has been loaded into the selected MC. This state is entered automatically from the D11 state when loading has been completed, or when the sequence operating status has changed from Off to Redy. End = Sequence execution has been completed as indicated by the execution of an END statement. This state is entered at the completion of a normal sequence or an abnormal (Hold, Shdn, or Emsd) sequence. Pause = Sequence execution has been suspended because of the execution of a Pause statement or the completion of a step when in the Semiauto or Snglstep sequence mode. Sequence can be resumed by operator command. Fail = Sequence execution is suspended because of the detection of a failure condition. Sequence execution can be resumed by operator command. Error = Sequence execution is suspended because of the detection of an error condition. Sequence execution can be resumed by operator command. Run = Sequence program is running.		

SEQMODE — Sequence Execution Mode (MC Process Module Data Point)

This parameter indicates the current execution mode of the CL/MC sequence in the selected process module data point.

Source:	System	Default Value:	N/A	Access Lock:	Supvr.
Value Type:	Enum. of SEQMODE	Value Range:	Auto = Semiauto = Snglstep =	Sequence runs from beginning to end without interruption. Sequence stops at all pause statements in the sequence. Sequence pauses at the end of every step. Snglstep is typically used for debugging a program.	

NOTE

Semiauto and Snglstep are not allowed when the operational state of a process module data point is Off or Emsd.

SEQNAME — Sequence Name (MC)

This parameter is the name of an CL/MC sequence that is executing in a process module data point.

Source:	System	Default Value:	All Blanks	Access Lock:	Read Only
Value Type:	String				

SEQSLTSZ — Sequence Slot Size (MC Process Module Data Point)

This parameter defines the size of the process module's sequence slot in terms of the number of sequence blocks. A sequence block is defined as 16 words of memory.

Source:	User	Default Value:	0	Access Lock:	DEB
Value Type:	Integer	Value Range:	0 to 264, 520, or 776, depending on whether the SOPL/Trend-memory options are installed as follows: Maximum number of sequence blocks is 264 if the memory option is not installed. Maximum number of sequence blocks is 520 if SOPL and Trend-memory options are installed. Maximum number of sequence blocks is 776 if only the SOPL-memory option is installed.		

NOTE

If MCREV contains Rev20, SOPL memory and trend memory can't coexist, so you can configure only one of the following: SOPLMEN containing Soplmem, or BOXTREND containing Trend.

SGNLTY12

SGNLTY12-SGNLTY78 — Signal Type Slots 1&2 - Signal Type Slots 7&8 (CB Regulatory Data Point)

These parameters define the types of signals that are applied to the inputs of the override selector (a CB regulatory data point that has been configured for the **Overhisl** or **Overlosl** algorithm). Signal type selection is accomplished separately for each pair of slots (slots 1&2, 3&4, 5&6, and 7&8).

Source:	User	Default Value:	Rv	Access Lock:	Engr.
Value Type:	Enum. of SGNLTY	Value Range:	Output = Output signals from the respective pair of slots are to be the input signals to the override selector. Rv = The remote variables at the respective pair of slots are to be the inputs to the override selector.		

SHEDMODE — Shed Mode

This parameter defines the mode to which this data point sheds. This parameter is not applicable to the following EC algorithms:

Automan	Incrmtl
Charact	Logic
Das	Massflow
Deadt	Switch
Extratbi	

Source:	User	Default Value:	Man	Access Lock:	Engr. for EC Regulatory; Read-Only for CB and MC Regulatory, and analog outputs/composites and digital outputs/composites.
Value Type:	Enum. of SHEDMODE	Value Range:	Noshed = This data point does not shed to another mode. Man = Manual Auto = Automatic Bcas = Backup Cascade		

NOTE

Because there is only one time-out gate per slot (see [BOXTOG1](#), [BOXTOG2](#), and [TOGINTSL](#)), for slots that have subslots, the results can be unexpected. For example, if two AMs are storing values in two different subslots in the same slot, one AM could fail while the other continues to operate. The slot would not go to the shed mode in this case, because the slot is still receiving updates from one of the AMs.

SHEDSTAT

Under certain conditions, the HG automatically stores a correct value in SHEDMODE. This chart indicates those conditions:

<u>ALGIDDAC</u>	<u>INITCONE</u>	<u>SPSIGNAL</u>	<u>SPSLTSRC</u>	<u>SHEDMODE</u>
Pidcm	(any value)	(any value)	(any value)	Man
Pidcma, Pidspc	Init	Notconfg	(any value)	Auto
Pidcma, Pidspc	Init	Lsp	Slotnum	Auto
Pidcma, Pidspc	Init	(any but Lsp)	(any value)	Bcas
Pidcma, Pidspc	Noinit	(any value)	(any value)	Auto

SHEDSTAT — Shed Status (Regulatory Data Points)

This parameter indicates whether the selected regulatory data point has shed to its shed mode.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Regulatory data point has shed to another mode. False = Regulatory data point has not shed to another mode.		

SIGCHNGE — Significant Change Limit

This parameter determines what the deadband should be in terms of the difference between the last reported value of the PV and the current value of the PV. An alarm is generated if the difference between the new current value and the last reported value is greater than the significant change limit established by this parameter.

Source:	User	Default Value:	0	Access Lock:	DEB
Value Type:	Integer	Value Range:	0 = No significant change alarming 1 = ± 126.0 (binary counts) or 3.5% of scale 2 = ± 36.0 (binary counts) or 1.0% of scale 3 = ± 18.0 (binary counts) or 0.5% of scale 4 = ± 6.0 (binary counts) or 0.1667% of scale 5 = ± 2.0 (binary counts) or 0.0556% of scale 6 = ± 1.0 (binary counts) or 0.0278% of scale 7 = ± 0.5 (binary counts) or 0.0139% of scale		

SLOTNUM

SLOTNUM — Slot Number or Input Slot Number

This parameter defines the slot number of the slot (card) where this particular data point resides in the hiway box. (For the DHP, it defines the software slot rather than the card slot.) For analog input and digital input and composite data points, regulatory data points, and counter data points, the slot number defines the slot where the input to the data point resides.

Source: User **Default Value:** 1 **Access Lock:** DEB

Value Type: Integer **Value Range:** The slot number ranges for the various data points are as follows:

Data Point Type	Slot Number Range
CB RV	1-8
CB Regulatory	1-8
DHP AI & AC	1-15, 17-31
DHP DI & DC	1-15, 17-31
DHP Counter	1-15, 17-31
EC Regulatory	1-16
EC DI	1-16
HLPIU AI & AC	1-32
HLPIU DI & DC	1-32
HLPIU Counter & Control Counter	1-32
LEPIU AI	1-32 (Slots 1 & 2 are in Remote Box 1, slots 3 & 4 are in Remote Box 2, etc.)
LLPIU AI	1-32
MC AI (W/Accum)	7-8
MC AI & AC	7-8
MC DI & DC	1-32
MC Counter	1-32
MC Process Module	1-16
MC Regulatory	1-16

AI = Analog Input DI = Digital Input RV = Remote Variable
 AC = Analog Composite DC = Digital Composite

SOAKRMN — Remaining Soak Time (EC Regulatory Data Point)

This parameter indicates the amount of time remaining (in minutes) in the current soak segment for an EC Regulatory Data Point that has been configured for the **Rampsoak** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Real	Value Range:	0.0 to 960.0 minutes		

SOAKT1-SOAKT3 — Soakt1- Soakt3 (Soak Times 1-3; EC Regulatory Data Point)

These parameters define the time intervals of the soak segments in minutes for an EC regulatory data point that has been configured for the **Rampsoak** algorithm. For hiway-based operator stations, these parameters are referred to as TS1-TS3.

Source:	User	Default Value:	0.0 minutes	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0 to 960 minutes		

SOAKV1-SOAKV3 — Soakv1- Soakv3 (Soak Values 1-3; EC Regulatory Data Point)

These parameters define the soak output values for an EC regulatory data point that has been configured for the **Rampsoak** algorithm. For hiway-based operator stations, these parameters are referred to as SV1-SV3.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-5.0% to 105.0%		

SOASIGNL — SOA Signal Type (EC Regulatory Data Point)

This parameter defines the type of signal that is provided to the SOA input of an EC regulatory data point by the data point specified by the SOASRC parameter.

Source:	User	Default Value:	None	Access Lock:	Engr.
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Value Type: Enum. of SOSSIGNL

Value Range:

- None = SOA signal type is not configured for this data point.
- S1 = SOA input is the S1-status signal of the data point specified by SOASRC.
- S2 = SOA input is the S2-status signal of the data point specified by SOASRC.
- S3 = SOA input is the S3-status signal of the data point specified by SOASRC.
- S4 = SOA input is the S4-status signal of the data point specified by SOASRC.
- S5 = SOA input is the S5-status signal (BAD PV) of the data point specified by SOASRC.
- S6 = SOA input is the S6-status signal (SHED) of the data point specified by SOASRC.
- S7 = SOA input is the S7-status signal (Soft Failure) of the data point specified by SOASRC.
- SI = SOA input is the SI (hardware input) of the data point specified by SOASRC.

NOTE

To update the value in this parameter, MODE must contain Man and EMS must contain Disable.

SOASRC — SOA Input Source (EC Regulatory Data Point)**SOBSRC — SOB Input Source (EC Regulatory Data Point)**

These parameters define the Tag Names of the data points that provide the input signals to the respective SOA and SOB inputs of an EC Regulatory Data Point. The type of signals that are supplied are determined by the SOASIGNL (SOA Signal Type) and SOBSIGNL (SOB Signal Type) parameters. Refer to the definition of NAME (Tag Name) for the valid character set.

Default Value: Tag Name from Line 1 of Configuration Form.

NOTE

To update the value in this parameter, MODE must contain Man and EMS must contain Disable.

SOBSIGNL — SOB Signal Type (EC Regulatory Data Point)

The entries for this parameter are exactly the same as those for SOASIGNL (SOA-Signal Type) except that this parameter identifies the type of signal that is to be applied to the SOB input of the data point. Refer to the definition of the SOASIGNL parameter.

NOTE

To update the value in this parameter, MODE must contain Man and EMS must contain Disable.

SOESYNCH — SOE Synchronization (Data Hiway Data Point)

This parameter defines whether the HG designated as Thishg is to be responsible for generating Sequence-of-Events (SOE) words for HLPIUs on the Data Hiway.

Source:	User	Default Value:	Disable	Access Lock:	DEB
Value Type:	Enum. of ENBLSTAT	Value Range:	Disable = HG designated as Thishg is not to broadcast SOE words to HLPIUs. Enable = HG designated as Thishg is to broadcast SOE words to HLPIUs.		

NOTE

Make an entry for this parameter if Thishg is entered for the HWYHTD (HTD Assignment) parameter on the *Data Hiway Data Point Configuration Form*.

SOPLMEM — Additional SOPL Memory (MC Box Data Point)

This parameter defines that additional SOPL memory is in this MC box. Make an entry for this parameter only if

- a. Rev20 and Notrend were entered for the MCREV and BOXTREND parameters, respectively, or
- b. Rev30 was entered for the MCREV parameter.

Source:	User	Default Value:	Nosopl	Access Lock:	DEB
Value Type:	Enum. of SOPLMEM	Value Range:	Nosopl = MC box does not contain additional SOPL memory. Sopl = MC box contains additional SOPL memory.		

SP — Target or Setpoint (CB, EC, MC)

This parameter defines the target value for a CB RV data point or the local setpoint (LSP) for a CB, EC, or MC regulatory data point. Target or local setpoint value must be in engineering units and can be anywhere from -6.86% to 106.86% of full scale.

Source:	User	Default Value:	0.0	Access Lock:	Oper.
Value Type:	Real	Value Range:	-9999 to 9999		

NOTE

To update the value in this parameter the conditions defined below must be met:

For timer data points, STATE must contain Stopped, and changes from a US require SPLOCK to contain Permit.

For regulatory data points:
The content of MODATTR affects access from a US and from a user-written program. Refer to 4.4.3 in *System Control Functions* in the *Implementation/Startup and Reconfiguration* binder.

Changes by continuous control functions require the following:

MODE = Cas and RCASENB = On.
ALGIDDAC defines the following SPC or DDC algorithm types:

CBs and MCs; DDCs are **Pidcma** and **Pidcm**; SPC is **Pidspc**.

ECs; DDCs are **Pidddc**, **Piddcsp**, **Leadlag**, **Leadlagm**, and **Selovddc**; SPCs are **Pidnom**, **Pidratio**, **Autratio**, **Autobias**, **Pidnonln**, **Pidgap**, **Pidexfb**, **Extratbi**, **Sqrtprod**, **Cbsumr**, **Multdiv**, **Incrmtl**, and **Rampsoak**.

PVTRACK = Notrack, if ALGIDDAC = **Pidcma**, **Pidcm**, **Pidddc**, **Leadlag**, **Leadlagm**, or **Selovddc**.

Change Permission Chart for CBs:

<u>ALGIDDAC</u>	(L)Man	(Init)Man	Man	Auto	Lcas	Rcas
Das Pidpl50, Pidpl50c, Leadlag, Overhisl, Overlosl Automan	0	0	0	0	0	0
Pidnorm, Pidratio, Autratio, Autobias Pidcm, Pidspc, Pidersgg, Pidersqi Pidgap	2	2	2	1	0	0
Pidcma	2	2	2	1	0	2
Switch	0	0	0	3	3	0
Sumrwman, Multwman, Cbsumr, Mult, Dividr Sqrtm, Sqrtprod, Sumsqrt, Hisel, Losel	0	0	0	1	0	0

SP (CB/EC/MC)

Key:

Abbrev.	Basic System Terms	LCN Terms
(L)Man	Local Manual	<u>MODE</u> = Man
(Init)Man	Initialization manual	<u>MODE</u> = Man, <u>INITMAN</u> = Init
Man	Manual mode	<u>MODE</u> = Man
Auto	Auto mode	<u>MODE</u> = Auto
Lcas	Local cascade	<u>MODE</u> = Cas, <u>RCASENB</u> = Off
Rcas	Remote Cascade	<u>MODE</u> = Cas, <u>RCASENB</u> = On

Values

- 0 = SP can't be changed or doesn't exist.
- 1 = SP can be changed.
- 2 = SP can be changed if PVTRACK = Notrack.
- 3 = SP can be changed if SPSIGNAL = Lsp and SPSLTSRC = (this Tag Name).

SP Change Permission Chart for MCs:

<u>ALGIDDAC</u>	(L)Man	(Init)Man	Man	Auto	Lcas	Rcas
Das	0	3	3	0	0	0
Pidnorm, Pidratio, Autratio, Autobias Pidcm, Pidspc, Pidersgg Pidersqi, Pidgap	2	2	2	1	0	0
Leadlag, Overhisl, Overlosl, Automan	0	0	0	0	0	0
Pidcma	2	2	2	1	0	2
Switch	0	0	0	1	1	0
Sumrwman, Multwman Cbsumr, Mult, Divdr Sqrt, Hisel, Losel	0	0	0	1	0	0

Key: Same as shown above for CB.

SP Change Permission Chart for ECs:

EC Primary Mode Chart

<u>ALGIDDAC</u>	L	I	O	M	A	C	R
Pidnorm, Pidratio, Autratio, Pidnonln, Pidgap, Pidexfb	9	9	7	3	1	0	0
Autobias	9	9	7	3	1	0	0
Piddc	9	9	7	3	1	0	3
Extratbi	6	8	7	2	1	0	0
Massflow	4	4	0	2	2	0	0
Sqrtprod	2	8	7	2	1	0	0

SP Change Permission Chart for ECs: (continued)

EC Primary Mode Chart

<u>ALGIDDAC</u>	L	I	O	M	A	C	R
Ecsumr, Multdiv	2	8	7	2	1	0	0
Rampsoak	1	1	7	1	1	0	0
Leadlags, Leadlagm with <u>CTLEQN</u> = Eqa	10	10	7	4	1	0	4
Leadlagm with <u>CTLEQN</u> = Eqb	9	9	7	3	1	0	3
Das, Selovddc, Automan, Switch, Midof3, Incrmtl, Logic, Deadtm, Charact	0	0	0	0	0	0	0

EC Secondary Mode Chart

<u>ALGIDDAC</u>	O Secondary Modes				L and I Secondary Modes			
	M	A	C	R	M	A	C	R
Pidnorm, Pidratio, Atratio, Pidnonln, Pidgap	3	1	0	0	3	3	0	0
Autobias	3	1	0	0	3	3	0	0
Pidddc	3	1	0	3	3	3	0	3
Extratbi	2	1	0	0	4	4	0	0
Massflow	2	2	0	0	0	0	0	0
Sqrtprod	2	1	0	0	4	4	0	0
Ecsumr, Multdiv	2	1	0	0	2	2	0	0
Rampsoak	1	1	0	0	0	0	0	0
Leadlags, Leadlagm with <u>CTLEQN</u> = Eqa	4	1	0	4	4	4	0	4
Leadlagm with <u>CTLEQN</u> = Eqb	3	1	0	3	3	3	0	3
Das, Selovddc, Automan, Switch, Midof3, Incrmtl, Logic Deadtm, Charact	0	0	0	0	0	0	0	0

Key:

Abbrev.	Current Point Condition	Operating Mode
L	LMAN = Local (any mode)	Local Manual
O	PRORST = Notsel	Override Initialization
I	INITMAN = Init	Initialization Manual
M	MODE = Man	Manual Mode
A	MODE = Auto	Automatic Mode
C	MODE = Cas, RCASENB = Off	Local Cascade Mode
R	MODE = Cas, RCASENB = On	Remote Cascade

Values

- 0 = SP can't be changed or doesn't exist.
- 1 = SP can be changed.
- 2 = SP can be changed if INITCONF = Noinit.
- 3 = SP can be changed if PVTRACK = Notrack.
- 4 = SP can be changed if INITCONF = Noinit and PVTRACK = Notrack.
- 5 = SP can be changed if INITCONF = Noinit or PVTRACK = Notrack.
- 6 = SP can be changed if INITCONF = Noinit and PVTRACK = Track
- 7 = Check the appropriate column of the Secondary Mode Chart.
- 8 = If INITCONF = Init, SP can't be changed. If INITCONF = Noinit, check the appropriate column of the Secondary Mode Chart.
- 9 = If PVTRACK = Track, SP can't be changed. If PVTRACK = Notrack, check the appropriate column of the Secondary Mode Chart.
- 10 = If INITCONF = Noinit and PVTRACK = Notrack, check the appropriate column of the Secondary Mode Chart. Otherwise, SP can't be changed.

SP — Preset Time Value (MC Timer Data Point)

This parameter defines the value that a timer data point is preset to whenever the timer is started or reset.

Source:	User	Default Value:	0.0	Access Lock:	Oper.
Value Type:	Real	Value Range:	0 to 9999		

SPARSEC — # of Store Parameters Requests (All Processor Status Data Points)

This parameter defines the number of store parameters requests per second to any LCN node.

Source:	System	Default Value:	0.0	Access Lock:	Engr.
Value Type:	Real	Value Range:	≥ 0.0		

SPCHAR — Characterization of Y (EC Regulatory Data Point)

The entries for this parameter are exactly the same as those for the PVCHAR (Characterization of X) parameter. Refer to the definition of PVCHAR.

SPDSPHI — SP Display Range High (EC Regulatory Data Point)**SPSDPLO — SP Display Range Low (EC Regulatory Data Point)**

These parameters define 0% and 100% of display range in engineering units for display of only this set point. These parameters exist only if the HG Point Display Ranges option is present. If the option is not present, or either of the parameters is set to the default value, the applicable EU range value is used to generate the display.

Source:	User	Default Value:	NaN (- - -)	Access Lock:	Engr.
Value Type:	Real	Value Range:	Within the range defined by <u>SPEULO</u> and <u>SPEUHI</u> . See the description of these parameters for further details.		

SPECIFI1 — Specifier Code or Input 1 Specifier Code

Used only with Modicon or Honeywell compatible PLCs, this parameter specifies whether the input PLC address is that of an analog input, digital input, or a counter. SPECIFI1 can be used as the specifier code for a digital data point with one input, or for the first input (Input 1) of a digital data point that has two inputs.

Source:	User	Default Value:	0	Access Lock:	Engr.
Value Type:	Integer	Value Range:	For Analog Input: 3 Read Only 4 Read and Write For Digital Input: 0 Read and Write 1 Read Only For Counter Input: 4 Read and Write		

SPECIFO1 (DHP)

SPECIFO1 — Specifier Code or Output 1 Specifier Code (DHP)

Used only with Modicon or Honeywell compatible PLCs, this parameter specifies whether the output PLC address is that of an analog composite, analog-output, digital composite, or digital output data point. SPECIFO1 is also used as the specifier code for a digital data point with one output, or for the first output (Output 1) of a digital data point that has two outputs.

Source:	User	Default Value:	0	Access Lock:	Engr.
Value Type:	Integer	Value Range:	For Analog Data Point: 4 For Digital Data Point: 0		

SPECIFO2 — Output 2 Specifier Code (DHP Digital Data Point)

Used only with Modicon or Honeywell compatible PLCs, this parameter specifies that the output PLC address is that of the second output (Output 2) of a digital output or digital composite data point that has two outputs.

Source:	User	Default Value:	0	Access Lock:	Engr.
Value Type:	Integer	Value Range:	For Digital Data Point: 0		

SPECIFI2 — Input 2 Specifier Code (DHP Digital Data Point)

This parameter specifies that the input PLC address is that of the second input (Input 2) of a digital data point that has two inputs.

Source:	User	Default Value:	0	Access Lock:	Engr.
Value Type:	Integer	Value Range:	For Digital Data Point: 1		

SPEUHI — SP Range High (EC Regulatory Data Point)
SPEULO — SP Range Low (EC Regulatory Data Point)

These parameters define the 0% and 100% of range (in engineering units) for display of this setpoint.

Source:	User	Default Value:	100.0 for SPEUHI 0.0 for SPEULO	Access Lock:	Engr.
Value Type:	Real	Value Range:	-9999 to 9999 when <u>SPCHAR</u> is Linear or Sqrroot. When <u>SPCHAR</u> is nonlinear, refer to the max. and min. temperatures for that characterization. See general note under <u>PVCHAR</u> entry in this manual for maximum allowable temperatures.		

NOTE

The values for SPEUHI and SPEULO are stored both in the HG and in the box. When a change in the SPEUHI or SPEULO value occurs, the values for the following parameters (also stored both in the HG and in the box) are updated in the HG: SPEUHI, SPEULO, and SPTMP or SPFORMAT (depending on the value in SPCHAR—linear types use SPFORMAT and temperature type use SPTMP).

These parameters represent the high and low ends of the EU range for the setpoint. The maximum, total range allowed is 1000.0 units. For example, if SPEULO contains -1000.0, SPEUHI can be no higher than 0.0 units. Similarly, if SPEULO contains 0.0, SPEUHI can be no higher than 1000.0 units.

Because SPEUHI represents the span of the setpoint range, a change to the value in SPEULO can affect the value in SPEUHI. For example, if SPEULO initially contains 0.0 and SPEUHI contains 1000.0, and then an engineer changes SPEULO to 100.0, SPEUHI is automatically adjusted to 1100.0, to retain the span of 1000.0 units. The span can be changed only by updating the value in SPEUHI.

SPFORMAT — Decimal Format of Y (EC Regulatory Data Point)

The entries for this parameter are exactly the same as those for the PVFORMAT (Decimal Format of X) parameter. Refer to the definition of PVFORMAT.

NOTE

To update the value in this parameter, MODE must contain Man and EMS must contain Disable.

The value for this parameter is stored both in the HG and in the box. When a change to this value occurs, the values in the following parameters are also updated: SPEUHI, SPEULO, and SPCHAR.

SPHIFL — Setpoint Hi Limit Exceeded Flag (EC Regulatory Data Point)**SPLOFL — Setpoint Lo Limit Exceeded Flag**

These parameters indicate whether the setpoint value has reached or exceeded the configured high or low limit.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Configured limit has been reached or exceeded. False = Configured limit has not been reached or exceeded.		

SPHILM — Setpoint High Limit (EC & MC Regulatory Data Points)**SPLOLM — Setpoint Low Limit**

These parameters define the high- and low-alarm limits of the setpoint in engineering units. Setpoint limits are from -6.86% to 106.86%.

Source:	User	Default Value:	0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-9999 to 9999 Allowable values depend on range established by <u>SPEUHI</u> and <u>SPEULO</u> in the EC, or by <u>PVEUHI</u> and <u>PVEULO</u> in the MC.		

SPLOCK — Operator Preset Change (MC Timer Data Point)

This parameter defines whether an operator is allowed to change the preset time value.

Source:	User	Default Value:	Permit	Access Lock:	Engr.
Value Type:	Enum. of MODEPERM	Value Range:	Permit = Allows operator to change preset time value. Nopermit = Does not allow operator to change preset time value.		

SPP — Setpoint Percent

This parameter provides the value of the setpoint in percent of the configured engineering unit range for the specified data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	-6.86% to 106.86%		

SPSIGNAL — SP Signal Type (CB, EC, & MC Regulatory Data Points)

The SP signal types are the same as those listed for the PVSIGNAL (PV Signal Type) for the CB, EC, and MC regulatory data points except that these signals are applied to the setpoint input rather than the PV input. Refer to the description of the PVSIGNAL parameter.

Default

Value: Lsp

NOTE

To update the value in this parameter, MODE must contain Man, INITCONF must contain Noinit; for EC points, EMS must contain Disable; and when ALGIDDAC contains Das, SPSIGNAL must contain Rv.

SPSLTSRC — SP Input Source (CB, MC, & EC Regulatory Data Points)

This parameter defines the Tag Name of the data point that supplies the signal to the SP input of this data point. The type of signal that is supplied is determined by the SPSIGNAL (SP-Signal Type) parameter. Refer to the definition of NAME for the valid character set.

Default

Value: Tag Name from Line 1 of Configuration Form.

NOTE

To update the value in this parameter, MODE must contain Man, INITCONF must contain Noinit; for EC points, EMS must contain Disable.

SPTEMP — Temp Display of Y (EC Regulatory Data Point)

The entries for this parameter are exactly the same as those for the PVTEMP (Temp Display of X) parameter. Refer to the definition of PVTEMP.

NOTE

To update the value in this parameter, MODE must contain Man, INITCONF must contain Noinit; for EC points, EMS must contain Disable.

The value for this parameter is stored both in the HG and in the box. When a change to this value occurs, the values in the following parameters are also updated: SPEUHI, SPEULO, and SPCHAR.

SPTV — Segment Variable (EC Regulatory Data Point)

This parameter is the value assigned to the current segment being performed by the **Rampsoak** algorithm as follows:

<u>Segment Type</u>	<u>SPTV Calculation</u>
Ramp1	$SPTV = SOAKV1$
Ramp2	$SPTV = SOAKV2$
Ramp3	$SPTV = SOAKV3$
Soak1	$SPTV = [(SOAKT1 - SOAKRMN) / SOAKT1] \times 100$
Soak2	$SPTV = [(SOAKT2 - SOAKRMN) / SOAKT2] \times 100$
Soak3	$SPTV = [(SOAKT3 - SOAKRMN) / SOAKT3] \times 100$

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	-5.0% to 105.0%		

Sqrt — Square Root (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Sqrt** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Sqrt** algorithm is that it is a 2-input auxiliary algorithm that takes the square root of a signal.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	DEVHITP	SUPPIO	NMODE
UNIT	PVCHAR	\$AUXUNIT	OUTIND	NMODATTR
PTDESC	INPTCOND	ALFMT	PVSIGNAL	PSTMODE
KEYWORD	PVFORMAT	PVLOTP	PVSLT SRC	MODEPERM
PRIMMOD	PVTEMP	PVHITP	SPSIGNAL	MODE
ASSOCDSP	PVEULO	DEVLOTP	SPSLT SRC	\$MODESEL
HWYNUM	PVEUHI	\$REALARM	INITCONF	
PNTBOXTY	PVDSPHI	DEVHIPR	OPLOLM	
BOXNUM	PVDSPLO	\$DHSELCT	OPHILM	
PTDISCL	OVERVAL	DEVLOPR	\$OPTOL	
SLOTNUM	PVRNGOP	\$DLSELCT	TD	
LOADDEST	PVCLAMP	PVHIPR	K	
ALGIDDAC	CALIBOFF	\$HISELCT	K1	
		PVLOPR	K2	
		\$LOSELCT		
		ALPRIOR		
		CCRANK		
		CCPRIPNT		
		EIPPCODE		
		EIPEVENT		
		EIPENB		

Sqrt (MC)

Sqrt — Square Root (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Sqrt** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Sqrt** algorithm is that it is a 2-input auxiliary algorithm that takes the square root of a signal.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSELECT PVROCPPR \$RPSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND PVSLTSRC PVSIGNAL SPSLTSRC SPSIGNAL INITCONF SPLOLM SPHILM \$SPTOL OPLOLM OPHILM \$OPTOL TD K K1 K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Sqrtprod (CB)

Sqrtprod — Double Square Root (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Sqrtprod** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Sqrtprod** algorithm is that it is a 2-input auxiliary algorithm whose output is the square root of the product of two inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME	EUDESC	DEVHITP	SUPPIO	NMODE
UNIT	PVCHAR	\$AUXUNIT	OUTIND	NMODATTR
PTDESC	INPTCOND	ALFMT	PVSIGNAL	PSTMODE
KEYWORD	PVFORMAT	PVLOTP	PVSLTSRC	MODEPERM
PRIMMOD	PVTEMP	PVHITP	SPSIGNAL	MODE
PRIMMOD	PVEULO	DEVLOTP	SPSLTSRC	\$MODESEL
HWYNUM	PVEUHI	\$REALARM	INITCONF	
PNTBOXTY	PVDSPHI	DEVHIPR	OPLOLM	
BOXNUM	PVDSPLO	\$DHSELCT	OPHILM	
PTDISCL	OVERVAL	DEVLOPR	\$OPTOL	
SLOTNUM	PVRNGOP	\$DLSELCT	TD	
ALGIDDAC	PVCLAMP	PVHIPR	K	
LOADDEST	CALIBOFF	\$HISELCT	K2	
		PVLOPR		
		\$LOSELCT		
		ALPRIOR		
		CCRANK		
		CCPRIPNT		
		EIPPCODE		
		EIPEVENT		
		EIPENB		

Sqrtprod — Square Root of Products (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Sqrtprod** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Sqrtprod** algorithm is that it is a 3-input auxiliary algorithm whose output is the square root of the product of the three inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELECT DEVLLPR DEVLOPR \$DLSELECT PVHHPR \$HHSELECT PVHIPR \$HISELECT PVLLPR \$LLSELECT PVLOPR \$LOSELECT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPCHAR SPFORMAT SPTEMP SPEULO SPEUHI SPDSPHI SPDSPLO SPLOLM SPHILM \$SPTOL SP SOASIGNL SOASRC SOBSIGNL SOBSRC ZSIGNAL ZSLTSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD KM BO	NMODE NMODATTR PSTMODE RCASENB MODEPERM SHEDMODE EMSENB MODE \$MODESEL

STATE — Accumulation/Counter/Timer State

This parameter indicates the current state of an analog input data point with accumulation, a counter data point, or a timer data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of STATE	Value Range:	Reset = Data point is in the reset condition. Stopped = Accumulation, counter, or timer has been stopped. Running = Accumulation, counter, or timer is running. Complete = Accumulation, counter, or timer has reached the maximum limit. Reserved = Reserved for future use.		

NOTE

The Reset, Complete, and Reserved states are not returned by the HG.

STATE1 — Lower Box Descriptor (Digital, Flag, and Logic Block Data Points, and EC Logic Algorithm)**STATE2 — Upper Box Descriptor**

These descriptors can be up to eight characters long and appear below the upper and lower boxes on Group and Detail Displays as shown in the following:

CAUTION

(Upper Box)

Upper Box Descriptor
(STATE2)



(Lower box)

Lower Box Descriptor
(STATE1)

In the TPS Enhanced Operator Station, STATE2 text descriptor is for the lower box while STATE1 text descriptor is for the upper box. Only the positions of the descriptors have changed; input action remains the same.

STATMENT

Source:	User	Default Value:	UPPER for upper box. LOWER for lower box.	Access Lock:	DEB
Value Type:	String	Value Range:	The permissible character set for the lower and upper box descriptors is as follows: Alphabetics A through Z (uppercase only), Numerics 0 through 9, Underscore (_)		

The colors of the boxes are specified by the UBOXCLR (Upper Box Color) and LBOXCLR (Lower Box Color) parameters. These boxes light depending on whether the digital data point has one or two inputs, and the states of the inputs (contact open or contact closed). Refer to the definition of the INPTDIR (Input Direction) parameter.

STATMENT — Current Statement Number (MC Process Module Data Point)

This parameter indicates the number of the current statement that is executing in a process module data point. If the statement number is 0, this means that no statement is executing; the first statement in a step has a statement number of 1.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 to 255		

STEP — Current Step (MC Process Module Data Point)

This parameter indicates the current step of the CL/MC sequence in a process module data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	String an HG Library.	Value Range:	The names of the steps are character strings stored by the user in		

STEPIDX — Current Step Index (MC Process Module Data Point)

This parameter indicates the text library index for the current step of the CL/MC sequence in a process module data point.

Source:	System	Default Value:	0	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0, and 256 through 511		

STIME — Sampling Interval (EC Regulatory Data Point; Deadtm Algorithm)

This parameter defines the sampling interval as measured in controller sampling-interval integers per deadtime sample.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 to 480		

STRAVGC – Parameter store rate (running average per second) in the current hour

This PSDP parameter indicates the rate at which parameters are stored in the current hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

STRAVGP – Parameter store rate (overall average per second) in the previous hour

This PSDP parameter indicates the rate at which parameters were stored in the previous hour.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

STRAVGS – Parameter store rate (per sec) in the last 15 seconds

This PSDP parameter indicates the rate at which parameters were stored in the last 15 seconds.

Source:	System	Default Value:	0.0	Access Lock:	View
Value Type:	Real	Value Range:	≥ 0.0		

STRTSTOP

STRTSTOP — Start/Stop Command

This parameter allows the operator to start and stop the MC counter, timer, or accumulator (at an MC analog data point w/accumulation) from a Group or Detail Display.

Source:	User	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Enum. of STRTSTOP	Value Range:	Start = Start the counter, timer, or accumulator. Stop = Stop the counter, timer, or accumulator.		

SUBSTEP — Sequence Subroutine Step (MC Process-Module Data Point)

This parameter indicates the current step of a CL/MC subroutine such as emergency shutdown, shutdown, hold, etc., that is currently executing in a process module data point. The names of the steps have been previously configured in the HG Library.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	String HG Library.	Value Range:	The names of the steps are character strings stored by the user in an		

SUBSTMT — Subroutine Statement (MC Process-Module Data Point)

This parameter is the number of the current statement in a CL/MC subroutine such as emergency shutdown, shutdown, hold, and so on, that is executing in a process module data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Integer	Value Range:	0 to 255		

Sumrwman (CB)

Sumrwman — Summer (with Manual) (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Sumrwman** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Sumrwman** algorithm is that it is a 2-input algorithm whose output is the sum of two independently scaled inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO OUTIND PVSIGNAL PVSLOTSRC SPSIGNAL SPSLOTSRC INITCONF OPLOLM OPHILM \$OPTOL TD K K1 K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Sumrwman (MC)

Sumrwman — Summer (with Manual) (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Sumrwman** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *MC Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Sumrwman** algorithm is that it is a 2-input algorithm whose output is the sum of two independently scaled inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSLCT PVROCPPR \$RPSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC INITCONF SPLOLM SPHILM \$SPTOL OPLOLM OPHILM \$OPTOL TD K K1 K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Sumsqrt — Sum of Square Roots (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Sumsqrt** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Sumsqrt** algorithm is that it is a 2-input auxiliary algorithm whose output is the sum of two square-rooted signals.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM ALGIDDAC LOADDEST	EUDESC PVCHAR INPTCOND PVFORMAT PVEULO PVTEMP PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO OUTIND PVSIGNAL PVSLT SRC SPSIGNAL SPSLT SRC INITCONF OPLOLM OPHILM \$OPTOL TD K K1 K2	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

SUPPIO — Display Suppression (CB, EC, and MC Regulatory Data Points)

This parameter defines whether the PV input and/or output from this regulatory data point is to be displayed on the Overview and Group Displays.

Source:	User	Default Value:	Nosuppr	Access Lock:	DEB
Value Type:	Enum. of SUPPIO	Value Range:	Nosuppr = Output and PV input is displayed. Insuppr = PV and SP inputs are suppressed, but the output is displayed. Outsuppr = Output and mode are suppressed, but the PV input is displayed.		

Switch (CB)

Switch — Switch (CB Regulatory Data Point)

The parameters that are applicable to a CB regulatory data point when it is configured for the **Switch** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *CB Regulatory Data Point Configuration Form* (for all algorithms except OVERHISL and OVERLOSL), and on the HG data entry displays. The outstanding characteristic of the **Switch** algorithm is that it is a single-pole double-throw switch for channeling signals.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	DEVHITP \$AUXUNIT ALFMT PVLOTP PVHITP DEVLOTP \$REALARM DEVHIPR \$DHSELECT DEVLOPR \$DLSELECT PVHIPR \$HISELECT PVLOPR \$LOSELECT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT	SUPPIO OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC INITCONF OPLOLM OPHILM \$OPTOL TD	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

Switch (EC)

Switch — Switch (EC Regulatory Data Point)

The parameters that are applicable to an EC regulatory data point when it is configured for the **Switch** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on *EC Regulatory Data Point Configuration Form*, and on the HG data entry displays. The outstanding characteristic of the **Switch** algorithm is that it is a 3-input auxiliary algorithm that connects the output to one of the inputs.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM LOADDEST ALGIDDAC	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$DLBADAL DLTIME \$REALARM BADPVPR DEVHHPR DEVHIPR \$DHSELECT DEVLLPR DEVLOPR \$DLSELECT PVHHPR \$HHSELECT PVHIPR \$HISELECT PVLLPR \$LLSELECT PVLOPR \$LOSELECT (Alarm Trip-point Values) ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB ALMASK INITCONF SPHILM SPLOLM \$SPTOL	SUPPIO OUTIND \$OPTOL PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC PVTRACK INITCONF SPCHAR SPFORMAT SPTEMP SPEULO SPEUHI SPDSPHI SPDSPLO SP SOASRC SOBSIGNL SOBSRC ZSIGNAL ZSLTSRC MSIGNAL MSLTSRC ASIGNAL ASLTSRC CSIGNAL CSLTSRC GREENSGL GREENSRC SENSSTOP OPLOLM OPHILM \$OPTOL TFCLLM TFCHLM TD	NMODE NMODATTR PSTMODE MODEPERM EMSENB MODE \$MODESEL

Switch (MC)

Switch — Switch (MC Regulatory Data Point)

The parameters that are applicable to an MC regulatory data point when it is configured for the **Switch** algorithm are contained in the following listing. These parameters are listed in the same order as they appear on the *MC Regulatory Data Point Configuration Form* and on the HG data entry displays. The outstanding characteristic of the **Switch** algorithm is that it is a single-pole double-throw switch for channeling signals.

POINT ASSIGNMENT DISPLAY	PV CONFIGURATION DISPLAY	ALARMING DISPLAY	OPERATING CONFIGURATION DISPLAY	MODE CONFIGURATION DISPLAY
NAME UNIT PTDESC KEYWORD PRIMMOD ASSOCDSP HWYNUM PNTBOXTY BOXNUM PTDISCL SLOTNUM ALGIDDAC LOADDEST	EUDESC PVCHAR INPTCOND PVFORMAT PVTEMP PVEULO PVEUHI PVDSPHI PVDSPLO OVERVAL PVRNGOP PVCLAMP CALIBOFF	\$AUXUNIT ALFMT \$ALDLOPT \$REALARM PVLOTP PVHITP DEVLOTP DEVHITP PVROCNTP PVROCPTP DEVHIPR \$DHSELCT DEVLOPR \$DLSELCT PVHIPR \$HISELCT PVLOPR \$LOSELCT PVROCNPR \$RNSELCT PVROCPPR \$RPSELCT ALPRIOR CCRANK CCPRIPNT EIPPCODE EIPEVENT EIPENB	SUPPIO ANLGCARD ALMDIS CONFTRND OUTIND PVSIGNAL PVSLTSRC SPSIGNAL SPSLTSRC INITCONF SPLOLM SPHILM \$SPTOL OPLOLM OPHILM \$PTOL TD	NMODE NMODATTR PSTMODE MODEPERM MODE \$MODESEL

-T-

T1 — T1 (Integral Time; CB, EC, & MC Regulatory Data Points)

This parameter defines the integral time for a regulatory data point that has been configured for a PID algorithm.

Source:	User	Default Value:	0.01 minutes per repeat for CB and EC. 0.02 min. per repeat for MC	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.01 to 9999 for the CB and EC. 0.02 to 9999 for the MC. Value greater than 91.02 turns off the integral action.		

NOTE

While the T1 value can be as high 91.02 minutes, it is normally used at much smaller values. The expression used to calculate T1 in the controller produces very "granular" results near the upper end of the value range. When such values are used, displayed values may vary significantly from entered values.

T1 — T1 (Lag Time Constant; CB, EC, & MC Regulatory Data Points)

This parameter defines the lag time constant in minutes for a regulatory data point that has been configured for the **Leadlag**, **Leadlagm**, or **Leadlags** algorithms. T2 divided by T1 should be less than or equal to 16. For the EC, T1 defines the first lag time constant.

Source:	User	Default Value:	0.01 minutes for CB and EC. 0.02 minutes for MC.	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.01 to 91.02 for CB. 0.0 to 91.01 for EC. 0.02 to 91.0 for MC. Value less than 0.02 turns T1 off.		

NOTE

While the T1 value can be as high 91.02 minutes, it is normally used at much smaller values. The expression used to calculate T1 in the controller produces very "granular" results near the upper end of the value range. When such values are used, displayed values may vary significantly from entered values.

T2 (CB/EC/MC)

T2 — T2 (Derivative Time; CB, EC, & MC Regulatory Data Points)

This parameter defines the derivative time for a regulatory data point that has been configured for a PID Algorithm. If the derivative time is specified as being less than 0.01 minutes (for the CB or EC) or 0.02 minutes (for the MC), the derivative action is deleted.

Source:	User	Default Value:	0 minutes for CB and EC. 0.03 minutes for MC.	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0 to 97.04 minutes for CB. 0.0 to 97.44 minutes for EC. 0.02 to 99.13 minutes for MC.		

T2 — T2 (Lead Time Constant; CB, EC, & MC Regulatory Data Points)

This parameter defines the lead time constant for a regulatory data point that has been configured for the **Leadlag**, **Leadlags**, or **Leadlagm** algorithms.

Source:	User	Default Value:	0.01 minutes for CB. 0.0 minutes for EC. 0.03 minutes for MC.	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.01 to 99.99 for CB. 0.0 to 99.99 for EC. 0.03 to 99.99 for MC.		

T3 — T3 (Second Lag Time Constant; EC Regulatory Data Point)

This parameter defines the second lag time constant for a regulatory data point that has been configured for the **Leadlagm** or **Leadlags** algorithm.

Source:	User	Default Value:	0.0 minutes	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0 to 91.01 minutes		

T5 — T5 (Transient Time; EC Regulatory Data Point)

This parameter defines the transient time in seconds for an EC regulatory data point that has been configured for the **Leadlagm** or **Leadlags** algorithm.

Source:	User	Default Value:	0.0 seconds	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0 to 272.9 seconds		

TD — Filter Constant (CB, EC, & MC Regulatory Data Points)

This parameter defines the filter time constant for a regulatory data point. If this parameter is set to 0, the digital-filter action is deleted.

Source:	User	Default Value:	0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	0.0 to 91.02 minutes for CB and EC. 0.02 to 91.0 minutes for MC.		

TFCHLM — Transmitter High Limit (EC Regulatory Data Point) TFCLLM — Transmitter Low Limit

These parameters define the failure limits in percent for the transmitter that is used for the PV.

Source:	User	Default Value:	0.0% for high and low	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-6.86% to 106.86%		

TIMEBASE — Time Base (MC Timer Data Point)

This parameter defines whether the timer data point is to be incremented each second or each minute.

Source:	User	Default Value:	Seconds	Access Lock:	Engr.
Value Type:	Enum. of TIMEBASE	Value Range:	Seconds or minutes		

NOTE

To update the value in this parameter, STATE must contain Stopped.

TIMELEFT — Time Remaining (MC Timer Data Point)

This parameter indicates the amount of time remaining for a timer data point as shown on a group or detail display.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	0.0 to 999.0 seconds or minutes, as configured.		

TIMOUTFL — Time Out Flag (MC Timer Data Point)

This parameter indicates when a timer has expired for a timer data point.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of TMAS	Value Range:	Normal = Timer is still running. Timeup = Timer has expired.		

NOTE

The CL/MC view of this parameter is Off/On. Refer to *CL/MC Reference Manual*, Table 4-9 plus notes.

TM-- — General Note (For Parameters Beginning with the Letters TM)

For all parameters beginning with the letters TM, a parameter index number must be specified to indicate to the HG which of the 32 timers is to be accessed. If the parameter index number exceeds 32, the values for all 32 timers are returned.

When MC Timer SP = 0, the timer becomes a “free running” timer which never times up. The PV goes from 0 to 9999 and then “rolls over” to 0 and counts up again.

TMAS — Timer Alarm State Array (MC Box Data Point)

This parameter indicates which timers on the timer display of the Process Module Detail Display have expired. Refer to the general note listed under the TM-- entry in this publication.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of TMAS	Value Range:	Timeup = EXP is displayed next to the elapsed time to indicate that the timer has expired. Normal = Timer is still running; blanks are displayed next to the elapsed time.		

TMPV — Timer PV Value Array (MC Box Data Point)

This parameter provides the user with the elapsed time of one or all timers on the timer display of a Process Module Detail Display. Refer to the general note listed under the TM-- entry in this publication.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	0.0 to 9999.0 seconds or minutes, as configured.		

TMRS — Timer Reset Array (MC Box Data Point)

This parameter allows the user to reset a timer (using the RESET TIMER target) on the timer display of a Process Module Detail Display. Refer to the general note listed under the TM-- entry in this publication.

Source:	System	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Enum. of RST	Value Range:	Reststrt = Timer is currently running. Reset = Timer has been reset using the RESET TIMER target; timer is stopped.		

TMRV — Timer Remaining Time Array (MC Box Data Point)

This parameter is an internal parameter and it is the difference between the set time and the elapsed time for one timer or all the timers as shown on the timer display of a Process Module Detail Display. Refer to the general note listed under the TM-- entry in this publication.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	0.0 to 9999.0 seconds or minutes, as configured.		

TMSP — Timer Preset Array (MC Box Data Point)

This parameter contains the preset time for one timer or for all timers as shown on the timer display of a Process Module Detail Display. Refer to the general note listed under the TM-- entry in this publication.

Source:	System	Default Value:	0.0	Access Lock:	Oper.
Value Type:	Real	Value Range:	0.0 to 9999.0 seconds or minutes, as configured.		

NOTE

To update the value in this parameter, the TMST value for this timer must be Stopped.

TMST — Timer State Array (MC Box Data Point)

This parameter indicates the current state of one timer or of all the timers as shown on the timer display of the Process Module Detail Display. Refer to the general note listed under the TM-- entry in this publication.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Enum. of STATE	Value Range:	Reset = Operator has reset the timer using the RESET TIMER target. Stopped = Operator has started the timer using the STOP TIMER target. Running = Timer is running. Complete = Timer has expired. Reserved = Reserved for future use.		

NOTE

Reset, Complete and Reserved are not returned by the HG.

TMTB — Timer Timebase Array (MC Box Data Point)

This parameter specifies the timebase for one timer or for all timers as shown on the timer display of a Process Module Detail Display. Refer to the general note listed under the TM-- entry in this publication.

Source:	System	Default Value:	N/A	Access Lock:	Engr.
Value Type:	Enum. of TIMEBASE	Value Range:	No = Timer data point has not been configured. Seconds = Timebase is in seconds Minutes = Timebase is in minutes		

NOTE

To update the value in this parameter, the TMST value for this timer must be Stopped.

TOGINTSL — Slot 1-n TOG Interval (Box Data Point)

This parameter defines which time-out interval (BOXTOG1 (TOG Interval 1) or BOXTOG2 (TOG Interval 2)) parameter is to be used for the specified slot in this box.

Source:	User	Default Value:	Interv1	Access Lock:	Engr.
Value Type:	Enum. of SLOTTOGS	Value Range:	Interv1 = Time-out interval specified by <u>BOXTOG1</u> parameter is to be used for this slot. Interv2 = Time-out interval specified by <u>BOXTOG2</u> parameter is to be used for this slot.		

TRACKING — Output Tracking (EC Regulatory Data Point)

This parameter applies to only the **Pidexfb** algorithm. It indicates whether the Z input is tracking the output value (OP).

Source:	System	Default Value:	N/A	Access Lock:	View
Value Type:	Enum. of TRACKING	Value Range:	Notrack = The M input is not configured or is "off," and the output is not tracked by the <u>Z</u> input. Track = The M input is "on" and the output is tracked by the <u>Z</u> input.		

-U-

UBOXCLR — Upper-Box Color

This parameter defines the color of the upper (STATE2) box that is used to show the status of the input(s) at a digital input, digital composite, flag, or logic block data point.

Source:	User	Default Value:	Red	Access Lock:	DEB
Value Type:	Enum. of BOXCOLOR	Value Range:	Red Cyan Green Yellow White Black Blue		

UDSIPRAT (I)— User Display Input Rate (CIU Processor Status Data Points)

This parameter defines the "User Display Input Rate." The User Display Input Rate is the number of data transfers from the US to the CIU during the most recent, full 1-minute sample period, where:

- N = 1, number of transfers in words/minute;
- N = 2, number of transfers in blocks/minute;
- N = 3, number of transfers in messages /minute; and
- N = 4 to 10, reserved for use by Honeywell IAC.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	ARRAY [1..10] of REAL	Value Range:	≥ 0.0		

UDSOPRAT (I)— User Display Output Rate (CIU Processor Status Data Points)

This parameter defines the "User Display Output Rate." The User Display Output Rate is the number of data transfers from the CIU to the US during the most recent, full 1-minute sample period, where:

- N = 1, number of transfers in words/minute;
- N = 2, number of transfers in blocks/minute;
- N = 3, number of transfers in messages /minute; and
- N = 4 to 10, reserved for use by Honeywell IAC.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	ARRAY [1..10] of REAL	Value Range:	≥ 0.0		

UNIT — Unit ID

This parameter defines the unit to which this data point is assigned. The unit identifier appears on the Detail Display for this data point as shown in Figure N-1, and in other displays and listings throughout the system. In the example shown in Figure N-1, the unit identifier is XX. (The unit identifiers are originally defined during network configuration.)

Source:	User	Default Value:	All Blanks	Access Lock:	DEB
Value Type:	String	Value Range:	The unit identifier can consist of one or two characters and the valid character set is as follows: Alphabetics A through Z (upper-case only). Numerics 0 through 9 (an all numeric identifier is permissible). Underscores are not permitted. No leading blanks (spaces). A single character with a trailing space is not permitted. Restriction: After a unit identifier has been defined, it should be written exactly the same way each time when the same identifier is to be used. As an example, if the unit identifier was defined as 03, it should be written as 03 and not as 3 for each usage of this unit identifier.		

NOTE

A Real number is returned for this parameter by the Picture Editor and by CL. This number is equivalent to the ordinal number in the enumeration list of units.

UNREASPR — Unreasonable Alarm Priority

This parameter defines the priority of an unreasonable alarm generated by a data point.

With R670, the default value is changed to NOACTION.

Source:	User	Default Value:	NOACTION	Access Lock:	Engr.
Value Type:	Enum. of ALPRIOR				
Value Range:	NOACTION =	Alarm is not reported to the system. Alarm is not annunciated.			
	JOURNAL =	Alarm is historized but is not annunciated. Alarm indicator does appear on Group and Detail displays.			
	LOW =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	HIGH =	Alarm is historized, annunciated, and does appear on Group and Detail displays.			
	EMERGNCY =	This alarm has emergency (first) priority, and is historized and reported to USs (this value not allowed for counter points). The Alarm is annunciated. There is no restriction on capacity.			
	PRINTER =	Alarm is reported to the US RTJ. Alarm is not historized or annunciated, and does appear on Group and Detail displays.			
NOTE					
		If PRINTER is chosen, the alarm is lost if the printer is not working.			
	JNLPRINT =	Alarm is historized and reported to the US RTJ. Alarm is not annunciated but appears on Group and Detail displays.			

-W-**WRITASC — Write ASCII Text**

This parameter is used to directly write 60 ASCII characters (30) words into the database of an HLPIU. When WRITASC is specified, an index must be specified along with the 60 characters. This index is then used as an offset into the HLPIU database.

The index can have a value from 0 to 994. Because the total number of words in the HLPIU database is 1023 and WRITASC is used to write 30 words, 994 is the highest index allowed where 30 words still follow in the HLPIU database.

To read the 60 ASCII characters that have been written into the database, use the READASC parameter.

WARNING

Because the first database word for each slot in the HLPIU stores the card type and card option for that slot (indexes 0, 64, 128, 192, . . . 960), the slot pointed to by the index must be unconfigured (PIUCRDY must contain Notconfg), to avoid overwriting valid information. When BOXSTATE for the HLPIU contains Run, the box prohibits overwriting this data, but when BOXSTATE contains Reset, the data can be inadvertently overwritten.

The relationship between the indexes and the slots is as follows:

<u>Index</u>	<u>Slot Number</u>	<u>Index</u>	<u>Slot Number</u>
0 to 63	1	512 to 575	9
64 to 127	2	576 to 639	10
128 to 191	3	640 to 703	11
192 to 255	4	704 to 767	12
256 to 319	5	768 to 831	13
320 to 383	6	832 to 895	14
384 to 447	7	896 to 959	15
448 to 511	8	960 to 1023	16

Source:	User	Default Value:	N/A	Access Lock:	Oper.
Value Type:	String	Value Range:	ASCII character set		

WRITINT

WRITINT — Write Integer

This parameter is used to directly write one word in integer form into the database of an HLPIU. When WRITINT is specified, an index must be specified along with the integer value. This index is then used as an offset into the HLPIU database.

The index can have a value from 0 to 1023, because this is the total number of words in the HLPIU database.

To read the integer that the user has written into the database using the WRITINT parameter, use the READINT parameter.

WARNING

Because the first database word for each slot in the HLPIU stores the card type and card option for that slot (indexes 0, 64, 128, 192, . . . 960), the slot pointed to by the index must be unconfigured (PIUCRDY must contain Notconfg), to avoid overwriting valid information. When BOXSTATE for the HLPIU contains Run, the box prohibits overwriting this data, but when BOXSTATE contains Reset, the data can be inadvertently overwritten.

The relationship between the indexes and the slots are as follows:

<u>Index</u>	<u>Slot Number</u>	<u>Index</u>	<u>Slot Number</u>
0 to 63	1	512 to 575	9
64 to 127	2	576 to 639	10
128 to 191	3	640 to 703	11
192 to 255	4	704 to 767	12
256 to 319	5	768 to 831	13
320 to 383	6	832 to 895	14
384 to 447	7	896 to 959	15
448 to 511	8	960 to 1023	16

Source:	User	Default Value:	N/A	Access Lock:	Oper.
Value Type:	Integer	Value Range:	-32766 to 32767		

WRITREAL

WRITREAL — Write Real Value

This parameter is used to directly write a floating-point value (2 words) into the database of an HLPIU. When WRITREAL is specified, an index must be specified along with the real value. This index is then used as an offset into the HLPIU database.

The index can have a value from 0 to 1022. Because the total number of words in the HLPIU database is 1022 and WRITREAL writes two words, 1022 is the highest index allowed where two words still follow in the database.

To read the real value that the user has written into the database using the WRITREAL parameter, use the READREAL parameter.

WARNING

Because the first data-base word for each slot in the HLPIU stores the card type and card option for that slot (indexes 0, 64, 128, 192, . . . 960), the slot pointed to by the index must be unconfigured (PIUCRDY must contain Notconfg), to avoid overwriting valid information. When BOXSTATE for the HLPIU contains Run, the box prohibits overwriting this data, but when BOXSTATE contains Reset, the data can be inadvertently overwritten.

The relationship between the indexes and the slots is as follows:

<u>Index</u>	<u>Slot Number</u>	<u>Index</u>	<u>Slot Number</u>
0 to 63	1	512 to 575	9
64 to 127	2	576 to 639	10
128 to 191	3	640 to 703	11
192 to 255	4	704 to 767	12
256 to 319	5	768 to 831	13
320 to 383	6	832 to 895	14
384 to 447	7	896 to 959	15
448 to 511	8	960 to 1023	16

Source:	User	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	There is no specific range for the real value		

-X-

X1-X7 — Coordinates for Charact Algorithm (EC Regulatory Data Point)

These parameters define the X coordinates that correspond to the OUTPT1-OUTPT7 coordinates for the **Charact** algorithm.

Source:	User	Default Value:	0.0	Access Lock:	Supvr.
Value Type:	Real	Value Range:	-6.86% to 106.86%		

XC — Configured X Input (Lead/Lag Algorithms)

This parameter is the current value of the X (PV) input at a regulatory data point that has been configured for a **Leadlag** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	-6.86% to 106.86%		

XDIGST — X-Input Digital State (EC Regulatory Data Point)

This parameter provides the current digital state of the X input for an EC regulatory data point that has been configured for the **Logic** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Input is true. False = Input is false.		

XMITRAT (N) — # of Data Transfers from CIU to ULP (CIU Processor Status Data Points)

This parameter defines the number of data transfers from the CIU to the ULP during the most recent, full 1-minute sampling period, where

- N = 1, number of transfers in words/minute;
- N = 2, number of transfers in blocks/minute;
- N = 3, number of transfers in messages /minute; and
- N = 4 to 10, reserved for use by Honeywell IAC.

Source:	System	Default Value:	0.0	Access Lock:	Read Only
Value Type:	ARRAY [1..10] of REAL	Value Range:	≥ 0.0		

Y

-Y-

Y — Y Input (Regulatory Data Points)

This parameter provides the current value of the Y input in engineering units.

Source:	System	Default	N/A	Access	Read Only	
		Value:		Lock:		
Value	Real	Value	-6.86% to 106.86% percent of scale in engineering units.			
Type:		Range:				

YDIGST — Y-Input Digital State (EC Regulatory Data Point)

This parameter provides the current digital state of the Y input for an EC regulatory data point that has been configured for the **Logic** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Input is true. False = Input is false.		

-Z-

Z — Z Input (EC Regulatory Data Point)

This parameter provides the current value of the Z input.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Real	Value Range:	For Midof3 algorithm: -6.86 to 106.86 percent of scale in engineering units. For all other applicable algorithms: -6.86% to 106.86% Note: NaN does not apply to this parameter. If you pull the wire, the value of Z goes to -6.9%		

ZDIGST — Z-Input Digital State (EC Regulatory Data Point)

This parameter provides the current digital state of the Z input for an EC regulatory data point that has been configured for the **Logic** algorithm.

Source:	System	Default Value:	N/A	Access Lock:	Read Only
Value Type:	Boolean	Value Range:	True = Input is true. False = Input is false.		

ZEUHI — Z Input (100% of Range) (EC Regulatory Data Point)**ZEULO — Z Input (0% of Range)**

These parameters indicate the engineering unit range (high and low) that is used for the Z input.

Source:	System	Default Value:	For ZEUI: 100 For ZEULO: 0.0	Access Lock:	Read Only
Value Type:	Real	Value Range:	For Midof3 algorithm: -9999 to 9999, depending on definition of of <u>PVEUHI</u> and <u>PVEULO</u>; they are forced to be the same. For all other applicable algorithms: <u>ZEUI</u> and <u>ZEULO</u> are predefined to be 100.0 and 0.0, respectively and are Read Only parameters.		

ZSIGNAL — Z Signal Type (EC Regulatory Data Point)

This parameter defines which of the signals at the data point (specified by ZSLTSRC) is to be applied to the Z input of this regulatory data point.

Source:	User	Default Value:	Notconfig	Access Lock:	Engr.
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Value Type: Enum. of SIGNAL

Value Range:

- Output = Z input to this data point is the output from the data point specified by ZSLTSRC.
- Pv = Z input to this data point is the PV at the data point specified by ZSLTSRC.
- Hwy = Z input to this data point is the output from a data point that resides in a TPS module as specified by ZSLTSRC.
- Lsp = Z input to this data point is the local setpoint at the data point specified by ZSLTSRC.
- S1 = Z input to this data point is the S1-status output from the data point specified by ZSLTSRC.
- S2 = Z input to this data point is the S2-status output from the data point specified by ZSLTSRC.
- S3 = Z input to this data point is the S3-status output from the data point specified by ZSLTSRC.
- S4 = Z input to this data point is the S4-status output from the data point specified by ZSLTSRC.
- Si = Z input to this data point is the SI (hardware input) at the data point specified by ZSLTSRC.
- S5 = Z input to this data point is the S5-status output from the data point specified by ZSLTSRC.
- Notconfig = Z input is not configured for this data point.

NOTE

To update the value in this parameter, MODE must contain Man, INITCONF must contain Noinit, and EMS must contain Disable.

ZSLTSRC — Z Input Source (EC Regulatory Data Point)

This parameter defines the tag name of the data point that provides the input signal to the Z input of an EC regulatory data point. The type of signal that is supplied is determined by the ZSIGNAL (Z Signal Type) parameter. Refer to NAME (Tag Name) for the valid character set.

Default

Value: Tag Name from Line 1 of Configuration Form.

NOTE

To update the value in this parameter, MODE must contain Man, INITCONF must contain Noinit, EMS must contain Disable, the Tag Name entered in this parameter must be that of a point in the same HG.

ZZTEXT — Non-Comp State 00 Text

This parameter defines the descriptors that are displayed when both inputs to a digital data point are noncomplementary. Text1 and Text2 are used to reference the descriptors that are entered in parameters NCMPTXT1 and NCMPTXT2 on the HG-Hiway Point Assignment display. This parameter is valid only for a digital data point that has two inputs.

Source:	User	Default Value:	Text1	Access Lock:	DEB
Value Type:	Enum. of ZZTEXT	Value Range:	Text1 = The Text1 descriptor is displayed when both inputs to this data point are false (00). The Text2 descriptor is displayed when both inputs to this data point are true (11). Text2 = The Text2 descriptor is displayed when both inputs to this data point are false (00). The Text1 descriptor is displayed when both inputs to this data point are true (11).		

NOTE

1. When neither the upper box or the lower box is lit for this data point on a Universal Station display, then the Text1 or Text2 descriptor that has been selected for this parameter is displayed.
2. When both the upper and lower boxes are lit on the display, then the Text1 or Text2 descriptor that is not selected as ZZTEXT is displayed.

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